

Block

1

OPTIMISATION TECHNIQUES-I

UNIT 1

Introduction to Operations Research	7
--	----------

UNIT 2

Linear Programming Problems	21
------------------------------------	-----------

UNIT 3

Simplex Method	41
-----------------------	-----------

UNIT 4

Transportation Problem	65
-------------------------------	-----------

Curriculum and Course Design Committee

Prof. K. R. Srivathsan
Former Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Parvin Sinclair
Former Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Geeta Kaicker
Former Director, School of Sciences
IGNOU, New Delhi

Prof. Jagdish Prasad
Department of Statistics
University of Rajasthan, Jaipur

Prof. R. M. Pandey
Department of Bio-Statistics
AIIMS, New Delhi

Dr. Manish Trivedi
School of Sciences, IGNOU

Dr. Nitin Gupta
School of Sciences, IGNOU

Dr. Neha Garg
School of Sciences, IGNOU

Mr. Rajesh Kaliraman
School of Sciences, IGNOU

Prof. Rahul Roy
Theoretical Statistics and Mathematics Unit
Indian Statistical Institute, New Delhi

Prof. Diwakar Shukla
Department of Mathematics and Statistics
Dr. Hari Singh Gaur University, Sagar

Prof. Rakesh Srivastava
Department of Statistics
M.S. University of Baroda, Vadodara

Prof. G. N. Singh
Department of Applied Mathematics
I.S.M., Dhanbad

Prof. Gulshan Lal Taneja
Department of Mathematics
M.D. University, Rohtak

Prof. Poornima Mital
School of Sciences, IGNOU

Prof. Sujatha Varma
School of Sciences, IGNOU

Dr. S. Venkataraman
School of Sciences, IGNOU

Dr. Deepika
School of Sciences, IGNOU

Block Preparation Team

Prof. Jayant Dubey (**Editor**)
Department of Business Studies
BTIRT, Sironja, Sagar (MP)

Prof. Vijayshri (**Language Editor**)
School of Sciences, IGNOU

Prof. Gulshan Lal Taneja (**Units 1 to 4**)
Department of Mathematics,
M. D. University, Rohtak

Formatted By
Dr. Manish Trivedi
School of Sciences, IGNOU

Programme and Course Coordinator: Dr. Manish Trivedi

Block Production

Mr. Sunil Kumar, AR (P)
School of Sciences, IGNOU

September, 2014

© Indira Gandhi National Open University, 2014

ISBN-978-81-266-6847-2

All rights reserved. No part of this work may be reproduced in any form, by mimeograph or any other means, without permission in writing from the Indira Gandhi National Open University

Further information on the Indira Gandhi National Open University may be obtained from the University's Office at Maidan Garhi, New Delhi-110068 or visit University's website <http://www.ignou.ac.in>

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by the Director, School of Sciences.

Laser Typeset by: Tessa Media & Computers, C-206, A.F.E.-II, Okhla, New Delhi

Printed at:

INDUSTRIAL STATISTICS-II

The elective MSTE-002 entitled **Industrial Statistics-II** is the second course for the specialisation in Industrial Statistics in the PG Diploma programme in Applied Statistics. This course has been designed to develop skills in applying the techniques and tools of statistics to industrial and business data. It deals with the concepts and techniques of Operations Research, Regression Modelling and Time Series Modelling. The purpose of this course is to provide you with a meaningful introduction to the concepts and applications of statistical techniques. Such statistical techniques and tools are sought in diverse fields including engineering, operations research, management science, computer science, systems analysis, business and economics. This course is designed to equip you with the ability to apply the techniques to both practical and theoretical problems. This knowledge will also enable you to study advanced techniques in this area. This course consists of four blocks.

In Blocks 1 and 2 of this course, you would be studying the concepts and techniques of Operations Research. In Block 1, we discuss the basics of Linear Programming Problem and the Simplex method of solving it. The Transportation Problem is also discussed in Block 1. In Block 2, we discuss various methods for solving the Assignment problems, Queueing problems, Sequential problems and Inventory problems.

In Block 3 of the course you shall study the concepts of Regression modelling. Regression analysis is a statistical tool for investigating and analysing the average relationship between two or more variables. Block 4 of this course describes the concepts of Time series analysis and forecasting techniques. The techniques discussed in Blocks 3 and 4 are useful in decision making and planning for future by governments, institutions, industries and even individuals.

BLOCK 1 OPTIMISATION TECHNIQUES-I

In many practical situations, we need to assist in solving strategic and tactical problems, i.e., to discuss, evolve and suggest ways and means to improve the execution of managerial and technological policy of any government, institution, industry or organisation. This new approach to systematic and scientific study of the operations of the system is called Operations Research.

Operations Research employs scientific methods for the purpose of solving problems and determining the best utilisation of limited resources. In this Block, we discuss these scientific methods, techniques and tools and their application to problems involving the operations of a system so as to provide those in control of the system with optimum solutions to the problem.

The Block is divided into four units.

Unit 1 entitled **Introduction to Operations Research** highlights the nature, scope, applications, uses and limitations of Operations Research. It also provides a broad classification of different O.R. models. We also discuss the concept of convex set and basic feasible solution of an LPP.

In Unit 2 entitled **Linear Programming Problems**, we discuss the linear programming problems and explain how these are formulated mathematically. The objective function is defined and graphical method of obtaining optimum value is also explained in this unit.

However, linear programming problems involving more than two decision variables cannot be solved by graphical method. Therefore, in Unit 3 entitled **Simplex Method**, we first discuss the trial and error method and then the Simplex method of solving the linear programming problems. We also explain the artificial variable technique and Big M method of solving the LPP.

Transportation problem is one of the special cases of LPP. Unit 4 entitled **Transportation Problem** describes different methods for obtaining the initial basic feasible solution and optimum solution of the transportation problem. In this unit, we explain various methods of solving the transportation problems with special cases such as the unbalanced transportation problem, degeneracy, alternative solutions, maximisation transportation problem and problems with prohibited routes.

Notations and Symbols

Z	:	Objective function to be maximised or minimised
X_i	:	i^{th} basic variable
C_i	:	Cost coefficient of i^{th} basic variable X_i
(a_{ij})	:	A $m \times n$ real matrix of $m \times n$ constants
$\{b_i\}$	:	A set of constants
A_i	:	i^{th} Artificial Variable
s_i	:	i^{th} slack/surplus variable
Max.	:	Maximise
LPP	:	Linear Programming Problem
$c_j - z_j$	:	Net evaluation
RR	:	Replacement Ratio
Qty	:	Quantity
a_{ij}	:	Key element/pivot element
M	:	An arbitrarily large number
D_j	:	j^{th} destination
C_{ij}	:	Cost of transporting one unit from origin i to destination j
O_i	:	i^{th} origin
a_i	:	Capacity/availability of items at origin i
b_j	:	Requirement/demand of the destination j
x_{ij}	:	Number of units transported from origin i to destination j
Sec.	:	Section/Subsection
Secs.	:	Sections/Subsections
Fig./Figs.	:	Figure/Figures

UNIT 1 INTRODUCTION TO OPERATIONS RESEARCH

Structure

- 1.1 Introduction
 - Objectives
- 1.2 Origin and Definition of Operations Research
- 1.3 Scope of Operations Research
- 1.4 Models in Operations Research
- 1.5 Uses and Limitations of Operations Research
- 1.6 Convex Set
- 1.7 Basic Feasible Solutions of Linear Programming Problems
- 1.8 Summary
- 1.9 Solutions/Answers

1.1 INTRODUCTION

Optimisation is the act of obtaining the best result under given circumstances. We come across many practical situations in which optimisation is observed. For example, suppose we are dealing with problems involving costs and want to minimise a cost. Then the least value of the cost is nothing but the optimised value. Similarly, maximisation of profit is the optimisation of profit. Further, suppose that different jobs are to be assigned to different persons and only one job is to be given to one person. We know that every person will complete each job in different times. Then the optimal solution is to assign each job to a person in such a way that the total time taken in completing all the jobs is the least.

We observe many other situations of optimisation around us. In design, construction and maintenance of any system, decision makers have to take many technological and managerial decisions at several stages. The ultimate goal of all such decisions is to either minimise the effort required or to maximise the desired profit/revenue. Optimisation can be defined as the process of finding the conditions that give the maximum or minimum value of a function. **Operations Research** is concerned with the application of scientific methods and techniques to decision making problems and with establishing the optimal solutions.

In this unit, we discuss the scope, applications, uses and limitations of Operations Research (O.R.). You will learn about various models in O.R. We also discuss the concept of convex set and basic feasible solutions. In the next unit, we deal with linear programming problems.

Objectives

After studying this unit, you should be able to:

- define Operations Research;
- describe the scope, uses and limitations of Operations Research;
- explain the models in Operations Research;
- define convex set; and
- determine basic feasible solutions of Linear Programming Problems.

1.2 ORIGIN AND DEFINITION OF OPERATIONS RESEARCH

You will agree that human beings have the tendency of getting maximum gains with minimum effort. The complexities in every aspect of life are increasing by the day. As a result, we would like to create/use scientific methods and techniques that help us attain optimum solutions in decision making problems. Operation Research (O.R.) is concerned with the applications of such scientific methods and techniques.

‘Operation’ implies some action applied in any area of interest. ‘Research’ means some organised process of getting and analysing information about the problem environment. The term Operations Research came into existence and gained prominence during the Second World War when military planners were faced with logistical tasks requiring prompt and effective solutions. Hence, a group of scientists with diverse educational backgrounds including mathematics, statistics and physics became involved in applying a scientific approach to deal with strategic and tactical problems of various military operations. This initial research on military operations soon found applications in other decision making problems in business and industry. Hence, the word ‘military’ was dropped and it was named as ‘Operations Research’. In India, Operations Research came into existence in 1949 with the opening of an Operations Research unit at the Regional Research Laboratory in Hyderabad and also in the Defence Science Laboratory.

Operations Research gained momentum in the market place in the nineteen fifties and sixties. Solutions to problems involving manufacturing processes, personnel and material planning, inventory, scheduling and others were sought through O.R. Though Operations Research had its roots in statistics, it has now acquired its own identity. Operations Research is used to solve the real world problems that involve a lot of mathematics, statistics, computer science, accounting, etc.

Operations Research has been defined in different ways by different scientists from time to time, but none of those definitions has been accepted by all. The more general and comprehensive definition of Operations Research is as follows:

“Operations Research is a branch of science which is concerned with the application of scientific methods and techniques to decision making problems and with establishing the optimal solutions.”

1.3 SCOPE OF OPERATIONS RESEARCH

O.R. has a wide scope in everyday life as it provides better solutions to various decision-making problems with great speed and competence. It finds applications in a wide range of areas including defence operations, planning, agriculture, industry (finance, marketing, personal management, production management), research and development. We now describe the applications briefly.

In Defence Operations

Since the Second World War, Operations Research techniques have been used for defence operations with the objective of getting maximum gains with minimum effort. It has been used for coordinating various activities of

Air Force, Army and Navy. Decisions regarding formulation and selection of strategies of the various available courses of action are taken by a team of scientists.

In Planning for Economic Development

Careful planning is necessary for economic development of any country. Operations Research is used to frame future economic and social policies.

In Agriculture

Agricultural output needs to be increased due to increasing needs for adequate quantity and quality of food for our increasing population. But there are a number of restrictions under which agricultural production is studied. Problems of agricultural production under various restrictions such as optimum allocation of land to various crops in accordance with the climatic conditions, optimum distribution of water from various resources for irrigation purposes can easily be solved by application of Operations Research techniques.

In Industry

Now-a-days, due to complexities of operations and huge sizes of industries, important decisions regarding various sections of the organisation, e.g., planning, procurement, marketing, finance, etc. have to be taken division wise. For example, the production department needs to minimise the cost of production, but maximise output; the finance department needs to optimise capital investment; the personnel department needs to appoint competent work force at minimum cost. Each department has to plan its own objectives which may be in conflict with the objectives of other departments and may not conform to the overall objectives of the organisation. For example, the sales department of an organisation may want to keep sufficient stocks in the inventory, whereas the finance department may want to have minimum investment. In that case, both departments would be in conflict with each other. The applications of O.R. techniques to such situations help in overcoming this difficulty by evolving an optimal strategy and serving efficiently the interest of the organisation as a whole.

Some of the problems faced by various divisions of an industry, which can be solved by the application of Operations Research techniques are as follows:

Finance department of an organisation needs to optimise capital investment, determine optimal replacement strategies, apply cash flow analysis for long range capital investments, formulate credit policies, credit risks, breakeven analysis. All these can be attained by applying Operations Research techniques.

The marketing department of any organisation has to face various problems like product selection, formulation of competitive strategies, sales forecasting, distribution strategies, selection of advertising media with respect to cost and time, finding the optimal number of salesmen, finding optimum time to launch a product. All such problems can be overcome using Operations Research methods.

Personnel Management of an organisation needs to find the best combination of workers in different categories with respect to costs, skills, age and nature of jobs. It also needs to forecast the work force requirement, frame recruitment policies, assign jobs to machines or workers, negotiate in

a bargaining situation, etc. This can be achieved very easily by the application of Operations Research techniques.

Problems related to **production management** of an organisation, i.e., determination of the optimal product mix, selection of the location and design of the sites for production plant, scheduling and sequencing the production run by proper allocation of machines, location and size of warehouse/new plant, etc. can very easily be solved by applying Operations Research techniques.

In Research and Development

Operations Research helps in planning and control of new research and development projects. It also helps in planning the launch of new products. Operations Research helps in solving many other problems faced by public as well as private sectors such as the ones in economic and social planning, management of natural resources, energy, housing, pollution control, waiting lines and administrative problems, insurance policies, and many more.

1.4 MODELS IN OPERATIONS RESEARCH

A model in Operations Research is a mathematical or theoretical description of the various variables of a system representing the basic aspects or the most important features of a typical problem under investigation. The objective of the model is to identify the significant factors and interrelationships. It helps in deciding how the changes in one or more variables of a model may affect other variables or the system as a whole. Operations Research models are broadly classified as follows:

- i) Mathematical and descriptive models, and
- ii) Static and dynamic models

In **mathematical models**, various variables/parameters explaining different operations of a system are expressed in mathematical terms and the relations are explained by means of mathematical equations or inequalities. The variables can be exact (deterministic) or probabilistic. In deterministic models, the variables and their relationships are stated exactly. The probabilistic models are developed for problems involving risk and uncertainty. Therefore, in these models, decision variables take the form of probability distributions. For example, the variables in linear programming, transportation and assignment problems, which you will study in Units 3 and 4 of this Block and Unit 5 of Block 2 are deterministic. The models discussed in Unit 6 of Block 2 of this course are probabilistic models. The models in which various operations are explained in non-mathematical language are called **descriptive models** and serve as preliminary models for the development of mathematical models.

A probabilistic mathematical model is called **static** or **dynamic** according as the distribution of parameters remains unchanged or changes with time. An inventory model is a static model wherein the re-ordering of goods is determined using average demand and average time and not on the basis of changes that take place in any particular period. However, an inventory model in which the re-order point is determined by a certain stock level at any point of time is **dynamic**. You will learn about inventory models in Unit 8 of Block 2 of this course.

Operations Research models are mainly of three kinds: **Iconic**, **Symbolic** and **Analogue**. We briefly explain them as under:

Iconic models are physical replicas of real life systems and are based on a smaller scale than the original. In many cases, they provide a pictorial presentation of various aspects of a system. Such models are designed for the purpose of understanding the behaviour of the operation of the system when conducting experiments on the real life systems is risky and/or costly affair. Flight simulators, missile firing simulators are examples of iconic models. Photographs, paintings, maps, drawings, clay, wooden or metallic models of systems are also iconic models. For example, a toy car can be considered as an iconic model of an automobile. Similarly, a blueprint representing the floor of a building is an iconic model, and the globe is an iconic model of earth.

Symbolic models reflect the structure of a system, denoting various components of a system and their interrelationships employing letters, numbers and various other types of mathematical symbols. For example, the buyers' behaviour at varying price level can be represented symbolically by the demand curve in Economics. A model for processing inventory cost in inventory problems is another example of symbolic model. Such models are very well suited for determination of various changes in the system.

Analogue models are helpful in representing all the significant properties of a system which are not represented by iconic or symbolic models. They are also physical, but not the exact replica of the system. They are used to explain the system by analogy. For example, the geological structure of earth cannot be represented by the globe, an iconic model of earth, if different colours are not used. If different colours are taken on the globe to represent the geological structure of the earth, it is known as Analogue model.

There is no unique method for solving all mathematical models. The nature of the method of solution depends on the type and complexity of the mathematical model. In Operations Research, the solutions are generally determined by algorithms which provide fixed computational rules. These rules are applied repetitively/iteratively to the problem, with each repetition/iteration moving the solution closer to the optimum. The three main methods for the solution of a mathematical model are:

- i) Analytic or deductive methods,
- ii) Numerical or inductive methods, and
- iii) Monte Carlo techniques or simulations.

Analytic methods involve graphs and elementary differential calculus. In **numerical methods**, numerical values are substituted for various variables involved in the model by trial and error. Then the set of values, which maximises the effectiveness of the system is taken as the required solution.

Monte Carlo techniques are applied on systems, which are not represented adequately by theoretical models. In these techniques, the knowledge of the important characteristics and rules along with random sampling is used to determine the probability distributions of various components of the model.

Some mathematical models may be so complex that it is impossible to solve them by any of the available optimisation algorithms. In such cases, it may be necessary to abandon the search for the optimal solution and simply seek a better solution using heuristics. **Heuristic** models use intuitive rules or

guidelines to find the solution to any problem. These models do not guarantee an optimum solution and give solutions depending on the assumptions based on past experience. These models, however, operate faster and are very useful for solving large size problems. However, they require a good amount of creativity and experience on the part of decision makers.

1.5 USES AND LIMITATIONS OF OPERATIONS RESEARCH

There are numerous advantages of Operations Research. Some of these are:

- Operations Research develops a model, which provides logical and systematic approach for understanding, solving and controlling a problem.
- It helps in optimum use of resources. For example, linear programming techniques in Operations Research suggest most effective methods and efficient ways of optimality. It also helps in finding the limitations and scope of an activity.
- Operations Research techniques can be used for improving the quality of decisions. A decision-maker can use a well-formulated mathematical model representing a real life situation to find the changes in the variables as per requirement. Such changes can be incorporated even without disturbing the system or problem under consideration.
- It helps, in preparing future managers. Operations Research methods constitute a means for improving the knowledge and skill of young managers.
- It modifies mathematical solutions before these are applied. Managers may accept or modify the mathematical solutions obtained using Operations Research techniques.
- It helps suggest alternative solutions for the same optimum profit if the management wants so.

To sum up, Operations Research is a very powerful method of getting the best out of limited resources. However, it does have some limitations, which emerge only due to the time and cost involved.

Some limitations of Operations Research are as follows:

- Formulation of mathematical models may take into account all possible factors for defining a real life problem and hence is difficult. As a result, the help of computers is required for the large number of cumbersome computations for such problems. This discourages small companies and other organisations from using O.R. techniques.
- Some problems may involve a large number of intangible factors such as human emotions, human relationship, etc. which cannot be quantified. Hence, the best solution cannot be determined for such problems because such factors have to be excluded.
- A specialist, who may be a mathematician or a statistician, is needed to understand the formulation of models, find solutions and recommend their implementation. Managers, who deal with such problems, may not have such specialisation and hence the results may not be optimal.
- Models are abstractions of real life situations and not the reality.

- Assumptions need to be made about the nature and importance of some factors in order to construct an Operations Research model.
- A reasonably good solution without the use of Operations Research may be preferred by the management as compared to a slightly better solution provided by using Operations Research since it is very expensive in terms of time and money.

So far we have introduced you to O.R. and its scope, uses, and limitations. From the next unit onwards, we shall introduce various O.R. techniques for obtaining optimal and feasible solutions. However, before you can learn these techniques, you must study certain related basic concepts such as the concepts of convex sets and basic feasible solutions. We discuss these concepts in the next two sections.

1.6 CONVEX SET

A region or a set R is convex if and only if for any two points on the set R , the line segment connecting these points lies entirely in R . Mathematically, set R is convex if and only if

$$(x_1, y_1) \in R \text{ and } (x_2, y_2) \in R$$

$$\Rightarrow (\lambda x_1 + (1-\lambda)x_2, \lambda y_1 + (1-\lambda)y_2) \in R, \quad 0 \leq \lambda \leq 1$$

where $(\lambda x_1 + (1-\lambda)x_2, \lambda y_1 + (1-\lambda)y_2)$ gives all the points which lie on the line segment joining (x_1, y_1) and (x_2, y_2) .

For example, let us consider Fig. 1.1.

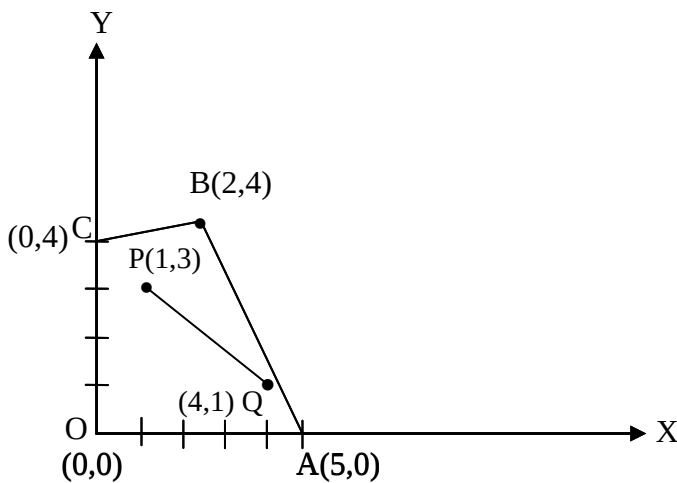


Fig. 1.1

Consider the region enclosed by OABC. Let us denote it by R . It is convex as the line segment joining any two points in this region lies wholly within it. As an example, let us take two points $P(1, 3)$ and $Q(4, 1)$. Then all points on the line segment joining P and Q are given by

$$\begin{aligned} &(\lambda(1) + (1-\lambda)4, \lambda(3) + (1-\lambda)1) \\ &= (\lambda + 4 - 4\lambda, 3\lambda + 1 - \lambda) \\ &= (4 - 3\lambda, 1 + 2\lambda), \quad 0 \leq \lambda \leq 1. \end{aligned}$$

Here $\lambda = 0$ gives the point $Q(4, 1)$ and $\lambda = 1$ gives the point $P(1, 3)$. Other points on the line segment PQ are given by

$$(4 - 3\lambda, 1 + 2\lambda), \text{ where } 0 < \lambda < 1$$

For example, let us take $\lambda = 0.1, 0.3, 0.5, 0.7, 0.9$; then the corresponding points (after substituting the values of $\lambda = 0.1, 0.3, 0.5, 0.7, 0.9$) are:

$$(4 - 3(0.1), 1 + 2(0.1)), (4 - 3 \times 0.3, 1 + 2 \times 0.3), (4 - 3 \times 0.5, 1 + 2 \times 0.5),$$

$$(4 - 3 \times 0.7, 1 + 2 \times 0.7), (4 - 3 \times 0.9, 1 + 2 \times 0.9)$$

$$\text{i.e., } (3.7, 1.2), (3.1, 1.6), (2.5, 2), (1.9, 2.4), (1.3, 2.8)$$

All these points clearly lie on the line and also in the region R.

Similarly, all other points on the line segment PQ also lie inside the region R. Hence, the line segment PQ lies in R.

Therefore, R is convex in this example. Other examples of convex sets are shown in Fig. 1.2.

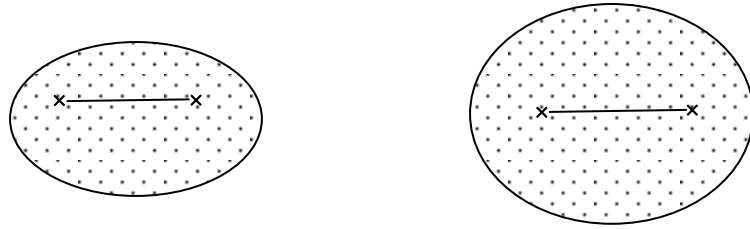


Fig. 1.2

Some examples of non-convex sets are the shaded regions in Figs. 1.3a and b.



Fig. 1.3

In each of the regions, in Figs. 1.3a and b, the line segment joining two points has been shown. Note that these do not lie wholly in the region, whereas the two points lie in the regions.

Example 1: Show that the set $S = \{(x, y) : x^2 + y^2 \leq 1\}$ is a convex set.

Solution: Let us take any two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in Fig. 1.4 such that:

$$x_1^2 + y_1^2 \leq 1, \text{ and } x_2^2 + y_2^2 \leq 1.$$

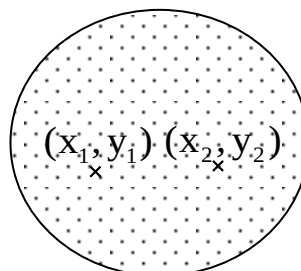


Fig. 1.4

Now, the line segment joining P and Q is the set

$$\{\lambda x_1 + (1-\lambda)x_2, \lambda y_1 + (1-\lambda)y_2\}; \quad 0 \leq \lambda \leq 1\}.$$

Let

$$u_1 = \lambda x_1 + (1-\lambda)x_2$$

$$u_2 = \lambda y_1 + (1-\lambda)y_2$$

Therefore, all points on the line segment PQ are given by (u_1, u_2) .

Now, the line segment PQ lies wholly in \$ if

$$u_1^2 + u_2^2 \leq 1$$

$$\begin{aligned} \text{Since } u_1^2 + u_2^2 &= [\lambda x_1 + (1-\lambda)x_2]^2 + [\lambda y_1 + (1-\lambda)y_2]^2 \\ &= \lambda^2 x_1^2 + (1-\lambda)^2 x_2^2 + 2\lambda(1-\lambda)x_1 x_2 + \lambda^2 y_1^2 + (1-\lambda)^2 y_2^2 + 2\lambda(1-\lambda)y_1 y_2 \\ &= \lambda^2 [x_1^2 + y_1^2] + (1-\lambda)^2 [x_2^2 + y_2^2] + 2\lambda(1-\lambda)[x_1 x_2 + y_1 y_2] \end{aligned}$$

We have

$$u_1^2 + u_2^2 \leq \lambda^2 + (1-\lambda)^2 + 2\lambda(1-\lambda)[x_1 x_2 + y_1 y_2] \quad \dots (1)$$

Now, consider

$$\begin{aligned} (x_1 x_2 + y_1 y_2)^2 &= x_1^2 x_2^2 + y_1^2 y_2^2 + 2x_1 x_2 y_1 y_2 \\ &= x_1^2 x_2^2 + y_1^2 y_2^2 + x_1^2 y_2^2 + x_2^2 y_1^2 - x_1^2 y_2^2 - x_2^2 y_1^2 + 2x_1 x_2 y_1 y_2 \\ &= (x_1^2 + y_1^2)(x_2^2 + y_2^2) - x_1 y_2 (x_1 y_2 - x_2 y_1) - x_2 y_1 (x_2 y_1 - x_1 y_2) \\ &= (x_1^2 + y_1^2)(x_2^2 + y_2^2) - (x_1 y_2 - x_2 y_1)^2 \\ &\leq (x_1^2 + y_1^2)(x_2^2 + y_2^2) \leq 1 \\ \Rightarrow (x_1 x_2 + y_1 y_2) &\leq 1 \quad \dots (2) \end{aligned}$$

$\therefore$ From (1) and (2), we have

$$u_1^2 + u_2^2 \leq \lambda^2 + (1-\lambda)^2 + 2\lambda(1-\lambda)$$

$$\text{or } u_1^2 + u_2^2 \leq [\lambda + (1-\lambda)]^2$$

$$\Rightarrow u_1^2 + u_2^2 \leq 1$$

$\therefore$ \$ is convex set.

Example 2: Show that the set \$ = $\{(x, y) : 0 \leq y \leq 5 \text{ when } 0 \leq x \leq 2 \text{ and } 3 \leq y \leq 5 \text{ when } 2 \leq x \leq 7\}$ is not a convex set.

Solution: Let us take two points P(1, 1) and Q(5, 4) in the given region \$ (see Fig. 1.5).

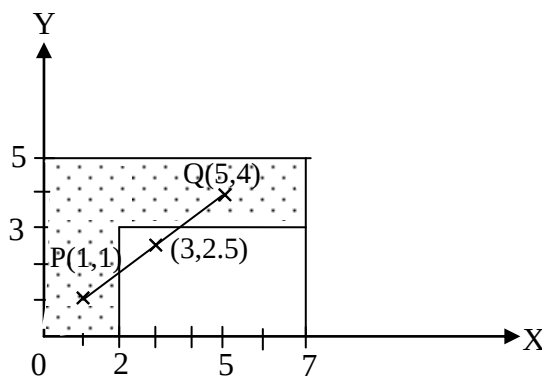


Fig. 1.5

Now, all the points on the line segment PQ are given as

$$\begin{aligned} &\{\lambda x_1 + (1-\lambda)x_2, \lambda y_1 + (1-\lambda)y_2, 0 \leq \lambda \leq 1\} \\ &= \{\lambda(1) + (1-\lambda)5, \lambda(1) + (1-\lambda)4, 0 \leq \lambda \leq 1\} \\ &= \{5-4\lambda, 4-3\lambda, 0 \leq \lambda \leq 1\} \end{aligned}$$

Let us take one of these points and put $\lambda = \frac{1}{2}$. So the point is

$$\left(5 - 4 \times \frac{1}{2}, 4 - 3 \times \frac{1}{2}\right) = \left(5 - 2, 4 - \frac{3}{2}\right) = (3, 2.5)$$

This point is on the line, but does not belong to the given set \$ since for $y = 2.5$, x should lie between 0 and 2 but here it is 3.

Therefore, the given set is not convex.

Extreme Points of a Convex Set

A point (x, y) in a convex set \$ is called an extreme point if it is not possible to locate two distinct points in or on R so that the line joining them will include (x, y) .

Mathematically, a point (x, y) is an extreme point of a convex set if it cannot be expressed as a convex combination of any two points (x_1, y_1) and (x_2, y_2) in \$, i.e., there do not exist other points $(x_1, y_1), (x_2, y_2)$ [for $(x_1, y_1) \neq (x_2, y_2)$] in the set such that

$$x = \lambda x_1 + (1-\lambda)x_2$$

$$\text{and } y = \lambda y_1 + (1-\lambda)y_2 \quad 0 < \lambda < 1.$$

For example,

- i) The vertices of the polygons, which are convex sets, are the extreme points.
- ii) Every point on the circumference of the region containing the portion in and on the circle is an extreme point.

Having explained briefly the concept of convex sets, we now discuss basic feasible solutions of linear programming problems. These will be used in Units 3 to 8.

1.7 BASIC FEASIBLE SOLUTIONS OF LINEAR PROGRAMMING PROBLEMS

We study feasible solution, basic solution, basic feasible solution of a system of equations less in number than the number of decision variables. Such solutions are required to be obtained for allocation problems which we discuss in Units 3 to 8.

Feasible Solution

Any non-negative value of the point corresponding to decision variables is a feasible solution for the system of equations representing the limited available resources for allocation problems.

Basic Solution

Suppose there are m equations representing constraints (limited available resources) containing $m + n$ variables in an allocation problem. The solution obtained by setting any n variables equal to zero and solving for the remaining m variables is called the basic solution. These m variables are the basic variables and the remaining n variables are non-basic variables.

The maximum number of possible basic solutions is given by the formula

$${}^{m+n}C_n$$

For example, if there are 2 equations in 3 variables, then the maximum number of possible basic solutions is

$${}^3C_2 = \frac{3!}{2!(3-2)!} = 3.$$

Basic Feasible Solution

A basic solution for which all the basic variables are non-negative is called the basic feasible solution.

Example 3: Determine all basic solutions and basic feasible solutions of the equations:

$$2x_1 + 6x_2 + 2x_3 + x_4 = 3$$

$$6x_1 + 4x_2 + 4x_3 + 6x_4 = 2$$

Solution: Here the maximum number of possible basic solutions is

$${}^4C_2 = \frac{|4|}{|2|2|} = \frac{4 \times 3 \times 2}{2 \times 2} = 6.$$

We obtain these as follows:

Setting $x_1 = 0, x_2 = 0$, we have $2x_3 + x_4 = 3$ and $4x_3 + 6x_4 = 2$

$$\text{P } x_3 = 2, x_4 = -1$$

Setting $x_1 = 0, x_3 = 0$, we have $6x_2 + x_4 = 3$ and $4x_2 + 6x_4 = 2$

$$\text{P } x_2 = \frac{1}{2}, x_4 = 0$$

Setting $x_1 = 0, x_4 = 0$, we have $6x_2 + 2x_3 = 3$ and $4x_2 + 4x_3 = 2$

$$\text{P } x_2 = \frac{1}{2}, x_3 = 0$$

Setting $x_2 = 0, x_3 = 0$, we have $2x_1 + x_4 = 3$ and $6x_1 + 6x_4 = 2$

$$\text{P } x_1 = \frac{8}{3} \text{ and } x_4 = -\frac{7}{3}$$

Setting $x_2 = 0, x_4 = 0$, we have $2x_1 + 2x_3 = 3$ and $6x_1 + 4x_3 = 2$

$$\text{P } x_1 = -2, x_3 = \frac{7}{2}$$

Setting $x_3 = 0, x_4 = 0$, we have $2x_1 + 6x_2 = 3$ and $6x_1 + 4x_2 = 2$

$$\text{P } x_1 = 0, x_2 = \frac{1}{2}$$

Thus, all basic solutions of the given system of equations are:

$$(0, 0, 2, -1), (0, \frac{1}{2}, 0, 0), (0, \frac{1}{2}, 0, 0), (\frac{8}{3}, 0, 0, -\frac{7}{3}), (-2, 0, \frac{7}{2}, 0), (0, \frac{1}{2}, 0, 0).$$

Here $(0, \frac{1}{2}, 0, 0)$ is repeated thrice and hence the basic solutions are

$$(0, 0, 2, -1), (0, \frac{1}{2}, 0, 0), (\frac{8}{3}, 0, 0, -\frac{7}{3}), (-2, 0, \frac{7}{2}, 0).$$

Of these solutions, the basic feasible solution is $(0, \frac{1}{2}, 0, 0)$ as in the other solutions, all decision variables are not positive.

Example 4: Find all basic solutions for the system of simultaneous equations:

$$2x_1 + 3x_2 + 4x_3 = 5 \quad \text{and} \quad 3x_1 + 4x_2 + 5x_3 = 6$$

Solution: The maximum number of possible basic solutions is

$${}^3C_2 = \frac{3!}{2!(3-2)!} = 3.$$

Now, putting $x_1 = 0$ and solving for x_2 and x_3 , we get

$$x_2 = -1, x_3 = 2$$

Again, putting $x_2 = 0$ and solving for x_1 and x_3 , we get

$$x_1 = -\frac{1}{2}, x_3 = \frac{3}{2}$$

Finally, putting $x_3 = 0$ and solve for x_1 and x_2 , we get

$$x_1 = -2, x_2 = 3.$$

This verifies that there are only three basic solutions but none of them are feasible solutions because in all solutions $(-2, 0, 3/2)$, $(-1, 0, 3/2)$ and $(-2, 3, 0)$, the values of the decision variables are not positive.

Now, you should try to solve the following exercise.

E1) Determine all the basic feasible solutions of the system of equations:

$$3x_1 + 5x_2 + x_3 = 15$$

$$5x_1 + 2x_2 + x_4 = 10$$

Match your answer to the above problem with the solution/answer given in Sec. 1.9. Let us now summarise the main points, which have been covered in this unit.

1.8 SUMMARY

1. Operations Research is a branch of science which is concerned with the application of scientific methods and techniques to decision making problems and establishing optimal solutions.
2. Operations Research has got a wide scope in everyday life as it provides better solutions to various decision-making problems with greater speed and competence. It finds applications in a wide range of areas including defence operations, planning, agriculture, industry (finance, marketing, personal management, production management), research and development.

3. A model in Operations Research is a mathematical or theoretical description of the various variables of a system representing the basic aspects or the most important features of a typical problem under investigation. The objective of the model is to identify the significant factors and their interrelationships. It helps in deciding how the changes in one or more variables of a model may affect other variables of the system as a whole. Operations Research models are broadly classified as mathematical and descriptive models, and also as static and dynamic models. The models are mainly of three kinds: Iconic, Symbolic and Analogue.
4. The three main methods for the solution of a mathematical model are: **Analytic or deductive method, numerical or inductive method and Monte Carlo techniques or simulation.**
5. Operations Research is a very powerful method of getting the best out of limited resources. However, it does have some limitations which emerge only due to time and cost involved.
6. A region or a set R is called **convex set** if and only if for any two points on the set R the line segment connecting these points lies entirely in R.
7. A point (x, y) in a convex set S is called an **extreme point** if it is not possible to locate two distinct points in or on R so that the line joining them will include (x, y).
8. Any non-negative value of the point corresponding to decision variables is a **feasible solution** for the system of equations representing the limited available resources for allocation problems.
9. Suppose there are m equations representing constraints (limited available resources) containing m + n variables in an allocation problem. The solution obtained by setting any n variables equal to zero and solving for the remaining m variables is called the **basic solution**. These m variables are the basic variables and the remaining n variables are non-basic variables.
10. A basic solution for which all the basic variables are non-negative is called a **basic feasible solution**.

1.9 SOLUTIONS/ANSWERS

E1) Here, the maximum number of possible basic solutions will be

$${}^4C_2 = \frac{|4|}{|2|2} = \frac{4 \times 3 \times 2}{2 \times 2} = 6.$$

These are obtained as follows:

Putting $x_1 = 0, x_2 = 0$, we have $x_3 = 15, x_4 = 10$.

Putting $x_1 = 0, x_3 = 0$, we have $5x_2 = 15$ and $2x_2 + x_4 = 10$

∴ $x_2 = 3$ and $x_4 = 4$.

For $x_1 = 0, x_4 = 0, 5x_2 + x_3 = 15$ and $2x_2 = 10$

∴ $x_2 = 5$ and $x_3 = -10$

For $x_2 = 0, x_3 = 0, 3x_1 = 15$ and $5x_1 + x_4 = 10$

∴ $x_1 = 5$ and $x_4 = -15$

For $x_2 = 0, x_4 = 0, 3x_1 + x_3 = 15$ and $5x_1 = 10$

$$\text{P } x_1 = 2 \text{ and } x_3 = 9$$

For $x_3 = 0, x_4 = 0, 3x_1 + 5x_2 = 15$ and $5x_1 + 2x_2 = 10$

$$\text{P } x_1 = \frac{50}{19}, x_2 = \frac{45}{19} \text{ (solving the equation)}$$

Thus, the possible solutions are

$(0, 0, 15, 10), (0, 3, 0, 4), (0, 5, -10, 0), (5, 0, 0, -15),$

$(2, 0, 9, 0)$ and $\left(\frac{50}{19}, \frac{45}{19}, 0, 0\right)$.

Of these, the basic feasible solutions are:

$(0, 0, 15, 10), (0, 3, 0, 4), (2, 0, 9, 0)$ and $\left(\frac{50}{19}, \frac{45}{19}, 0, 0\right)$.

UNIT 2 LINEAR PROGRAMMING PROBLEMS

Structure

- 2.1 Introduction
 - Objectives
- 2.2 Linear Programming Problem (LPP)
- 2.3 Mathematical Formulation of LPP
- 2.4 Graphical Solution of Linear Programming Problems
- 2.5 Canonical and Standard Forms of LPP
 - General Linear Programming Problem
 - Slack and Surplus Variables
 - Canonical Form
 - Standard Form
- 2.6 Summary
- 2.7 Solutions/Answers

2.1 INTRODUCTION

In Unit 1, you have learnt about the origin, scope, uses and limitations of Operations Research. We have also discussed the concept of optimisation and explained the basic feasible solution of linear programming problem.

In this unit, we discuss linear programming problems and explain how they are formulated mathematically in Secs. 2.2 and 2.3, respectively. We also define the objective function and explain how its optimum value is obtained graphically in Sec. 2.4. The objective function may be profit, cost, revenue, production capacity, etc. Linear programming can be applied to a variety of problems such as production, transportation, advertising and problems in public and private organisations, e.g., business, industry, hospitals, libraries as also in education. In order to solve linear programming problems, we need to convert them into a canonical or standard form, as explained in Sec. 2.5.

In the next unit, you will learn how to solve an LPP using trial and error method and the Simplex method. We shall also discuss the artificial variable technique and the Big-M method.

Objectives

After studying this unit, you should be able to:

- describe a linear programming problem and its mathematical formulation;
- discuss the applications and limitations of linear programming problems;
- formulate the linear programming problems;
- explain how linear programming problems are solved graphically; and
- express the linear programming problems to their canonical and standard form.

2.2 LINEAR PROGRAMMING PROBLEM (LPP)

In Unit 1, you have learnt about the concept of optimisation. Usually, the requirements of any agency, industry or country far exceed their limited resources of land, workforce, capital, organisational facilities, etc. Since the resources at their disposal are limited, the problem is to use them in such a way as to obtain the maximum production or profit, or to minimise the cost of production. You have learnt that such problems are called **optimisation problems**. A problem wherein the objective is to allot the limited available resources to the jobs in such a way as to optimise the overall effectiveness, minimising the total cost or maximising the total profit, is called mathematical programming. Mathematical programming in which constraints are expressed as linear equalities/inequalities is called **linear programming**.

Linear programming deals with the optimisation of the total effectiveness expressed as a linear function of **decision variables**, known as the objective function, subject to a set of linear equalities/inequalities known as **constraints**. Decision variables are the variables in terms of which the problem is defined.

Every organisation, big or small, has at its disposal four Ms, i.e., Men, Machines, Money and Materials. The supply of each of these four Ms is limited. If the supply of these resources is unlimited, the need for management tools such as linear programming will not arise at all. Since it is limited, there is a need to find the best allocation of resources in order to optimise the objective function. Linear programming can be used if the following conditions are satisfied:

1. Objective function, to be optimised, should be well defined and can be expressed as a linear function of the decision variables.
2. Resources leading to constraints should be finite and can be expressed as linear equalities or inequalities in terms of variables.
3. There must be alternative and finite courses of action.
4. Decision variables should be non-negative.

Applications

Linear programming methods are used in various fields including business and industry by almost all their departments such as production, marketing, finance, personnel. For example, using linear programming methods, we can:

- i) maximise profits or the number of effective exposures or returns or revenue; or
- ii) minimise costs or times of assembling the parts or the number of personnel for a job or the transportation cost or the travelling distance of a salesman.

There are many more applications. However, there are some limitations as well.

Limitations

1. Linear programming method cannot be applied if relationships are not linear.
2. It may give non-integral values even for those decision variables which have only integral values.

3. Constraints in the linear programming methods are written assuming all parameters are known. However, in real problems, sometimes these are not known and hence the full set of constraints cannot be written.

2.3 MATHEMATICAL FORMULATION OF LPP

The mathematical formulation of linear programming problem (LPP) is described in the following steps:

1. Identify the decision variables of the problem.
2. Express the objective function, which is to be optimised, i.e., maximised or minimised, as a linear function of the decision variables.
3. Identify the limited available resources, i.e., the constraints and express them as linear inequalities or equalities in terms of decision variables.
4. Since negative values of the decision variables do not have any valid physical interpretation, introduce non-negative restrictions.

Let us take an example to illustrate these steps.

Example 1: A small scale industry manufactures two products P and Q which are processed in a machine shop and assembly shop. Product P requires 2 hours of work in a machine shop and 4 hours of work in the assembly shop to manufacture while product Q requires 3 hours of work in machine shop and 2 hours of work in assembly shop. In one day, the industry cannot use more than 16 hours of machine shop and 22 hours of assembly shop. It earns a profit of ₹3 per unit of product P and ₹4 per unit of product Q. Give the mathematical formulation of the problem so as to maximise profit.

Solution: Let x and y be the number of units of product P and Q, which are to be produced. Here, x and y are the **decision variables**. Suppose Z is the profit function.

Since one unit of product P and one unit of product Q gives the profit of ₹3 and ₹4, respectively, the objective function is

$$\text{Maximise } Z = 3x + 4y$$

The requirement and availability in hours of each of the shops for manufacturing the products are tabulated as follows:

	Machine Shop	Assembly Shop	Profit
Product P	2 hours	4 hours	₹3 per unit
Product Q	3 hours	2 hours	₹4 per unit
Available hours per day	16 hours	22 hours	

Total hours of machine shop required for both types of product = $2x + 3y$

Total hours of assembly shop required for both types of product = $4x + 2y$

Hence, the constraints as per the limited available resources are:

$$2x + 3y \leq 16$$

and
$$4x + 2y \leq 22$$

Since the number of units produced for both P and Q cannot be negative, the non-negative restrictions are:

$$x \geq 0, y \geq 0$$

Thus, the mathematical formulation of the given problem is

Maximise $Z = 3x + 4y$
subject to the constraints

$$2x + 3y \leq 16$$

$$4x + 2y \leq 22$$

and non-negative restrictions

$$x \geq 0, y \geq 0$$

Now, you should try to formulate the following problem mathematically.

g is the symbol of gram.

E1) A company produces two types of items P and Q that require gold and silver. Each unit of type P requires 4g silver and 1g gold while that of Q requires 1g silver and 3g gold. The company can produce 8g silver and 9g gold. Suppose each unit of type P brings a profit of `44 and that of type Q, `55. Give the mathematical formulation for the problem to determine the number of units of each type that the company should produce to maximise the profit.

Before studying the next section, match your solution with the solution given in Sec. 2.7.

2.4 GRAPHICAL SOLUTION OF LINEAR PROGRAMMING PROBLEMS

In Sec. 2.3, you have learnt the mathematical formulation of a linear programming problem (LPP). In this section, we discuss how to solve this linear programming problem graphically using the method of graphs of the inequalities.

The graphical method is used to solve linear programming problems having two decision variables. For solving LPPs involving more than two decision variables, we use another method called the Simplex method. We shall discuss it in Unit 3. But you need to learn the graphical method to acquire the necessary grounding for learning the Simplex method.

The graphical method of solving a linear programming problem comprises the following steps:

1. First of all, the graphs are plotted for the equalities corresponding to the given inequalities for constraints as well as restrictions. That is, we first draw the straight lines. For example, suppose one of the given inequalities is $2x + 3y \leq 6$. Then, we first plot the graph for the equation $2x + 3y = 6$, which is a straight line. For this, we take any two points on it as follows and join them:

x	0	3
y	2	0

For example, for $x = 0$, $y = 6/3 = 2$ and for $y = 0$, $x = 6/2 = 3$. So we get the straight line shown in Fig. 2.1.

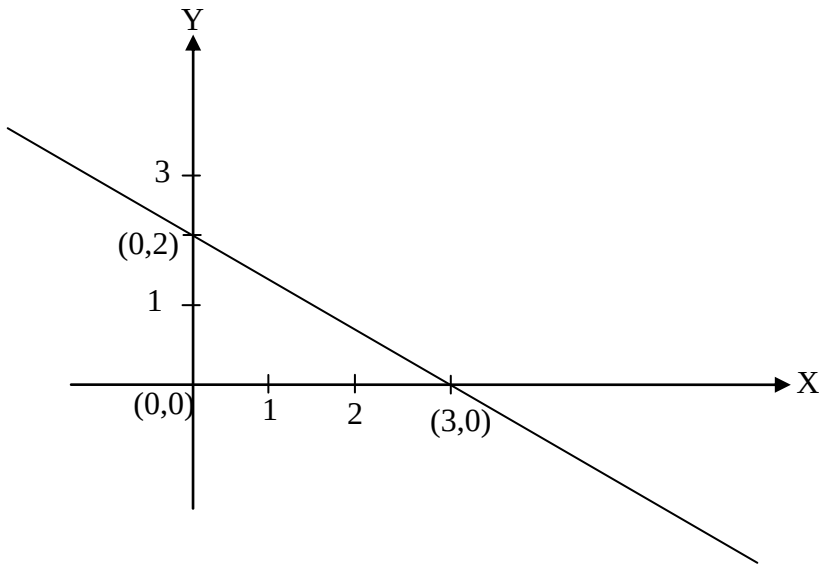


Fig. 2.1

2. Then we determine the region corresponding to each inequality. Let us consider the inequality $2x + 3y \leq 6$ again. We can find the region on the graph satisfied by this inequality by substituting $x = 0$ and $y = 0$ in it. We get

$$2(0) + 3(0) \leq 6 \Rightarrow 0 \leq 6$$

which is correct. So it is the region containing the point $(0, 0)$. Hence, the half plane shown in Fig. 2.2 by arrows starting from the line towards the point $(0, 0)$ is the graph of the given inequality.

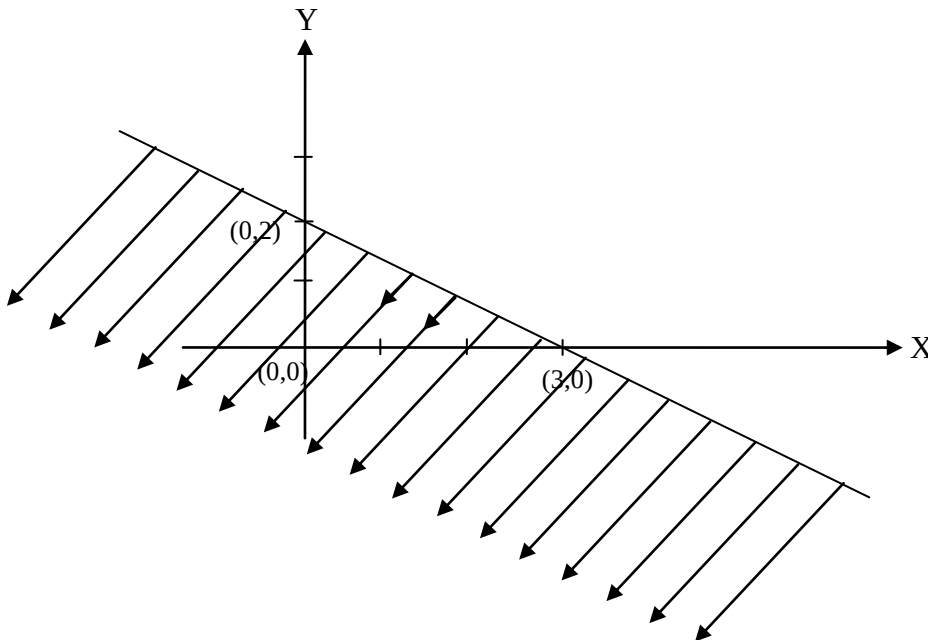


Fig. 2.2

Had the given inequality been $2x + 3y \geq 6$, then we would have drawn the arrows on the opposite side of the line. This is because on putting $x = 0, y = 0$ in the inequality, we get

$$2(0) + 3(0) \geq 6 \Rightarrow 0 \geq 6$$

which is not correct.

Thus the point $(0, 0)$ does not satisfy the inequality and hence does not lie in this region. The graph for the inequality $2x + 3y \geq 6$ would, therefore, be as shown in Fig. 2.3.

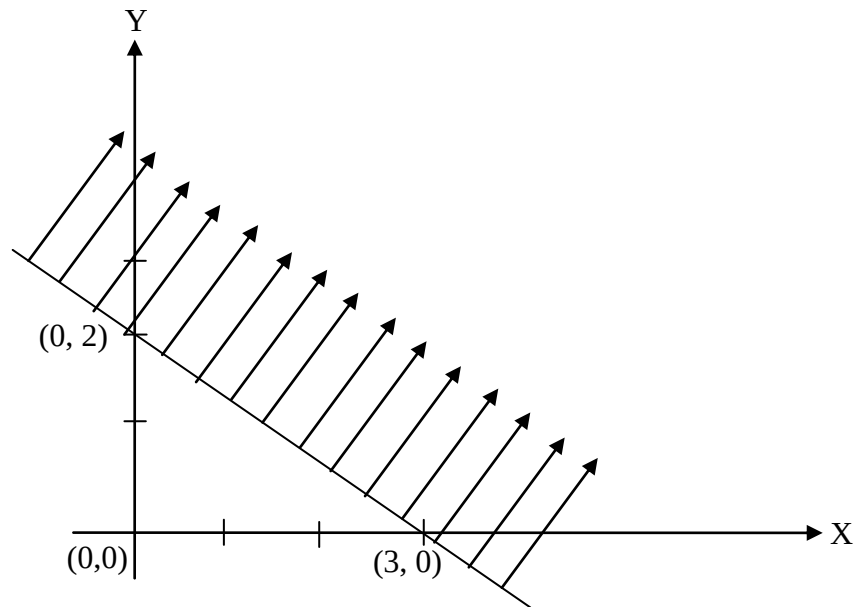


Fig. 2.3

In this example, we have used the point $(0, 0)$ to determine which half plane corresponds to the given inequality. However, you can take any other point. But using $(0, 0)$ is far easier. If the right hand side of the given inequality is zero, using the point $(0, 0)$ in it is meaningless. For example, suppose the given inequality is $2x - 3y \geq 0$. The plot of $2x - 3y = 0$ is given in Fig. 2.4. It is a straight line passing through the origin. Using the point $(0, 0)$ in the inequality $2x - 3y \geq 0$, we get $2(0) - 3(0) \geq 0$, i.e., $0 \geq 0$. So we cannot decide which half plane is the region of the given inequality. Therefore, in this case, we use any other point, say $(2, 0)$. On putting $x=2, y=0$ in the given inequality, we get

$$2(2) - 3(0) \geq 0 \quad \Rightarrow \quad 4 \geq 0$$

which is true. Therefore, the half plane containing $(2, 0)$ is the required region as shown in Fig. 2.4.

$$2x - 3y = 0 \Rightarrow 3y = 2x \Rightarrow y = \frac{2x}{3}$$

x	0	3
y	0	2

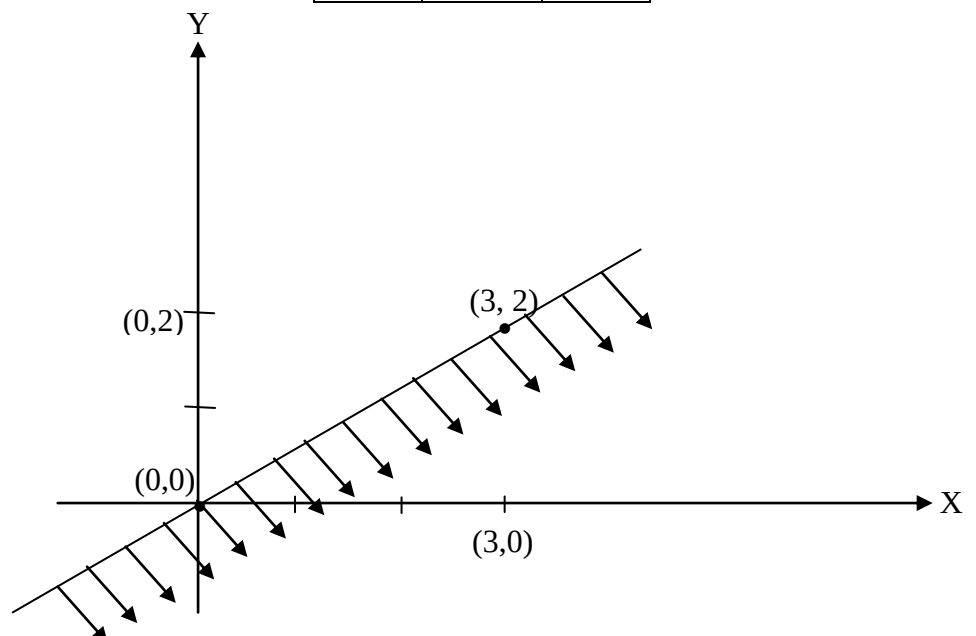


Fig. 2.4

3. After determining the regions for each inequality, we find their common region, i.e., the region obtained on superimposing their regions of inequality. This is the region where all the given inequalities and the non-negative restrictions are satisfied. This common region is known as the **feasible region** or the **solution set** or the **polygonal convex set** (Recall that you have learnt about the convex set in Sec. 1.6 of Unit 1).
4. Next we determine each of the corner points (vertices) of the polygon obtained in step 3. This is done either by plotting the graphs on graph paper or by solving the two equations of the lines intersecting at that point.
5. a) The objective function at each corner point is evaluated. If the feasible region is bounded, then
 - i) the maximum of the values obtained for the objective function at the corner points is the optimum value when the objective function is of the maximisation form. The point corresponding to this maximum value gives the required values of the decision variables.
 - ii) the minimum of the values obtained for the objective function at the corner points is the optimum value when the objective function is of the minimisation form. The point corresponding to this minimum value gives the required values of the decision variables.
- b) The feasible region may not be bounded. Then either there are additional hidden conditions which can be used to bound the region or there is no solution to the problem.
- c) If the same optimum value exists at two of the vertices, then there are multiple solutions to the problem. Suppose these two points are (x_1, y_1) and (x_2, y_2) . Then other solutions are given by the points as follows:
 [First ordinate of first point $\times t$ + First ordinate of second point $\times (1-t)$,
 Second ordinate of first point $\times t$ + Second ordinate of second point $\times (1-t)$]
 For the points (x_1, y_1) and (x_2, y_2) , the other solutions exist at the points: $[(x_1 \times t + x_2 \times (1-t), y_1 \times t + y_2 \times (1-t))]$
 where t is any real number lying between 0 and 1.

For example, let the objective function be $Z = 3x - y$ and let A (2, 1) and B(3, 4) be the points which give the same optimum value of the objective function, i.e., $Z = 5$. Then other solutions which give the same value of the objective function are:

$$\begin{aligned} & (2 \times t + 3 \times (1-t), 1 \times t + 4 \times (1-t)) \\ \text{or} & (2t + 3 - 3t, t + 4 - 4t) \\ \text{or} & (3 - t, 4 - 3t), 0 \leq t \leq 1 \end{aligned}$$

Here $t = 0$ gives the point (3, 4), which is point B and $t = 1$ gives the point (2, 1), which is point A. The real values of t between 0 and 1 give other points which give the same optimum solution. One such point other than A and B is

$$\left(3 - \frac{1}{2}, 4 - 3 \times \frac{1}{2}\right), \text{ for } t = \frac{1}{2}, \text{ i.e., } \left(\frac{5}{2}, \frac{5}{2}\right)$$

You can verify that $Z = 5$ at the point $\left(\frac{5}{2}, \frac{5}{2}\right)$. We illustrate this method in

Example 2.

$$\begin{aligned} Z &= 3x - y \\ &= 3 \times \frac{5}{2} - \frac{5}{2} \\ &= \frac{15}{2} - \frac{5}{2} = 5 \end{aligned}$$

Example 2: A company produces two types of items P and Q that require gold and silver. Each unit of type P requires 4g silver and 1g gold while that of type Q requires 1g silver and 3g gold. The company can produce 8g silver and 9g gold. If each unit of type P brings a profit of `44 and that of type Q `55, determine the number of units of each type that the company should produce to maximise the profit. What is the maximum profit?

Solution: Let x be the number of units of type P to be produced and y be the number of units of type Q to be produced. It is given that:

	Silver	Gold	Profit
Type P	4g	1g	`44 per unit
Type Q	1g	3g	`55 per unit
Available (at the most)	8g	9g	

Let Z be the profit function. The mathematical formulation of the given problem is

$$\text{Max. } Z = 44x + 55y$$

subject to the constraints:

$$\begin{aligned} 4x + y &\leq 8, \\ x + 3y &\leq 9, \\ x &\geq 0, y \geq 0. \end{aligned}$$

First of all, we plot the graphs for the following equations:

$$4x + y = 8, \quad x + 3y = 9, \quad x = 0, y = 0.$$

Since these equations are of straight lines, only two points are sufficient to plot the graphs (see Fig. 2.5). For the line $4x + y = 8$, we take the following two points:

x	0	2
y	8	0

Similarly, for the line $x + 3y = 9$, we take

x	0	9
y	3	0

Now, plotting the above lines, we get Fig. 2.5.

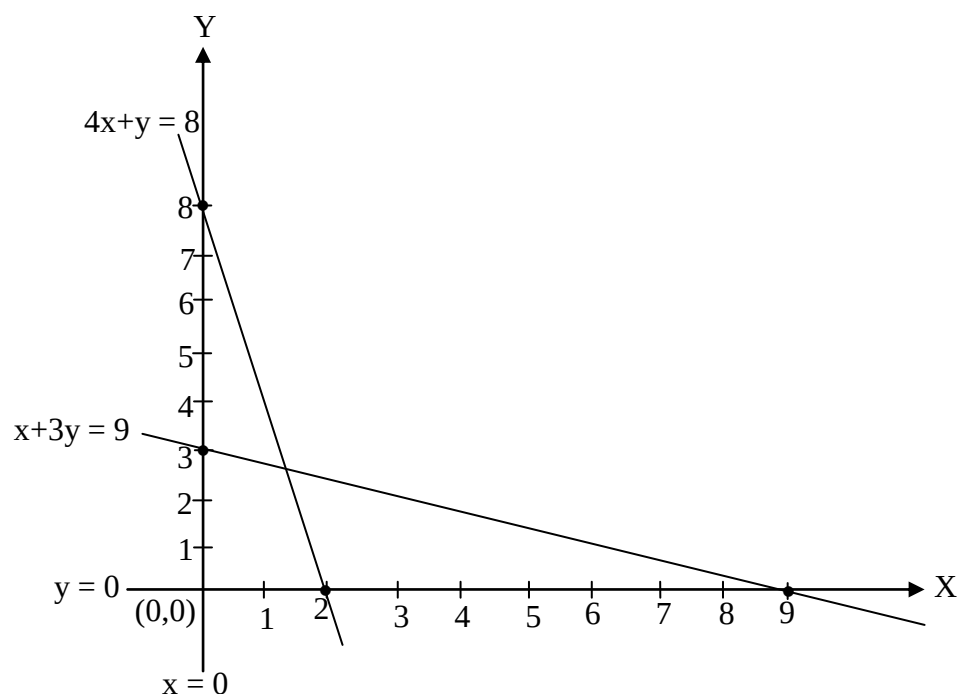


Fig. 2.5

Note that $x = 0$ is the y-axis and $y = 0$ is the x-axis.

For plotting the graph of the inequality $4x + y \leq 8$ we put $(0, 0)$ in it. We get $0 \leq 8$, which is true. Therefore, starting from the line $4x + y \leq 8$, we shall shade towards origin. Similarly, for the graph $x + 3y \leq 9$, we shall shade towards origin. For the graph $x \geq 0$, we shall shade towards right side of $x = 0$ and for the graph $y \geq 0$, the region above $y = 0$ will be shaded.

Thus, the regions for the inequalities are shown in Fig. 2.6 by arrows:

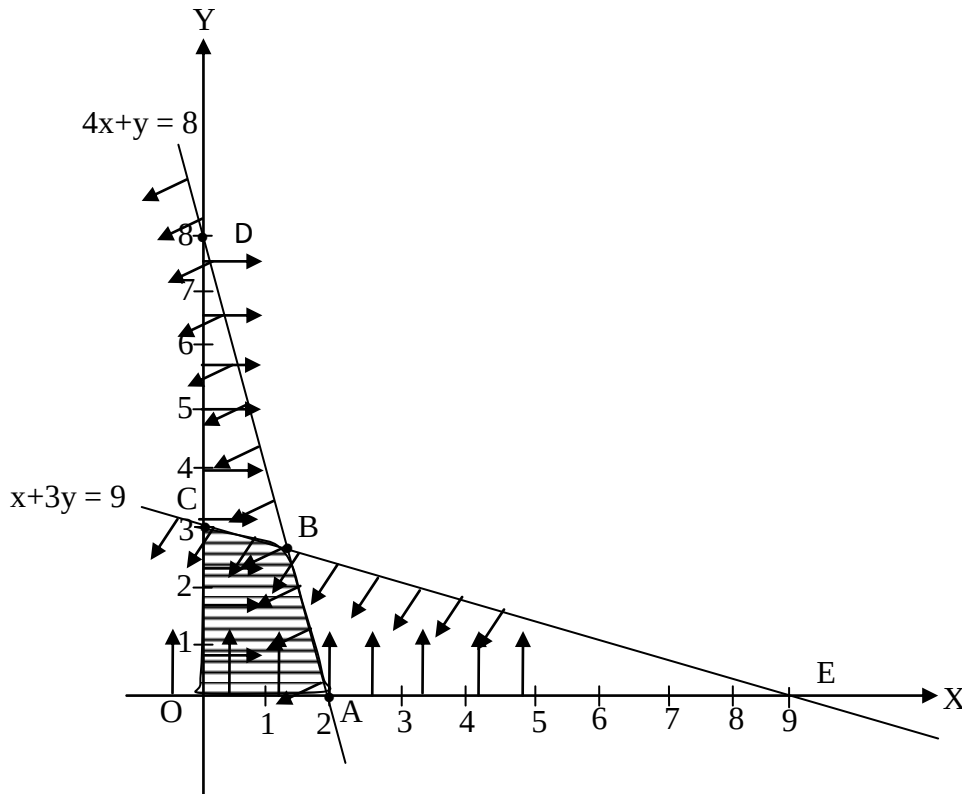


Fig. 2.6

The most common shaded portion is the region inside and on the polygon OABC shown in Fig. 2.6. You can see from Fig. 2.6 that the coordinates of O, A and C are $(0, 0)$, $(2, 0)$ and $(0, 3)$, respectively.

The coordinates of the point B are obtained by solving the equations $4x + y = 8$ and $x + 3y = 9$ as it is the point of intersection of the two lines represented by them. The solution of $4x + y = 8$ and $x + 3y = 9$ is given as

$$x = \frac{15}{11} \text{ and } y = \frac{28}{11}$$

So the vertices of OABC are O $(0, 0)$, A $(2, 0)$, B $\left(\frac{15}{11}, \frac{28}{11}\right)$ and C $(0, 3)$.

We now obtain the values of $Z = 44x + 55y$ at each of the vertices of OABC as follows:

$$\text{At O } (0, 0), Z = 44(0) + 55(0) = 0$$

$$\text{At A } (2, 0), Z = 44(2) + 55(0) = 88$$

$$\begin{aligned} \text{At B } \left(\frac{15}{11}, \frac{28}{11}\right), Z &= 44\left(\frac{15}{11}\right) + 55\left(\frac{28}{11}\right) \\ &= 60 + 140 = 200 \end{aligned}$$

$$\text{At C } (0, 3), Z = 44(0) + 55(3) = 165$$

$$4x + y = 8 \quad (i)$$

$$x + 3y = 9 \quad (ii)$$

We multiply equation (i) by 3:

$$12x + 3y = 24 \quad (iii)$$

We subtract equation (ii) from equation (iii):

$$11x = 15$$

$$\text{So we get } x = \frac{15}{11}$$

$$\text{Putting } x = \frac{15}{11} \text{ in equation (ii),}$$

$$\text{we get } \frac{15}{11} + 3y = 9 \Rightarrow y = \frac{28}{11}$$

Thus, the value of Z is maximum at $B\left(\frac{15}{11}, \frac{28}{11}\right)$

and the optimum solution is

$$\text{Max. } Z = 200 \text{ when } x = \frac{15}{11} \text{ and } y = \frac{28}{11}.$$

Now, you should try to solve the following exercises.

E2) Maximise $Z = 6X + 3Y$

subject to the constraints

$$2X + 5Y \leq 120$$

$$4X + 2Y \leq 80$$

$$X \geq 0, Y \geq 0$$

E3) Maximise $z = 3x_1 + 2x_2$

subject to the constraints

$$x_1 - x_2 \leq 1$$

$$x_1 + x_2 \geq 3$$

$$x_1 \geq 0, x_2 \geq 0$$

E4) Maximise $Z = x_1 + x_2$

subject to the constraints

$$x_1 + x_2 \leq 1$$

$$-3x_1 + x_2 \geq 3$$

$$x_1 \geq 0, x_2 \geq 0$$

E5) A company manufactures two products X and Y, each of which requires three types of processing. The length of time for processing each unit and the profit per unit are given in the following table:

	Product X (hr/unit)	Product Y (hr/unit)	Available capacity per day (hr)
Process I	12	12	840
Process II	3	6	300
Process III	8	4	480
Profit per unit (₹)	5	7	

How many units of each product should the company manufacture per day in order to maximise profit?

E6) A company produces soft drinks and has a contract requiring that a minimum of 80 units of chemical A and 60 units of chemical B go into each bottle of the drink. The chemicals are available in a prepared mix from two different suppliers. The supplier X_1 has a mix of 4 units of A and 2 units of B that costs ₹10, and the supplier X_2 has a mix of 1 unit of A and 1 unit of B that costs ₹4. How many mixes from the company X_1 and company X_2 should the company purchase to honour contract requirement and yet minimise cost?

Before studying the next section, match your solution with the solution given in Sec. 2.7.

2.5 CANONICAL AND STANDARD FORMS OF LPP

After formulating a linear programming problem, our next step is to solve it. You have learnt in Secs. 2.3 and 2.4 that linear programming problems can be presented as problems of maximisation or minimisation with constraints such as $\leq$, $=$ or $\geq$. In order to develop a standard procedure for solving LPPs, we need to convert them into well known forms, namely, the Canonical form and the Standard form. We now discuss the General LPP along with these two forms. The canonical form is especially used in the duality theory and the standard form is used to develop the general procedure for solving any linear programming problem. In order to understand these forms you also need to learn about slack and surplus variables.

2.5.1 General Linear Programming Problem

Let us formulate the general linear programming problem. Let Z be a linear function of n basic variables $X_1, X_2, \dots, X_n$, which is to be maximised (or minimised). We write the problem as

$$\text{Maximise (or Minimise) } Z = C_1X_1 + C_2X_2 + \dots + C_nX_n \quad \dots (1)$$

where $C_1, C_2, \dots, C_n$ are known constants termed as **cost coefficients** of basic variables.

Let (a_{ij}) be an $m \times n$ real matrix of $m \times n$ constants a_{ij} 's and let $\{b_1, b_2, \dots, b_m\}$ be a set of constants such that

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &\leq \text{ or } \geq \text{ or } = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &\leq \text{ or } \geq \text{ or } = b_2 \\ \cdot &\quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \\ \cdot &\quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &\leq \text{ or } \geq \text{ or } = b_m \end{aligned} \quad \dots (2)$$

$$\text{and } x_j \geq 0 \quad \text{for all } j=1, 2, \dots, n \quad \dots (3)$$

The linear function Z in equation (1) is called the **objective function**. The set of inequalities given in (2) is called **constraints** of a general LPP and the set of inequalities given in (3) are known as **non-negative restrictions** of a general LPP.

2.5.2 Slack and Surplus Variables

In general, if in any linear programming problem, we have a constraint of the type

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1, \quad \text{where } b_1 \geq 0$$

then this inequality can be converted into an equation by **adding** one non-negative variable s_1 to the left hand side. This new variable is called a **slack variable** and the constraints are transformed into the following equation:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n + s_1 = b_1, \quad \text{where } s_1 \geq 0, b_1 \geq 0$$

Thus, a **non-negative variable added to the left-hand side** of a less than or equal to ($\leq$) type of constraint that converts it into an equation is called a **slack variable**. The value of this variable can be interpreted as the amount of unused resource.

Similarly, if in any linear programming problem, we have a constraint of the type

$$a_{11}y_1 + a_{12}y_2 + \dots + a_{1n}y_n \leq b_1, \quad b_1 \geq 0$$

then this inequality can be converted into an equation by **subtracting** one non-negative variable s_1 from the left hand side. This new variable is called a **surplus variable** and the constraints are transformed into an equation as

$$a_{11}y_1 + a_{12}y_2 + \dots + a_{1n}y_n - s_1 = b_1, \quad \text{where } s_1 \geq 0, b_1 \geq 0$$

Thus, a **non-negative variable subtracted from the left-hand side** of a greater than or equal to ($\geq$) type of constraint that converts it into an equation is called a **surplus variable**. The value of this variable can be interpreted as the amount over and above the required minimum level.

Let us now obtain the canonical form of the LPP.

2.5.3 Canonical Form

The characteristics of the canonical form are explained in the following steps:

- i) The objective function should be of maximisation form. If it is given in the minimisation form, it should be converted into maximisation form. Suppose $Z = ax + by$ is the given objective function which is to be minimised. Then, it is equivalent to maximising its negative function, i.e., $-Z$. The reason is that if a function $f(x)$ is minimum at the same point $x = a$ (say), then $-f(x)$ will be maximum at $x = a$ as shown in Fig. 2.7:

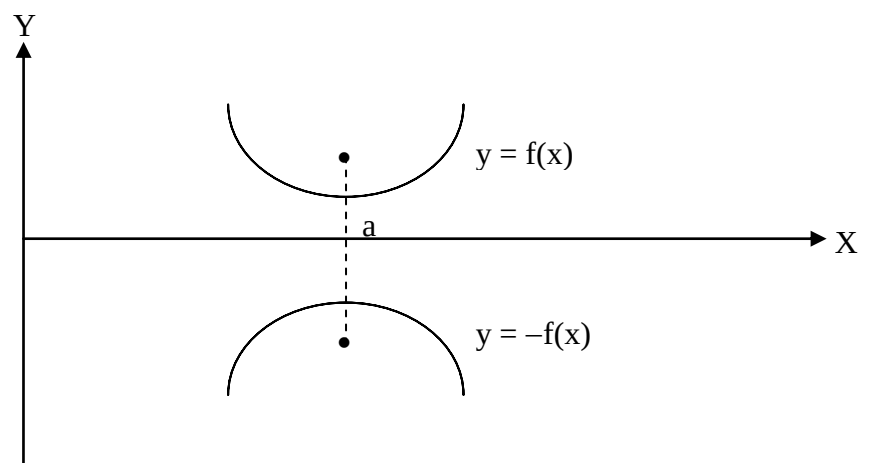


Fig. 2.7

Note: The graph of $y = f(x)$ is the mirror image of the graph of $y = -f(x)$ about the x -axis as shown Fig. 2.7 and $x = a$ is the point where $f(x)$ is minimum and $-f(x)$ is maximum.

- ii) All constraints should be of “ $\leq$ ” type, except for non-negative restrictions. Inequality of “ $\geq$ ” type, if any, should be changed to an inequality of the “ $\leq$ ” type by multiplying both sides of the inequality by -1 . For example, suppose a given inequality is

$$2x + 3y \geq 5$$

Then it can be written as

$$-2x - 3y \leq -5$$

- iii) All the variables should be non-negative. If a given variable is unrestricted in sign (i.e., positive, negative or zero), it can be written as a difference of two non-negative variables. Suppose x is unrestricted in sign, then x can be written as

$$x = x' - x'', \text{ where } x' \geq 0, x'' \geq 0.$$

Let us explain this point further: Suppose $x = -5$. Then it can be written as the difference of two non-negative numbers: $2 - 7$ (say). Here 2 and 7 are non-negative.

Example 3: Rewrite the following linear programming problem in canonical form:

$$\text{Minimise } Z = 2x_1 + x_2 + 4x_3$$

subject to the constraints:

$$-2x_1 + 4x_2 \leq 4$$

$$x_1 + 2x_2 + x_3 \geq 5$$

$$2x_1 + 3x_3 \leq 2$$

$$x_1, x_2 \geq 0 \text{ and } x_3 \text{ is unrestricted in sign}$$

Solution: Here, the objective function is of the minimisation form. We rewrite it in the maximisation form as follows:

$$\text{Minimise } Z = 2x_1 + x_2 + 4x_3$$

Thus, we have to maximise $-Z = -2x_1 - x_2 - 4x_3$. So the problem becomes,

$$\text{Maximise } Z' = -2x_1 - x_2 - 4x_3, \text{ where } Z' = -Z$$

Now, the second constraint is of the type “ $\geq$ ”. Hence, to convert it into type “ $\leq$ ”, we multiply the inequality by -1 and write

$$-x_1 - 2x_2 - x_3 \leq -5$$

Other constraints are already in the desired form. But x_3 is unrestricted in sign. So we write

$$x_3 = x'_3 - x''_3, \text{ where } x'_3 \geq 0, x''_3 \geq 0.$$

The canonical form of the given problem, therefore, is

$$\text{Maximise } Z' = -2x_1 - x_2 - 4(x'_3 - x''_3), \text{ where } Z' = -Z$$

subject to the constraints:

$$-2x_1 + 4x_2 \leq 4$$

$$-x_1 - 2x_2 - (x'_3 - x''_3) \leq -5$$

$$2x_1 + 3(x'_3 - x''_3) \leq 2$$

$$x_1 \geq 0, x_2 \geq 0, x'_3 \geq 0, x''_3 \geq 0$$

We now discuss the standard form.

2.5.4 Standard Form

The characteristics of the standard form are explained in the following steps:

- The objective function should be in the maximisation form as already explained in Sec. 2.5.3.
- The right side element of each constraint should be non-negative. If it is negative, we multiply the inequality by -1 .
- All constraints should be expressed in the form of equations, except for the non-negative restrictions. The inequalities should be converted into equations by augmenting the non-negative variables to the left side of each constraint. For inequalities of the “ $\leq$ ” type, we add the slack variables. If the inequalities are of the “ $\geq$ ” type, we subtract surplus variables.

Example 4: Rewrite the following linear programming problem in the standard form:

$$\text{Minimise } Z = 2x_1 + x_2 + 4x_3$$

subject to the constraints:

$$-2x_1 + 4x_2 \leq 4$$

$$x_1 + 2x_2 + x_3 \geq 5$$

$$2x_1 + 3x_3 \leq -2$$

$$x_1 \geq 0, \quad x_2 \geq 0 \quad x_3 \geq 0.$$

Solution: The objective function should be of maximisation form, i.e., we have to

$$\text{Maximise } Z' = -2x_1 - x_2 - 4x_3, \quad \text{where } Z' = -Z$$

subject to the constraints:

$$-2x_1 + 4x_2 \leq 4$$

$$x_1 + 2x_2 + x_3 \geq 5$$

$$-2x_1 - 3x_3 \leq 2 \quad [\because \text{Right side should be non-negative}]$$

$$x_1 \geq 0, \quad x_2 \geq 0 \quad x_3 \geq 0.$$

Now, the inequalities are to be converted to equations. Note that the first and third inequalities are of the type “less than or equal to ($\leq$)”.

Therefore, a slack variable is to be added to the left side of each of these inequalities. The second inequality is of the type “more than or equal to ($\geq$)”. So a surplus variable is to be subtracted from the left side of this inequality.

Thus, the standard form of the given LPP is

$$\text{Max. } Z' = -2x_1 - x_2 - 4x_3 + 0s_1 + 0s_2 + 0s_3, \quad \text{where } Z' = -Z$$

subject to the constraints:

$$-2x_1 + 4x_2 + s_1 = 4$$

$$x_1 + 2x_2 + x_3 - s_2 = 5$$

$$-2x_1 - 3x_3 + s_3 = 2$$

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0, \quad s_1 \geq 0, \quad s_2 \geq 0, \quad s_3 \geq 0$$

Now, you should try to solve the following exercises.

E7) Express the following LPP in canonical form:

$$\text{Minimise } Z = x_1 - 2x_2 + x_3$$

subject to the constraints:

$$2x_1 + 3x_2 + 4x_3 \geq -4$$

$$3x_1 + 5x_2 + 2x_3 \geq 7$$

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0$$

E8) Express the following LPP in standard form:

$$\text{Minimise } Z = x_1 - 2x_2 + x_3$$

subject to the constraints:

$$2x_1 + 3x_2 + 4x_3 \geq -4$$

$$3x_1 + 5x_2 + 2x_3 \geq 7$$

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

You may like to match your solution with the solution given in Sec. 2.7.

Let us now summarise the main points which have been covered in this unit.

2.6 SUMMARY

1. A problem wherein the objective is to allot the limited available resources to the jobs in such a way as to optimise the overall effectiveness, i.e., minimise the total cost or maximise the total profit, is called **mathematical programming**. Mathematical programming wherein constraints are expressed as linear equalities/inequalities is called **linear programming**. Linear programming deals with the optimisation of the total effectiveness expressed as a linear function of decision variables, known as the **objective function**, subject to a set of linear equalities/inequalities known as **constraints**.
2. Steps involved in the **mathematical formulation** of a linear programming problem (LPP) are: i) Identification of the decision variables of the problem; ii) expressing the objective function as a linear function of the decision variables; iii) identifying the limited available resources to write the constraints as linear inequalities or equalities in terms of decision variables; iv) introducing the non-negative restrictions.
3. **Graphical method** is used to solve linear programming problems having **two decision variables**. The graphical method of solving linear programming programs comprises the following steps: i) The graphs are plotted for the equations corresponding to the given inequalities for constraints as well as restrictions; ii) The region corresponding to each inequality is shaded; iii) After shading the regions for each inequality, the most common shaded portion, i.e., the region obtained on superimposing all the shaded regions is determined. This is the region where all the given inequalities, including non-negative restrictions are

satisfied. This common region is known as the **feasible region** or the **solution set** or the **polygonal convex set**; iv) Each of the corner points (vertices) of the polygon is then determined; v) The objective function at each corner point is evaluated.

4. If the feasible region is bounded, then the **greatest value** obtained for the objective function at the corner points is the **optimum** value when the objective function is of **maximisation** form; and the **least value** obtained for the objective function at the corner points is the **optimum** value when the objective function is of **minimisation** form.
5. The feasible **region may not be bounded**. Then either there are additional hidden conditions which can be used to bound the region or there is no solution to the problem.
6. If the same optimum value occurs at two vertices, then there are **multiple optimal solutions** to the problem.
7. The characteristics of the **canonical form** are: i) Objective function should be of maximisation form. If it is given in minimisation form, it should be converted into maximisation form; ii) All the constraints should be of “ $\leq$ ” type, except for non-negative restrictions. Inequality of “ $\geq$ ” type, if any, should be changed to an inequality of the “ $\leq$ ” type; iii) All variables should be non-negative.
8. The characteristics of **standard form** are: i) The objective function should be of maximisation form; ii) The right side element of each constraint should be non-negative; iii) All constraints should be expressed in the form of equations, except for the non-negative restrictions by augmenting slack or surplus variables.

2.7 SOLUTIONS/ANSWERS

- E1)** Let x be the number of units of type P to be produced and y be the number of units of type Q to be produced. It is given that:

	Silver	Gold	Profit
Type P	4g	1g	`44 per unit
Type Q	1g	3g	`55 per unit
Available (at the most)	8g	9g	

Let Z be the profit function.

$\therefore$ The mathematical formulation of the given problem is

$$\text{Max. } Z = 44x + 55y$$

subject to the constraints:

$$4x + y \leq 8$$

$$x + 3y \leq 9$$

$$x \geq 0, y \geq 0.$$

E2) Max. $Z = 5x + 7y$

subject to the constraints:

$$12x + 12y \leq 840 \Rightarrow x + y \leq 70$$

$$3x + 6y \leq 300 \Rightarrow x + 2y \leq 100$$

$$8x + 4y \leq 480 \Rightarrow 2x + y \leq 120$$

$$x \geq 0, y \geq 0$$

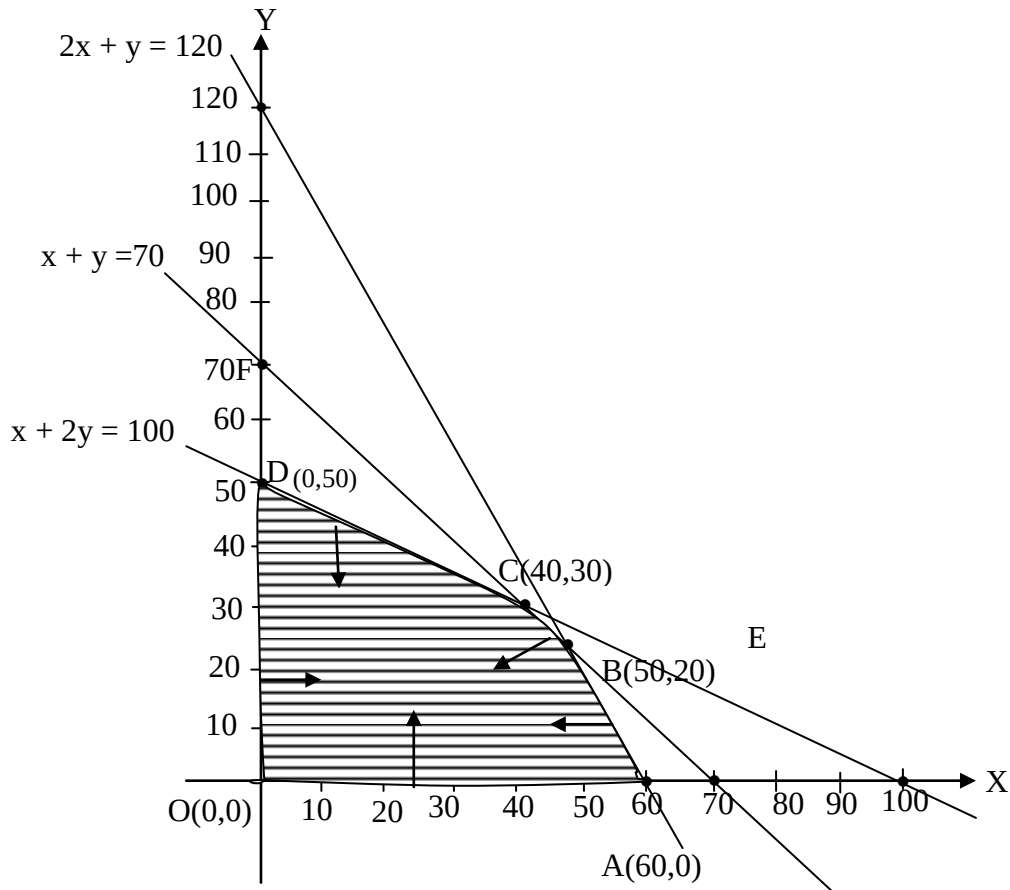


Fig. 2.8

At O(0, 0), $Z = 5(0) + 7(0) = 0$

At A(60, 0), $Z = 5(60) + 7(0) = 300$

At B(50, 20), $Z = 5(50) + 7(20)$
 $= 250 + 140 = 390$

At C(40, 30), $Z = 5(40) + 7(30)$
 $= 200 + 210 = 410$

At D(0, 50), $Z = 5(0) + 7(50) = 350$

The maximum value of Z is 410 at C(40, 30), i.e., at $x = 40$, $y = 30$.

E3) Minimise $Z = 10x_1 + 4x_2$

subject to the constraints

$$4x_1 + x_2 \geq 80$$

$$2x_1 + x_2 \geq 60$$

$$x_1 \geq 0, x_2 \geq 0$$

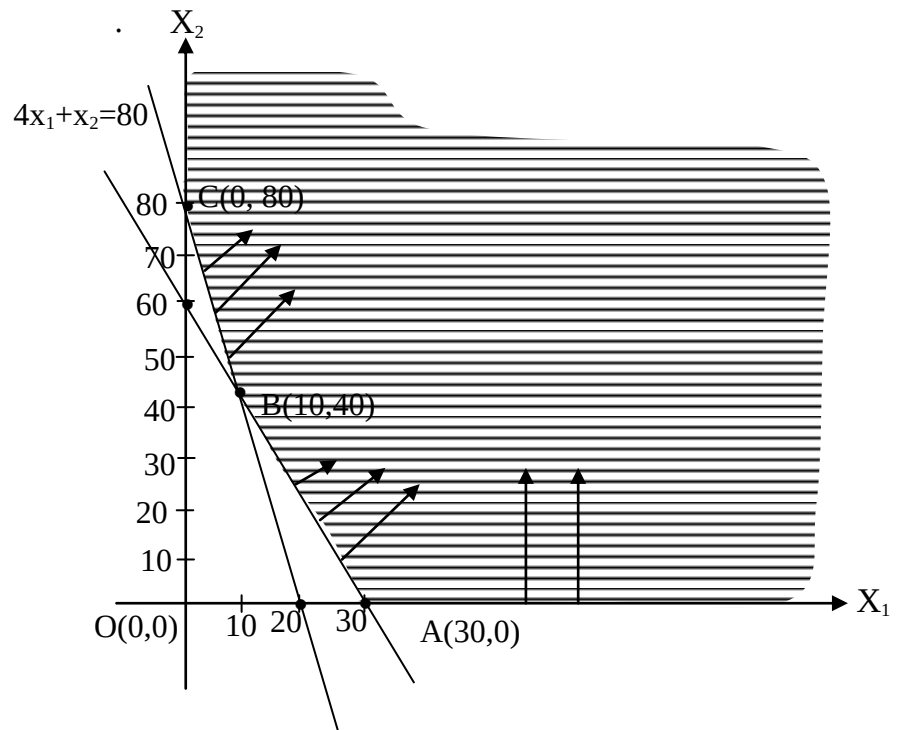


Fig. 2.9

At $A(30, 0)$, $Z = 10(30) + 4(0) = 300$

At $B(10, 40)$, $Z = 10(10) + 4(40)$
 $= 100 + 160 = 260$

At $C(0, 80)$, $Z = 10(0) + 4(80) = 320$

The minimum value of Z is 260 at $B(10, 40)$, i.e., when $x_1 = 10$ and $x_2 = 40$.

Note: Had the objective function been of maximisation form, the problem would have the unbounded solution. This is because the values of x_1 and x_2 could be increased beyond any limit, which would result in higher and higher value of Z with no upper bound.

E4) At $O(0, 0)$, $Z = 6(0) + 3(0) = 0$

At $A(20, 0)$, $Z = 6(20) + 3(0) = 120$

At $B(10, 20)$, $Z = 6(10) + 3(20) = 60 + 60 = 120$

At $C(0, 24)$, $Z = 6(0) + 3(24) = 0 + 72 = 72$

The maximum value of Z is 120 at $A(20, 0)$ as well as at $B(10, 20)$. Therefore, the given LPP has multiple optimal solutions (Fig. 2.10).

Other points at which Z gives the same maximum value are given as

$$(20 \times t + 10 \times (1 - t), 0 \times t + 20 \times (1 - t))$$

$$\text{or } (20t + 10 - 10t, 0 + 20 - 20t)$$

$$\text{or } (10 + 10t, 20 - 20t), 0 \leq t \leq 1.$$

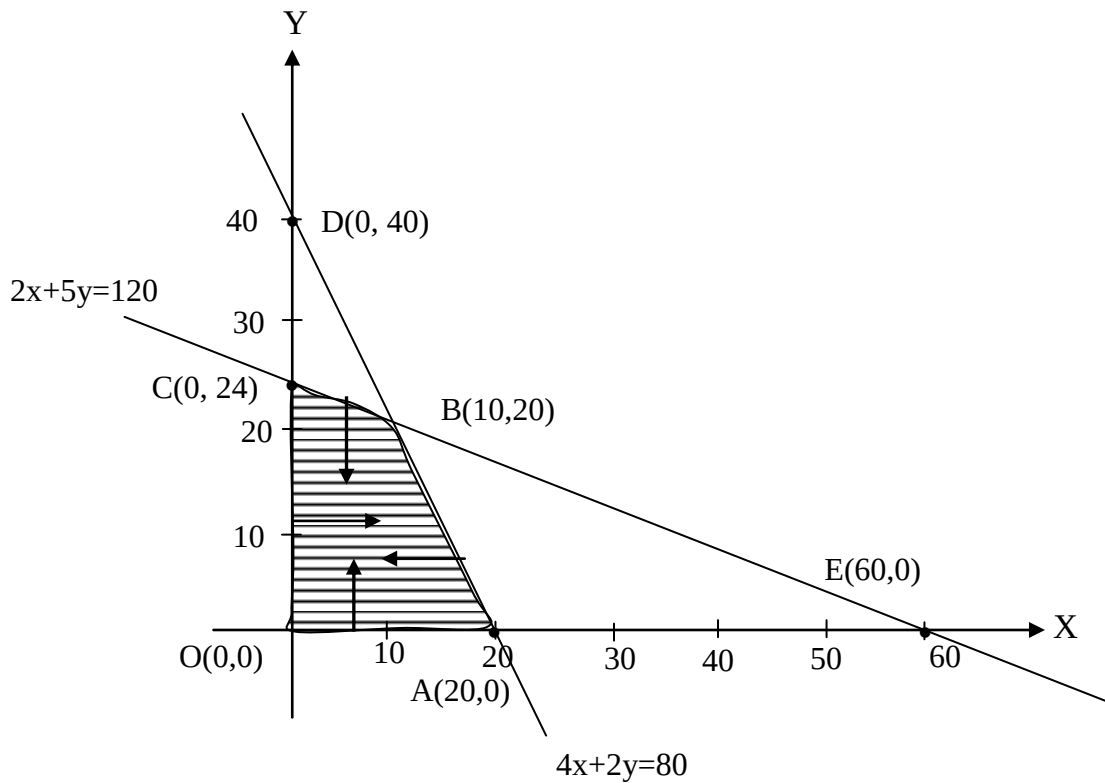


Fig. 2.10

- E5)** Here, we have to maximise the objective function. But the values of decision variables can be increased infinitely without violating the feasibility condition and hence the value of the objective function can be increased infinitely. Thus, the solution in this case is unbounded (Fig.2.11).

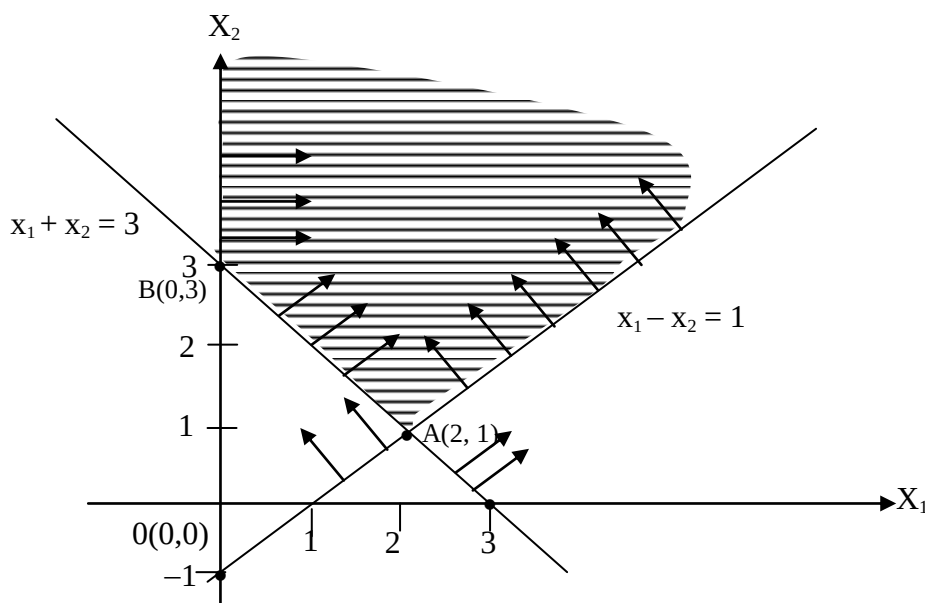


Fig. 2.11

Note: Had it been a minimisation problem, then it would have the optimum solution at one of the points A (2, 1) and B (3, 0) where it has the minimum value.

$$\text{At } A(2, 1), Z = 3(2) + 2(1) = 8$$

$$\text{At } B(3, 0), Z = 3(3) + 2(0) = 9$$

The minimum value is at $A(2, 1)$, i.e., when $x_1 = 2$ and $x_2 = 1$.

- E6)** We find no common region for all the four graphs and hence the given LPP has no solution (Fig. 2.12).

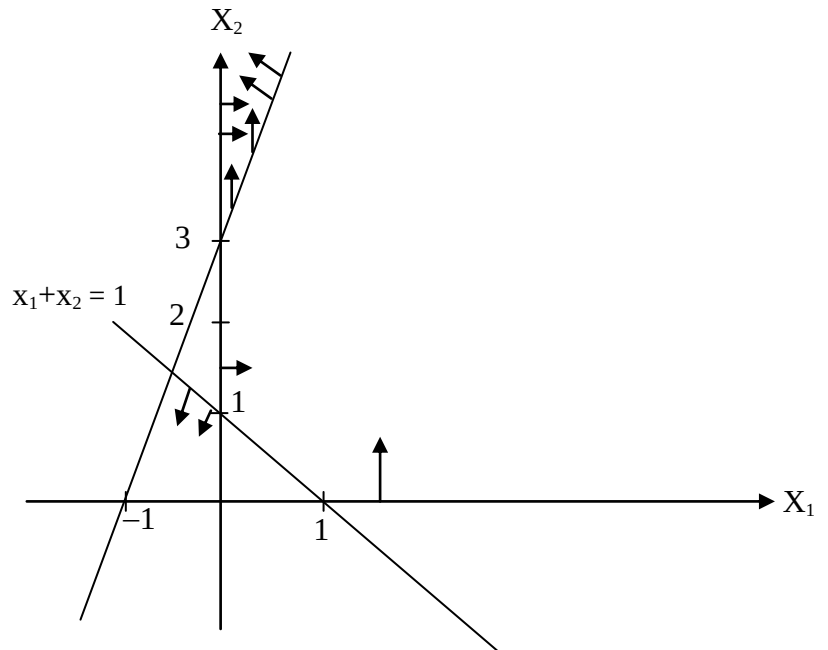


Fig. 2.12

- E7)** Maximise $Z' = -x_1 + 2x_2 - x_3$, where $Z' = -Z$

subject to the constraints

$$-2x_1 - 3x_2 - 4x_3 \leq 4$$

$$-3x_1 - 5x_2 - 2x_3 \leq -7$$

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

- E8)** Maximise $Z' = -x_1 + 2x_2 - x_3 + 0s_1 + 0s_2$, where $Z' = -Z$

subject to the constraints

$$-2x_1 - 3x_2 - 4x_3 + s_1 = 4$$

$$3x_1 + 5x_2 + 2x_3 - s_2 = 7$$

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0, \quad s_1 \geq 0, \quad s_2 \geq 0.$$

UNIT 3 SIMPLEX METHOD

Structure

- 3.1 Introduction
 - Objectives
 - 3.2 Trial and Error Method of Solving a Linear Programming Problem
 - 3.3 Simplex Method of Solving a Linear Programming Problem
 - 3.4 Artificial Variable Technique and the Big M Method
 - 3.5 Degeneracy
 - 3.6 Summary
 - 3.7 Solutions/Answers
-

3.1 INTRODUCTION

In Unit 2, you have studied the graphical method of solving linear programming problems and learnt how to express a linear programming problem in canonical and standard forms. You know that the graphical method cannot be used for problems involving more than two decision variables. For more than two decision variables, methods based on the concept of slack variables, i.e., trial and error method and Simplex method, are used. The trial and error method is extremely cumbersome and time-consuming and successive solutions may not culminate into the optimal one.

The Simplex method is an improvement over the trial and error method and it overcomes its inefficiencies. In this unit, we shall first discuss the trial and error method and then the Simplex method of solving the linear programming problems.

In the next unit, you will learn how to obtain an initial basic feasible and an optimum solution of a transportation problem.

Objectives

After studying this unit, you should be able to:

- explain the trial and error method of solving Linear Programming problems;
 - apply the Simplex method to solve Linear Programming problems;
 - describe the Artificial Variable Technique;
 - apply the Big M method to solve Linear Programming problems; and
 - explain the concept of degeneracy while solving an LPP by Simplex method.
-

3.2 TRIAL AND ERROR METHOD OF SOLVING A LINEAR PROGRAMMING PROBLEM

In the trial and error method, we first write the given LPP in the standard form as discussed in Unit 2 of this block. Suppose there are m constraints with n variables (where $n > m$) in the standard form of LPP. Before proceeding further, we state the Extreme Point theorem and Basic theorem.

Extreme Point Theorem: An optimal solution to an LPP occurs at one of the vertices of the feasible region.

Basic Theorem: For a system of m equations in n variables (where $n > m$), a solution in which at least $(n - m)$ variables have the value zero is called a **basic solution**.

Now, the objective function is optimal at least at one of the basic solutions. Some of the vertices may be infeasible if they have negative coordinates. Such solutions are dropped in view of the non-negativity condition on all variables including slack and surplus variables.

We consider the following LPP to explain the **trial and error method**:

$$\text{Max. } Z = 3x_1 + 2x_2$$

subject to the constraints:

$$x_1 + x_2 \leq 4$$

$$x_1 - x_2 \leq 2$$

$$x_1 \geq 0, x_2 \geq 0$$

Introducing slack variables s_1 and s_2 , we write the standard form of the LPP as:

$$\text{Max. } Z = 3x_1 + 2x_2 + 0s_1 + 0s_2$$

subject to the constraints:

$$x_1 + x_2 + s_1 = 4$$

$$x_1 - x_2 + s_2 = 2$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Here, we have $n = 4$ variables and $m = 2$ equations. Thus, $n - m = 4 - 2 = 2$. So, at least 2 variables have the value zero as a vertex.

According to the Basic theorem, we set $n - m = 2$ variables equal to zero at a time. Thus, we have the following sets of solutions:

$$\text{i) } x_1 = 0, x_2 = 0$$

$$\Rightarrow s_1 = 4, s_2 = 2$$

$$\text{ii) } x_1 = 0, s_1 = 0$$

$$\Rightarrow x_2 = 4, -x_2 + s_2 = 2$$

$$\Rightarrow x_2 = 4, s_2 = 2 + 4 = 6$$

$$\text{iii) } x_1 = 0, s_2 = 0$$

$$\text{P } x_2 + s_1 = 4, x_2 = -2$$

$$\Rightarrow s_1 = 6, x_2 = -2$$

$$\text{iv) } x_2 = 0, s_1 = 0$$

$$\text{P } x_1 = 4, x_1 + s_2 = 2$$

$$\Rightarrow x_1 = 4, s_2 = -2$$

$$v) \quad x_2 = 0, s_2 = 0$$

$$P \quad x_1 + s_1 = 4, x_1 = 2$$

$$\Rightarrow s_1 = 2, x_1 = 2$$

$$vi) \quad s_1 = 0, s_2 = 0$$

$$\Rightarrow x_1 + x_2 = 4, x_1 - x_2 = 2$$

$$\Rightarrow x_1 = 3, x_2 = 1$$

Since solutions (iii) and (iv) yield a negative coordinate each, these are infeasible and are dropped.

Since the optimal solution lies at one of the vertices, we find the value of the objective function $Z = 3x_1 + 2x_2$ for each of the above sets of solutions for x_1 and x_2 except (iii) and (iv).

$$\text{For Set (i),} \quad Z = 3(0) + 2(0) = 0$$

$$\text{For Set (ii),} \quad Z = 3(0) + 2(4) = 8$$

$$\text{For Set (v),} \quad Z = 3(2) + 2(0) = 6$$

$$\text{For Set (vi),} \quad Z = 3(3) + 2(1) = 11$$

Thus, solution (vi) is optimal with a profit of 11.

This is how a problem is solved by the trial and error method. But this method becomes cumbersome and time consuming if m and n are large. In such cases we shall have to solve numerous sets of simultaneous equations. Hence, we use another method known as the Simplex method to solve the linear programming problems.

3.3 SIMPLEX METHOD OF SOLVING A LINEAR PROGRAMMING PROBLEM

The Simplex method is an improvement over the trial and error method. It is an iterative optimising technique where we first find an initial basic solution (a vertex) and then proceed to an adjacent vertex and continue moving from adjacent vertex to another adjacent vertex till an optimal solution is attained. In this method, the value of the objective function improves with each solution and the optimum solution is achieved in a finite number of steps.

Let us explain the steps for solving the LPP by the Simplex method.

Steps for Computation of an Optimum Solution by Simplex Method

Suppose we have to maximise (or minimise) Z , a linear function of n basic variables $X_1, X_2, \dots, X_n$. The LPP is written as:

$$\text{Maximise } Z = C_1X_1 + C_2X_2 + \dots + C_nX_n \quad \dots (1)$$

subject to the constraints:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &\leq b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &\leq b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &\leq b_m \end{aligned} \quad \dots (2)$$

$$\text{and} \quad x_j \geq 0 \text{ for all } j=1, 2, \dots, n \quad \dots (3)$$

You can watch the video lecture series of Prof. G. Srinivasan on Simplex method on the following link: <http://youtube/qxls3cYg8to?list=PLD8C5517267A8D5>. In these video lectures, all steps are explained in detail.

where the constants $C_1, C_2, \dots, C_n$ are the **cost coefficients** of decision variables. Let (a_{ij}) be an $m \times n$ real matrix and $\{b_1, b_2, \dots, b_m\}$ be a set of constants.

The linear function Z given in equation (1) is called the **objective function**. The set of inequalities given in equation (2) is called **constraints** of LPP and the inequalities given in equation (3) are known as **non-negative restrictions** of LPP (which means that all x_j values are non-negative).

We convert the LPP into standard form by adding slack variables $s_1, s_2, \dots, s_m$ (refer to Sec. 2.5 of Unit 2):

$$\text{Maximise } Z = C_1X_1 + C_2X_2 + \dots + C_nX_n + 0s_1 + 0s_2 + \dots + 0s_m \quad \dots (4)$$

subject to the given constraints:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n + s_1 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n + s_2 &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n + s_m &= b_m \end{aligned} \quad \dots (5)$$

$$\text{and } x_j \geq 0 \text{ and } s_i \geq 0 \text{ for all } i = 1, 2, \dots, m \text{ and } j = 1, 2, \dots, n \quad \dots (6)$$

The initial simplex table is formed as follows:

Table 1: Initial Simplex Table

$C_j \rightarrow$			C_1	C_2	$\dots C_r$	$\dots 0$	$0 \dots$	
Basic variables	Profit/unit (C_B)	Quantity	X_1	X_2	$\dots X_r$	$\dots S_1$	$S_2 \dots$	Replacement Ratios
s_1	0	b_1	a_{11}	a_{12}	$\dots a_{1r}$	1	0	
s_2	0	b_2	a_{21}	a_{22}	$\dots a_{2r}$	.	1	
.	.	.	.	.	.	.	.	
.	.	.	.	.	.	.	.	
.	.	.	.	.	.	.	.	
$Z \quad z_j \rightarrow$			$C_B X_1$	$C_B X_2$	$\dots C_B X_r$	$C_B S_1$	$C_B S_2$	
$c_j - z_j \rightarrow$								

Note that in the initial simplex table (Table 1), the slack variables are kept in the column of basic variables. Once we obtain the values of $c_j - z_j$, the nature of the solution could be any one of the following:

1. If all $c_j - z_j \leq 0$, the solution under test is an **optimal** solution.
2. If at least one value of $c_j - z_j$ is positive and corresponding to the most positive $c_j - z_j$, all elements of the column X_j are negative or zero, the solution under test is an **unbounded** solution.

3. Suppose at least one value of $c_j - z_j$ is positive. Suppose the most positive value is, say, $c_2 - z_2$ and at least one entry in the column of X_2 is positive. Then the solution under test is **not optimal**. The most positive entry could be any of $(c_j - z_j)$.

Then we proceed as follows to obtain the optimal solution:

- Let X_r be the variable which corresponds to the most positive value of $c_j - z_j$. This variable is called the **incoming variable**.
- We divide the values of the Quantity (Qty) column by the corresponding positive values in the column of X_r . These ratios are called **Replacement Ratios (RR)**. Note that we do not consider the negative values in the column of X_r for calculating RR. Then we select the minimum RR. The basic variable corresponding to this value of the RR is called the **outgoing variable**. It is called outgoing variable because it is removed (goes out) from the next simplex table. The element at the intersection of the row corresponding to the outgoing variable and the column corresponding to the incoming variable is called the **key element** or **leading element** or **pivotal element**.
- We convert the key element to unity by dividing all entries in the row by the key element itself.
- In the next step, we would like that the values of all other elements in the column corresponding to key element are zero. For this we carry out suitable operations on each row using the row containing the key element.
- We repeat the procedure until either an optimal solution is obtained or there is an indication of unbounded solution.

For understanding the Simplex method, you should carefully go through the solution of the following example:

Example 1: Maximise $Z = 3x_1 + 2x_2$

subject to the constraints:

$$x_1 + x_2 \leq 4$$

$$x_1 - x_2 \leq 2$$

$$x_1 \geq 0, x_2 \geq 0$$

Solution: Let us re-write the given LPP in the standard form as follows:

$$\text{Max. } Z = 3x_1 + 2x_2 + 0s_1 + 0s_2$$

subject to the constraints:

$$x_1 + x_2 + s_1 = 4$$

$$x_1 - x_2 + s_1 = 2$$

$$x_1, x_2, s_1, s_2 \geq 0.$$

The variables s_1 and s_2 are the slack variables.

1. We now explain **how to construct the initial simplex table** in the following steps:

Note that in this problem,
 $a_{11}=1, a_{12}=1, a_{13}=1, a_{14}=0,$
 $a_{21}=1, a_{22}=-1, a_{23}=0,$
 $a_{24}=1, b_1=4$ and $b_2=2$.

- i) First of all, we write the basic variables s_1 and s_2 in the first column and the variables x_1, x_2, s_1, s_2 in the second row as shown in Table 2. Then we write the values of coefficients a_{ij} (read margin remark). We also write the cost coefficients (c_j) of the objective function in the first row above the corresponding basic variables at the top of the table.

Table 2: Initial Simplex Table

			$c_j \rightarrow$	3	2	0	0
Basic Variables	Profit/ Unit	Qty	x_1	x_2	s_1	s_2	
s_1			1	1	1	0	
s_2			1	-1	0	1	
	$Z =$	$z_j \rightarrow$ $c_j - z_j \rightarrow$					

- ii) Note that the columns corresponding to the basic variables in initial simplex tables in the Simplex method form an identity matrix. For example, in Table 2, the basic variables are s_1 and s_2 and their coefficients in the constraints are:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

which form an identity matrix.

- iii) In the column with the caption "Profit/Unit", we write the coefficients c_j of the basic variables in the objective function. These coefficients have the value zero for basic variables s_1 and s_2 . In the column with the caption "Qty (Quantity)", we write the values of the basic variables obtained from the constraints by setting $x_1 = 0, x_2 = 0$, i.e., $s_1 = 4, s_2 = 2$. Note that these are just the constants in the right hand side of the constraint equations. So, these constants are written in "Qty" column. Thus, we get the complete initial simplex table (Table 3).

Table 3: Complete Initial Simplex Table

			$c_j \rightarrow$	3	2	0	0
Basic Variables	Profit/ Unit	Qty	x_1	x_2	s_1	s_2	
s_1	0	4	1	1	1	0	
s_2	0	2	1	-1	0	1	
	$Z =$	$z_j \rightarrow$ $c_j - z_j \rightarrow$					

- iv) The value of Z in column of Table 3 is obtained from the equation

$$Z = \sum c_j b_j$$

Since $c_1 = 0, b_1 = 4, c_2 = 0$ and $b_2 = 2$ in the initial simplex table, we get

$$Z = 0 \times 4 + 0 \times 2 = 0$$

The values of z_j 's are obtained from the equation

$$z_j = \sum_{i=1}^m c_i a_{ij}$$

where m is the number of rows. In this example $m = 2$ and

$$\therefore z_1 = c_1 a_{11} + c_2 a_{21} = 0 \times 1 + 0 \times 1 = 0$$

Similarly,

$$z_2 = 0 \times 1 + 0 \times (-1) = 0$$

$$z_3 = 0 \times 1 + 0 \times 0 = 0$$

$$z_4 = 0 \times 0 + 0 \times 1 = 0$$

Thus, z_j 's are 0, 0, 0, 0.

- v) Next we obtain $c_j - z_j$, known as **net-evaluations**, on subtracting z_j from c_j . These are $3 - 0$, $2 - 0$, $0 - 0$, $0 - 0$, or 3, 2, 0, 0.

Thus, the table takes the following form (Table 4):

Table 4: Simplex Table

	$c_j \rightarrow$		3	2	0	0
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2
s_1	0	4	1	1	1	0
s_2	0	2	1	-1	0	1
	$Z = 0$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	0 3	0 2	0 0	0 0

- vi) Now, we select the most positive value of $c_j - z_j$, which is 3 in this case. This corresponds to the variable x_1 which becomes the **incoming variable**. We shall enter it as a basic variable in the next simplex table. So, one of the variables s_1, s_2 will now be the outgoing variable which would be replaced by x_1 . To find out which one of these variables (s_1 or s_2) is the outgoing variable, we determine the replacement ratio (RR). Recall that RR for any row is obtained by dividing the value of the Qty column by the corresponding value of the Incoming variable for that particular row. For example, consider the first row in Table 4. The value of Qty in the first row is 4 and the value of the incoming variable x_1 in this row is 1.

Therefore, RR for the first row is $\frac{4}{1}$. For the second row, Qty is 2

and the value of x_1 is 1. Hence, RR is $\frac{2}{1}$. RR is shown in the last column of the next simplex table (Table 5).

The outgoing variable is the variable for which RR minimum. In Table 5, RR in the second row is minimum. It corresponds to s_2 and hence s_2 is the outgoing variable.

The element which lies at the intersection of the column of incoming variable (x_1) and the row of outgoing variable (s_2) is called the key element. Here it is 1 and is enclosed by the rectangle as shown in Table 5.

Table 5: Simplex Table

	$c_j \rightarrow$		3	2	0	0	
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2	R.R.
s_1	0	4	1	1	1	0	$4/1 = 4$
(outgoing variable) s_2	0	2	1	-1	0	1	$2/1 = 2$
	$Z = 0$	$z_j \rightarrow$	0	0	0	0	
		$c_j - z_j \rightarrow$	3	2	0	0	

Key (or Pivotal or Leading) element

The objective function $Z = 0$ at $x_1 = 0$ and $x_2 = 0$ (since x_1 and x_2 do not appear in the column of basic variables, these are non-basic variables. The values of non-basic variables are taken as zero). This is the initial solution, which we shall improve.

2. Now, we form the next simplex table to find the adjacent vertex, i.e., the improved solution. The steps for forming this table are explained below:

- i) The initial simplex table (Table 3) has revealed that x_1 is an incoming variable which will enter in place of s_2 , the outgoing variable. The cost coefficient of s_2 will also be replaced by the cost coefficient of x_1 in "Profit/Unit" column. In this case its value is 3. Therefore, now the simplex table takes the form of Table 6:

Table 6: Simplex Table

	$c_j \rightarrow$		3	2	0	0
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2
s_1	0					
x_1	3					

Other entries will be filled up as explained below:

- ii) Since x_1 has entered as a basic variable, the coefficients of x_1 along with s_1 should form an identity matrix; i.e., the column corresponding to x_1 should be $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Thus, we have to make the key element unity and

the other element zero. Note that it is already unity in this case (see Table 5). Had it been any number other than unity, we would have divided the row containing leading element by the leading element itself, excluding the elements of the column "profit/unit".

So, the Table takes the form:

Table 7: Simplex Table

	$c_j \rightarrow$		3	2	0	0
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2
s_1	0	4	1	1	1	0
x_1	3	2	1	-1	0	1

- iii) Now, we have to make the other element in the column of key element (x_1) zero. In this case, its value is 1 ($a_{11} = 1$). For this, we multiply the row of the key element (excluding profit/unit) by negative of the element a_{11} (in this case) and add it to the first row. This row operation is shown below:

First row of Table 7	→	4	1	1	1	0
Second row of Table 7	→	-2	-1	1	0	-1
(on multiplying by -1)						
Sum of the rows		2	0	2	1	-1

Thus, the sum of the rows is the new first row (excluding profit/unit), which replaces the first row of Table 7. We get Table 8 as follows:

Table 8: Simplex Table

	$c_j \rightarrow$		3	2	0	0
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2
s_1	0	2	0	2	1	-1
x_1	3	2	1	-1	0	1

Note that now the matrix for the basic variables s_1 and x_1 in Table 8 is the identity matrix.

- iv) Next we calculate z_j and $c_j - z_j$ again as explained in Steps 1(iii) and 1(iv) of this section. The resulting simplex table is given below:

Table 9: Simplex Table

	$c_j \rightarrow$		3	2	0	0	
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2	RR
$\leftarrow s_1$	0	2	0	2	1	-1	$2/2=1 \leftarrow$
x_1	3	2	1	-1	0	1	-
	$Z = 0 \times 2 + 3 \times 2 = 6$	$z_j \rightarrow$	$0 \times 0 + 3 \times 1 = 3$	$0 \times 2 + 3 \times (-1) = -3$	$0 \times 1 + 3 \times 0 = 0$	$0 \times (-1) + 3 \times 1 = 3$	
		$c_j - z_j \rightarrow$	0	5 ↑	0	-3	

Here $Z = 6$ at $x_1 = 2$ (see the value in the Qty column) and $x_2 = 0$ (x_2 being non-basic variable).

In Table 9, the incoming variable is x_2 corresponding to the most positive value of $c_j - z_j = 5$. The key element is 2. We find the RR by dividing the elements in the Qty column by the corresponding elements in the column of the incoming variable x_2 and ignore the negative or zero values of the incoming variable. We have to divide by only the positive values. So, in this case, there will be only one RR and that will be considered as minimum RR. This implies that s_1 is the outgoing variable. If none of the elements in the column of incoming variable is positive, then the given LPP has an unbounded solution and we will stop there.

3. For the next simplex table, x_2 will enter in place of s_1 as a basic variable and accordingly we shall write the cost coefficient of x_2 in the LPP as the value for the column of profit/unit, i.e., 2. The element 2 enclosed in a rectangle in Table 9 is the key element. So we shall divide the row containing the key element by the key element itself, i.e., by 2, excluding the values in the column of profit/unit. Thus, we get Table 10:

Table 10: Simplex Table

	$c_j \rightarrow$		3	2	0	0
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2
x_2	2	1	0	1	1/2	-1/2
x_1	3	2	1	-1	0	1

Now the coefficients of x_2 and x_1 have to form an identity matrix, i.e., the column corresponding to x_2 should be $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. We have already made the

key element unity. Now, to make the second element, (i.e., -1) in its column as zero, we just add the first row corresponding to x_2 , to the second row of Table 10 excluding the values in the column of profit/unit as follows:

First row of Table 10	$\rightarrow$	1	0	1	1/2	-1/2
Second row of Table 10 (on multiplying by 1)	$\rightarrow$	2	1	-1	0	1
Sum of the rows		3	1	0	1/2	1/2

We write the sum above as the second row, excluding profit/unit as shown in Table 11. Then we can obtain Z , z_j , and $c_j - z_j$ as explained in Steps 1(iii) and (iv) and also in Step 2(iii). The complete resulting simplex table is given below:

Table 11: Simplex Table

	$c_j \rightarrow$		3	2	0	0	RR
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2	
x_2	2	1	0	1	1/2	-1/2	
x_1	3	3	1	0	1/2	1/2	
	$Z = 2 \times 1 + 3 \times 3 = 11$	$z_j \rightarrow$	$2 \times 0 + 3 \times 1 = 3$	$2 \times 1 + 3 \times 0 = 2$	$2 \times 1/2 + 3 \times 1/2 = 5/2$	$2 \times (-1/2) + 3 \times (1/2) = 1/2$	
	$c_j - z_j \rightarrow$		0	0 $\uparrow$	-5/2	-1/2	

Now, in Table 11 none of the net-evaluations, i.e., the values of $c_j - z_j$ are positive. Therefore, the optimum solution is attained for $x_1 = 3$ and $x_2 = 1$, the values of x_1 and x_2 in the Qty column of Table 11.

At this stage, it is necessary to check whether any of the non-basic variables (other than those appearing in the first column, i.e., the column with caption "Basic Variables", i.e., x_1 and x_2 in Table 11) has value 0 in the net-evaluation row. If "yes", then the LPP has multiple optimum

solutions. If “not”, then we stop here concluding that the LPP has the unique solution. In this example, it is given by

Max. $Z = 11$ at $x_1 = 3$ and $x_2 = 1$ (see the values in the Qty column).

We discuss the case of multiple optimum solutions in the next example.

Example 2: Max. $Z = 6x + 3y$

subject to the constraints:

$$2x + 5y \leq 120$$

$$2x + y \leq 40$$

$$x \geq 0, y \geq 0$$

Solution: Rewriting the given LPP in the standard form, we have

$$\text{Max. } Z = 6x + 3y + 0s_1 + 0s_2$$

subject to the constraints:

$$2x + 5y + s_1 = 120$$

$$2x + y + s_2 = 40$$

$$x, y, s_1, s_2 \geq 0$$

We form the initial simplex table (Table 1) as explained in Example 1:

Table 1: Initial Simplex Table

	$c_j \rightarrow$		6	3	0	0	
Basic Variables	Profit/Unit	Qty	x	y	s_1	s_2	RR
s_1	0	120	2	5	1	0	$120/2=60$
$\leftarrow s_2$	0	40	2	1	0	1	$40/2 = 20$
	$Z = 0$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	0 6 $\uparrow$	0 3	0 0	0 0	

Note from Table 1 that $c_j - z_j = 6$ is maximum for x and RR is minimum for s_2 . Therefore, x is the incoming variable and s_2 is the outgoing variable. Then in the second simplex table, x will enter in place of s_2 as a basic variable. Profit/unit will be written accordingly. The key element is 2 and enclosed by the rectangle in Table 1. To make the key element unity, we divide the second row in Table 1 by 2 excluding the values of the column of profit/unit. So, we get Table 2 as follows:

Table 2: Simplex Table

	$c_j \rightarrow$		6	3	0	0
Basic Variables	Profit/Unit	Qty	x	y	s_1	s_2
s_1	0	120	2	5	1	0
x	6	20	1	$1/2$	0	$1/2$

Now we have to make the other element in column of x zero so that the coefficients of s_1 and x form the identity matrix. So we multiply the second row of Table 2 (excluding the elements in the column of profit/unit) by the

negative of the element (a_{ij}) in the row of s_1 and column of x in Table 1, i.e., by -2 and then add it to the first row of the Table 2 as follows:

Second row of Table 2	→	-40	-2	-1	0	-1
(after multiplying by -2)						
First row of the Table 2	→	120	2	5	1	0
Sum of the rows	→	80	0	4	1	-1

We also calculate Z , z_j , and $c_j - z_j$. So the resulting completed simplex table is as follows:

Table 3: Simplex Table

	$c_j \rightarrow$		6	3	0	0	RR
Basic Variables	Profit/Unit	Qty	x	y	s_1	s_2	
s_1	0	80	0	4	1	-1	
x	6	20	1	1/2	0	1/2	
	$Z = 120$	$z_j \rightarrow$	6	3	0	3	
		$c_j - z_j \rightarrow$	0	0	0	-3	

As none of the net-evaluations is positive, the optimum solution is attained. The optimum solution is

$$\text{Max. } Z = 120, \quad \text{when } x = 20 \text{ (see the value in the Qty column),}$$

$$y = 0 \text{ (since } y \text{ is non-basic variable)}$$

But we also note that the non-basic variable y has value zero in its net-evaluation row in Table 3. Therefore, the given LPP has multiple optimal solutions. To find another optimal solution, let us find another vertex at which $\text{Max. } Z = 120$.

So, another simplex table has to be formed. Here, instead of selecting the column corresponding to the most positive element in the net-evaluation row, we select the column of non-basic variable which has zero in the net-evaluation row, i.e., the column of y shown in Table 4.

Table 4: Simplex Table

	$c_j \rightarrow$		6	3	0	0	RR
Basic Variables	Profit/Unit	Qty	x	y	s_1	s_2	
$\leftarrow s_1$	0	80	0	4	1	-1	$80/4=20 \leftarrow$
x	6	20	1	1/2	0	1/2	$20/(1/2)=40$
	$Z = 120$	$z_j \rightarrow$	6	3	0	3	
		$c_j - z_j \rightarrow$	0	0	0	-3	

↑

Note that in Table 4, the key element is 4 and the minimum RR corresponds to s_1 . So s_1 is the outgoing variable and y is the incoming variable. We form the next simplex table (Table 5) following the steps explained for forming Table 3. The resulting simplex table (Table 5) is given as follows:

Table 5: Simplex Table

	$c_j \rightarrow$		6	3	0	0	RR
Basic Variables	Profit/Unit	Qty	x	y	s_1	s_2	
y	3	20	0	1	1/4	-1/4	
x	6	10	1	0	-1/8	5/8	
	$Z = 120$	$z_j \rightarrow$	6	3	0	3	
		$c_j - z_j \rightarrow$	0	0	0	-3	

You should verify all entries of Table 5 before studying further. Note that for forming Table 5, we have first divided the first row by the value of key element to make the key element unity. Then we obtain the second row of Table 5 as follows:

First row of the Table 5 $\rightarrow$ -10 0 -1/2 -1/8 1/8
(after multiplying by -1/2)

Second row of the Table 4 $\rightarrow$ 20 1 1/2 0 1/2

Sum of the rows $\rightarrow$ 10 1 0 -1/8 5/8

From the above simplex table (Table 5), we find that Max. $Z = 120$ at (10, 20) also (see the values of $x = 10$, $y = 20$ in the Qty column).

So, we have two vertices at which the maximum value of Z is the same, i.e., 120. So, the other solutions of the LPP are obtained as follows:

First ordinate of other solutions

$$= t \times (\text{First ordinate of first vertex}) \\ + (1 - t) \times (\text{First ordinate of second vertex})$$

Second ordinate of other solutions

$$= t \times (\text{second ordinate of first vertex}) \\ + (1 - t) \times (\text{second ordinate of second vertex})$$

So, the other solutions are given as

$$\text{First ordinate} = t \times 10 + (1 - t) \times 20 = 10t + 20 - 20t = 20 - 10t$$

$$\text{Second ordinate} = t \times 20 + (1 - t) \times 0 = 20t$$

The other solutions are $(20 - 10t, 20t)$, $0 \leq t \leq 1$.

Now, you should try to solve the following exercises.

E1) Maximise $Z = 2x_1 + 4x_2$

subject to the constraints:

$$x_1 + 2x_2 \leq 5$$

$$x_1 + x_2 \leq 4$$

$$x_1 \geq 0, x_2 \geq 0$$

E2) Solve the following LPP by the Simplex method:

$$\text{Maximise } Z = 100x_1 + 60x_2 + 40x_3$$

subject to the constraints:

$$x_1 + x_2 + x_3 \leq 100$$

$$10x_1 + 4x_2 + 5x_3 \leq 500$$

$$x_1 + x_2 + 3x_3 \leq 150$$

$$x_1, x_2, x_3 \geq 0$$

Before studying next section, match your answers with the solutions given in Sec. 2.7.

3.4 ARTIFICIAL VARIABLE TECHNIQUE AND THE BIG M METHOD

After converting the given LPP into standard form, sometimes we observe that some or all the variables whose coefficients in the constraints should form the identity matrix are missing in the initial simplex table. Such a situation is usually observed if the constraints equation(s) is (are) of the type “ $\geq$ ”. In such a situation, we introduce new type of variable(s) known as artificial variable(s). These are added to act as basic variables. They have no physical meaning and are used only to initiate the solution so that the simplex procedure may be adopted as usual till the optimal solution is obtained. The artificial variables are eliminated from the simplex table as and when they become non-basic variables. This technique is called the **artificial variable technique** for solving LPP. LPPs involving artificial variables are solved by either of the following methods:

- i) Big-M or Penalty Method
- ii) Two-phase Technique

In this unit, we discuss only the Big-M method for solving an LPP.

Big-M Method

In this method, the objective function coefficients impose a huge and hence unacceptable penalty. In case of maximisation, the objective function is modified by adding $-MA_1$, where M is arbitrarily large and A_1 is an artificial variable. If there are two artificial variables A_1 and A_2 , then $-MA_1 - MA_2$ is added to the objective function. Similar treatment is done for more artificial variables. The logic behind taking the coefficient as $-M$ is that we should never get the net-evaluation positive in the column of the artificial variable, i.e., the artificial variable should not enter again as a basic variable. M is very big and hence adding $-MA_1$ is the penalty to the objective function. Hence this method is called the **penalty method**.

Though $-M$ is a big penalty, it does not affect the objective function. This is because the value of the artificial variable should come out to be zero so that $-MA_1$ becomes zero. If the artificial variable remains as a basic variable till the final simplex table, then its value in the Qty column should be zero for the solution of LPP to exist. Otherwise, if in the final simplex table, an artificial variable appears as a basic variable and is non-zero, the LPP does not possess any feasible solution.

Example 3: Maximise $Z = x_1 + 2x_2$

subject to the constraints:

$$x_1 - x_2 \geq 3$$

$$2x_1 + x_2 \leq 10$$

$$x_1 \geq 0, x_2 \geq 0$$

Solution: The standard form of the LPP is

$$\text{Max } Z = x_1 + 2x_2 + 0s_1 + 0s_2$$

subject to the constraints:

$$x_1 - x_2 - s_1 = 3$$

$$2x_1 + x_2 + s_2 = 10$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Now, let us try to form simplex table as follows:

Table 1: Simplex Table

	$c_j \rightarrow$		1	2	0	0
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2
			1	-1	-1	0
			2	1	0	1

We cannot form the initial simplex table with the variables x_1, x_2, s_1, s_2 as there is only one variable s_2 that has a column of identity matrix. The

column $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is missing. Therefore, one more variable, i.e., artificial variable

A_1 (say) needs to be introduced in the first constraint so as to get $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Its

coefficient in the objective function will be taken as $-M$.

Thus, the objective function is

$$\text{Max. } Z = x_1 + 2x_2 + 0s_1 + 0s_2 - MA_1$$

subject to the constraints

$$x_1 - x_2 - s_1 + A_1 = 3$$

$$2x_1 + x_2 + s_2 = 10$$

$$x_1, x_2, s_1, s_2, A_1 \geq 0$$

The initial simplex table, therefore, is as follows:

Table 2: Initial Simplex Table

	$c_j \rightarrow$		1	2	0	0	-M	
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2	A_1	R.R.
$\leftarrow A_1$	-M	3	1	-1	-1	0	1	$3/1 = 3 \leftarrow$
s_2	0	10	2	1	0	1	0	$10/2 = 5$
	$Z = -3M$	$z_j \rightarrow$	-M	M	M	0	-M	
		$c_j - z_j \rightarrow$	1+M	2-M	-M	0	0	

↑

Here, since M is big, $1 + M$ is most positive and x_1 is the incoming variable.

The least replacement ratio is $\frac{3}{4}$ which corresponds to A_1 and hence it is the outgoing variable. Thus, the resulting simplex table is as follows:

Table 3: Simplex Table

	$c_j \rightarrow$		1	2	0	0	$-M$	
Basic Variables	Profit/unit	Qty	x_1	x_2	s_1	s_2	A_1	R.R.
x_1	1	3	1	-1	-1	0	$\times$	-
$\leftarrow s_2$	0	4	0	3	2	1	$\times$	$4/3 \leftarrow$
	$Z = 3$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	1	-1	-1	0	$\times$	
			0	3	1	0		

↑

We obtain the second row (excluding profit/unit) in Table 3 as follows:

First row of the Second Simplex table (Table 3) $\rightarrow -6 \quad -2 \quad 2 \quad 2 \quad 0 \quad \times$
(after multiplying by -2)

Second row of First Simplex table (Table 2) $\rightarrow 10 \quad 2 \quad 1 \quad 0 \quad 1 \quad \times$

Second row of the Third Simplex table $\rightarrow 4 \quad 0 \quad 3 \quad 2 \quad 1 \quad \times$

Once the artificial variable is removed from the basic variables, there is no need to do any computational work for it.

Thus, we get the resulting simplex table.

Table 4: Simplex Table

	$c_j \rightarrow$		1	2	0	0	$-M$	
Basic Variables	Profit/unit	Qty	x_1	x_2	s_1	s_2	A_1	
x_1	1	$13/3$	1	0	$-1/3$	$1/3$	$\times$	
x_2	2	$4/3$	0	1	$2/3$	$1/3$	$\times$	
	$Z = 21/3$ $= 7$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	1	2	1	1	$\times$	
			0	0	-1	-1	$\times$	

The first row (excluding profit/unit) of Table 4 is obtained as follows:

Second row of Third Simplex table (Table 4) $\rightarrow 4/3 \quad 0 \quad 1 \quad 2/3 \quad 1/3 \quad \times$
(after multiplying by 1)

First row of Second Simplex table (Table 3) $\rightarrow 3 \quad 1 \quad -1 \quad -1 \quad 0 \quad \times$

Sum of the rows $\rightarrow 13/3 \quad 1 \quad 0 \quad -1/3 \quad 1/3 \quad \times$

Since none of the net-evaluations is positive, the optimum solution is attained and is given by

$$\text{Max. } Z = 7 \quad \text{at} \quad x_1 = \frac{13}{3} \quad \text{and} \quad x_2 = \frac{4}{3}.$$

Now, you should try to solve the following exercises.

E3) Minimise $Z = 4x_1 + 2x_2$

subject to the constraints:

$$3x_1 + x_2 \geq 27$$

$$x_1 + x_2 \geq 21$$

$$x_1 + 2x_2 \geq 30$$

$$x_1, x_2 \geq 0$$

E4) Maximise $Z = x_1 + 2x_2$

subject to the constraints:

$$x_1 + x_2 \leq 4$$

$$x_1 + x_2 \geq 6$$

$$x_1, x_2 \geq 0$$

E5) Maximise $Z = 10x_1 + 4x_2$

subject to the constraints:

$$-x_1 + x_2 \leq 2$$

$$x_1 + x_2 \geq 4$$

$$x_1, x_2 \geq 0$$

Before going to the next section, match your solutions with the solutions given in Sec. 3.7.

3.5 DEGENERACY

Degeneracy is the process of obtaining a degenerate basic feasible solution in an LPP. Degeneracy may occur at the initial stage while forming the initial simplex table if the values of some basic variables are zero.

Degeneracy may also occur at any iteration stage if in any of the simplex tables there is a tie in the minimum replacement ratios. In case of tie, if we select any of the rows arbitrarily, it may be possible that the subsequent iterations may not produce improvements in the value of the objective function. This concept is known as cycling. Then we have to simply go back and select another row.

The situation of cycling may be avoided if in case of tie, the row is not selected arbitrarily but is selected adopting the following procedure:

Compute the non-negative ratios of the first column of the identity matrix (and not that of Qty column) to the entries of the entering variable. Then choose the minimum of the values occurring at the places of a tie. If there is a tie again, we compute the ratios of the second column of the identity matrix to the entries of the entering variable. We continue the process till the ratios do not break the tie.

Example 4: Maximise $Z = 3x_1 + 9x_2$

subject to the constraints:

$$x_1 + 4x_2 \leq 8$$

$$x_1 + 2x_2 \leq 4$$

$$x_1 \geq 0, x_2 \geq 0$$

Solution: The standard form of the LPP is

$$\text{Maximise } Z = 3x_1 + 9x_2 + 0s_1 + 0s_2$$

subject to the constraints:

$$x_1 + 4x_2 + s_1 = 8$$

$$x_1 + 2x_2 + s_2 = 4$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Therefore, the initial simplex table is as follows:

Table 1: Initial Simplex Table

	$c_j \rightarrow$		3	9	0	0	
Basic Variables	Profit/unit	Qty	x_1	x_2	s_1	s_2	R.R.
s_1	0	8	1	4	1	0	$8/4=2$
s_2	0	4	1	2	0	1	$4/2=2$
	$Z = 0$	$z_j \rightarrow$	0	0	0	0	
		$c_j - z_j \rightarrow$	3	9	0	0	

↑

In Table 1, there is tie in the minimum replacement ratio as it is 2 in each of the two rows. Now, we should not select any row arbitrarily. We will proceed as follows:

We divide the non-negative ratios of the first column of identity matrix,

i.e., $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ by the entries of the entering variable, i.e., $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$. So we have

replacement ratios as $\begin{pmatrix} 1/4 \\ 0/2 \end{pmatrix}$, i.e., $\begin{pmatrix} 1/4 \\ 0 \end{pmatrix}$. So here we have found the

minimum RR to be 0 and the tie has been broken. Therefore, we choose the second row. Thus, the completed initial simplex table is as follows:

Table 2: Complete Initial Simplex Table

	$c_j \rightarrow$		3	9	0	0	
Basic Variables	Profit/unit	Qty	x_1	x_2	s_1	s_2	R.R.
s_1	0	8	1	4	1	0	$2, \frac{1}{4}$
$\leftarrow s_2$	0	4	1	2	0	1	$2, 0 \leftarrow$
	$Z = 0$	$z_j \rightarrow$	0	0	0	0	
		$c_j - z_j \rightarrow$	3	9	0	0	

↑

Now, you can proceed yourself to form the next simplex table(s) and the answer you should get is Max. $Z = 18$ at $x_1 = 0, x_2 = 2$.

Let us now summarise the main points which have been covered in this unit.

3.6 SUMMARY

1. In **trial and error method**, the given LPP is first re-written in the standard form. After writing it in standard form, we set $n - m = 2$ variables equal to zero at a time if there are m constraints in n variables

- (where $n > m$). Some of the solutions so obtained may be infeasible if they have negative coordinates. Such solutions are dropped in view of non-negativity condition on all variables, including slack and surplus variables. The optimal solution lies at one of the vertices and hence we find the value of the objective function for each set of the solutions except the dropped ones.
2. The **Simplex method** is an improvement over the trial and error method. It is an iterative optimizing technique where we first find an initial basic solution (a vertex) and then proceed to an adjacent vertex and continue moving from adjacent vertex to another adjacent vertex till an optimal solution is attained. In this method, the value of the objective function improves with each solution and the optimum solution is achieved in a finite number of steps.
 3. If at least one of the net evaluations is positive and corresponding to the most positive net evaluation, all the elements of the column are negative or zero, the solution under test is **unbounded solution**.
 4. At the stage of final simplex table, it is necessary to check whether any of the non-basic variables (other than those appearing in the first column, i.e., the column with the caption "Basic Variables") has 0 value in the net-evaluation row. If "yes", then the LPP has **multiple optimum solutions**.
 5. After converting the given LPP into standard form, sometimes we observe that some or all of the variables whose coefficients in the constraints should form the identity matrix are missing in the initial simplex table. Such a situation is usually observed if the constraints equation(s) is (are) of the type " $\geq$ ". In such a situation, we introduce new type of variable(s) known as **artificial variable(s)**. These are added to act as basic variables. They have no physical meaning and are used only to initiate the solution so that the simplex procedure may be adopted as usual till the optimal solution is obtained. The artificial variables are eliminated from the simplex table as and when they become non-basic variables.
 6. In **Big M Method**, the objective function coefficients impose a huge and hence unacceptable penalty. In case of maximization, the objective function is modified by adding $-MA_1$, where M is arbitrarily large and A_1 is an artificial variable. If there are two artificial variables A_1 and A_2 , then $-MA_1 - MA_2$ is added to the objective function. Similar treatment is done for more artificial variables. While solving such a problem, the value of the artificial variable should come out to be zero in the final simplex table. If in the final simplex table, an artificial variable appears as a basic variable and is not at zero level, the LPP does not possess any **feasible solution**.
 7. While forming the initial simplex table, if the values of some basic variables are zero or if in any table, there is a tie in the minimum replacement ratio, then **degeneracy** is said to occur in a linear programming problem. In this case, the non-negative ratios of the first column of the identity matrix (and not that of Qty column) to the entries of the entering variable are computed and then the minimum of the values occurring at the places of tie is chosen. If there is a tie again, the ratios of the second column of the identity matrix to the entries of the entering variable are computed. The process is continued till the ratios do not break the tie.

3.7 SOLUTIONS/ANSWERS

- E1)** Proceeding in the same way as explained in Example 2, we obtain
 Max. $Z = 10$ at the two vertices $(0, 5/2)$ and $(3, 1)$.
 Max. $Z = 10$ at many other points also which are given as
 $(3 - 3t, 1 + 3t/2)$, $0 \leq t \leq 1$.

- E2)** Standard form of the LPP is

$$\text{Max. } Z = 100x_1 + 60x_2 + 40x_3 + 0s_1 + 0s_2 + 0s_3$$

subject to the constraints:

$$x_1 + x_2 + x_3 + s_1 = 100$$

$$10x_1 + 4x_2 + 5x_3 + s_2 = 600$$

$$x_1 + x_2 + 3x_3 + s_3 = 150$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

The initial simplex table is as follows:

Table 1: Initial Simplex Table

	$c_j \rightarrow$		100	60	40	0	0	0	
Basic Variables	Profit/Unit	Qty	x_1	x_2	x_3	s_1	s_2	s_3	RR
s_1	0	100	1	1	1	1	0	0	100
$\leftarrow s_2$	0	600	10	4	5	0	1	0	60
s_3	0	150	1	1	3	0	0	1	150
	$Z = 0$	$z_j \rightarrow$	0	0	0	0	0	0	
		$c_j - z_j \rightarrow$	100	60	40	0	0	0	



The resulting simplex table is

Table 2: Simplex Table

	$c_j \rightarrow$		100	60	40	0	0	0	
Basic Variables	Profit/Unit	Qty	x_1	x_2	x_3	s_1	s_2	s_3	RR
$\leftarrow s_1$	0	40	0	3/5	1/2	1	-1/10	0	200/3
x_1	100	60	1	2/5	1/2	0	1/10	0	150
s_3	0	90	0	3/5	5/2	0	-1/10	1	150
	$Z = 6000$	$z_j \rightarrow$	100	40	50	0	10	0	
		$c_j - z_j \rightarrow$	0	20	-10	0	-10	0	



The resulting simplex table:

Table 3: Simplex Table

	$c_j \rightarrow$		100	60	40	0	0	0	
Basic Variables	Profit/Unit	Qty	x_1	x_2	x_3	s_1	s_2	s_3	RR
x_2	60	200/3	0	1	5/6	5/3	-1/6	0	
x_1	100	100/3	1	0	1/6	-2/3	1/6	0	
s_3	0	50	0	0	1	-1	0	1	
	$Z = 22000/3$	$z_j \rightarrow$	100	60	200/3	100/3	20/3	0	
		$c_j - z_j \rightarrow$	0	0	-80/3	-	-20/3	0	
						100/3			

Explanation to get the second row:

First row of Table 3

$$\begin{array}{ccccccc} -\frac{80}{3} & 0 & -\frac{2}{5} & -\frac{1}{3} & -\frac{2}{3} & -\frac{1}{15} & -\frac{2}{5} \end{array}$$

Second row of Table 2

$$\begin{array}{ccccccc} 60 & 1 & \frac{2}{5} & \frac{1}{2} & 0 & \frac{1}{10} & 0 \end{array}$$

$$\begin{array}{ccccccc} \text{Sum of the rows} & 100/3 & 1 & 0 & 1/6 & -2/3 & 1/6 & 0 \end{array}$$

Explanation to get the third row:

$$\begin{array}{ccccccc} \text{First row of Table 3} & -40 & 0 & -\frac{3}{5} & -\frac{1}{2} & -1 & -\frac{1}{10} & 0 \end{array}$$

$$\begin{array}{ccccccc} \text{Third row of Table 3} & 90 & 0 & \frac{3}{5} & \frac{5}{2} & 0 & -\frac{1}{10} & 1 \end{array}$$

$$\begin{array}{ccccccc} \text{Sum of the rows} & 50 & 0 & 0 & 1 & -1 & 0 & 1 \end{array}$$

As none of the net-evaluations in the simplex table given in Table 3 is positive, therefore optimal solution is attained and is given by

$$Z = \frac{22000}{3} \quad \text{at} \quad x_1 = \frac{100}{3}, \quad x_2 = \frac{200}{3}, \quad x_3 = 0$$

E3) Standard form of the LPP is

$$\text{Max. } -Z = -4x_1 - 2x_2 + 0s_1 + 0s_2 + 0s_3$$

subject to the constraints:

$$3x_1 + x_2 - s_1 = 27$$

$$x_1 + x_2 - s_2 = 21$$

$$x_1 + 2x_2 - s_3 = 30$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

But, if we form the initial simplex table using the above equations for constraints, we do not find any basic variable as no column of identity

matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ exists. Hence, we will introduce three artificial

variables to initiate the solution. Thus, we have

$$\text{Max. } Z' = -4x_1 - 2x_2 + 0s_1 + 0s_2 + 0s_3 - MA_1 - MA_2 - MA_3, \quad Z' = -Z$$

subject to the constraints:

$$3x_1 + x_2 - s_1 + A_1 = 27$$

$$x_1 + x_2 - s_2 + A_2 = 21$$

$$x_1 + 2x_2 - s_3 + A_3 = 30$$

$$x_1, x_2, s_1, s_2, s_3, A_1, A_2, A_3 \geq 0$$

Thus, the initial simplex table is

Table 1: Initial Simplex Table

	$c_j \rightarrow$		-4	-2	0	0	0	-M	-M	-M	
Basic Variable s	Profit/ Unit	Qty	x_1	x_2	s_1	s_2	s_3	A_1	A_2	A_3	RR
$\leftarrow A_1$	-M	27	3	1	-1	0	0	1	0	0	$27/3=9 \leftarrow$
A_2	-M	21	1	1	0	-1	0	0	1	0	$21/1=21$
A_3	-M	30	1	2	0	0	-1	0	0	1	$30/1=30$
	$Z = -78M$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	-5M -4+5M	-4M -2+4M	M -M	M -M	M -M	-M 0	-M 0	-M 0	



Now, the second and subsequent simplex tables can be formed and at the end, the answer you get is

$$\text{Max } Z' = -48, \quad \text{i.e., Max } -Z = -48$$

$$\text{i.e., Min. } Z = 48 \quad \text{when } x_1 = 3, x_2 = 18.$$

E4) Standard form of the linear programming problem is

$$\text{Max. } Z = x_1 + 2x_2 + 0s_1 + 0s_2$$

subject to the constraints:

$$x_1 + x_2 + s_1 = 4$$

$$x_1 + x_2 - s_2 = 6$$

$$x_1, x_2, s_1, s_2 \geq 0$$

But, with these equations for constraints we have only one basic variable with the column $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and hence we have to introduce an

artificial variable in the second equation so as to get the column $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Thus, we have

$$\text{Max. } Z = x_1 + 2x_2 + 0s_1 + 0s_2 - MA_1$$

subject to the constraints:

$$x_1 + x_2 + s_1 = 4$$

$$x_1 + x_2 - s_2 + A_1 = 6$$

$$x_1, x_2, s_1, s_2, A_1 \geq 0$$

Hence, the initial simplex table is

Table 1: Initial Simplex Table

	$c_j \rightarrow$		1	2	0	0	-M	
Basic Variables	Profit/ Unit	Qty	x_1	x_2	s_1	s_2	A_1	RR
$\leftarrow s_1$	0	4	1	1	1	0	0	$4 \leftarrow$
A_1	-M	6	1	1	0	-1	1	6
	$Z = -6M$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	-M 1+5M	-M 2+M	0 0	M -M	-M 0	



The resulting simplex table, therefore, is

Table 2: Simplex Table

	$c_j \rightarrow$		1	2	0	0	-M	RR
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2	A_1	
x_2	2	4	1	1	1	0	0	
A_1	-M	2	0	0	-1	-1	1	
	$Z = 8 - 2M$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	2 -1	2 0	$2 + M$ $-M - M$	M $-M$	$-M$ 0	

As none of the net-evaluations is positive, we stop here. We notice that the artificial variable appears as a basic variable and that too not at the zero level as $A_1 = 2$ and is non-zero. Therefore, the given LPP does not possess any feasible solution.

E5) Standard form of the given LPP is

$$\text{Max. } Z = 10x_1 + 4x_2 + 0s_1 + 0s_2$$

subject to the constraints:

$$4x_1 + x_2 - s_1 = 80$$

$$2x_1 + x_2 - s_2 = 60$$

$$x_1, x_2, s_1, s_2 \geq 0$$

In this problem also we do not get the columns of identity matrix without introducing artificial variables. Introducing artificial variables A_1, A_2 in the two equations, respectively, we have the objective function as

$$\text{Max. } Z = 10x_1 + 4x_2 + 0s_1 + 0s_2 - MA_1 - MA_2$$

subject to the constraints:

$$4x_1 + x_2 - s_1 + A_1 = 80$$

$$2x_1 + x_2 - s_2 + A_2 = 60$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

Thus, the initial simplex table is

Table 1: Initial Simplex Table

	$c_j \rightarrow$		10	4	0	0	-M	-M	
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2	A_1	A_2	RR
$\leftarrow A_1$	-M	80	4	1	-1	0	1	0	$20 \leftarrow$
A_2	-M	60	2	1	0	-1	0	1	30
	$Z = -140M$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	-6M $10 + 6M$	-2M $4 + 2M$	M $-M$	M $-M$	$-M$ 0	$-M$ 0	



The resulting simplex table, therefore, is as follows:

Table 2: Simplex Table

	$c_j \rightarrow$		10	4	0	0	-M	-M	
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2	A_1	A_2	RR
$\leftarrow x_1$	10	20	1	1/4	-1/4	0	$\times$	0	-
A_2	-M	20	0	1/2	1/2	-1	$\times$	1	40 $\leftarrow$
	$Z = 200 - 20M$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	10 0	$5/2 - M(1/2)$ $3/2 + (1/2)M$	$-5/2 - (1/2)M$ $5/2 + (1/2)M$	M $-M$	$\times$ $\times$	$-M$ 0	



The resulting simplex table is given below:

Table 3: Simplex Table

	$c_j \rightarrow$		10	4	0	0	-M	-M	RR
Basic Variables	Profit/Unit	Qty	x_1	x_2	s_1	s_2	A_1	A_2	
$\leftarrow x_1$	10	30	1	1/2	0	-1/2	$\times$	$\times$	-
s_1	0	40	0	1	1	-2	$\times$	$\times$	-
	$Z = 300$	$z_j \rightarrow$ $c_j - z_j \rightarrow$	10 0	5 -1	0 0	-5 5	$\times$ $\times$	$\times$ $\times$	



Note that corresponding to the most positive net-evaluation, i.e., 5, the values in the column of s_2 are $\begin{pmatrix} -1/2 \\ -2 \end{pmatrix}$. Since none of these values is positive, the solution of the given LPP is unbounded.

UNIT 4 TRANSPORTATION PROBLEM

Structure

- 4.1 Introduction
 - Objectives
- 4.2 Mathematical Formulation of the Transportation Problem
- 4.3 Methods of Finding Initial Basic Feasible Solution
 - North-West Corner Rule
 - Least Cost Method
 - Vogel's Approximation Method
- 4.4 Methods of Finding Optimal Solution
 - Stepping Stone Method
 - Modified Distribution (MODI) Method
- 4.5 Some Special Cases
 - Unbalanced Transportation Problem
 - Degeneracy
 - Alternative Optimal Solutions
 - Maximisation Transportation Problem
 - Prohibited Routes
- 4.6 Summary
- 4.7 Solutions / Answers

4.1 INTRODUCTION

In Units 2 and 3 of this block, we have discussed linear programming problems and methods of solving them. You have studied the graphical and simplex methods of solving LPPs in Unit 2 and Unit 3, respectively. Transportation problem is also one of the sub-classes of Linear Programming problems in which the objective is to transport various quantities (goods) initially stored at various origins/plants/factories to different destinations/distribution centres/warehouses in such a way that the total transportation cost is minimised. To achieve this objective, we must know the quantity of available supplies and the quantities demanded. Also, we must know the costs of shipping a unit from various origins to various destinations. Solving LPPs by either of the methods discussed in Units 2 and 3 in this block involves a lot of computational work and is quite cumbersome and difficult for solving the transportation problems. So in this unit, we discuss the methods, which are specifically applied for solving transportation problems.

Objectives

After studying this unit, you should be able to:

- define a transportation problem;
- determine the basic feasible solution of a transportation problem using North-West Corner Rule, Least Cost and Vogel's Approximation methods;
- state the conditions for performing optimality test;
- apply the Stepping Stone and Modified Distribution (MODI) methods of obtaining the optimal solution of a transportation problem; and

- solve the transportation problems for special cases such as unbalanced transportation problem, case of degeneracy, case of alternative solutions, maximisation transportation problem, problems with prohibited routes.

4.2 MATHEMATICAL FORMULATION OF THE TRANSPORTATION PROBLEM

Suppose a manufacturer of an item has m origins (plants/factories) situated at places $O_1, O_2, \dots, O_i, \dots, O_m$ and suppose there are n destinations (warehouses/distribution centres) at $D_1, D_2, \dots, D_j, \dots, D_n$. The aggregate of the capacities of all m origins is assumed to be equal to the aggregate of the requirements of all n destinations. Let C_{ij} be the cost of transporting one unit from origin i to destination j . Let a_i be the capacity/availability of items at origin i and b_j , the requirement/demand of the destination j . Then this transportation problem can be expressed in a tabular form as follows:

Origin	Destinations					Availability / Capacity
	D ₁	D ₂	...	D _j ...	D _n	
O ₁	C ₁₁	C ₁₂	...	C _{1j} ...	C _{1n}	a ₁
O ₂	C ₂₁	C ₂₂	...	C _{2j} ...	C _{2n}	a ₂
.	.	.		.	.	.
.	.	.		.	.	.
.	.	.		.	.	.
O _i	C _{i1}	C _{i2}	...	C _{ij} ...	C _{in}	a _i
.	.	.		.	.	.
.	.	.		.	.	.
.	.	.		.	.	.
O _m	C _{m1}	C _{m2}	...	C _{mj} ...	C _{mn}	a _m
Requirement/ Demand	b ₁	b ₂	...	b _j ...	b _n	Total

The condition for the existence of a feasible solution to a transportation problem is given as

$$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j \quad \dots(1)$$

Equation (1) tells us that the total requirement equals the total capacity. If it is not so, a dummy origin or destination is created to balance the total capacity and requirement.

Now let x_{ij} be the number of units to be transported from origin i to destination j and C_{ij} the corresponding cost of transportation. Then the total transportation cost is $\sum_{i=1}^m \sum_{j=1}^n C_{ij} x_{ij}$. It means that the product of the number of units allocated to each cell with the transportation cost per unit for that cell is calculated and added for all cells. Our objective is to allocate the units in such a way that the **total transportation cost is least**. The problem can also be stated as a linear programming problem as follows:

We are to minimise the total transportation cost

$$Z = \sum_{i=1}^m \sum_{j=1}^n C_{ij} x_{ij} \quad \dots(2)$$

subject to the constraints:

$$\begin{aligned} \sum_{j=1}^n x_{1j} &= a_1, & \sum_{j=1}^n x_{2j} &= a_2, \dots, & \sum_{j=1}^n x_{mj} &= a_m \\ \sum_{i=1}^m x_{i1} &= b_1, & \sum_{i=1}^m x_{i2} &= b_2, \dots, & \sum_{i=1}^m x_{in} &= b_n \end{aligned} \quad \dots(3)$$

and $x_{ij} \geq 0$ for all $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$.

We can solve this transportation problem by the Simplex method. However, the solution is very lengthy. Moreover, it is a cumbersome process since a large number of decision variables and artificial variables are involved. Hence, we use an alternative method of solving a transportation problem known as the **transportation method**. It requires far less computational effort as compared to the Simplex method. In the transportation method, we first obtain the initial basic feasible solution and then perform the optimality test as explained in Secs. 4.3 and 4.4.

Solution by transportation method involves making a transportation model in the form of a matrix, finding an initial basic feasible solution, performing optimality test and moving towards an optimal solution.

4.3 METHODS OF FINDING INITIAL BASIC FEASIBLE SOLUTION

We can use any one of the following methods to determine the initial basic feasible solution:

- i) **North-West Corner Rule**
- ii) **Least Cost Method**
- iii) **Vogel's Approximation Method**

Vogel's Approximation method generally gives a solution closer to the optimum solution. Hence, it is preferred over other methods.

We first discuss the North West Corner Rule.

4.3.1 North-West Corner Rule

The algorithm involved in the North-West Corner Rule (NWC) consists of the following steps.

We start with the north-west corner of the given problem and proceed as follows:

If the quantity needed at First Distribution Centre/Destination (b_1) is **less** than the quantity available at First Plant/Origin (a_1), we allocate a quantity equal to the requirement at First Distribution Centre to the cell (1, 1). At this stage, Column 1 is exhausted, so we cross it out. Since the requirement b_1 is fulfilled, we reduce the availability a_1 by b_1 and proceed to north-west corner of the resulting matrix, i.e., cell (1, 2).

If the quantity needed at First Distribution Centre/Destination (b_1) is **greater** than the quantity available at First Plant/Origin (a_1), we allocate a quantity equal to the quantity available at First Plant/Origin (a_1) to cell (1, 1). At this stage, Row 1 is exhausted, so we cross it out and proceed to north-west corner of the resulting matrix, i.e., cell (2, 1).

If the quantity needed at First Distribution Centre/Destination (b_1) is **equal** to the quantity available at First Plant/Origin (a_1), we allocate a quantity

equal to the requirement at First Distribution Centre/Destination (or the quantity available at First Plant/Origin). At this stage, both Column 1 as well as Row 1 is exhausted. We cross them out and proceed to the north-west corner of the resulting matrix, i.e., cell (2, 2).

We continue in this manner, until we reach the south-east corner of the original matrix.

Let us take up an example to further explain NWC rule.

Example 1: Apply the North-West Corner rule for finding the basic feasible solution of the following transportation problem:

Warehouse Factory	D	E	F	G	Capacity
A	42	48	38	37	160
B	40	49	52	51	150
C	39	38	40	43	190
Requirement	80	90	110	220	500

Solution: We start from the North West corner, i.e., the Factory A and Warehouse D. The quantity needed at the First Warehouse (Warehouse D) is 80, which is less than the quantity available (160) at the First Factory A. Therefore, a quantity equal to the requirement at Warehouse D is to be allocated to the cell (A, D). Thus, the requirement of Warehouse D is met by Factory A. So we cross out column 1 and reduce the capacity of Factory A by 80. Then we go to cell (A, E), which is the North-West corner of the resulting matrix.

Warehouse Factory	D	E	F	G	Capacity
A	42 (80)	48	38	37	160 (80)
B	40	49	52	51	150
C	39	38	40	43	190
Requirement	80 (0)	90	110	220	500

The resulting matrix is

Warehouse Factory	E	F	G	Capacity
A	48	38	37	80
B	49	52	51	150
C	38	40	43	190
Requirement	90	110	220	

Now, the quantity needed at the Second Warehouse (Warehouse E) is 90, which is greater than the quantity available (80) at the First Factory A. Therefore, we allocate a quantity equal to the capacity at Factory A, i.e., 80 to the cell (A, E). The requirement of warehouse E is reduced to 10. The capacity of Factory A is exhausted and has to be removed from the matrix. Therefore, we cross out row 1 and proceed to cell (B, E). This is the North-West corner of the resulting matrix.

Warehouse Factory	E	F	G	Capacity
A	48 (80)	38	37	80 (0)
B	49	52	51	150
C	38	40	43	190
Requirement	90 (10)	110	220	

The resulting matrix is

Warehouse Factory	E	F	G	Capacity
B	49	52	51	150
C	38	40	43	190
Requirement	10	110	220	

Now, the quantity needed at the Second Warehouse (Warehouse E) is 10, which is less than the quantity available at the Second Factory B, which is 150. Therefore, the quantity 10 equal to the requirement at Warehouse E is allocated to the cell (B, E). Hence, the requirement of Warehouse E is met and we cross out column 1. We reduce the capacity of Factory B by 10 and proceed to cell (B, F), which is the new North-West corner of the resulting matrix.

Warehouse Factory	E	F	G	Capacity
B	49 (10)	52	51	150 (140)
C	38	40	43	190
Requirement	10 (0)	110	220	

The resulting matrix is

Warehouse Factory	F	G	Capacity
B	52 (110)	51	140 (30)
C	40	43	190
Requirement	110 (0)	220	

Again, the quantity needed at the Third Warehouse (Warehouse F) is 110. It is less than the quantity available at the Second Factory (Factory B), which is 140. Therefore, a quantity equal to the requirement at Warehouse F is allocated to the cell (B, F). Since, the requirement of warehouse F is met, we cross out Column 1 and reduce the capacity of Factory B by 110. Then we proceed to cell (B, G), which is the new North-West corner of the resulting matrix given below:

Warehouse Factory	G	Capacity
B	51	30
C	43	190
Requirement	220	

Now, the quantity needed at the Fourth Warehouse (Warehouse G) is 220, which is greater than the quantity available at the Second Factory (Factory B). Therefore, we allocate the quantity equal to the capacity at Factory B to

the cell (B, G) so that the capacity of Factory B is exhausted and the requirement of Warehouse G is reduced to 190. Hence, we cross out Row 1 and proceed to cell (C, G).

Warehouse \ Factory	G	Capacity
B	51 (30)	30 (0)
C	43	190
Requirement	220 (190)	

The resulting matrix is

Warehouse \ Factory	G	Capacity
C	43	190
Requirement	190	

Thus, the allocations given using North-West corner rule are as shown in the following matrix along with the cost per unit of transportation:

Warehouse \ Factory	D	E	F	G	Capacity
A	42 (80)	48 (80)	38	37	160
B	40	49 (10)	52 (110)	51 (30)	150
C	39	38	40	43 (190)	190
Requirement	80	90	110	220	500

Thus, the total transportation cost for these allocations

$$\begin{aligned}
 &= 42 \times 80 + 48 \times 80 + 49 \times 10 + 52 \times 110 + 51 \times 30 + 43 \times 190 \\
 &= 3360 + 3840 + 490 + 5720 + 1530 + 8170 = 23110
 \end{aligned}$$

Now, you should try to solve the following exercises to practice the NWC rule.

E1) Find the basic feasible solution of the following problem using North-West Corner Rule:

Origin / Distribution Centre	1	2	3	4	5	6	Availability
1	4	6	9	2	7	8	10
2	3	5	4	8	10	0	12
3	2	6	9	8	4	13	4
4	4	4	5	9	3	6	18
5	9	8	7	3	2	14	20
Requirements	8	8	16	3	8	21	

Before studying the next section, match your answer with the answer given in Sec. 4.7.

4.3.2 Least Cost Method

The least cost method is also known as the Matrix Minimum method or Inspection method. It starts by making the first allocation to the cell for which the shipping cost (or transportation cost) per unit is lowest. The row

or column for which the capacity is exhausted or requirement is satisfied is removed from the transportation table. The process is repeated with the reduced matrix till all the requirements are satisfied. If there is a tie for the lowest cost cell while making any allocation, the choice may be made for a row or a column by which maximum requirement is exhausted. If there is a tie in making this allocation as well, then we can arbitrarily choose a cell for allocation.

Let us explain this method with the help of an example.

Example 2: Apply the Least Cost method for finding the Basic Feasible solution of the transportation problem of Example 1.

Solution: Here, the least cost is 37 in the cell (A, G). The requirement of the Warehouse G is 220 and the capacity of Factory A is 160. Hence, the maximum number of units that can be allocated to this cell is 160. Thus, Factory A is exhausted and the row has to be removed from the next matrix. The requirement of Warehouse G is reduced by 160.

Warehouse Factory	D	E	F	G	Capacity
A	42	48	38	37 160	160 0
B	40	49	52	51	150
C	39	38	40	43	190
Requirement	80	90	110	220 60	500

The reduced matrix, therefore, is

Warehouse Factory	D	E	F	G	Capacity
B	40	49	52	51	150
C	39	38	40	43	190
Requirement	80	90	110	60	

Now, the least cost is 38, which is in the cell (C,E). The requirement of the Warehouse E is 90 and the capacity of Factory C is 190. Hence, the maximum number of units that can be allocated to this cell is 90. Thus, Warehouse E is exhausted and has to be removed for the next matrix. So we cross out this column. Moreover, we reduce the capacity of factory C by 90.

Warehouse Factory	D	E	F	G	Capacity
B	40	49	52	51	150
C	39	38 90	40	43	190 100
Requirement	80	90 0	110	60	

The reduced matrix, therefore, is

Warehouse Factory	D	F	G	Capacity
B	40	52	51	150
C	39	40	43	100
Requirement	80	110	60	

The least cost in this matrix is 39, which is in the cell (C, D). The requirement of Warehouse D is 80 and the capacity of Factory C is 100. Hence, the maximum number of units that can be allocated to this cell is 80. Thus, the requirement of Warehouse D is exhausted and this column is removed for the next matrix. The capacity of Factory C is also reduced by 80.

Warehouse \ Factory	D	F	G	Capacity
B	40	52	51	150
C	39 (80)	40	43	100 (20)
Requirement	80 (0)	110	60	

Thus, the reduced matrix is

Warehouse \ Factory	F	G	Capacity
B	52	51	150
C	40	43	20
Requirement	110	60	

The least cost in this matrix is 40 which is in the cell (C, F). The requirement of Warehouse F is 110 and the capacity of Factory C is 20. Hence, the maximum number of units that can be allocated to this cell is 20. Thus, Factory C is exhausted and we remove the row for the next matrix. The requirement of Warehouse F is reduced by 20. It is now 90 in the reduced matrix.

Warehouse \ Factory	F	G	Capacity
B	52 (20)	51	150
C	40	43	20 (0)
Requirement	110 (90)	60	

The reduced matrix is

Warehouse \ Factory	F	G	Capacity
B	52	51 (60)	150 (90)
Requirement	90	60 (0)	

The least cost is 51 in the cell (B, G) and the requirement of warehouse G is 60 units. So we allocate 60 units to cell (B, G) and the remaining 90 units to the cell (B, F). Thus, the allocations given using Least Cost method are as shown in the following matrix along with the cost per unit of transportation:

Warehouse \ Factory	D	E	F	G	Capacity
A	42	48	38	37 (160)	160
B	40	49	52 (90)	51 (60)	150
C	39 (80)	38 (90)	40 (20)	43	190
Requirement	80	90	110	220	500

$$\begin{aligned}
 \text{Thus, the total transportation cost} &= 37 \times 160 + 52 \times 90 + 51 \times 60 + 39 \times 80 \\
 &\quad + 38 \times 90 + 40 \times 20 \\
 &= 5920 + 4680 + 3060 + 3120 + 3420 \\
 &\quad + 800 = 21000
 \end{aligned}$$

Note that this method has reduced the total transportation cost in comparison to the NWC rule.

Now, you should try to solve the following exercise and apply the Least Cost method.

E2) Find the basic feasible solution of the following problem using the Least Cost method:

Origin \ D. Centre	1	2	3	4	5	6	Availability
1	4	6	9	2	7	8	10
2	3	5	4	8	10	0	12
3	2	6	9	8	4	13	4
4	4	4	5	9	3	6	18
5	9	8	7	3	2	14	20
Requirement	8	8	16	3	8	21	

Before studying the next section, match your answer with the answer given in Sec. 4.7.

4.3.3 Vogel's Approximation Method (VAM)

We describe the procedure for finding the initial basic feasible solution by Vogel's Approximation method in the following steps:

1. For each row of the transportation table, we identify the least and second least costs. We find the difference between these two and display it to the right of that row in a new column formed by extending the table on the right. Likewise, we find such differences for each column and display it below that column in a new row formed for the purpose by extending the table at the bottom. The new column and row formed by extending the table at the right and bottom are labelled as **penalty column** and **penalty row**, respectively. If two cells in a row (or column) contain the same least costs then the difference is taken as zero.
2. From amongst these difference values displayed in the penalty column and the penalty row, we select the largest value (largest difference). We, allocate the maximum possible units to the least cost cell in the selected column or row. If a tie occurs amongst the largest differences, the choice may be made for that row or column, which has the least cost. In case there is a tie in such least cost as well, choice may be made from that row or column by which maximum requirements are exhausted. The cell so chosen is allocated the units and the corresponding exhausted row (or column) is removed (or ignored) from further consideration.
3. Next, we compute the column and row differences for the reduced transportation table and repeat the procedure until all column and row totals are exhausted.

This method is also called the **Penalty method**. Let us illustrate it with the help of an example.

Example 3: Apply the Vogel's Approximation Method for finding the Basic Feasible Solution for the transportation problem of Example 1.

Solution: In the first row, the least and the second least costs are 37 and 38 and their difference is 1. We write 1 in a new column created on the right. It is labelled **Penalty**. Similarly, the differences between the least and the second least costs in the second and third row, respectively, are $49-40=9$ and $39-38=1$. So we write the values (differences), i.e., 9 and 1 in the penalty column.

Next we find the differences of the least and second least elements of each of the columns D, E, F and G. These are $40-39=1$, $48-38=10$, $40-38=2$ and $43-37=6$, respectively. We write them in a newly created **penalty row** at the bottom of the table.

We now select the largest of these differences in the penalty row and column, which is 10 in this case. This value (10) corresponds to the second column (Column E) and the least cost in this column is 38. Hence the allocation of 90 units (the maximum requirement of Warehouse E) is to be made in the cell (C, E) from Factory C. Since the column corresponding to E is exhausted, it is removed for the next reduced matrix and the capacity of C is reduced by 90.

Warehouse Factory	D	E	F	G	Capacity	Penalty
A	42	48	38	37	160	1
B	40	49	52	51	150	9
C	39	38 (90)	40	43	190 (100)	1
Requirement	80	90 (0)	110	220	500	
Penalty	1	10	2	6		

The reduced matrix, therefore, is

Warehouse Factory	D	F	G	Capacity	Penalty
A	42	38	37	160	1
B	40	52	51	150	11
C	39	40	43	100	1
Requirement	80	110	220	410	
Penalty	1	2	6		

We now take the differences between the least and second least cost for each row and column of the reduced matrix. In the first row, the least and the second least costs are 37 and 38 and their difference is 1. We write it in the newly created penalty column. Similarly, we write the second difference element $51-40=11$ and third difference element $40-39=1$ in the second and third row of this column. Likewise, the differences of the smallest and second smallest elements of each of the columns D, F and G are $40-39=1$, $40-38=2$ and $43-37=6$, respectively. We write these in a newly created penalty row at the bottom of the table.

Now, we select the largest of these differences in the penalty row and column, which is 11 in this case. This value (11) corresponds to Row B. Since the least cost in row B is 40, we allocate 80 units (the maximum requirement of Warehouse D) to the cell (B, D). Thus, the requirement of

Warehouse D is exhausted and we can remove it. We also reduce the capacity of Factory B by 80 in the next reduced matrix.

Warehouse \ Factory	D	F	G	Capacity
A	42	38	37	160
B	(80) 40	52	51	160 (70)
C	39	40	43	100
Requirement	80 (0)	110	220	330

The reduced matrix is

Warehouse \ Factory	F	G	Capacity	Penalty
A	38	37	160	1
B	52	51	70	1
C	40	43	100	3
Requirement	110	220	330	
Penalty	2	6		

Again, in the first row, the least and the second least costs are 37 and 38 and their difference is 1. We write it to the right of this row in the newly created penalty column. Similarly, the second and third elements in the second and third rows of this column are $52-51=1$ and $43-40=3$, respectively. Likewise, the differences of the least and second least elements of each of the columns F and G are $40-38=2$ and $43-37=6$, respectively. We write these in a newly created penalty row at the bottom of the table. Now, we select the largest of these differences, which is 6 in this case. It corresponds to Column G and the least cost in this column is 37. Hence, we allocate 160 units (the maximum capacity of Factory A) to the cell (A, G). Since Row A is exhausted, it is removed for the next reduced matrix. We also reduce the requirement of Warehouse G by 160 units.

Warehouse \ Factory	F	G	Capacity
A	38	37 (160)	160 (0)
B	52	51	70
C	40	43	100
Requirement	110	220 (60)	170

The reduced matrix is

Warehouse \ Factory	F	G	Capacity	Penalty
B	52	51	70	1
C	40	43	100	3
Requirement	110	60	170	
Penalty	12	8		

Once again, the difference of the least and the second least costs in the first row is $52-51=1$. We write it in the newly created penalty column. Similarly, for the second row, the difference is $43-40=3$. Likewise, the differences of the least and second least elements of each of the columns F and G are $52-40=12$ and $51-43=8$, respectively. We write them in the newly created

penalty row at the bottom of the table. The largest of these differences is 12 in this case. It corresponds to Column F and the least cost in this column is 40. Hence, we allocate 100 units from the row (the maximum capacity of Factory C) to the cell (C, F). Since Row C is exhausted, it is removed and the requirement of Warehouse F is reduced to 10 for the next reduced matrix.

Warehouse \ Factory	F	G	Capacity
B	52	51	70
C	40 (100)	43	100 (0)
Requirement	110 (10)	60	70

The reduced matrix is

Warehouse \ Factory	F	G	Capacity
B	52	(60) 51	70 (10)
Requirement	10	(0) 60	10

At the end, of the 70 units available in Factory B, we allocate 60 units to the lower cost (51), i.e., to the cell (B, G) and the remaining 10 units to the cell (B, F).

The entire procedure of allocating units by Vogel's Approximation Method can be done in a single table also, as shown below:

Warehouse \ Factory	D	E	F	G	Capacity	Diff ₁	Diff ₂	Diff ₃	Diff ₄
A	42	48	38	37 (160)	160	1	1	1	-
B	40 (80)	49	52 (10)	51 (60)	150	9	11*	1	1
C	39	38 (90)	40 (100)	43	190	1	1	3	3
Requirement	80	90	110	220	500				
Diff ₁	1	10*	2	6					
Diff ₂	1	-	2	6					
Diff ₃	-	-	2	6*					
Diff ₄	-	-	12*	8					

Thus, the total transportation cost

$$\begin{aligned}
 &= 40 \times 80 + 38 \times 90 + 52 \times 10 + 40 \times 100 + 37 \times 160 + 51 \times 60 \\
 &= 3200 + 3420 + 520 + 4000 + 5920 + 3060 \\
 &= 20120
 \end{aligned}$$

This is the lowest total transportation cost among the three methods.

Now, you should apply Vogel's Approximation method to solve the following exercise.

E3) Obtain the basic feasible solution of the following problem using Vogel's Approximation method:

Origin / Distribution Centre	1	2	3	4	5	6	Availability
1	4	6	9	2	7	8	10
2	3	5	4	8	10	0	12
3	2	6	9	8	4	13	4
4	4	4	5	9	3	6	18
5	9	8	7	3	2	14	20
Requirement	8	8	16	3	8	21	

Before studying the next section, match your answer with the answer given in Sec. 4.7.

4.4 METHODS OF FINDING OPTIMAL SOLUTION

Once the initial basic feasible solution is determined, the optimality test is performed to find whether the obtained feasible solution is optimal or not. The optimality test is performed by applying one of the following methods:

- Stepping Stone Method**
- Modified Distribution (MODI) Method**

These methods reveal whether the initial basic feasible solution is optimal or not. They also improve the solution until the optimal solution is obtained. The Stepping Stone method is applied to a problem of small dimension as its application to a problem of large dimension is quite tedious and cumbersome. The MODI method is usually preferred over the Stepping Stone method.

Before discussing the methods of performing optimality test and finding optimal solutions, we explain the concept of independent allocations. We also state the conditions for performing optimality test.

Independent and Non-Independent Allocations

If it is not possible to form any closed loop through the allocations under consideration then the allocations are said to be **independent**. Here, formation of closed loop means that it is possible to travel from any allocation, back to itself by a series of horizontal and vertical jumps from one occupied cell (i.e., the cell containing allocation) to another, without a direct reversal of route (see Fig. 4.1a) . If such a loop can be formed using some or all of the allocations under consideration, the allocations are known as **non-independent**.

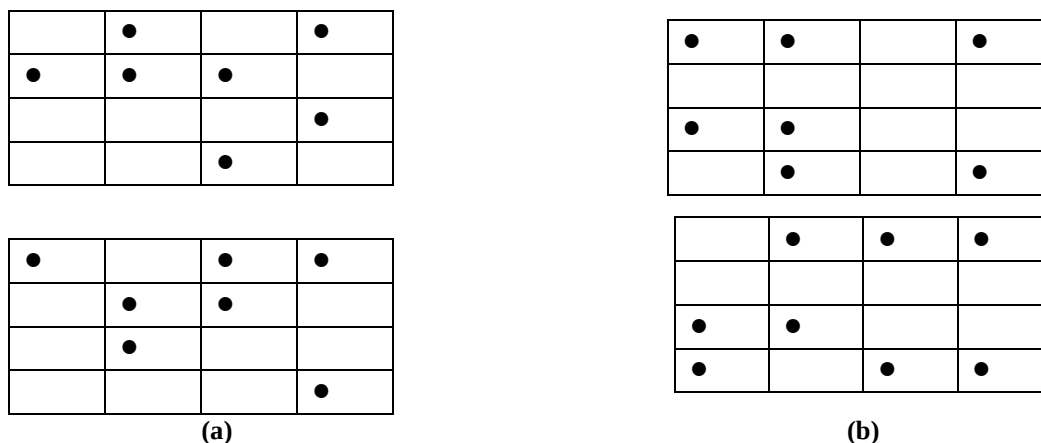


Fig. 4.1: Example of a) Independent Allocations; b) Non-Independent Allocations.

Note: Every loop has an even number of cells.

Conditions for Performing Optimality Test

An optimality test can be applied to that feasible solution which satisfies the following conditions:

1. It contains exactly $m + n - 1$ allocations where m and n represent the number of rows and columns, respectively, of the transportation table.
2. These allocations are independent.

Let us now discuss the methods of performing optimality test and hence finding optimal solutions for each of these methods.

4.4.1 Stepping Stone Method

In the Stepping Stone method for obtaining the optimal solution of a transportation problem, you should follow the steps given below:

1. Determine an initial basic feasible solution.
2. Evaluate all unoccupied cells for the effect of transferring one unit from an occupied cell to the unoccupied cell as follows:
 - a) Select an unoccupied cell to be evaluated.
 - b) Starting from this cell, form a closed path (or loop) through at least three occupied cells. The direction of movement is immaterial because the result will be the same in both directions. Note that except for the evaluated cell, all cells at the corners of the loop have to be occupied.
 - c) At each corner of the closed path, assign plus (+) and minus (−) sign alternatively, beginning with the plus sign for the unoccupied cell to be evaluated.
 - d) Compute the net change in cost with respect to the costs associated with each cell traced in the closed path.
 - e) Repeat steps 2(a) to 2(d) until the net change in cost has been calculated for all occupied cells.
3. If all net changes are positive or zero, an optimal solution has been arrived at. Otherwise go to step 4.
4. If some net changes are negative, select the unoccupied cell having the most negative net change. If two negative values are equal, select the one that results in moving more units into the selected unoccupied cell with the minimum cost.
5. Assign as many units as possible to this unoccupied cell.
6. Go to Step 2 and repeat the procedure until all unoccupied cells are evaluated and the value of net change, i.e., net evaluation is positive or zero.

Let us now take up an example of transportation problem and apply the Stepping Stone method for finding the optimal solution.

Example 4: A company is spending `1000 on transportation of its units from three plants to four distribution centres. The availability of unit per plant and requirement of units per distribution centre, with unit cost of transportation are given as follows:

D. Centres Plants	D ₁	D ₂	D ₃	D ₄	Availability
P ₁	19	30	50	12	7
P ₂	70	30	40	60	10
P ₃	40	10	60	20	18
Requirement	5	8	7	15	

What is the maximum possible saving by optimum distribution? Use the Stepping Stone method to solve the problem.

Solution: Applying the Vogel's Approximation Method to the problem, the initial feasible solution is given as follows:

D Cs Plants	D ₁	D ₂	D ₃	D ₄	Av.	diff ₁	diff ₂	diff ₃	diff ₄
P ₁	19(5)	30	50	12(2)	7	7	18	38	38
P ₂	70	30	40(7)	60(3)	10	10	10	20	20
P ₃	40	10(8)	60	20(10)	18	10	10	40*	-
Requirements	5	8	7	15					
diff ₁	21*	20	10	8					
diff ₂	-	20*	10	8					
diff ₃	-	-	10	8					
diff ₄	-	-	10	48*					

In the above table, the encircled values are the allocations. The total transportation cost associated with this initial basic feasible solution is given as:

$$\begin{aligned} \text{Total Transportation Cost} &= 19 \times 5 + 12 \times 2 + 40 \times 7 + 60 \times 3 + 10 \times 8 + 20 \times 10 \\ &= 95 + 24 + 280 + 180 + 80 + 200 = \text{`859} \end{aligned}$$

This solution has $3+4-1 = 6$ occupied cells. These are independent as it is not possible to form any closed loop through these allocations (see Fig. 4.2).

•			•
		•	•
	•		•

Fig. 4.2: The allocations are Independent as a closed loop cannot be formed in this case.

Hence, the optimality test can be performed. Let us apply the optimality test using the Stepping Stone method.

Optimality Test using the Stepping Stone Method

We evaluate the effect of allocating one unit to each of the unoccupied cells making closed paths. Note that the unoccupied cells are (P₁, D₂), (P₁, D₃), (P₂, D₁), (P₂, D₂), (P₃, D₁) and (P₃, D₃). We have to make closed paths so that each path contains at least three occupied cells. We also have to evaluate the net change in cost for each and every unoccupied cell. Then we have to select the one unoccupied cell, which has the most negative opportunity cost and allocate as many units as possible to reduce the total transportation cost. The computations are shown below:

Unoccupied Cell	Closed Path	Net Change in Cost ()
(P ₁ , D ₂)	(P ₁ , D ₂) → (P ₁ , D ₄) → (P ₃ , D ₄) → (P ₃ , D ₂)	30 - 12 + 20 - 10 = 28
(P ₁ , D ₃)	(P ₁ , D ₃) → (P ₁ , D ₄) → (P ₂ , D ₄) → (P ₂ , D ₃)	50 - 12 + 60 - 40 = 58
(P ₂ , D ₁)	(P ₂ , D ₁) → (P ₁ , D ₁) → (P ₁ , D ₄) → (P ₂ , D ₄)	70 - 19 + 12 - 60 = 3
(P ₂ , D ₂)	(P ₂ , D ₂) → (P ₃ , D ₂) → (P ₃ , D ₄) → (P ₂ , D ₄)	30 - 10 + 20 - 60 = -20
(P ₃ , D ₁)	(P ₃ , D ₁) → (P ₁ , D ₁) → (P ₁ , D ₄) → (P ₃ , D ₄)	40 - 19 + 12 - 20 = 13
(P ₃ , D ₃)	(P ₃ , D ₃) → (P ₂ , D ₃) → (P ₂ , D ₄) → (P ₃ , D ₄)	60 - 40 + 60 - 20 = 60

The cell (P₂, D₂) has the most negative opportunity cost (net change in cost). Therefore, transportation cost can be reduced by making allocation to this unoccupied cell. This means that if one unit is shifted to this unoccupied cell through the loop shown in the fourth row of the table, then `20 can be saved (see Fig. 4.3a). Hence, we shall shift as many units as possible to the cell (P₂, D₂) through this loop. The maximum number of units that can be allocated to (P₂, D₂) through this loop is 3.

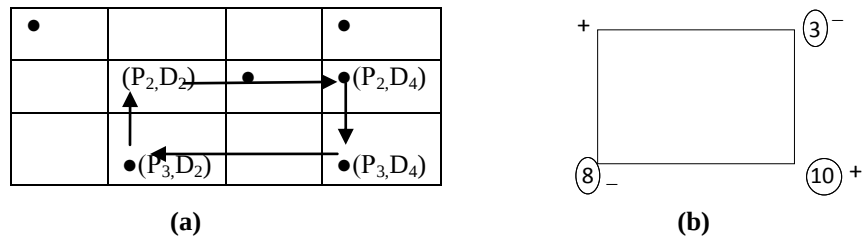


Fig. 4.3: Closed Loop.

This is because the shifting can be done only from the corners of the loop and more than 3 units cannot be shifted to (P₂, D₂) as explained below:

The allocations at other corners of the loops are 3, 10 and 8. If we try to shift more than 3 units, say, 4 units from the corner (P₂, D₄), then 4 units will have to be subtracted from the corner (P₃, D₂) so that the total of the column D₄ remains unchanged. But this will give 3 - 4 = -1 allocations to the cell (P₂, D₄), which is impossible as negative allocations cannot be made.

We obtain the maximum number of units that can be allocated to the cell (P₂, D₂) through the loop as follows:

1. First we assign (+) sign to the unoccupied cell (P₂, D₂) to be evaluated and then (-) and (+) signs alternatively to other corners of the closed loop (moving in one direction) as shown in Fig. 4.3b.
2. Then we take the minimum of the values at the corners that have been assigned the negative sign. In this case, the maximum number of units that can be allocated to the cell (P₂, D₂) through the mentioned loop is the minimum of 3 and 8. It is 3. So we write it as:

$$\min. \begin{cases} \text{the no. of units in } (P_2, D_4) = 3 \\ \text{the no. of units in } (P_3, D_2) = 8 \end{cases} = 3$$

The new table with these changes becomes:

	D ₁	D ₂	D ₃	D ₄
P ₁	19 (5)	30	50	(2) 12
P ₂	70	(3) 30	(7) 40	60
P ₃	40	(5) 10	60	(13) 20

In the above table, note that we have also allocated 3 units from (P₃, D₂) to (P₃, D₄) so that column D₄ remains unchanged. This leaves 5 units in the cell (P₃, D₂) and there are 13 units in (P₃, D₄). The total transportation cost associated with this solution is

$$\begin{aligned}\text{Total cost} &= 19 \times 5 + 12 \times 2 + 30 \times 3 + 40 \times 7 + 10 \times 5 + 20 \times 13 \\ &= 95 + 24 + 90 + 280 + 50 + 260 = \text{₹}799\end{aligned}$$

Now, we repeat the optimality test to see if further allocation can be made to reduce the total transportation cost. The computation for the unoccupied cells is as follows:

Unoccupied Cell	Closed Path	Net Change in Cost (₹)
(P ₁ , D ₂)	(P ₁ , D ₂) → (P ₁ , D ₄) → (P ₃ , D ₄) → (P ₃ , D ₂)	30 – 12 + 20 – 10 = 28
(P ₁ , D ₃)	(P ₁ , D ₃) → (P ₁ , D ₄) → (P ₃ , D ₄) → (P ₃ , D ₂) (P ₂ , D ₂) → (P ₂ , D ₃)	50 – 12 + 20 – 10 + 30 – 40 = 38
(P ₂ , D ₁)	(P ₂ , D ₁) → (P ₁ , D ₁) → (P ₁ , D ₄) → (P ₃ , D ₄) (P ₃ , D ₂) → (P ₂ , D ₂)	70 – 19 + 12 – 20 + 10 – 30 = 23
(P ₂ , D ₄)	(P ₂ , D ₄) → (P ₃ , D ₄) → (P ₃ , D ₂) → (P ₂ , D ₂)	60 – 20 + 10 – 30 = 20
(P ₃ , D ₁)	(P ₃ , D ₁) → (P ₁ , D ₁) → (P ₁ , D ₄) → (P ₃ , D ₄)	40 – 19 + 12 – 20 = 13
(P ₃ , D ₃)	(P ₃ , D ₃) → (P ₂ , D ₃) → (P ₂ , D ₂) → (P ₃ , D ₂)	60 – 10 + 30 – 40 = 40

Since all opportunity costs in the unoccupied cells are non-negative, the current solution is an optimal solution with total transportation cost ₹799. Hence the maximum saving by optimum distribution is ₹(1000 – 799) = ₹201.

Now, you should try to solve the following exercise:

E4) Perform the optimality test using the Stepping Stone method on the solution of the problem given in Example 1.

Before studying the next method, match your answer with the answer given in Sec. 4.7.

Note that the Stepping Stone method should be applied only to problems of small dimensions. For large dimensions, this method becomes quite tedious. For instance, if a problem involving five factories and seven warehouses has to be solved by the Stepping Stone method, an initial solution may involve $5+7-1=11$ occupied cells and hence $5 \times 7 - 11 = 24$ unoccupied cells. To check whether this solution is an optimal solution or not, we shall have to find the opportunity cost for each of the 24 unoccupied cells by making 24 separate loops: one for each case. Then we will have to proceed as in the above problem. This will be very tedious and cumbersome. So, for problems of large dimensions, we use another method known as the Modified Distribution method (MODI). This method may be conveniently applied to problems of small dimension as well. We now describe the MODI method.

4.4.2 Modified Distribution Method (MODI)

The modified distribution method (MODI) is an improved form of the Stepping Stone method for obtaining an optimal solution of a transportation problem. The difference between the two methods is that in the Stepping Stone method, closed loops are drawn for all unoccupied cells for determining their opportunity costs. However, in the MODI method, the opportunity costs of all the unoccupied cells are calculated and the cell with

the highest negative opportunity cost is identified without drawing any closed loop. Then only one loop is drawn for the highest negative opportunity cost. The procedure for determining the optimal solution of a transportation problem with the help of the MODI method is as follows:

1. Determine an initial feasible solution using a suitable method.
2. For the current basic feasible solution with exactly $(m+n-1)$ independent allocations, write the cost matrix for only the allocated cells.
3. Using this cost matrix, determine the set of m row numbers u_i ($i = 1, 2, \dots, m$) and n column numbers v_j ($1, 2, \dots, n$) such that $C_{ij} = u_i + v_j$, taking one of u_i or v_j as zero.
4. Fill the vacant cells using these numbers, i.e., using $u_i + v_j$.
5. Compute the net evaluations $\Delta_{ij} = C_{ij} - (u_i + v_j)$ by subtracting the values so obtained in **Step 4** from the corresponding values of the original cost matrix.
 - i) If all $\Delta_{ij} \geq 0$, an optimal solution has been arrived at.
 - ii) If some of Δ_{ij} s are negative, the current solution is not optimal. Then select the cell having the most negative Δ_{ij} and tick it.
6. Construct a closed path for the unoccupied cell ticked in **Step 5(ii)** using the already allocated cells. At each corner of the closed path, assign plus (+) and minus (-) signs alternatively, beginning with the plus sign for the unoccupied cell. Then θ units are to be allocated to this cell and the same numbers of units are to be added and subtracted alternatively at the corners assigned plus and minus signs, respectively. The value of θ is the maximum number of units that can be allocated to this cell through the loop. It is obtained by equating the allocations at the negative sign corners to zero. In this way, allocations have been improved.
7. Write the cost matrix for only those cells which have improved allocations. Go to **Step 3** and repeat the procedure until all $\Delta_{ij} \geq 0$. Calculate the associated total transportation cost.

Example 5: Perform optimality test by applying the MODI method for the problem taken in Example 4.

Solution: The basic solution of the problem as already obtained by VAM is

D Centre Plants	D ₁	D ₂	D ₃	D ₄	Availability
P ₁	19 (5)	30	50	12 (2)	7
P ₂	70	30	40 (7)	60 (3)	10
P ₃	40	10 (8)	60	20 (10)	18
Requirement	5	8	7	15	35

Now we perform optimality test by applying the MODI method. First of all, we write the cost matrix for only allocated cells:

19			12	u_1
		40	60	u_2
	10		20	u_3
v_1	v_2	v_3	v_4	

Let us denote the row numbers by u_1, u_2, u_3 and column numbers by v_1, v_2, v_3 and v_4 such that

$$u_1 + v_1 = 19, \quad u_1 + v_4 = 12, \quad u_2 + v_3 = 40,$$

$$u_2 + v_4 = 60, \quad u_3 + v_2 = 10, \quad u_3 + v_4 = 20.$$

Taking $u_1 = 0$, we have $v_1 = 19$ and $v_4 = 12$ from the first two equations.

Putting the value of v_4 in the fourth equation, we have $u_2 = 60 - 12 = 48$.

Similarly, $u_3 = 20 - 12 = 8$, $v_2 = 10 - 8 = 2$ and $v_3 = 40 - 48 = -8$.

Using these values, we fill the vacant cells of the above table using $C_{ij} = u_i + v_j$ and put dots in the already filled cells so that these cells are not considered again.

	$v_1 = 19$	$v_2 = 2$	$v_3 = -8$	$v_4 = 12$
$u_1 = 0$	•	2	-8	•
$u_2 = 48$	67	50	•	•
$u_3 = 8$	27	•	0	•

Now, subtracting these values from the corresponding values of the original cost matrix, we have the net evaluations, i.e., $\Delta_{ij} = C_{ij} - (u_i + v_j)$

•	$30 - 2 = 28$	$50 - (-8) = 58$	•
$70 - 67 = 3$	$30 - 50 = -20$	•	•
$40 - 27 = 13$	•	$60 - 0 = 60$	•

Note that the cell (P_2, D_2) has the most negative opportunity cost (net change in cost). Therefore, the transportation cost can be reduced by making allocation to this unoccupied cell. This means that if one unit is shifted to this unoccupied cell through the closed loop formed, beginning from this cell and using allocated cells, '20 can be saved. We form the loop beginning from this cell, i.e., the cell (P_2, D_2) (see Fig. 4.4). We shift θ units to this unoccupied cell through the loop and add and subtract θ from the cells at the other corners of the loop which are assigned '+' and '-' signs. So we get Fig. 4.4:



Fig. 4.4: Closed loop for cell (P_2, D_2) .

The maximum number of units that can be allocated to the cell (P_2, D_2) through this loop is given by the minimum of the solution of the equations $3 - \theta = 0$ and $8 - \theta = 0$, i.e.,

$$\theta = \min. \begin{cases} \text{the no. of units in } (P_2, D_4) = 3 \\ \text{the no. of units in } (P_3, D_2) = 8 \end{cases} = 3$$

So with the improved allocations, the table now becomes:

19 (5)	30	50	12 (2)
70	30 (3)	40 (7)	60
40	10 (5)	60	20 (13)

Thus, the total cost of transportation for this set

$$= 19 \times 5 + 12 \times 2 + 30 \times 3 + 40 \times 7 + 10 \times 5 + 20 \times 13$$

$$= 95 + 24 + 90 + 280 + 50 + 260 = 799$$

Now, let us apply the optimality test to the improved solution. Proceeding in the same way as in the first iteration, first of all, we write the cost matrix for only allocated cells:

19			12	u_1
	30	40		u_2
	10		20	u_3
v_1	v_2	v_3	v_4	

Let us denote the row numbers by u_1, u_2, u_3 and column numbers by v_1, v_2, v_3 and v_4 such that

$$u_1 + v_1 = 19, \quad u_1 + v_4 = 12, \quad u_2 + v_3 = 40,$$

$$u_2 + v_2 = 30, \quad u_3 + v_2 = 10, \quad u_3 + v_4 = 20.$$

Taking $u_1 = 0$, we have $v_1 = 19, v_4 = 12$ from the first two equations.

Putting the value of v_4 in the sixth equation, we have $u_3 = 20 - 12 = 8$.

Similarly, $u_2 = 30 - 02 = 28, \quad v_2 = 10 - 8 = 2, \quad v_3 = 40 - 28 = 12$.

Using these values, we fill all the vacant (unoccupied) cells of the table using $c_{ij} = u_i + v_j$ for each unoccupied cell and put dot in already filled cells so that these cells are not considered again.

	$v_1 = 19$	$v_2 = 2$	$v_3 = 12$	$v_4 = 12$
$u_1 = 0$	•	2	12	•
$u_2 = 28$	47	•	•	40
$u_3 = 8$	27	•	20	•

Now, subtracting these values from the corresponding values of the original cost matrix, we have the net evaluations $\Delta_{ij} = C_{ij} - (u_i + v_j)$ as:

•	28	38	•
23	•	•	20
13	•	40	•

Since none of the net evaluations is negative, this solution is optimal. Thus, the total minimum transportation cost is `799 and the maximum saving

$$= `(1000 - 799) = `201.$$

Now, you should try to solve the following exercise.

E5) Determine the optimum distribution for the transportation problem given in exercise E1 using the MODI method.

Before studying the next section, match your answer with the answer given in Sec. 4.7.

4.5 SOME SPECIAL CASES

We now consider some special cases of the transportation problem such as unbalanced transportation problem, case of degeneracy, case of alternative solution, maximisation transportation problem and problems with prohibited routes.

4.5.1 Unbalanced Transportation Problem

The transportation problems wherein the total capacity of all sources and total requirement (demand) of all destinations are not equal is called the **unbalanced transportation problem**. If the total capacity of sources is greater than (less than) the total requirement at destination, we add a dummy destination (source) in the transportation table with zero transportation cost so that the problem becomes balanced. The augmented problem is then solved by the methods explained earlier.

Example 6: A company has factories at A, B and C which supply warehouses at D, E, F and G. The monthly factory capacities are 160, 150 and 190 units, respectively. Monthly warehouse requirements are 80, 90, 110 and 160, respectively. Unit shipping costs (in rupees) are as follows:

From	To				
		D	E	F	G
	A	42	48	38	37
	B	40	49	52	51
	C	39	38	40	43

Determine the optimum distribution for this company to minimise shipping costs.

Solution: Here the total capacity of sources (Factories) is $160+150+190=500$. It is greater than the total requirement of all the destinations (Warehouses), which is $80+90+110+160=440$. Therefore, we add a dummy destination in the transportation table with zero transportation cost and take 60 as its requirement. Thus, the problem becomes balanced, i.e., the total capacity and total requirement are equal. The balanced problem is as follows:

From	To						Capacity
		D	E	F	G	H	
	A	42	48	38	37	0	160
	B	40	49	52	51	0	150
	C	39	38	40	43	0	190
Requirement		80	90	110	160	60	500

We can solve this problem using VAM to determine the basic feasible solution and then use the MODI method to find the optimal solution. We have solved this problem in Example 3. You may like to try the remaining solution yourself.

The optimum allocation is:

160 to (A, F), 80 to (B, D), 10 to (B, E), 60 to (B, G), 80 to (C, E) and 110 to (C, F).

Note that if the number of allocations in the basic feasible solution is less than $m+n-1$, we first go through the special case of degeneracy and then apply the test of optimality. Let us explain how the problem of degeneracy is dealt with in the transportation problem.

4.5.2 Degeneracy

It may happen sometimes that the number of occupied cells is less than $m+n-1$. Such a solution is called a **degenerate solution**. We handle such a situation by introducing an infinitesimally small allocation e (say) in the least cost and independent cell. This means that if ' e ' is added to or subtracted from any quantity, the quantity remains unaltered, i.e., the value of ' e ' is nothing but zero. But we do not write 0 in place of e as it is one of the allocations and will not come into counting if it is written as zero. So we write e and not zero in its place and count it as one of the allocations.

Degeneracy occurs in **Example 6**. We resolve this problem as follows:

Using Vogel Approximation method, we get the basic feasible solution as:

Warehouse Factory	D	E	F	G	H	Av.	D ₁	D ₂	D ₃	D ₄			
A	42	48	38	37	160	0	e	160	37	1	1	1	
B	40	80	49	52	10	51	0	60	150	40*	9	11*	1
C	39	38	90	40	100	43	0	190	38	1	1	3	
Requirement	80	90	110	160	60	500							
D ₁	1	10	2	6	0								
D ₂	1	10*	2	6	-								
D ₃	1	-	2	6	-								
D ₄	-	-	2	6*	-								

Here the number of allocations is 6 but the number of allocations should be $3+5-1=7$ to perform optimality test. Hence it is the case of degeneracy. To resolve such a problem, we introduce an infinitesimally small allocation e (say) in the least cost and independent cell, i.e., the cell (A, H). Another cell (C, H) also has the same least cost as the cell (A, H). But this cell is not independent because a loop (C, H) → (C, F) → (B, F) → (B, H) can be formed beginning with the cell (C, H) as shown in Fig. 4.5. Now the number of allocations is 7, i.e., as many as required for performing the optimality test.

42	48	38	•	0
•	49	• (B, F)	51	• (B, H)
39	•	• (C, F)	43	0 (C, H)

Fig. 4.5: Closed loop for cell (C, H).

Thus, applying optimality test by the MODI method, the improved allocations after first iteration will be:

42	48	38 (e)	37 (160)	0
40 (80)	49	52 (10)	51	0 (60)
39	38 (90)	40 (100)	43	0

Again, applying optimality test, the improved solution is

	D	E	F	G	H
A	42	48	38 (e)	37 (160)	0
B	40 (80)	49 (10)	52	51	0 (60)
C	39	38 (80)	40 (110)	43	0

Again applying the optimality test, you will see that these allocations give the optimal solution and the total transportation cost = `17050

4.5.3 Alternative Optimal Solutions

Such solutions exist if all net-evaluations, i.e., $\Delta_{ij} = C_{ij} - (u_i + v_j) \geq 0$ but one or more of them are equal to zero. To determine the alternative optimal solution, we tick the cell having zero value of net-evaluation and make a closed loop beginning from this cell. The transportation allocation might not be the same, yet the total transportation cost is same.

We illustrate this special case with the help of Example 7.

Example 7: Solve the following transportation problem:

	D	E	F	G	Capacity
A	5	3	6	2	19
B	4	7	9	1	37
C	3	4	7	5	34
Requirement	16	18	31	25	

Solution: Using VAM, the initial basic feasible solution is obtained as follows:

	D	E	F	G	Capacity	Diff. ₁	Diff. ₂	Diff. ₃	Diff. ₄
A	5	3 (18)	6 (1)	2	19	1	2	2*	1
B	4 (12)	7	9	1 (25)	37	3*	3*	-	-
C	3 (4)	4	7 (30)	5	34	1	1	1	4*
Requirement	16	18	31	25	90				
Diff. ₁	1	1	1	1					
Diff. ₂	1	1	1	-					
Diff. ₃	1	1	1	-					
Diff. ₄	2	-	1	-					

Now, performing the optimality test using the MODI method, you can get the following optimal solution:

5	3	6 (19)	2
4 (12)	7	9	1 (25)
3 (4)	4 (18)	7 (12)	5

But the net evaluations, while performing optimality test for this solution, will be

	D	E	F	G
A	3	0	•	3
B	•	2	1	•
C	•	•	•	5

Since one net evaluation is zero, the problem has an alternative optimal solution.

We form the closed loop beginning with the cell containing zero, i.e., the cell (A, E). The loop is (A, E)→(A, F)→(C, F)→(C, E). We perform the optimality test by the MODI method again. The alternative allocations are:

5	3 (18)	6 (1)	2
4 (12)	7	9	1 (25)
3 (4)	4	7 (30)	5

Here, the first set of allocations gives the transportation cost as

$$6 \times 19 + 4 \times 12 + 1 \times 25 + 3 \times 4 + 4 \times 18 + 7 \times 12 = 114 + 48 + 25 + 12 + 72 + 84 = 355$$

The second set of allocations also gives the same transportation cost:

$$3 \times 18 + 6 \times 1 + 4 \times 12 + 1 \times 25 + 3 \times 4 + 7 \times 30 = 54 + 6 + 48 + 25 + 12 + 210 = 355$$

Both these different sets of allocations give the same total transportation cost. Hence, any of these alternative solutions may be chosen.

4.5.4 Maximisation Transportation Problem

The data given in the transportation problems may be such that we have to maximise it. This may be possible if instead of costs, the profits (or anything else like revenue which needs maximisation) are given in the cells. To handle such a problem, we find the opportunity loss matrix by subtracting the value of each cell from the largest value chosen from amongst all the given cells. When the value of a cell is subtracted from the highest value, it gives the loss of amount caused by not getting the opportunity which would have given the highest value. The matrix so obtained is known as the **opportunity loss matrix**. It is handled in the same way as the minimisation problem.

Example 8: A company has 3 factories manufacturing the same product and 5 agencies in different parts of the country. Production costs differ from factory to factory and the sales prices from agency to agency. The shipping cost per unit product from each factory to any agency is known. Given the following data, find the production and distribution schedules most profitable to the company.

Production cost/unit	Max. Capacity No. of Units	Factory i
20	150	1
22	200	2
18	125	3

1	1	1	5	9	4	
2	9	7	8	3	6	Shipping costs
3	4	5	3	2	7	
	1	2	3	4	5	Agency j
	80	100	75	45	125	Demand to be met
	30	32	31	34	29	Sales price

Solution: We first obtain the profit matrix by finding the excess of sales price over the various costs, i.e.,

$$\text{Profit/unit} = \text{Sales price/unit} - \text{Production cost/unit} - \text{Shipping cost/unit}$$

Thus, the profit matrix is

Agency Factory	1	2	3	4	5
1	$30-20-1=9$	$32-20-1=11$	$31-20-5=6$	$34-20-9=5$	$29-20-4=5$
2	$30-22-9=-1$	$32-22-7=3$	$31-22-8=1$	$34-22-3=9$	$29-22-6=1$
3	$30-18-4=8$	$32-18-5=9$	$31-18-3=10$	$34-18-2=14$	$29-18-7=4$

Note that this is the profit matrix and we want to maximise the profit. But the methods which we have discussed deal with minimisation problems. Therefore, we convert this profit matrix into the opportunity loss matrix used for minimisation problems or to minimise the total cost. The maximum value of the profit in this table is 14 and hence the opportunity loss matrix is obtained on subtracting each of the values of the above matrix from 14. The opportunity loss matrix, therefore, is

$14 - 9$	$14 - 11$	$14 - 6$	$14 - 5$	$14 - 5$
$14 - (-1)$	$14 - 3$	$14 - 1$	$14 - 9$	$14 - 1$
$14 - 8$	$14 - 9$	$14 - 10$	$14 - 14$	$14 - 4$

Here, the problem is also unbalanced since demand (425) and capacity (475) are not equal. So, an agency with requirement of 50 units (the difference of demand and capacity) also needs to be introduced to make the problem balanced. We solve the resulting matrix by VAM and then by the MODI method as follows.

If you have attempted E3, you must have obtained the following basic feasible solution using VAM as:

								D ₁	D ₂	D ₃	D ₄	D ₅	D ₆
	5 (50)	3 (100)	8	9	9	0	150	3	2	2	2*	4*	-
	15	11	13	5 (45)	13 (105)	0 (50)	200	5*	6*	2	2	2	2
	6 (30)	5	4 (75)	0	10 (20)	0	125	4	4	1	1	4	4
	80	100	75	45	125	50	475						
D ₁	1	2	4	5	1	0							
D ₂	1	2	4	5	1	-							
D ₃	1	2	4*	-	1	-							
D ₄	1	2	-	-	1	-							
D ₅	1	-	-	-	1	-							
D ₆	9*	-	-	-	3	-							

If you have attempted E5, you can perform the optimality test by using the MODI method. The allocations which give the optimal solution are:

5 (50)	3	8	9	9	0
15	11 (100)	13	5 (25)	13	0
6 (30)	5	4 (75)	0 (20)	10 (125)	0 (50)

The maximum profit (on multiplying the allocations with the corresponding profit values)

$$\begin{aligned}
 &= 5 \times 50 + 11 \\
 &\quad \times 100 + 5 \times 25 + 10 \times 125 + 0 \times 50 + 6 \times 30 + 4 \times 75 + 0 \times 20 \\
 &= 250 + 1100 + 125 + 125 + 0 + 180 + 300 + 0 = 2080
 \end{aligned}$$

4.5.5 Prohibited Routes

Sometimes, in a transportation problem, some routes may not be available. There could be several reasons for this such as bad road conditions or strike, etc. To handle such a situation, a very large cost (or a negative profit for maximisation problem) represented by ∞ or M is assigned to each such route which is not available. Due to assignment of very large cost, such routes would automatically be eliminated in the final solution.

Now, you should try to solve the following exercise.

E6) Solve the following transportation problem:

		Godowns						Stock Available
		1	2	3	4	5	6	
Factory	1	7	5	7	7	5	3	60
	2	9	11	6	11	-	5	20
	3	11	10	6	2	2	8	90
	4	9	10	9	6	9	12	50
		60	20	40	20	40	40	

Before going further, match your answer with the answer given in Sec. 4.7.

Let us now summarise the main points, which have been covered in this unit.

4.6 SUMMARY

1. A **transportation problem** is one of the sub-classes of Linear Programming Problems in which the objective is to transport various quantities (initially stored at various origins/plants/factories) to different destinations/distribution centres/warehouses in such a way that the total transportation cost is minimum. These problems can be solved by the Simplex method but this method is cumbersome and hence another method known as **transportation method** is preferred.

It involves making a transportation model in the form of a matrix, finding initial basic feasible solution, performing optimality test and moving towards optimal solution. Methods of finding initial basic feasible solution are **North-West Corner Rule**, **Least Cost Method** and **Vogel's Approximation Method**. Vogel's approximation method generally gives the solution closest to the optimum solution and hence is preferred over other methods of finding the initial basic feasible solutions. After finding the initial basic feasible solution, optimality test

is performed to find whether the obtained feasible solution is optimal or not.

2. **Conditions for applying optimality test** are:

- i) It contains exactly $(m + n - 1)$ allocations where m and n represent the number of rows and columns, respectively, of the transportation table.
- ii) These allocations are independent.

If it is not possible to form any closed loop through the allocations under consideration then the allocations are said to be **independent**. If such a loop can be formed using some or all of the allocations under consideration, the allocations are known as **non-independent**.

3. Optimality test is performed by applying the **Stepping Stone Method** or **Modified Distribution (MODI) Method**.
4. The transportation problems wherein the total capacity of all the sources and total requirement (demand) of all the destinations are not equal is said to be an **unbalanced problem**. If the total capacity of sources is greater than (less than) the total requirement at destination, we add a dummy destination (source) in the transportation table with zero transportation cost so that the problem becomes balanced.
5. It may happen sometimes that the number of occupied cells is less than $m + n - 1$. Such a solution is called a **degenerate solution**. We handle such a situation by introducing an infinitesimally small allocation ϵ (say) in the least cost and independent cell.
6. If all net-evaluations, i.e., $\Delta_{ij} = c_{ij} - (u_i + v_j) \geq 0$ but one or more of them are equal to zero then there is an **alternative optimal solution** for the transportation problem.
7. To handle a **Maximisation Transportation Problem**, we find the opportunity loss matrix by subtracting the value of each cell from the largest value chosen from amongst all the given cells.
8. Sometimes, in a transportation problem, some routes may not be available. There could be several reasons for this such as bad road conditions or strike, etc. To handle such a situation, a very large cost (or a negative profit for maximisation problem) represented by ∞ or M is assigned to each of such routes, which are not available.

4.7 SOLUTIONS/ANSWERS

- E1)** Using North-West Corner Rule, the allocations are to be made as under:

8 units to cell (1,1), 2 units to cell (1,2), 6 units to cell (2,2), 6 units to cell (2,3), 4 units to cell (3,3), 6 units to cell (4,3), 3 units to cell (4, 4), 8 units to cell (4, 5), 1 unit to cell (4, 6) and 20 units to cell (5,6).

The transportation cost for this solution will be:

$$4 \times 8 + 6 \times 2 + 5 \times 6 + 4 \times 6 + 9 \times 4 + 5 \times 6 + 9 \times 3 + 3 \times 8 + 6 \times 1 + 14 \times 20 = 32 + 12 + 30 + 24 + 36 + 30 + 27 + 24 + 6 + 280 = 501$$

- E2)** Using Least Cost method, the allocations are to be made as under:

4 units to cell (1,1), 3 units to cell (1,4), 3 units to cell (1,6), 12 units to cell (2,6), 4 units to cell (3,1), 8 units to cell (4,2), 10 units to cell (4,3), 6 units to cell (5, 3), 8 units to cell (5,5) and 6 units to cell (5,6).

The transportation cost for this solution will be:

$$4 \times 4 + 2 \times 3 + 8 \times 3 + 0 \times 12 + 2 \times 4 + 4 \times 8 + 5 \times 10 + 7 \times 6 + 2 \times 8 + 14 \times 6 = 16 + 6 + 24 + 0 + 8 + 32 + 50 + 42 + 16 + 84 = 278$$

E3) Using Vogel's Approximation method, the allocations are to be made as under:

D. Centre Factory	1	2	3	4	5	6	Capacity	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	D ₈	
1	4(4)	6	9		2(3)	7	8(3)	10	2	2	2*	2	2	4*	1	1
2	3	5	4		8	10	0(12)	12	3	-	-	-	-	-	-	-
3	2(4)	6	9		8	4	13	4	2	2*	-	-	-	-	-	-
4	4	4	(8)	5	(4)	9	3	6(6)	1	1	1	1	1	1	1	1
5	9	8		7	(12)	3	2	(8)	1	1	1	5*	1	2	7*	-
Requirement	8	8	16	3	8	21	64									
D ₁	1	1	1	1	1	6*										
D ₂	2	2	2	1	1	2										
D ₃	0	2	2	1	1	2										
D ₄	0	2	2	-	1	2										
D ₅	0	2*	2	-	-	2										
D ₆	0	-	2	-	-	2										
D ₇	-	-	2	-	-	2										
D ₈	-	-	4*	-	-	2										

In the above table D_i (i=1, 2, 3, 4, 5, 6, 7, 8) means the differences of the least and next to least costs. Thus, the total transportation cost

$$= 16 + 6 + 24 + 0 + 8 + 32 + 20 + 36 + 84 + 16 = 242$$

E4) The allocations which give optimal solutions are given in the following table:

19 (5)	30	50	12 (2)
70	30 (3)	40 (7)	60
40	10 (5)	60	20 (13)

Thus, the total cost of transportation for this set

$$= 19 \times 5 + 12 \times 2 + 30 \times 3 + 40 \times 7 + 10 \times 5 + 20 \times 13$$

$$= 95 + 24 + 90 + 280 + 50 + 260 = 799$$

and the maximum saving = `(1000 – 799) = `201

E5) After obtaining the basic feasible solution by VAM as in E3, we apply the MODI method and write the cost matrix for only the allocated cells as follows:

4			2		8
					0
2					
	4	5			6
		7		2	

Defining row and column numbers as described in the solution by MODI method for Example 1 and then filling the vacant cells, we get

	$v_1 = 4$	$v_2 = 6$	$v_3 = 7$	$v_4 = 2$	$v_5 = 2$	$v_6 = 8$
$u_1 = 0$		6	7		2	
$u_2 = -8$	-4	-2	-1	-6	-6	
$u_3 = -2$		4	5	0	0	6
$u_4 = -2$	2			0	0	
$u_5 = 0$	4	6		2		8

Now subtracting these from the corresponding costs of original cost matrix i.e. $\Delta_{ij} = c_{ij} - (u_i + v_j)$, we get

	0	2		5	
7	7	5	14	16	
	2	4	8	4	7
2			9	3	
5	2		1		6

Since all $\Delta_{ij} \geq 0$, this solution is the optimal solution. Thus, the optimal solution = 242

- E6)** In this problem, the route from Factory 2 to Godown 5 is prohibited and hence to handle this situation, a very large cost represented by ∞ or M is assigned to this cell. Due to assignment of very large cost, this route would automatically be eliminated in the final solution as the routes with maximum values are never selected while solving transportation problems. Total transportation cost to be obtained by you will be `1120.

Block

2

OPTIMISATION TECHNIQUES-II

UNIT 5

Assignment Problems	5
----------------------------	----------

UNIT 6

Queueing Theory	23
------------------------	-----------

UNIT 7

Sequencing Problems	43
----------------------------	-----------

UNIT 8

Inventory Models	57
-------------------------	-----------

Curriculum and Course Design Committee

Prof. K. R. Srivathsan
Former Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Parvin Sinclair
Former Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Geeta Kaicker
Former Director, School of Sciences
IGNOU, New Delhi

Prof. Jagdish Prasad
Department of Statistics
University of Rajasthan, Jaipur

Prof. R. M. Pandey
Department of Bio-Statistics
AIIMS, New Delhi

Dr. Manish Trivedi
School of Sciences, IGNOU

Dr. Nitin Gupta
School of Sciences, IGNOU

Dr. Neha Garg
School of Sciences, IGNOU

Mr. Rajesh Kaliraman
School of Sciences, IGNOU

Prof. Rahul Roy
Theoretical Statistics and Mathematics Unit
Indian Statistical Institute, New Delhi

Prof. Diwakar Shukla
Department of Mathematics and Statistics
Dr. Hari Singh Gaur University, Sagar

Prof. Rakesh Srivastava
Department of Statistics
M.S. University of Baroda, Vadodara

Prof. G. N. Singh
Department of Applied Mathematics
I.S.M., Dhanbad

Prof. Gulshan Lal Taneja
Department of Mathematics
M.D. University, Rohtak

Prof. Poornima Mital
School of Sciences, IGNOU

Prof. Sujatha Varma
School of Sciences, IGNOU

Dr. S. Venkataraman
School of Sciences, IGNOU

Dr. Deepika
School of Sciences, IGNOU

Block Preparation Team

Prof. Jayant Dubey (**Editor**)
Department of Business Studies
BTIRT, Sironja, Sagar (MP)

Prof. Vijayshri (**Language Editor**)
School of Sciences, IGNOU

Prof. Gulshan Lal Taneja (**Units 5 to 8**)
Department of Mathematics,
M. D. University, Rohtak

Formatted By
Dr. Manish Trivedi
School of Sciences, IGNOU

Programme and Course Coordinator: Dr. Manish Trivedi

Block Production

Mr. Sunil Kumar, AR (P)
School of Sciences, IGNOU

September, 2014

© Indira Gandhi National Open University, 2014

ISBN-978-81-266-

All rights reserved. No part of this work may be reproduced in any form, by mimeograph or any other means, without permission in writing from the Indira Gandhi National Open University

Further information on the Indira Gandhi National Open University may be obtained from the University's Office at Maidan Garhi, New Delhi-110068 or visit University's website <http://www.ignou.ac.in>

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by the Director, School of Sciences.

Laser Typeset by: Tessa Media & Computers, C-206, A.F.E.-II, Okhla, New Delhi

Printed at:

BLOCK 2 OPTIMISATION TECHNIQUES-II

In Block 1 of this course, you have learnt the graphical method and the simplex method of solving linear programming problems. You have also learnt some methods of finding the initial feasible solutions as well as the methods of obtaining the optimum solution of transportation problems.

With the rapid and continuous progress of mechanisation and automation, problems of machine maintenance have become of particular interest to production and operations management. To the modern-day management, maintenance is more than a mere synonym for repair. Instead, maintenance embraces a new idea of keeping the plant and equipment competitive and productive. For example, in a textile mill, a maintenance incharge with limited resources has to allocate maintenance staff to various jobs in such a way that the breakdown time is minimised. In this block, we discuss various techniques to optimise the processing/service time or total cost while dealing with queueing systems, sequential problems and inventory problems, respectively. The block is divided into four units.

In Unit 5 entitled **Assignment Problems**, we discuss various types of assignment problems, including travelling salesman problem and apply the Hungarian method for solving these problems.

Unit 6 entitled **Queueing Theory** explains the basic concepts, which would help you understand the techniques of queueing models thoroughly. We describe the fundamental structure and operating characteristics of a queueing system alongwith the M/M/1 queueing model and its applications in this unit.

Sequencing problems are quite common in real life. They arise whenever there is a choice of determining the order in which a number of jobs can be performed. Therefore, in Unit 7 entitled **Sequencing Problems**, we discuss some procedures for determining the optimum order or sequence of jobs for processes with n number of jobs to be completed through 2 machines and with 2 jobs to be completed through m number of machines, in some pre-assigned order, so as to optimise the total time involved.

In Unit 8 entitled **Inventory Models**, we discuss inventory control and various factors involved in inventory analysis. We also describe the models for determining the economic order quantity considering the situations: i) when demand is uniform, ii) when rates of demand are different in different cycles, iii) when shortages are allowed, iv) when replenishment is uniform, and v) when price (or quantity) discounts are given.

Notations and Symbols

Z	: Objective function to be maximised or minimised
x_{ij}	: j^{th} job assigned to the i^{th} person
c_{ij}	: Amount of time taken by the i^{th} person to complete the j^{th} job
$N(t)$	: Total number of occurrences of an event E in an interval of time t
λ	: Average arrival rate
μ	: Average departure rate
L_s	: Average number of customers in the queueing system
L_q	: Average number of customers in queue waiting to be served
W_q	: Average time each customer spends waiting in the queue to get serviced
W_s	: Average time each customer spends in the queueing system from entry into the queue to completion of the service
ρ	: Traffic intensity or the utilisation factor
$P_n(t)$	: Probability of n customers in an interval t
FCFS	: First come, first served
LIFO	: Last in, first out
Q	: Number of units ordered (supplied) per order
D	: Demand in units of inventory per year
N	: Number of orders placed per year
TC	: Total inventory cost
C_o	: Ordering cost per order
C	: Purchase or manufacturing price per unit inventory
C_h	: Carrying or holding cost per unit per period of time the inventory is kept
C_s	: Shortage cost per unit of inventory
t	: Elapsed time between placement of two successive orders
r_p	: Replenishment rate at which lot size Q is added to inventory.

UNIT 5 ASSIGNMENT PROBLEMS

Structure

- 5.1 Introduction
 - Objectives
- 5.2 Assignment Problem and Its Solution
- 5.3 Some Special Cases
 - Maximisation Problem
 - The Unbalanced Assignment Problem
 - Alternative Optimal Solutions
 - Restriction on Assignments
- 5.4 Travelling Salesman Problem
- 5.5 Summary
- 5.6 Solutions/Answers

5.1 INTRODUCTION

In Block 1 of this course, we have discussed the basic concepts related to Linear Programming Problems and the Simplex method for solving them. The Transportation Problem was also discussed in Block 1. In this unit, we explain the Assignment problem and discuss various methods for solving it.

The assignment problem deals with allocating various resources (items) to various activities (receivers) on a one to one basis, i.e., the number of operations are to be assigned to an equal number of operators where each operator performs only one operation. For example, suppose an accounts officer has 4 subordinates and 4 tasks. The subordinates differ in efficiency and take different time to perform each task. If one task is to be assigned to one person in such a way that the total person hours are minimised, the problem is called an **assignment problem**. Though the assignment problem is a special case of transportation problem, it is not solved using the methods described in Unit 4. We use another method called the Hungarian method for solving an assignment problem. It is shorter and easier compared to any method of finding the optimal solution of a transportation problem. In this unit, we discuss various types of assignment problems, including travelling salesman problem and apply the Hungarian method for solving these problems.

In the next unit, we shall discuss the fundamental structure and operating characteristics of a queueing system and explain a single server M/M/1 queueing model with Poisson input and exponential service time.

Objectives

After studying this unit, you should be able to:

- formulate an assignment problem;
- determine the optimal solutions of assignment problems using the Hungarian method;
- obtain the solutions for special cases of assignment problems, i.e, maximisation problem, unbalanced assignment problem, alternative optimal solutions and restriction on assignments; and
- solve the travelling salesman problem as an assignment problem.

5.2 ASSIGNMENT PROBLEM AND ITS SOLUTION

An assignment problem may be considered as a special type of transportation problem in which the number of sources and destinations are equal. The capacity of each source as well as the requirement of each destination is taken as 1. In the case of an assignment problem, the given matrix must necessarily be a square matrix which is not the condition for a transportation problem.

Suppose there are n persons and n jobs and the assignment of jobs has to be done on a one-to-one basis. This assignment problem can be stated in the form of an $n \times n$ matrix of real numbers (known as the **cost matrix**) as given in the following table:

Person	Job					
	1	2	...	j	...	n
1	C_{11}	C_{12}	...	C_{1j}	...	C_{1n}
2	C_{21}	C_{22}	...	C_{2j}	...	C_{2n}
.	.					
.	.					
.	.					
i	C_{i1}	C_{i2}	...	C_{ij}	...	C_{in}
.	.					
.	.					
.	.					
n	C_{n1}	C_{n2}	...	C_{nj}	...	C_{nn}

where c_{ij} represents the amount of time taken by i^{th} person to complete the j^{th} job. Let x_{ij} denote the j^{th} job assigned to the i^{th} person. Then, mathematically, the assignment problem can be stated as follows:

$$\text{Minimise } Z = \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij}, \quad \text{where } i = 1, 2, \dots, n \text{ and } j = 1, 2, \dots, n.$$

subject to

$$x_{ij} = \begin{cases} 1 & \text{if the } i^{\text{th}} \text{ person is assigned the } j^{\text{th}} \text{ job} \\ 0 & \text{if the } i^{\text{th}} \text{ person is not assigned the } j^{\text{th}} \text{ job} \end{cases},$$

$$x_{i1} + x_{i2} + \dots + x_{in} = 1, \quad i = 1, 2, \dots, n \text{ (one job is done by the } i^{\text{th}} \text{ person)}$$

$$x_{1j} + x_{2j} + \dots + x_{nj} = 1, \quad j = 1, 2, \dots, n \text{ (only one person is assigned the } j^{\text{th}} \text{ job)}$$

The constant c_{ij} in the above problem represents time. It may be cost or some other parameter which is to be minimised in the assignment problem under consideration.

Note that an assignment problem is a special type of transportation problem and may be solved as one. However, we use another method known as the Hungarian method for solving it. This method is shorter and easier compared to any other method of finding the optimal solution of a transportation problem. Let us explain the Hungarian method of finding the optimal solution of an assignment problem.

The steps for obtaining an optimal solution of an assignment problem are as follows:

1. Check whether the given matrix is square. If not, make it square by adding a suitable number of dummy rows (or columns) with 0 cost/time elements.
2. Locate the smallest cost element in each row of the cost matrix. Subtract the smallest element of each row from every element of that row.
3. In the resulting cost matrix, locate the smallest element in each column and subtract the smallest element of each column from every element of that column.
4. In the resulting matrix, search for an optimum assignment as follows:
 - i) Examine the rows successively until a row with exactly one zero is found. Draw a rectangle around this zero (as $\boxed{0}$) and cross out all other zeroes in the corresponding column. Proceed in this manner until all the rows have been examined. If there is more than one zero in any row, do not touch that row; pass on to the next row.
 - ii) Repeat step (i) above for the columns of the resulting cost matrix.
 - iii) If a row or column of the reduced matrix contains more than one zeroes, arbitrarily choose a row or column having the minimum number of zeroes. Arbitrarily select any zero in the row or column so chosen. Draw a rectangle around it and cross out all the zeroes in the corresponding row and column. Repeat steps (i), (ii) and (iii) until all the zeroes have either been assigned (by drawing a rectangle around them) or crossed.
 - iv) If each row and each column of the resulting matrix has one and only one assigned 0, the optimum assignment is made in the cells corresponding to $\boxed{0}$. The optimum solution of the problem is attained and you can stop here.

Otherwise, go to the next step.

5. Draw the minimum number of horizontal and/or vertical lines through all the zeroes as follows:
 - i) Tick mark ($\checkmark$) the rows in which assignment has not been made.
 - ii) Tick mark ($\checkmark$) columns, which have zeroes in the marked rows.
 - iii) Tick mark ($\checkmark$) rows (not already marked) which have assignments in marked columns. Then tick mark ($\checkmark$) columns, which have zeroes in newly marked rows, if any. Tick mark ($\checkmark$) rows (not already marked), which have assignments in these newly marked columns.
 - iv) Draw straight lines through all unmarked rows and marked columns.
6. Revise the cost matrix as follows:
 - i) Find the smallest element not covered by any of the lines.
 - ii) Subtract this from all the uncovered elements and add it to the elements at the intersection of the two lines.
 - iii) Other elements covered by the lines remain unchanged.
7. Repeat the procedure until an optimum solution is attained.

We now illustrate the procedure with the help of an example.

Example 1: A computer centre has four expert programmers and needs to develop four application programmes. The head of the computer centre, estimates the computer time (in minutes) required by the respective experts to develop the application programmes as follows:

		Programmes			
		A	B	C	D
Programmers	1	120	100	80	90
	2	80	90	110	70
	3	110	140	120	100
	4	90	90	80	90

Find the assignment pattern that minimises the time required to develop the application programmes.

Solution: Let us subtract the minimum element of each row from every element of that row. Note that the minimum element in the first row is 80. So 80 is to be subtracted from every element of the first row, i.e., from 120, 100, 80 and 90, respectively. As a result, the elements of the first row of the resulting matrix would be 40, 20, 0, 10, respectively. Similarly, we obtain the elements of the other rows of the resulting matrix. Thus, the resulting matrix is:

	A	B	C	D
1	40	20	0	10
2	10	20	40	0
3	10	40	20	0
4	10	10	0	10

Let us now subtract the minimum element of each column from every element of that column in the resulting matrix. The minimum element in the first column is 10. So 10 is to be subtracted from every element of the first column, i.e., from 40, 10, 10, and 10, respectively. As a result, the elements of the first column of the resulting matrix are 30, 0, 0, 0, respectively. Similarly, we obtain the elements of the other columns of the resulting matrix. Thus, the resulting matrix is:

	A	B	C	D
1	30	10	0	10
2	0	10	40	0
3	0	30	20	0
4	0	0	0	10

Now, starting from first row onward, we draw a rectangle around the 0 in each row having a single zero and cross all other zeroes in the corresponding column. Here, in the very first row we find a single zero. So, we draw a rectangle around it and cross all other zeroes in the corresponding column. We get

	A	B	C	D
1	30	10	0	10
2	0	10	40	0
3	0	30	20	0
4	0	0	0	10

In the second, third and fourth row, there is no single zero. Hence, we move column-wise. In the second column, we have a single zero. Hence, we draw a rectangle around it and cross all other zeroes in the corresponding row. We get

	A	B	C	D
1	30	10	0	10
2	0	10	40	0
3	0	30	20	0
4	0	0	0	10

In the matrix above, there is no row or column, which has a single zero. Therefore, we first move row-wise to locate the row having more than one zero. The second row has two zeroes. So, we draw a rectangle arbitrarily around one of these zeroes and cross the other one. Let us draw a rectangle around the zero in the cell (2, A) and cross the zero in the cell (2, D). We cross out the other zeroes in the first column. Note that we could just as well have selected the zero in the cell (2, D), drawn a rectangle around it and crossed all other zeroes. This would have led to an alternative solution.

In this way, we are left with only one zero in every row and column around which a rectangle has been drawn. This means that we have assigned only one operation to one operator. Thus, we get the optimum solution as follows:

	A	B	C	D
1	30	10	0	10
2	0	10	40	0
3	0	30	20	0
4	0	0	0	10

Note that the assignment of jobs should be made on the basis of the cells corresponding to the zeroes around which rectangles have been drawn. Therefore, the optimum solution for this problem is:

$$1 \rightarrow C, 2 \rightarrow A, 3 \rightarrow D, 4 \rightarrow B$$

This means that programmer 1 is assigned programme C, programmer 2 is assigned programme A, and so on. The minimum time taken in developing the programmes is

$$= 80 + 80 + 100 + 90 = 350 \text{ min.}$$

Example 2: A company is producing a single product and selling it through five agencies situated in different cities. All of a sudden, there is a demand for the product in five more cities that do not have any agency of the company.

The company is faced with the problem of deciding on how to assign the existing agencies to dispatch the product to the additional cities in such a way that the travelling distance is minimised. The distances (in km) between the surplus and deficit cities are given in the following distance matrix.

Deficit City Surplus city	I	II	III	IV	V
A	160	130	175	190	200
B	135	120	130	160	175
C	140	110	155	170	185
D	50	50	80	80	110
E	55	35	70	80	105

Determine the optimum assignment schedule.

Solution: Subtracting the minimum element of each row from every element of that row, we have

	I	II	III	IV	V
A	30	0	45	60	70
B	15	0	10	40	55
C	30	0	45	60	75
D	0	0	30	30	60
E	20	0	35	45	70

Subtracting the minimum element of each column from every element of that column, we have

	I	II	III	IV	V
A	30	0	35	30	15
B	15	0	0	10	0
C	30	0	35	30	20
D	0	0	20	0	5
E	20	0	25	15	15

We now assign zeroes by drawing rectangles around them as explained in Example 1. Thus, we get

	I	II	III	IV	V
A	30	0	35	30	15
B	15	✗	0	10	✗
C	30	✗	35	30	20
D	0	✗	20	✗	5
E	20	✗	25	15	15

Since the number of assignments is less than the number of rows (or columns), we proceed from Step 5 onwards of the Hungarian method as follows:

- we tick mark (✓) the rows in which the assignment has not been made. These are the 3rd and 5th rows.

- ii) we tick mark ($\checkmark$) the columns which have zeroes in the marked rows.
This is the 2nd column.
- iii) we tick mark ($\checkmark$) the rows which have assignments in marked columns.
This is the 1st row.
- iv) again we tick mark ($\checkmark$) the column(s) which have zeroes in the newly marked row. This is the 2nd column, which has already been marked.
There is no other such column. So, we have

	I	II	III	IV	V	
A	30	0	35	30	15	$\checkmark$
B	15	0	0	10	0	
C	30	0	35	30	20	$\checkmark$
D	0	0	20	0	5	
E	20	0	25	15	15	$\checkmark$
		$\checkmark$				

We draw straight lines through unmarked rows and marked columns as follows:

	I	II	III	IV	V	
A	30	0	35	30	15	$\checkmark$
B	15	0	0	10	0	
C	30	0	35	30	20	$\checkmark$
D	0	0	20	0	5	
E	20	0	25	15	15	$\checkmark$
		$\checkmark$				

We proceed as follows, as explained in step 6 of the Hungarian method:

- i) We find the smallest element in the matrix not covered by any of the lines. It is 15 in this case.
- ii) We subtract the number '15' from all the uncovered elements and add it to the elements at the intersection of the two lines.
- iii) Other elements covered by the lines remain unchanged.

Thus, we have

	I	II	III	IV	V
A	15	0	20	15	0
B	15	15	0	10	0
C	15	0	20	15	5
D	0	15	20	0	5
E	5	0	10	0	0

We repeat Steps 1 to 4 of the Hungarian method and obtain the following matrix:

	I	II	III	IV	V
A	15	✗	20	15	0
B	15	15	0	20	✗
C	15	0	20	15	5
D	0	15	20	✗	5
E	5	✗	10	0	0

Since each row and each column of this matrix has one and only one assigned 0, we obtain the optimum assignment schedule as follows:

$$A \rightarrow V, B \rightarrow III, C \rightarrow II, D \rightarrow I, E \rightarrow IV$$

Thus, the minimum distance is $200 + 130 + 110 + 50 + 80 = 570$ km.

You should pause here and try to solve the following assignment problem to check your understanding.

-
- E1)** A solicitor's firm employs typists on an hourly piece-rate basis for their daily work. There are five typists for service and their charges and speeds are different. According to the contract, only one job is given to one typist. Find the least cost allocation for the following data:

	P	Q	R	S	T
A	85	75	65	85	75
B	90	180	66	90	78
C	75	66	57	75	69
D	80	72	60	80	72
E	76	64	56	72	68

5.3 SOME SPECIAL CASES

In this section, we consider some special cases of the assignment problem such as the maximisation problem, unbalanced assignment problem, alternative optimal solutions and restriction on assignments and discuss the techniques to solve them.

5.3.1 Maximisation Problem

There may be an assignment problem in the form of maximisation problem. For example, profits (or anything else like revenues), which need maximisation may be given in the cells instead of costs/times. To solve such a problem, we find the **opportunity loss matrix** by subtracting the value of each cell from the largest value chosen from amongst all the given cells. When the value of a cell is subtracted from the highest value, it gives the **loss** of amount caused by not getting the opportunity which would have given the highest value. The matrix so obtained is known as the **opportunity loss matrix** and is handled in the same way as the minimisation problem. Let us explain this case with the help of an example.

Example 3: Five salesmen are to be assigned to five districts. Estimates of sales revenue (in thousands) for each salesman are given as follows:

	A	B	C	D	E
1	32	38	40	28	40
2	40	24	28	21	36
3	41	27	33	30	37
4	22	38	41	36	36
5	29	33	40	35	39

Find the assignment pattern that maximises the sales revenue.

Solution: Since we are to maximise the sales revenue, we need to convert it into minimisation form before applying the Hungarian method. For this, we obtain the opportunity loss matrix by subtracting every element in the given table from the largest element in it. In this case, the largest element is 41.

Thus, we obtain the following opportunity loss matrix:

9	3	1	13	1
1	17	13	20	5
0	14	8	11	4
19	3	0	5	5
12	8	1	6	2

Now, we apply the Hungarian method (Steps 1 to 4) and finally obtain the following result matrix:

	A	B	C	D	E
1	8	0	8	7	8
2	0	14	12	14	4
3	8	12	8	6	4
4	19	1	0	8	5
5	11	5	8	0	1

Since the number of assigned zeroes is less than the number of rows, we apply Step 5 of the Hungarian method and draw the minimum number of horizontal/vertical lines that cover all the zeroes as shown in the following table:

	A	B	C	D	E	
1	8	0	8	7	8	
2	0	14	12	14	4	√
3	8	12	8	6	4	√
4	19	1	0	8	5	
5	11	5	8	0	1	
	√					

Let us now, select the minimum element from amongst the uncovered elements, which is 4 in this case. We subtract the element 4 from each of the uncovered elements and add it to the elements which lie at the intersection of the horizontal and vertical lines. Other covered elements remain unaltered. Then applying the Hungarian method to the resulting matrix, we get

	A	B	C	D	E
1	12	0	∞	7	∞
2	0	10	8	10	∞
3	∞	8	4	2	0
4	23	1	0	∞	5
5	15	5	∞	0	1

Since the number of assigned zeroes is equal to the number of rows, the optimum assignment has been attained and is given as:

$$1 \rightarrow B, 2 \rightarrow A, 3 \rightarrow E, 4 \rightarrow C, 5 \rightarrow D$$

Thus, the maximum sales revenue = $38 + 40 + 37 + 41 + 35$ thousand rupees
= 191 thousand rupees.

5.3.2 The Unbalanced Assignment Problem

The assignment problem wherein the number of rows is not equal to the number of columns is said to be an **unbalanced assignment problem**. Such a problem is handled by introducing dummy row(s) if the number of rows is less than the number of columns and dummy column(s) if the number of columns is less than the number of rows. All the elements of such a dummy row/column are taken as zero. The augmented problem is then solved by the Hungarian method as explained earlier.

Example 4: To stimulate interest and provide an atmosphere for intellectual discussion, the faculty of mathematical sciences in an institute decides to hold special seminars on four contemporary topics – Statistics, Operations Research, Discrete Mathematics, Matrices. Each such seminar is to be held once a week. However, scheduling these seminars (one for each topic and not more than one seminar per day) has to be done carefully so that the number of students unable to attend is kept to a minimum. A careful study indicates that the number of students who cannot attend a particular seminar on a specific day is as follows:

	Statistics	Operations Research	Discrete Mathematics	Matrices
Monday	50	40	60	20
Tuesday	40	30	40	30
Wednesday	60	20	30	20
Thursday	30	30	20	30
Friday	10	20	10	30

Find an optimal schedule for the seminars. Also find the number of students who will be missing at least one seminar.

Solution: Here the number of rows is 5 and the number of columns is 4. Therefore, the given assignment problem is unbalanced. As the number of columns is one less than the number of rows, we introduce one dummy column to convert the given assignment problem into a balanced problem. The number of students in each cell of this column is taken as zero. Thus, the problem takes the following form:

	Statistics	Operations Research	Discrete Mathematics	Matrices	Dummy
Monday	50	40	60	20	0
Tuesday	40	30	40	30	0
Wednesday	60	20	30	20	0
Thursday	30	30	20	30	0
Friday	10	20	10	30	0

Now, on applying the Hungarian method (Steps 1 to 4), we get

	Statistics	Operations Research	Discrete Mathematics	Matrices	Dummy
Monday	40	20	50	0	✗
Tuesday	30	10	30	10	0
Wednesday	50	0	20	✗	✗
Thursday	20	10	10	10	✗
Friday	0	✗	✗	10	✗

Since the number of assigned zeroes < number of rows, we apply Step 5 of the Hungarian method and draw the minimum number of horizontal/vertical lines that cover all the zeroes as shown in the following table:

	Statistics	Operations Research	Discrete Mathematics	Matrices	Dummy
Monday	40	20	50	0	✗
Tuesday	30	10	30	10	0
Wednesday	50	0	20	✗	✗
Thursday	20	10	10	10	✗
Friday	0	✗	✗	10	✗

We select the minimum element from amongst the uncovered elements, which is 10 in this case. We subtract this element, i.e., 10 from each uncovered element and add it to the elements which lie at the intersection of the horizontal/vertical lines. Other covered elements will remain unaltered. Thus, the resulting matrix is:

40	20	50	0	10
20	0	20	0	0
50	0	20	0	10
10	0	0	0	0
0	0	0	10	10

Now, on applying the Hungarian method, we have

40	20	50	0	10
20	✗	20	✗	0
50	0	20	✗	10
10	✗	0	✗	✗
0	✗	✗	10	10

Since each row and each column of the matrix has one and only one assigned 0, optimum assignment is made in the cells containing those zeroes around which rectangles have been drawn as

Monday → Matrices, Wednesday → Operations Research,
Thursday → Discrete Mathematics, Friday → Statistics

The total number of students who will be missing at least one seminar
 $= 20 + 20 + 20 + 10 = 70$

5.3.3 Alternative Optimal Solutions

Such solutions exist if while assigning zeroes, i.e., while carrying out Step 4 of the Hungarian method, we neither find any row nor any column, which has a single zero. Then we first move row-wise and then column-wise to locate a row/column having two zeroes. We draw a rectangle arbitrarily around any one of these zeroes and cross the other. Alternatively, the zero around which rectangle has been drawn could have been crossed and the rectangle could have been drawn around the other zero. This will lead to two alternative optimum solutions. While assigning zeroes, i.e., while carrying out Step 4 of the Hungarian method, we may neither find any row nor any column, which has single or two zeroes. Then we first move row-wise and then column-wise to locate a row/column having three or more zeroes. This will lead to three or more alternative solutions.

Note that there are two alternative solutions in **Example 1**.

5.3.4 Restriction on Assignments

Sometimes, in an assignment problem, there may be a case when a particular resource (say, a person) cannot be assigned a particular activity (say, a job). To handle such a problem, we assign a very large cost (or time or anything else which is to be minimised) to that case and represent it by ∞ or M , where M denotes a very high cost. This is done to restrict the entry of this pair of resource-activity in the final solution.

Example 5: A company has taken the third floor of a multi-storied building for rent to locate one of its zonal offices. There are five main rooms in this floor to be assigned to five managers. Each room has its own advantages and disadvantages. Some have two cupboards, some are closer to the washrooms or to the canteens, some are of big sizes and are on different floors, etc. Each of the five managers were asked to rank their room preferences amongst the rooms 201, 302, 103, 304 and 205. Their preferences were recorded in a table as indicated below:

		MANAGERS				
		M_1	M_2	M_3	M_4	M_5
ROOMS	302	302	103	302	201	
	103	304	201	205	302	
	304	205	304	304	304	
	201	205	103			
		302				

Most of the managers did not include all five rooms in the list since they were not satisfied with some of them. Assuming that their preferences can be quantified by numbers, find out which manager should be assigned which room so that their total preference ranking is minimum.

Solution: Let us give the ranks 1, 2, 3, 4 and 5 to the first, second, third, fourth and fifth preferences. We assign ∞ to the cells for which no preference is given. Thus, the given problem can be represented by the following assignment table:

	201	302	103	304	205
M_1	∞	1	2	3	∞
M_2	4	1	∞	2	3
M_3	2	5	1	3	4
M_4	∞	1	4	3	2
M_5	1	2	∞	3	∞

Let us now solve this assignment problem using the Hungarian method. Following Steps 1 to 4, we get

	201	302	103	304	205
M_1	∞	0	1	1	∞
M_2	3	∞	∞	0	1
M_3	1	4	0	1	2
M_4	∞	∞	3	1	0
M_5	0	1	∞	1	∞

Now, since each row and each column has one and only one assigned zero, the optimum assignment, i.e., the assignment with maximum satisfaction is made and is given by:

$$M_1 \rightarrow 302, M_2 \rightarrow 304, M_3 \rightarrow 103, M_4 \rightarrow 205, M_5 \rightarrow 201.$$

5.4 TRAVELLING SALESMAN PROBLEM

Consider a travelling salesman who has to visit a certain number of cities in his assigned territory. For each city of his territory, he wishes to visit each city once and only once and arrive back in the city from where he started. He knows the distances (or cost or time) of journey between every pair of cities, and wishes to determine the tour schedule that represents the least distance/cost/time. Such types of problems can be solved by the assignment algorithm. The difference between a travelling salesman problem and an assignment problem is that in an assignment problem, different destinations are assigned to different sources but in a travelling salesman problem, a destination is assigned to a source. Then this destination becomes another source to which we assign another destination, which in turn becomes another source, and so on. Let us explain this point further with the help of an example.

Example 6: A travelling salesman has to visit five cities. He wishes to start from a particular city, visit each city once and then return to his starting point. The travelling time (in hours) for each city from a particular city is given below:

To From	A	B	C	D	E
A	∞	5	8	4	5
B	5	∞	7	4	5
C	8	7	∞	8	6
D	4	4	8	∞	8
E	5	5	6	8	∞

What is the sequence of visits of the salesman so that the total travelling time is minimised?

Solution: Applying the Hungarian method to this problem, we get

	A	B	C	D	E
A	∞	5	2	<u>0</u>	5
B	<u>0</u>	∞	1	4	5
C	2	1	∞	3	<u>0</u>
D	4	<u>0</u>	3	∞	4
E	5	5	<u>0</u>	4	∞

As per the above assignment, the salesman should travel from A to D, D to B, B to A, i.e., $A \rightarrow D \rightarrow B \rightarrow A$.

The above solution is not a complete solution of the travelling salesman problem as the salesman returns to A without travelling through all the cities. So, we proceed as follows:

Since the assignment of zeroes has not given the solution of the travelling salesman problem, we bring the next minimum non-zero element into the solution. Thus, we obtain the next best solution by bringing 1 into the solution. Had there not been 1 in any cell, we would have taken the minimum, but greater than 1 value from amongst all the values of the table. Here, we have 1 and it appears in two places in this problem. One of these is chosen arbitrarily. Let us choose the cell (B, C) and form a rectangle around the value in this cell and cross out the zeroes in its row and column. Now, we apply the Hungarian method for the assignment of zeroes. Thus, we have

	A	B	C	D	E
A	∞	5	2	<u>0</u>	5
B	5	∞	<u>1</u>	4	5
C	2	1	∞	3	<u>0</u>
D	<u>0</u>	4	3	∞	4
E	5	<u>0</u>	6	4	∞

Alternatively, we get

	A	B	C	D	E
A	∞	5	2	4	<u>0</u>
B	5	∞	1	<u>0</u>	5
C	2	<u>1</u>	∞	3	6
D	<u>0</u>	4	3	∞	4
E	5	5	<u>0</u>	4	∞

In the case of the first alternative, the optimum assignment is $A \rightarrow D \rightarrow A$, but this is not the solution of the travelling salesman problem.

In the case of the second alternative, the optimum assignment is

$$A \rightarrow E \rightarrow C \rightarrow B \rightarrow D \rightarrow A$$

This is the complete solution for the problem as starting from A, the salesman returns to A visiting all the other cities. The minimum time taken by him to travel to all the cities is $5 + 6 + 7 + 4 + 4 = 26$ hrs.

You may like to solve the following exercise to check your understanding.

E2) Solve the following travelling salesman problem so as to minimise the cost per cycle:

To From	A	B	C	D	E
A	∞	3	6	2	3
B	3	∞	5	2	3
C	6	5	∞	6	4
D	2	2	6	∞	6
E	3	3	4	6	∞

What is the sequence of visits of the salesman so that total travelling time is minimised?

Let us now summarise the main points which have been covered in this unit.

5.5 SUMMARY

1. In an **assignment problem**, the number of operations is assigned to an equal number of operators where each operator performs only one operation. So, an assignment problem may be considered as a special type of transportation problem in which the capacity of each of the sources as well as the requirement of each of the destinations is taken as 1. In an assignment problem, the given matrix must necessarily be a square matrix, which is not the condition for a transportation problem. Such problems are solved using the **Hungarian method**, which is shorter and easier compared to any other method of finding the optimal solution.
2. There may be assignment problems where **maximisation** is required to be done instead of minimisation. To handle such a problem, we find the opportunity loss matrix by subtracting the value of each cell from the largest value chosen from amongst all the given cells. When the value of a cell is subtracted from the highest value, it gives the loss of amount caused by not getting the opportunity which would have given the highest value. The matrix so obtained is known as the opportunity loss matrix and is handled in the same way as the minimisation problem.
3. The assignment problem wherein the number of rows is not equal to the number of columns is said to be an **unbalanced problem**. Such a problem is handled by introducing dummy row(s) if the number of rows is less than the number of columns and dummy column(s) if the number of columns is less than the number of rows. All elements for such a dummy row/column are taken as zero. The augmented problem is then solved by the Hungarian method.

4. While assigning zeroes in an assignment problem, if we neither find any row nor any column which has single zero, then we first move row-wise and then column-wise to locate a row/column having two (if not two, then three or more) zeroes. Then a rectangle is formed arbitrarily around one of these zeroes and the others are crossed. Alternatively, the zero around which the rectangle has been made could have been crossed and the rectangle could have been formed around any of the other zeroes. This leads to **alternative optimum solutions**.
5. Sometimes, there may be **restriction on assigning** a particular activity to a particular resource. Then a very large cost (or time or anything else which is to be minimised) is considered and represented by ∞ or M for such a restricted pair.
6. The **travelling salesman problem** wherein there are a certain number of cities to be visited by a salesman in his assigned territory can also be solved as an assignment problem. The only difference between the travelling salesman problem and an assignment problem is that in an assignment problem different destinations are assigned to different sources.

But in a travelling salesman problem, a destination is assigned to a source and then that destination becomes another source to which another destination is assigned, and so on. This is because, in a travelling salesman problem, for each city of his territory, the salesman wishes to visit each city once and only once and arrive back at the city from where he started. He wishes to determine the tour schedule that represents the least distance/cost/time.

5.6 SOLUTIONS /ANSWERS

- E1)** Applying the Hungarian method of solving an assignment problem, we finally get

	P	Q	R	S	T
A	2	4	2	4	0
B	4	106	0	6	∞
C	∞	0	2	2	2
D	0	4	0	2	0
E	2	2	∞	0	2

Thus, the least cost allocation is given by :

$$A \rightarrow T, B \rightarrow R, C \rightarrow Q, D \rightarrow P, E \rightarrow S$$

$$\begin{aligned} \text{and the total minimum cost is } & (75 + 66 + 66 + 80 + 72) \\ & = \text{`}359. \end{aligned}$$

- E2)** Applying the Hungarian method of solving an assignment problem, we reduce the cost matrix and make assignments in rows and columns having single zeroes.

	A	B	C	D	E	
A	∞	$\times$	3	0	$\times$	$\checkmark$
B	1	∞	2	$\times$	$\times$	$\checkmark$
C	2	1	∞	2	0	
D	$\times$	0	3	∞	4	
E	$\times$	$\times$	0	3	∞	
				$\checkmark$		

Now, we draw the minimum number of lines to cover all the zeroes. Then we subtract the lowest element from all the elements not covered by any of lines and add the same at the intersection of two lines. We have

	A	B	C	D	E	
A	∞	$\times$	2	0	$\times$	$\checkmark$
B	0	∞	1	$\times$	$\times$	$\checkmark$
C	2	1	∞	3	0	
D	$\times$	0	3	∞	4	
E	$\times$	$\times$	0	4	∞	
				$\checkmark$		

As per the above assignment, the salesman should travel from A to D, D to B, B to A, i.e., $A \rightarrow D \rightarrow B \rightarrow A$.

The above solution is not a complete solution of the travelling salesman problem as the salesman returns to A without travelling through all the cities. So, we proceed as follows:

Since the assignment of zeroes has not given the solution of the travelling salesman problem, we bring the next minimum non-zero element in the solution. Thus, we obtain the next best solution by bringing 1 into the solution. Had there not been 1 in any cell, we would have taken the minimum, but greater than 1 value from amongst all the values of the table. Here, we have 1 and it appears in two places in this problem. One of these is chosen arbitrarily.

Let us choose the cell (B, C) and form a rectangle around the value in this cell and cross out the zeroes in its row and column. Now, we apply the Hungarian method for the assignment of zeroes. Thus, we have

	A	B	C	D	E
A	∞	$\times$	2	0	$\times$
B	$\times$	∞	1	$\times$	$\times$
C	2	1	∞	3	0
D	0	$\times$	3	∞	4
E	$\times$	0	$\times$	4	∞

Alternatively, we get

	A	B	C	D	E
A	∞	∞	2	∞	0
B	∞	∞	1	∞	∞
C	2	1	∞	3	∞
D	0	∞	3	∞	4
E	∞	∞	0	4	∞

In the case of the first alternative, the optimum assignment is $A \rightarrow D \rightarrow A$, but this is not the solution of the travelling salesman problem.

In the case of the second alternative, the optimum assignment is

$A \rightarrow E \rightarrow C \rightarrow B \rightarrow D \rightarrow A$, i.e., $3 + 4 + 5 + 2 + 2 = 16$ hr

UNIT 6 QUEUEING THEORY

Structure

- 6.1 Introduction
 - Objectives
- 6.2 Basic Concepts of Queueing Theory
 - Poisson Process
 - Birth and Death Process
- 6.3 Fundamental Structure of a Queueing System
- 6.4 Operating Characteristics of a Queueing System
 - Operating Characteristics
 - Classification of Queueing Systems
- 6.5 M/M/1 Queueing Model
 - Arrival-Departure Equations for M/M/1 Queueing Model
 - Operating Characteristics for the M/M/1 Queueing Model
- 6.6 Summary
- 6.7 Solutions /Answers

6.1 INTRODUCTION

Queueing is a common phenomenon in everyday life. We wait in queues in post offices, banks, restaurants, railways and airline reservation counters. Vehicles wait at traffic lights and aeroplanes circle around airports while waiting to land. You can think of many more examples. In all such cases, there are customers who require some sort of services after waiting in a queue for some time. The customer may be a person, machine, vehicle or anything else which requires service. In fact, waiting for service is an integral part of our daily life and that too at considerable cost most of the times.

We would like to find ways of reducing the time spent in waiting by the customer and at the same time optimising the cost to the service provider. This is where the **queueing theory**, also known as the waiting line theory helps us. It was developed in 1909 when A. K. Erlang made an effort to analyse telephone traffic congestion. The purpose of queueing analysis is to provide information to determine an acceptable level of service and service capacity since providing too much service capacity is costly (owing to idle employees or equipment). However, providing too little service capacity is also costly (owing to waiting members in the queue). For example, when a hospital consistently has a long queue in its emergency room, a large number of waiting patients may aggravate the injury or illness.

In this unit, we first discuss the basic concepts of queueing theory in Sec. 6.2, which would help you understand the techniques of queueing models. In Secs. 6.3 and 6.4, we explain the fundamental structure and operating characteristics of a queueing system. We describe the M/M/1 queueing model and its applications in Sec. 6.5.

In the next unit, we shall discuss the sequencing problems and explain the n-jobs, 2-machine and 2-jobs, m-machines, sequencing problems.

Objectives

After studying this unit, you should be able to:

- explain the Poisson process and the birth and death process in queueing theory;

- explain the fundamental structure of a queueing system;
- describe the operating characteristics of a queueing system;
- explain a single server queueing model with Poisson input and exponential service time; and
- solve problems based on the M/M/1 queueing model.

6.2 BASIC CONCEPTS OF QUEUEING THEORY

For understanding queueing systems, you have to be familiar with the probability theory. The concept of random variable and its probability distribution such as Poisson distribution, Exponential distribution and Poisson process play a significant role in describing queueing systems. You have studied probability theory, random variable, Poisson and Exponential distributions in the course MST-003 on Probability Theory. However, the Poisson process is a new concept for you and we explain it, in brief. Let us quickly recall a few basic definitions.

Consider an experiment whose outcome is not uniquely determined. In such a situation an observed outcome of the experiment is one from a set of possible outcomes. An outcome of an experiment is called a **sample point** and the set of all possible outcomes of a random experiment is called a **sample space**. Subsets of the sample space are called **events**. **A random variable is a function that associates a point of the sample space with a real number. A random process or stochastic process** is a family (or collection) of random variables. For example, if a die with six faces numbered 1, 2, ..., 6 is thrown, the set of all possible outcomes is $S = \{1, 2, 3, 4, 5, 6\}$. The set S is a sample space. A function X that assigns to an outcome or sample point, the number written on it, is the random variable. The event that an even number is observed corresponds to the set $\{2, 4, 6\}$ which is a subset of S .

Let us consider another simple experiment such as throwing a fair die.

1. Suppose X_n is the outcome of the n^{th} throw, $n \geq 1$. Then $\{X_n, n \geq 1\}$ is a family of random variables, such that for a distinct value of n , one gets a distinct random variable X_n . The sequence $\{X_n, n \geq 1\}$ constitutes a random (stochastic) process known as the **Bernoulli process**.
2. Suppose X_n is the number of sixes in the first n throws. For a distinct value of $n = 1, 2, \dots$, we get a distinct **Binomial** random variable. The sequence $X_n: \{X_n, n \geq 1\}$, which gives a family of random variables, is a random or stochastic process.
3. Suppose a telephone call is received at a switchboard. Let X_t be the random variable, which represents the number of incoming calls in an interval $(0, t)$. Then X_t is a random variable and the family $\{X_t, t \in T\}$ constitutes a stochastic process, where T is the interval $0 \leq t \leq \infty$.

In Examples 1 and 2 above we saw that the subscript n of X_n was restricted to non-negative integers $n = 0, 1, 2, \dots$. In these examples we observe the outcome of a random variable at distinct time points $n = 0, 1, 2, \dots$. Here, the word 'time' is used in a wider sense. You can visualise an infinite family of random variables $\{X_t, t \in T\}$ such that the state of the system is characterised at every instant over a finite or infinite interval. The process (or collection) is then defined for a continuous range of time and we say that we have a family of random variables. On the other hand, the family $\{X_n, n = 0, 1, 2, \dots\}$ is called a discrete parameter (or discrete time) stochastic process. The value of X_n for a specific realisation of the process is called the state of the process at the n^{th} step. If the random variables X_n are discrete random variables, i.e., if

they take only integer values, it is called a discrete state process. Let us now explain the Poisson process used in the M/M/1 model being discussed in Sec. 6.5.

6.2.1 Poisson Process

Suppose $X(t)$ represents the maximum temperature at a particular place. Here we deal with discrete state, continuous time stochastic processes, i.e., $X(t)$ is a discrete random variable. The Poisson process is one of the representatives of this type of stochastic processes. It plays an important role in the study of a large number of phenomena. The Poisson process may be described as follows: Let E be a random event such as (i) incoming telephone calls at a switchboard, (ii) arrival of patients for treatment at a clinic, (iii) occurrence of accidents at a certain place.

We consider the total number $N(t)$ of occurrences of an event E in an interval of time ' t '. Suppose $P_n(t)$ is the probability that the random variable $N(t)$ assumes the value n , i.e.,

$$P_n(t) = P[N(t) = n] \quad \dots (1)$$

for $n = 0, 1, 2, 3, \dots$. We have

$$\sum_{n=0}^{\infty} P_n(t) = 1, \text{ for each fixed } t. \quad \dots (2)$$

We can thus say from equation (2) that $P_n(t)$ is a probability mass function of the random variable $N(t)$ and the family of random variables $\{N_t, t > 0\}$ is a stochastic process. From our earlier discussion, you may understand that this family is a continuous parameter (in this case, time) stochastic process with a discrete state space. This is called a Poisson process. Under certain conditions, $N(t)$ follows a Poisson distribution with mean λt (λ being constant). This is true for most practical situations.

Assumptions in Poisson Process

- i) The process has independent increments. Future changes in $N(t)$ are independent of past changes in it, i.e., the number of customers which arrive in disjoint time intervals are statistically independent.
- ii) The probability of more than one occurrence of event E between time t and $t+\Delta t$ is $o(\Delta t)$, i.e., the probability of two or more arrivals of customers during the small interval of time Δt is negligible. Thus,

$$P_0(\Delta t) + P_1(\Delta t) + o(\Delta t) = 1 \quad \dots (3)$$

- iii) The probability that event E occurs between time t and Δt is equal to $\lambda \Delta t + o(\Delta t)$. Thus,

$$P_1(\Delta t) = \lambda \Delta t + o(\Delta t),$$

i.e., $P_1(\Delta t)$ is approximately proportional to the length of the interval where λ is the expected average number of arrivals of customers per unit time and $\lambda > 0$. Here λ is constant and Δt is an incremental element.

Under the assumptions stated above, $N(t)$ follows a Poisson distribution with mean λt , i.e., $P_n(t)$ is given by

$$P_n(t) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}, \quad n = 0, 1, 2, 3, \dots \quad \dots (4)$$

To formulate a queueing model, we have to specify the assumed form of the probability distributions of both inter-arrival times and service times. You have learnt that inter-arrival times follow the Poisson distribution. Similarly, most of the times, the service times in a queueing system follow the exponential distribution. Hence, the exponential distribution is the most important distribution in queueing theory, which we now discuss.

Suppose a random variable T represents either inter-arrival or service times. T is said to have an exponential distribution with parameter α if its probability density function is given by

$$f(t) = \begin{cases} \alpha e^{-\alpha t} & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases} \quad \dots(5)$$

Let $N(t)$ denote the number of occurrences of an event E (say arrival or number of telephone calls) in duration Δt . There is an associated variable: the interval T between two successive occurrences of event E . The interval T between two successive occurrences of a Poisson process $\{N_t, t \geq 0\}$, having parameter λ , has an exponential distribution with parameter λ . If the intervals between successive occurrences of an event E are independently distributed with a common exponential distribution with parameter λ , the events follow a Poisson process with mean λt , i.e., the probability distribution of the number of times these kinds of events occur over a specified duration of time has a Poisson distribution with parameter λt .

The basic methodology discussed here was developed initially by the Russian mathematician, A. A. Markov, around the beginning of the 20th century. The Markov Process forms a sub-class of the set of all random processes. It is a sub-class with enough simplifying assumptions to make them easy to handle. Mathematically, we define the Markov Process as follows:

Let $\{X(t), t \geq 0\}$ be a continuous time stochastic process taking on values in the set of non-negative integers. If, for all $t_n, t_{n-1}, \dots, t_0$, satisfying $t_n > t_{n-1} > \dots > t_0$, and non-negative integer j ,

$$P[X(t_n) = j_n / X(t_{n-1}) = j_{n-1}, \dots, X(t_0) = j_0] = P[X(t_n) = j_n / X(t_{n-1}) = j_{n-1}],$$

the process has the Markov property and is called a **continuous time Markov process**. Now we confine to Markov processes in which time is measured discretely. The process is described by a sequence of random variables, $\{X_1, X_2, X_3, \dots\}$. Here the probability distribution of each X_i must be specified. But it is not enough to specify a stochastic process, since X_i s are not independent. At this point we need Markov's simplifying assumption: A **Markov chain** is a discrete time stochastic process in which each random variable, X_i depends on the previous one, X_{i-1} and affects only the subsequent one, X_{i+1} . The term **chain** suggests the linking of the random variable to neighbours in the sequence.

6.2.2 Birth and Death Process

The birth and death process is a special case of the general continuous time Markov process. A birth and death process is characterised as a Markov process in which all transitions are to be to the next state, immediately above (a "birth") or immediately below (a "death") in the natural integer ordering states, i.e., a birth and death process does not 'jump' states. The Poisson process is a special case of the birth-death process. It might be more appropriate to call it a pure birth process since the death rates (μ_i) are all zero. The birth rates, λ_i , are all equal to a constant value λ . The Poisson process is

often used to model the kind of situation in which a count is made of the number of events occurring in a given time. For example, it will be used in the discussion of queueing models to represent the arrivals of customers to a service facility. The state at time t would correspond to the number of arrivals by time t . $P_n(t)$ denotes the probability that there are n customers in the system at any time t , both waiting and being served and P_n stands for time independent probability that there are n customers in the system, both waiting and being served. Arrivals can be considered as births. If the system is in state E_n and an arrival occurs, the state is changed to E_{n+1} . Similarly, a departure can be looked upon as death. A departure occurring while the system is in state E_n changes the system to the state E_{n-1} . This type of process is generally referred to as a birth-death process.

Assumptions in the Birth-Death Process

The assumptions of the birth-death process are as follows:

- i) If the system is in state E_n , the current probability distribution of the time t until the next arrival (birth) is exponential with parameter λ_n , where $n = 0, 1, 2, \dots$.
- ii) If the system is in state E_n , the current probability distribution of the time t until the next departure or service completion (death) is exponential with parameter μ_n , where $n = 0, 1, 2, \dots$.
- iii) Only one birth or death can occur in a small interval of time, i.e., Δt .

Having discussed the basic concepts and techniques for studying queueing systems, we now explain the fundamental structure of a queueing system.

6.3 FUNDAMENTAL STRUCTURE OF A QUEUEING SYSTEM

Queueing theory is concerned with the mathematical study of queues or waiting lines (seen in banks, post offices, hospitals, airports, etc.). The formation of waiting lines usually occurs whenever the current demand for a service exceeds the current capacity to provide that service.

In many cases, the customer's arrival and his or her service time are not known in advance or cannot be predicted accurately. Otherwise, the operation of the service facility could be scheduled in a manner that would eliminate waiting completely. Both arrival and departure phenomena are random. This necessitates mathematical modelling or queueing systems/ models to alleviate waiting. It involves reducing excessive costs that result from creating excess service capacity and at the same time ensuring that the system has enough service capacity to avoid long waiting lines. There has to be a balance between service capacity and waiting time. Therefore, an industry or an agency would like to provide such services and also maintain balance between the cost of service and the cost associated with waiting for the service.

A simple queueing system has the following fundamental structure:

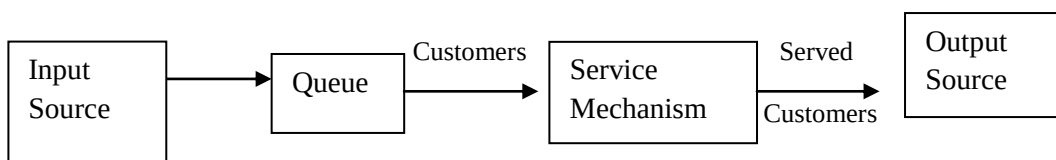


Fig. 6.1: Queueing system.

A queueing system is described by the following elements:

1. Input or arrival process of customers;
2. Service mechanism; and
3. Queue discipline.

We now give a brief description of each of these components.

1. Input or Arrival Process of Customers

This is the element concerned with the pattern in which the customers arrive and join the system. An input process is characterised by its size, the arrival time distribution, and the attitude of the customers. We describe these, in brief.

- i) **Size:** It may be finite or infinite according as the arrival rate is affected or not affected by the number of customers in the service system.
- ii) The **arrival time distribution:** In most cases, the arrivals occur in accordance with a Poisson distribution.
- iii) **Customer behaviour:**
 - The customer may stay in the system until served. Such a customer is known as **Patient Customer**.
 - The customer may wait for a certain time and leave the system if service is not commenced by that time. This kind of departure from the queue without receiving the service is known as **impatient or reneging behaviour**.
 - When a customer arrives at a queueing system and perceives that the number of customers already in the queue is too large, s/he does not join the queue. This behaviour of the customers is known as **balking behaviour**.
 - When a customer moves from one waiting line to another because he/she thinks that his/her queue is moving slower, the behaviour of the customer is known as **queue jockeying**.

Generally, it is assumed that the customers arrive into the system one at a time. But, sometimes, customers may arrive in groups and such arrival is called **bulk arrival**.

2. Service Mechanism

Service time distributions are generally exponential distributions. A service facility can be any one of the following types:

- i) Single channel facility, i.e., one queue-one service station facility;
- ii) One queue-several station facilities (e.g., booking at a service station that has several service mechanisms);
- iii) Several queues-one service station (e.g., railway station ticket counters);
- iv) Multi-channel facility (e.g., several counters); and
- v) Multistage Channel facility (e.g., various medical tests of a patient).

3. Queue Discipline

Queue discipline refers to the order in which the service station selects the next customer from the waiting line to be served. It may be any one of the following:

- i) First In, First Out (FIFO) or in other words First Come First Served (FCFS);

- ii) Last In, First Out (LIFO); and
- iii) Service In Random Order (SIRO).

In this unit, we consider the FCFS queue discipline.

So far you have learnt the fundamental structure of a queueing system. We now describe its operating characteristics.

6.4 OPERATING CHARACTERISTICS OF A QUEUEING SYSTEM

We have set up a set of equations to study the queueing system. We have to solve these equations to determine the operating characteristics. There are two types of solutions of these equations: transient and steady state. The time dependent solutions are known as **transient solutions**. For a complete description of a queueing process, we need transient solutions. However, it is difficult to obtain such solutions. Moreover, in many practical situations, we usually need to know the system's behaviour in steady state, i.e., the behaviour of the system when it reaches its equilibrium state after being in operation for a long time. The steady state solution is independent of time and represents the probability of the system being in a particular state in the long run.

A birth and death process gradually attains steady state after a sufficiently large time has elapsed, provided that the parameters of the process permit reaching the steady state. For example, queues with arrival rate λ higher than the departure rate μ will not reach a steady state as the queue size will increase with time. The ultimate objective of analysing queueing situations is to develop measures of performance for evaluating the real system. Here we shall be analysing the system under the steady state condition since the time dependent analysis is beyond the scope of this course. Let us now list the operating characteristics of a queueing system.

6.4.1 Operating Characteristics

The operating characteristics of a queueing systems are determined by two statistical properties, namely, the probability distribution of inter-arrival times and the probability distribution of service times. In the analysis of a queueing system, we would like to compute various operating characteristics, i.e., the measures of performance. These include:

1. The average number of customers in the queueing system (L_s), i.e., those waiting to be served and those being served.
2. The average time each customer spends in the queueing system from entry into the queue to completion of the service (W_s), i.e., the time spent waiting in the queue and during the service.
3. The average number of customers in the queue waiting to get service (L_q), i.e., this excludes customers undergoing service.
4. The average time each customer spends waiting in the queue to get service (W_q), i.e., this excludes time spent during the service.
5. The relative frequency with which the service system is idle is called **service idle time**. Though the idle time is directly related to the cost, its reduction may have adverse effects on other characteristics.

These equations can be used to study the time dependent behaviour as well. We now state the equations for each of the operating characteristics listed above under steady state condition.

By definition of expectation, we have

$$L_s = \sum_{n=1}^{\infty} nP_n \quad \dots (6)$$

$$L_q = \sum_{n=c+1}^{\infty} (n - c)P_n \quad \dots (7)$$

Suppose that arrivals follow the Poisson distribution, and the inter-arrival times follow exponential distributions [Chapter 4 of Stochastic Process by J. Medhi]. The exponential distribution has memoryless property, which means that the future is independent of the past. It depends only on the present and hence the Markov property [Unit 15, Block 4 of MST-003: Probability Theory and Chapter 3 of Stochastic Process by J. Medhi] is satisfied by the inter-arrival times. Therefore, symbol 'M' has been used in place of 'a' to represent the Markov property. Also, if service times follow the exponential distribution, then the service time distribution too has the Markov property and hence symbol 'M' can be used in place of 'b'.

where there are c parallel servers so that c customers can be served simultaneously. With λ arrival rates for those who join the system, we have

$$L_s = \lambda W_s \quad \text{and} \quad L_q = \lambda W_q \quad \dots (8)$$

If μ is the service rate, the expected service time is $\frac{1}{\mu}$ and we have

$$W_s = W_q + \frac{1}{\mu} \quad \dots (9)$$

Multiplying both sides of equation (9) with λ , we get

$$L_s = L_q + \frac{\lambda}{\mu} \quad \dots (10)$$

6.4.2 Classification of Queueing Systems

A queueing system is usually described by five symbols and denoted as **a / b / c : d / e** or separated by slashes as **a / b / c / d / e**. The first symbol 'a' describes the arrival process. The second symbol 'b' describes the service time distribution. The third symbol 'c' stands for the number of servers. The symbols 'd' and 'e' stand for system capacity and queue discipline, respectively. If the system has FCFS queue discipline, the queueing system may be described as **a / b / c / d**, i.e., the fifth symbol may be omitted in this case. If a system has infinite capacity with FCFS queue discipline, the queueing system may be described as **a/b/c : ∞ / FCFS** or simply **a/b/c**.

Note: If arrivals follow the Poisson distribution, the symbol **M** is used in place of 'a'. If departures are exponential, the symbol **M** is used in place of 'b'. The explanation for using the symbol **M** is given in the margin for the sake of interest.

So far you have learnt about the operating characteristics of a queueing system. We now discuss the M/M/1 queueing model and determine its operating characteristics.

6.5 M/M/1 QUEUEING MODEL

As the symbol indicates, the M/M/1 queueing model deals with a queueing system having a single service channel with Poisson input, exponential distribution for services and there is no limit on the system capacity while the customers are served on a "First Come First Served" basis.

For this model, arrivals follow Poisson distribution and hence inter-arrival times have an exponential distribution. Let λ denote the average arrival rate. Therefore, $1/\lambda$ is the mean arrival time. For example, if customers arrive at a rate of 15 per hour, it means that on an average, 1 customer arrives in every 4 minutes, i.e., the mean arrival time is $\frac{1}{15}$ hr or 4 minutes (the reciprocal of the arrival rate). The service times for this model follow an exponential distribution. Let μ denote the average service rate. Hence, $1/\mu$ is the mean service time.

Then the ratio

$$\rho = \frac{\lambda}{\mu} \quad \dots (11)$$

is called the **traffic intensity** or the **utilisation factor**. It is a measure of the degree to which the capacity of the service station is utilised. For example, if customers arrive at a rate of 15 per minute and the service rate is 20 per minute, the utilisation of the service facility is $\frac{15}{20} = 75\%$, i.e., the service facility is kept busy 75% of the time and remains idle 25% of the time.

6.5.1 Arrival-Departure Equations for M/M/1 Queueing Model

In this section, we obtain the arrival-departure equations for M/M/1 queueing model. But before doing so, it is necessary to state the postulates of the Poisson process, which are as follows:

- i) The number of occurrences of an event in an interval of duration t are independent of the number of occurrences in an interval prior to the interval $(0, t)$, i.e., future changes are independent of past changes.
- ii) The probability of n customers in ' t ' depends only on the duration ' t ' of the interval and is independent of where this interval is situated. Let $P_n(t)$ be the probability of n customers in an interval t . Then $P_n(t)$ also gives the probability of a number of occurrences of an event in the interval $(a, t + a)$, which is of duration t for every ' a '. For example, the interval may be $(5, 5 + t)$ $(7, 7 + t)$ or any other interval of duration t . For all these intervals, the probability will have homogeneity in time, i.e., it will remain $P_n(t)$ for all intervals of duration t .
- iii) In an interval of infinitesimal duration Δt , i.e., for very small Δt , the probability of exactly one occurrence is $\lambda\Delta t + o(\Delta t)$ and that of more than one occurrences is $o(\Delta t)$, where $o(\Delta t)$ is a function of Δt which tends to 0 more rapidly than Δt . The function $o(\Delta t)$ contains the terms of squares and higher powers of Δt . Thus, as $\Delta t \rightarrow 0$, $\frac{o(\Delta t)}{\Delta t} \rightarrow 0$. In other words,

$$P_1(\Delta t) = \lambda\Delta t + o(\Delta t), \quad \dots (12)$$

$$\text{and } P_2(\Delta t) + P_3(\Delta t) + P_4(\Delta t) + P_5(\Delta t) + \dots = o(\Delta t) \quad \dots (13)$$

Since the sum of probabilities for occurrence/non-occurrence of an event is 1, we have

$$P_0(\Delta t) + P_1(\Delta t) + P_2(\Delta t) + P_3(\Delta t) + P_4(\Delta t) + \dots = 1 \quad \dots (14)$$

Therefore,

$$\begin{aligned} P_0(\Delta t) &= 1 - P_1(\Delta t) - \{P_2(\Delta t) + P_3(\Delta t) + P_4(\Delta t) + \dots\} \\ &= 1 - \{\lambda\Delta t + o(\Delta t)\} - \{o(\Delta t)\} \\ &= 1 - \lambda\Delta t + o(\Delta t) \end{aligned} \quad \dots (15)$$

Since $o(\Delta t)$ is a function containing squares and higher powers of (Δt) , $o(\Delta t) + o(\Delta t)$ is very small and has been written as $o(\Delta t)$ in equation (15).

Now, we can write the arrival-departure equations for the M/M/1 queueing model as follows:

If $n \geq 1$, the probability of n customers in the system at time $t + \Delta t$

= [{Probability of $n - 1$ customers in the system at time t and (1 arrival, no departure in time Δt or 2 arrivals, one departure in time Δt or 3 arrivals, two departures in time Δt or ...)} + {Probability of $n - 2$ customers in the system at time t and (2 arrivals, no departure in time Δt or 3 arrivals, one departure in time Δt or 4 arrivals, two departures in time Δt or ...)} + {Probability of $n - 3$ customers in the system at time t and (3 arrivals, no departure in time Δt or 4 arrivals, one departure in time Δt or 5 arrivals, two departures in time Δt or ...)} + ...]

+ [{Probability of $n + 1$ customers in the system at time t and (1 departure, no arrival in time Δt or 2 departures, one arrival in time Δt or 3 departures, two arrivals in time Δt or ...)} + {Probability of $n + 2$ customers in the system at time t and (2 departures, no arrival in time Δt or 3 departures, one arrival in time Δt or 4 departures, 2 arrivals in time Δt or ...)} + {Probability of $n + 3$ customers in the system at time t and (3 departures, no arrival in time Δt or 4 departures, one arrival in time Δt or 5 departures, two arrivals in time Δt or ...)} + ...] + [{Probability of n customers in the system at time t and (no arrival, no departure in time Δt or 1 arrival, one departure in time Δt or 2 arrivals, 2 departures in time Δt or ...)}]

$$\begin{aligned}
 &= P_{n-1}(t) \cdot [\{\lambda \Delta t + o(\Delta t)\} \{1 - \mu \Delta t + o(\Delta t)\} + \text{The terms equal to } o(\Delta t)] \\
 &\quad + P_{n-2}(t) \cdot [\{o(\Delta t)\} \{1 - \mu \Delta t + o(\Delta t)\} + \text{The terms equal to } o(\Delta t)] \\
 &\quad + P_{n-3}(t) \cdot o(\Delta t) + \text{The terms equal to } o(\Delta t) \\
 &\quad + P_{n+1}(t) \cdot [\{\mu \Delta t + o(\Delta t)\} \{1 - \lambda \Delta t + o(\Delta t)\} + \text{The terms equal to } o(\Delta t)] \\
 &\quad + P_{n+2}(t) \cdot [\{o(\Delta t)\} \{1 - \lambda \Delta t + o(\Delta t)\} + \text{The terms equal to } o(\Delta t)] \\
 &\quad + P_{n+3}(t) \cdot o(\Delta t) + \text{The terms equal to } o(\Delta t) \\
 &\quad + P_n(t) \cdot [\{1 - \mu \Delta t + o(\Delta t)\} \{1 - \lambda \Delta t + o(\Delta t)\} + \{\mu \Delta t + o(\Delta t)\} \{\lambda \Delta t + o(\Delta t)\} \\
 &\quad + \text{The terms which ultimately become equal to } o(\Delta t)]
 \end{aligned}$$

Notice that $\Delta t \cdot o(\Delta t)$, $o(\Delta t) \cdot o(\Delta t)$, $o(\Delta t) + o(\Delta t)$, $o(\Delta t) - o(\Delta t)$, $\Delta t \cdot \Delta t$ are very small and may be taken as equal to $o(\Delta t)$.

We deal with the case $n = 0$ separately because there cannot be any possibility of less than zero customers. However, in the above equation, the case of less than n customers is included.

The probability of 0 customers in the system at time $t + \Delta t$

= [{Probability of 1 customer in the system at time t and (1 departure, no arrival in time Δt or 2 departures, one arrival in time Δt or 3 departures, two arrivals in time Δt)} + {Probability of 2 customers in the system at time t and (2 departures, no arrival in time Δt or 3 departures, one arrival in time Δt or 4 departures, 2 arrivals in time Δt)} + {Probability of 3 customers in the system at time t and (3 departures, no arrival in time Δt or 4 departures, one arrival in time Δt or 5 departures, two arrivals in time Δt)} + ...] + [{Probability of 0 customers in the system at

time t and (no arrival, no departure in time Δt or 1 arrival, one departure in time Δt or 2 arrivals, 2 departures in time Δt)]

$$\begin{aligned}
&= P_1(t) \cdot [\{\mu\Delta t + o(\Delta t)\} \{1 - \lambda\Delta t + o(\Delta t)\} \\
&\quad + \text{The terms which ultimately become equal to } o(\Delta t)] \\
&+ P_2(t) \cdot [\{o(\Delta t)\} \{1 - \lambda\Delta t + o(\Delta t)\} \\
&\quad + \text{The terms which ultimately become equal to } o(\Delta t)] \\
&+ P_3(t) \cdot o(\Delta t) + \text{The terms which ultimately become equal to } o(\Delta t) \\
&+ P_0(t) \cdot [\{\text{Probability of no departure which is 1 as there is no customer} \\
&\quad \text{in system and hence no chance of departure}\} \cdot \{1 - \lambda\Delta t + o(\Delta t)\} \\
&\quad + \{\mu\Delta t + o(\Delta t)\} \{\lambda\Delta t + o(\Delta t)\} + \text{The terms which ultimately become equal to } o(\Delta t)]
\end{aligned}$$

Therefore, the arrival-departure equations for the M/M/1 queueing model are:

$$\begin{cases} P_n(t + \Delta t) = P_{n-1}(t) \cdot [\lambda\Delta t + o(\Delta t)] + P_{n+1}(t) [\mu\Delta t + o(\Delta t)] \\ \quad + P_n(t) [\lambda\Delta t + o(\Delta t)] [\mu\Delta t + o(\Delta t)] \\ \quad + P_n(t) [1 - \lambda\Delta t] [1 - \mu\Delta t] + o(\Delta t), n \geq 1 \\ P_0(t + \Delta t) = P_1(t) [\mu\Delta t + o(\Delta t)] + P_0(t) [\lambda\Delta t + o(\Delta t)] \\ \quad + P_0(t) [1 - \lambda\Delta t] \cdot 1 + o(\Delta t), n = 0 \end{cases} \quad \dots(16)$$

$$\Rightarrow \begin{cases} P_n(t + \Delta t) = P_{n-1}(t) \cdot \lambda\Delta t + P_{n+1}(t) \mu\Delta t \\ \quad + P_n(t) [1 - \lambda\Delta t - \mu\Delta t] + o(\Delta t), n \geq 1 \\ P_0(t + \Delta t) = P_1(t) \mu\Delta t + P_0(t) [1 - \lambda\Delta t] + o(\Delta t) \end{cases} \quad \dots(17)$$

Dividing equation (16) by Δt and taking the limit as Δt tends to zero, we have

$$\begin{aligned} p'_n(t) &= \lambda P_{n-1}(t) + \mu P_{n+1}(t) - (\lambda + \mu) P_n(t) \\ [\because \lim_{n \rightarrow \infty} \frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} &= p'_n(t) [\text{derivative of } P_n(t)] \end{aligned} \quad \dots(18)$$

$$\text{and} \quad p'_0(t) = -\lambda P_0(t) + \mu P_1(t) \quad \dots(19)$$

When steady state, i.e., the equilibrium state, is reached, $p'_n(t)$ becomes independent of time, say p_n , and the rate of its change with respect to time becomes zero, i.e.,

$$p'_n(t) = 0$$

Therefore, the steady-state solution is given by

$$\begin{cases} \lambda P_{n-1} + \mu P_{n+1} - (\lambda + \mu) P_n = 0, n \geq 1 \\ \mu P_1 = \lambda P_0 \end{cases} \quad \dots(20)$$

Now, from equation (20), we have

$$\mu P_{n+1} - \lambda P_n = \mu P_n - \lambda P_{n-1} \quad \dots (21)$$

This implies that

$$\mu P_n - \lambda P_{n-1} = \mu P_{n-1} - \lambda P_{n-2} \text{ (changing } n \text{ to } n-1)$$

$$\mu P_{n-1} - \lambda P_{n-2} = \mu P_{n-2} - \lambda P_{n-3} \text{ (again changing } n \text{ to } n-1)$$

$$\mu P_{n-2} - \lambda P_{n-3} = \mu P_{n-3} - \lambda P_{n-4} \text{ (again changing } n \text{ to } n-1)$$

...

...

...

$$\mu P_2 - \lambda P_1 = \mu P_1 - \lambda P_0$$

$$\text{But, } \mu P_1 - \lambda P_0 = 0 \quad [\because \mu P_1 = \lambda P_0]$$

$$\begin{aligned} \text{Therefore, } \mu P_{n+1} - \lambda P_n &= \mu P_n - \lambda P_{n-1} \\ &= \mu P_{n-1} - \lambda P_{n-2} \\ &= \mu P_1 - \lambda P_0 = 0 \end{aligned}$$

This implies that

$$\begin{aligned} P_n &= \frac{\lambda}{\mu} P_{n-1} \\ &= \frac{\lambda}{\mu} \left(\frac{\lambda}{\mu} P_{n-2} \right) \\ &= \frac{\lambda}{\mu} \frac{\lambda}{\mu} \left(\frac{\lambda}{\mu} P_{n-3} \right) \\ &\dots \\ &= \frac{\lambda \cdot \lambda \dots \lambda (n \text{ times})}{\mu \cdot \mu \dots \mu (n \text{ times})} P_0 \\ &= \frac{\lambda^n}{\mu^n} P_0 = (\lambda / \mu)^n P_0 = \rho^n P_0 \end{aligned}$$

$$\text{Since } \sum_{n=0}^{\infty} P_n = 1, \text{ it follows that } \sum_{n=0}^{\infty} P_0 \rho^n = 1$$

$$\Rightarrow P_0 (1 + \rho + \rho^2 + \dots + \rho^n) = 1$$

$$\Rightarrow P_0 \left(\frac{1}{1-\rho} \right) = 1 \Rightarrow P_0 = 1 - \rho$$

Therefore, the probability of n customers (units) in the system is given by

$$P_n = (1 - \rho) \rho^n, \quad n \geq 0 \quad \dots (22)$$

We can now determine the operating characteristics for the M/M/1 queueing model.

6.5.2 Operating Characteristics for the M/M/1 Queueing Model

1. The average number of customers in the system is given by

$$L_s = E(n) = \sum_{n=0}^{\infty} nP_n = \sum_{n=0}^{\infty} n\rho^n(1-\rho)$$

$$= \frac{\rho}{1-\rho} \quad (\text{on simplification}) \quad \dots (23)$$

$$\text{or } L_s = \frac{\lambda}{\mu - \lambda} \quad \dots (24)$$

2. The average queue length is

$$L_q = L_s - \text{Traffic intensity}$$

$$= L_s - r = \frac{r}{1-r} - r$$

$$\text{or } L_q = \frac{r^2}{1-r} = \frac{\lambda^2}{\mu(\mu - \lambda)} \quad \dots (25)$$

3. The average time an arrival spends in the system is

$$W_s = \frac{L_s}{\lambda} = \frac{1}{\mu - \lambda} \quad \dots (26)$$

4. The average waiting time of an arrival in the queue is

$$W_q = \frac{L_q}{\lambda} = \frac{\lambda}{\mu(\mu - \lambda)} \quad \dots (27)$$

W_q can also be obtained using the following result:

$$W_q = W_s - \frac{1}{\mu} \quad \dots (28)$$

5. The probability that the number of units waiting in the queue and the number of units being serviced is greater than k is

$$P[n > k] = \rho^{k+1} \quad \dots (29)$$

6. The probability of having a queue, i.e. $[1 - P_0] = \rho$... (30)

We now illustrate the M/M/1 queueing model with some examples.

Example 1: A TV repairman finds that the time spent on his job has an exponential distribution with mean 30 minutes. If he repairs sets in the order in which these come in, and if the arrival of sets is approximately Poisson with an average rate of 10 per 8-hour day, what is the repairman's expected idle time each day? How many jobs are ahead of the average set just brought in?

Solution: It is given that

$$\text{Arrival rate } (\lambda) = \frac{10}{8} \text{ arrivals/hr}$$

$$\text{Service rate } (\mu) = \frac{60}{30} = 2 \text{ services/hr}$$

∴ The probability of the repairman being idle is $P_0 = 1 - \rho$

$$\text{or } P_0 = 1 - \frac{\lambda}{\mu} = 1 - \frac{5}{8} = \frac{3}{8}$$

Hence, the idle time in the 8-hour day = $\frac{3}{8} \times 8 \text{ hours} = 3 \text{ hours}$

Now, the average number of jobs that are ahead of the set just brought in

= Average number of units in the system

$$= \frac{\lambda}{\mu - \lambda} = \frac{\frac{10}{8}}{2 - \frac{10}{8}} = 1.67$$

Example 2: Customers arrive at a window in a bank, according to a Poisson distribution with mean 10 per hour. Service time per customer is exponential with mean 5 minutes. The space in front of the window including that for the serviced customers can accommodate a maximum of three customers. Other customers can wait outside this space.

- What is the probability that an arriving customer can go directly to the space in front of the window?
- What is the probability that an arriving customer will have to wait outside the indicated space?
- How long is an arriving customer expected to wait before being served?

Solution: Arrival rate $\lambda = 10$ per hour

Service rate $\mu = \frac{60}{5}$ per hour = 12 per hour

$$\rho = \frac{10}{12} = \frac{5}{6}$$

- From equation (22), the required probability = $P_0 + P_1 + P_2$

$$= (1 - \rho) + (1 - \rho)\rho + (1 - \rho)\rho^2 \approx 0.42$$

- The required probability = $P_3 + P_4 + P_5 + \dots$

$$= 1 - (P_0 + P_1 + P_2)$$

$$= 1 - 0.42 = 0.58$$

- Expected waiting time before being served = $W_q = \frac{\lambda}{\mu(\mu - \lambda)}$

$$= \frac{10}{12(12 - 10)} = 0.417 \text{ hours}$$

Example 3: Arrivals of machinists at a tool crib are considered to be Poisson distributed at an average rate 6 per hour. The length of time the machinists must remain at the tool crib is exponentially distributed with average time of 0.05 hours.

- What is the probability that a machinist arriving at the tool crib will have to wait?
- What is the average number of machinists at the tool crib?

- c) The company will install a second tool crib when convinced that a machinist would have to spend 6 minutes in waiting and being served at the tool crib. At what rate should the arrival of machinists to the tool crib increase to justify the addition of a second crib?

Solution: a) Here the arrival rate $\lambda = 6/\text{hr}$ and the service rate $\mu = 20/\text{hr}$.

Therefore, the probability of zero customer in queue is

$$P_0 = 1 - \rho \quad \text{where } \rho = \frac{\lambda}{\mu} = \frac{6}{20}$$

$$\text{or } P_0 = 1 - \frac{6}{20} = \frac{14}{20} = \frac{7}{10} = 0.7$$

The probability that a machinist arriving at the tool crib will have to wait

= Probability that there is at least one machinist at the tool crib

= $1 - \text{Probability that there is no machinist at the tool crib}$

$$= 1 - P_0 = 1 - 0.7 = 0.3$$

- b) The average number of machinists at the tool crib is given by

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{6}{20 - 6} = \frac{6}{14} = 0.428 \approx 0.43$$

- c) The company is ready to install a second tool crib when convinced that a machinist would have to spend 6 min. in waiting and being served. Let the increased arrival rate be λ' .

Waiting time in the system = 6 min. = $\frac{1}{10}$ hr. We have

$$\frac{1}{\mu - \lambda'} = \frac{1}{10}$$

$$\text{or } \frac{1}{20 - \lambda'} = \frac{1}{10} \quad \text{or } 10 = 20 - \lambda'$$

$$\Rightarrow \lambda' = 20 - 10 = 10/\text{hr.}$$

The increase is, therefore, $(10 - 6)/\text{hr} = 4/\text{hr}$.

Example 4: A repairman is to be hired to repair machines which break down at an average rate of 3 per hour. The breakdown follows a Poisson distribution. Nonproductive time of a machine is considered to cost ₹10 per hour. Two repairmen have been interviewed of whom one is slow but charges less and the other is fast but more expensive. The slow repairman charges ₹5 per hour and services breakdown machines at the rate of 4 per hour. The fast repairman demands ₹7 per hour, but services breakdown machines at an average rate of 6 per hour. Which repairman should be hired?

Solution: The data given is summarised below:

Slow/Less expensive Repairman	Fast/ More expensive Repairman
$\lambda = 3/\text{hr}$, $\mu = 4/\text{hr}$, Labour cost = ₹5/hr	$\lambda = 3/\text{hr}$, $\mu = 6/\text{hr}$, Labour cost = ₹7/hr

Case of Slow/Less expensive Repairman

Cost of engaging slow repairman for 8 hours

= Breakdown cost + Labour cost for 8-hour working day.

$$= (\text{No. of breakdowns per hr}) \times 8 \text{ hours} \times \text{Av. time spent in the system} \\ \times (\text{Breakdown cost/ cost for non-productive time}) \\ + \text{Labour cost per hr} \times 8 \text{ hours}$$

$$= (\lambda) \times 8 \times W_s \times 10 + 5 \times 8$$

$$\text{Since the average time spent in the system is } W_s = \frac{1}{\mu - \lambda} = \frac{1}{4 - 3} = 1,$$

the total cost for engaging the slow repairman is

$$3 \times 8 \times 1 \times 10 + 5 \times 8 = 240 + 40 = \text{'280}$$

Case of Fast/More expensive Repairman

The cost of engaging fast repairman for 8 hours

$$= \text{Breakdown cost} + \text{Labour cost in 8-hour working day.}$$

$$= (\text{No. of breakdowns per hr}) \times 8 \text{ hours} \times \text{Av. time spent in the system} \\ \times (\text{Breakdown cost/ cost for non-productive time}) \\ + \text{Labour cost per hr} \times 8 \text{ hours}$$

$$= (\lambda) \times 8 \times W_s \times 10 + 7 \times 8$$

$$\text{Since the average time spent in the system is } W_s = \frac{1}{\mu - \lambda} = \frac{1}{6 - 3} = \frac{1}{3},$$

the total cost for engaging the fast repairman is

$$3 \times 8 \times \frac{1}{3} \times 10 + 7 \times 8 = 80 + 56 = \text{'136}$$

Since the total cost of engaging the fast repairman is less, we should engage the fast repairman.

You should now apply the M/M/1 model to solve the following exercises.

-
- E1)** Customers arrive at a box office window being served by a single individual according to a Poisson input process with a mean rate of 30 per hour. The time required to serve a customer has an exponential distribution with a mean of 90 seconds. Find the average waiting time of a customer in the queue.
- E2)** Arrivals at a telephone booth are considered to be Poisson with an average time of 10 minutes between one arrival and the next. The length of a phone call is assumed to be distributed exponentially with mean 3 minutes.
- What is the probability that a person arriving at the booth will have to wait?
 - The telephone company will install a second booth when convinced that an arrival would be expected to wait at least 3 minutes for the phone. By how much should the flow arrivals increase in order to justify the setting up of a second booth?
- E3)** A fertiliser company distributes its products by trucks loaded at its only loading station. Both company trucks and contractor's trucks are used for this purpose. It was noticed that on an average one truck arrived every 5 minutes and the average loading time was 3 minutes. Forty percent of the trucks belong to the contractor. Determine the expected waiting time of contractor's trucks per day.
- E4)** In a railway marshalling yard, goods trains arrive at a rate of 36 trains per day. Assuming that the inter-arrival time follows exponential

distribution and the service time distribution is also exponential with an average of 30 minutes, calculate the following:

- a) The mean line length (mean length of the system).
- b) The probability that the queue size exceeds 10.
- c) If the input increases to an average of 42 per day, what will the change in (a) and (b) be?

E5) A road transport company has one reservation clerk on duty at a time. She handles information of bus schedules and makes reservations. Customers arrive at a rate of 8 per hr and on an average the clerk can service 12 customers per hr.

- i) What is the average waiting time of a customer in the system?
- ii) The management is contemplating to install a computer system to handle the information and reservations. This is expected to reduce the service time from 5 to 3 minutes. The additional cost of having the new system works out to be ₹50/- per day. If the cost of the goodwill of having a customer to wait is estimated to be 12 paise per minute spent waiting before being served, should the company install the computer system? Assume an 8-hour working day.
- iii) What is the expected waiting time of all the customers in an 8-hour day?

Let us summarise the concepts that we have discussed in this unit.

6.6 SUMMARY

1. The purpose of queueing theory is to provide information for determining an acceptable level of service since providing too much service capacity is costly (owing to idle employees or equipment) and providing too little service capacity is also costly (owing to waiting members of the queue). A queueing system is described by three elements: (1) Input or arrival process of customers, (2) Service mechanism, and (3) Queue discipline.
2. **Input process** is concerned with the pattern in which the customers arrive and join the system. An input process is characterised by its size, the arrival time distribution, and the attitude of the customers.
3. **Service time** distributions are generally exponential distributions. Service facility can be 'Single channel facility' or 'One queue-several station facilities' or 'Several queues-one service station', 'Multi-channel facility' or 'Multi-Stage Channel facility'.
4. **Queue discipline** refers to the order in which the service station selects the next customer from the waiting line to be served. It may be FCFS, LIFO or SIRO. In this unit, the FCFS queue discipline has been considered.
5. The time dependent solutions are known as **transient solutions**. **Steady state solution** is independent of time and represents the probability of the system being in the equilibrium state.
6. A queueing system is usually described by five symbols such as **a/b/c : d/e** or **a/b/c/d/e**. The first symbol 'a' describes the arrival process. The second symbol 'b' describes the service time distribution. The third symbol 'c' stands for the number of servers. The symbols 'd' and 'e' stand for system

capacity and queue discipline, respectively. If arrivals are Poisson, symbol **M** is used in place of ‘a’. If departures are exponential, symbol **M** is used in place of ‘b’.

7. The **M/M/1 queueing model** deals with a queueing system having single service channel, Poisson input, and exponential services. There is no limit on the system capacity while the customers are served on a “First Come First Served” basis. The probability of n customers (units) in the system for this model is

$$P_n = (1 - \rho)\rho^n, \quad n \geq 0,$$

where $\rho = \frac{\lambda}{\mu}$, λ is the average arrival rate and μ , the average service rate.

Some important characteristics of this model are:

- Average number of customers in the system (L_s) = $\frac{\lambda}{\mu - \lambda}$
- Average queue length (L_q) = $L_s - \rho = \frac{\lambda^2}{\mu(\mu - \lambda)}$
- Average time an arrival spends in the system (W_s) = $\frac{L_s}{\lambda} = \frac{1}{\mu - \lambda}$
- Average waiting time of an arrival in the queue

$$(W_q) = \frac{L_q}{\lambda} = \frac{\lambda}{\mu(\mu - \lambda)} = W_s - \frac{1}{\mu}$$

6.7 SOLUTIONS/ANSWERS

E1) Arrival rate (λ) = 30/hr

$$\text{Service rate } (\mu) = \frac{1}{90} / \text{seconds} = \frac{60 \times 60}{90} / \text{hr}$$

Therefore, the average waiting time of a customer in a queue is

$$\begin{aligned} W_q &= \frac{\lambda}{\mu(\mu - \lambda)} = \frac{30}{40(40 - 30)} = \frac{3}{40} \text{ hours} \\ &= \frac{3}{4} \times 60 \text{ minutes} = 4.5 \text{ minutes} \end{aligned}$$

E2) We are given that

$$\text{Arrival rate } (\lambda) = \frac{1}{10} \text{ per minute}$$

$$\text{Service rate } (\mu) = \frac{1}{3} \text{ person/minute}$$

- a) The probability that a person arriving at a booth will have to wait
- $$\begin{aligned} &= \text{Probability that there is at least one customer} \\ &= P(n > 0) = 1 - P_0 = \lambda/\mu \\ &= \frac{1}{10} \times \frac{3}{1} = 0.3 \end{aligned}$$

- b) The installation of the second booth will be justified only if the arrival rate is more than the waiting time. Let λ' be the increased arrival rate.

Then the expected waiting time in queue is

$$W_q = \frac{\lambda'}{\mu(\mu - \lambda')} \Rightarrow 3 = \frac{\lambda'}{\frac{1}{3}(\frac{1}{3} - \lambda')}$$

Thus $\lambda' = \frac{1}{6}$ per minute

Hence, the increase in the arrival rate is

$$= \frac{1}{6} - \frac{1}{10} = \frac{1}{15} \text{ per minute}$$

E3) We are given that

$$\text{Arrival rate } (\lambda) = 1/5 \text{ per minute} = \frac{60}{5} = 12 \text{ per hour}$$

$$\text{Service rate } (\mu) = \frac{1}{3} \text{ per minute} = \frac{60}{3} = 20 \text{ per hour}$$

The expected waiting time of contractor's trucks per day

= Expected waiting time for a truck $\times$ Number of contractor's trucks per day

$$= \frac{\lambda}{\mu(\mu - \lambda)} \times (40 \text{ per cent of the total number of trucks per day})$$

$$= \frac{\lambda}{\mu(\mu - \lambda)} \times [40 \text{ per cent of (Arrival rate per hr} \times 24 \text{ hours)}]$$

$$= \frac{\lambda}{\mu(\mu - \lambda)} \times \frac{40}{100} \times (12 \times 24)$$

$$= \frac{12}{20(20 - 12)} \times \frac{40}{100} \times (12 \times 24) = 8.64 \text{ hours}$$

E4) We are given that

Arrival rate (λ) = 36 trains/day

Service rate (μ) = 1 per 30 minutes

$$= \frac{60}{30} \text{ per hour} = \frac{60}{30} \times 24 \text{ per day} = 48 \text{ trains/day}$$

Therefore, traffic intensity $\rho = \frac{\lambda}{\mu} = 0.75$

a) Mean length of system

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{36}{48 - 36} = \frac{36}{12} = 3 \text{ trains}$$

b) Probability that the queue size exceeds 10

= Probability that the system size exceeds 11

$$\text{i.e., } P(n > 11) = \rho^{11+1} = (0.75)^{12}$$

- c) If the input increases to an average of 42 per day, then the arrival rate (λ') = 42/day.

$$\text{Hence, the new traffic intensity } \rho' = \frac{42}{48} = \frac{7}{8}$$

$$\text{Therefore, in this case, } L_s = \frac{\lambda'}{\mu - \lambda'} = \frac{42}{48 - 42} = \frac{42}{6} = 7 \text{ trains}$$

$$\text{and the probability that the queue size exceeds 10 is } (\rho')^{11+1} = \left(\frac{7}{8}\right)^{12}$$

E5) We are given that

$$\text{Arrival rate } (\lambda) = 8/\text{ hr,}$$

$$\text{Service rate } (\mu) = 12/\text{ hr}$$

- i) Average waiting time of customers in the system is

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{12 - 8} = \frac{1}{4} \text{ hr}$$

- ii) An arrival waits for W_q hours before being served and there are λ arrivals per hour. Thus, the expected waiting time for all customers in an 8-hour day with one system

$$= 8\lambda \times W_q$$

With the existing system, the waiting time for the customers, therefore, is

$$\begin{aligned} &= 8 \times 8 \times \frac{1}{12 - 8} = 8 \times 8 \times \frac{8}{12(12 - 8)} \\ &= \frac{64}{6} \text{ hours} = 640 \text{ min.} \end{aligned}$$

The goodwill cost per day with the existing system

$$= 640 \times \frac{12}{100} = ₹ 76.80$$

If a computer system is installed, the service rate is 1 per 3 minutes or 20 per hr. Hence, the expected waiting time in this case = $8\lambda \times W_q'$

$$= 8 \times 8 \times \frac{8}{20(20 - 8)} = 8 \times 8 \times \frac{1}{30} \text{ hr} = 128 \text{ min.}$$

The goodwill cost per day when the computer system is installed

$$= 128 \times \frac{12}{100} = ₹ 15.36$$

The additional cost of computer system = ₹ 50

Therefore, the total cost of having the computer system

$$= ₹ (50 + 15.36) = ₹ 65.36$$

Since this amount is less than the goodwill cost to be incurred with the existing system, the management should install a computer system.

UNIT 7 SEQUENCING PROBLEMS

Structure

- 7.1 Introduction
 - Objectives
- 7.2 Sequencing Problems
- 7.3 Processing of n Jobs through 2 Machines
- 7.4 Processing of 2 Jobs through m Machines
- 7.5 Summary
- 7.6 Solutions/Answers

7.1 INTRODUCTION

In Unit 6, we have discussed the queueing system wherein customers are served after waiting in a queue for some time. In fact, waiting for service is an integral part of our daily life and that too at considerable cost most of the time. The purpose of queueing analysis is to provide information to determine an acceptable level of service. However, the appropriate order in which the waiting customers should be served also needs to be studied to minimise the total elapsed time. This aspect of the problem is not considered in queueing theory. It is covered in the study of sequencing problems, which are discussed in this unit.

Sequencing problems are quite common in real life. They arise whenever there is a choice of determining the order in which a number of jobs can be performed. These problems play a very important role in manufacturing set-ups for the optimal use of resources and/or the customer's satisfaction. Many-a-times, manufacturing firms produce goods made up of several components, which are sourced from other manufacturing firms. The final product undergoes many processes before it is taken to the market for sale. Since several machines are used in the production process, it is important to sequence the production process optimally so that the performance measures of the firm are satisfied.

In Sec.7.2, we discuss some procedures that help in determining the optimum order or sequence of jobs for a process. In Secs.7.3 and 7.4, we discuss how to solve the sequencing problems with n number of jobs to be completed through 2 machines and with 2 jobs to be completed through m machines, in some pre-assigned order, so as to optimise the total time involved.

In the next unit, we shall discuss inventory control and various factors, involved in inventory analysis. We shall also describe the models for determining the economic order quantity considering the situations: i) when demand is uniform; ii) when rates of demand are different in different cycles; iii) when shortages are allowed; iv) when replenishment is uniform, and v) when price (or quantity) discounts are given.

Objectives

After studying this unit, you should be able to:

- explain what a sequencing problem is;
- determine the appropriate order for the jobs to be done so as to minimise the elapsed time for 2-Machine, n -Job problems;

- determine the appropriate order for the jobs to be done so as to minimise the elapsed time for 2-Job, m-Machine problems; and
- obtain the minimum elapsed times for sequencing problems.

7.2 SEQUENCING PROBLEMS

In this section, we discuss the sequencing problems, which occur commonly in our real life whenever a number of tasks are to be performed in a predefined time schedule. The aim is to determine the best available possible sequences of jobs. A sequencing problem is the determination of the best sequence among all possible sequences. Here 'best' may be defined with respect to set objectives or performance measure for the concerned problem.

Let us consider a common experience that most grown-up men face everyday: whether to shave first and then take a bath, or to shave after taking bath. Clearly, the problem is to decide on the sequence (or order) of the two tasks: shaving and bathing. Now the choice lies between two alternative sequences, i.e., (1. Shaving, 2. Bathing) and (1. Bathing, 2. Shaving). The first option ensures more cleanliness, whereas the second one gives rise to ease in shaving. The solution to the problem depends on the objective or performance measure, i.e., cleanliness or ease in shaving, of the concerned person. Once the performance measure is specified, the solution is easy.

The selection of the appropriate order in which a number of jobs (customers) are to be performed (served) is called sequencing. In sequencing problems, there are two or more customers (jobs) to be served and one or more facilities (machines) available for this purpose. We wish to know when each job is to begin and what its due date or time is. We also wish to know which facilities are required to do each job, in which order these facilities are required and how long each operation is to take.

The following assumptions are usually made while dealing with sequencing problems:

- i) Only one operation is carried out on a machine at a time.
- ii) Processing times are known and do not change.
- iii) Processing times are independent of the order of processing the jobs.
- iv) The time involved in moving jobs from one machine to another is negligible.
- v) Each operation, once started, must be completed.
- vi) An operation must be completed before its succeeding operation can start.
- vii) Only one machine of each type is available.
- viii) A job is processed as soon as possible, but only in the order specified.
- ix) No passing rule is followed strictly, i.e., the same order of jobs is maintained over each machine.

Sequencing problems have been most commonly encountered in production shops where different products are to be processed over various combinations of machines. A general sequencing problem may be defined as follows:

Let there be n jobs (1, 2, 3, . . . , n), each of which has to be processed, one at a time, on each of m machines (A, B, C, . . .). The time required for processing each job on each machine is also given. The order of processing each job through these machines may be any of the $(n!)^m$ possible sequences. But, we are interested in finding that technologically feasible sequence for processing the jobs which gives the minimum total elapsed time for all the jobs. For example, suppose we have a problem of 2 jobs (J_1, J_2) and 2 machines (A, B) with the following processing times (in hours):

Job	:	J_1	J_2
Machine A	:	2	4
Machine B	:	3	6

The possible sequences for processing of the jobs will be $(2!)^2 = 4$, which are:

- First process J_1, J_2 in this order on machines A and B, respectively;
- First process J_1, J_2 in this order on machines B and A, respectively;
- First process J_2, J_1 in this order on machines A and B, respectively;
- First process J_2, J_1 in this order on machines B and A, respectively.

For the **first sequence**, the various times can be calculated as in the following table:

Jobs	Machine A		Machine B	
	Time in	Time out	Time in	Time out
J_1	0	2	2	$2 + 3 = 5$
J_2	2	$2 + 4 = 6$	6	$6 + 6 = 12$

Here, the total elapsed time is 12 hours.

The idle time for machine A

$$= \text{Total elapsed time} - \text{Time when the last job is out of the machine A,}$$

$$= 12 - 6 = 6 \text{ hrs}$$

The idle time for machine B

$$= \text{Time of starting the first job at B}$$

$$+ (\text{Time of starting the second job at B}$$

$$- \text{time of finishing the first job at B})$$

$$= 2 + (6 - 5) = 3 \text{ hrs}$$

For the **second sequence**, the various times can be calculated as in the following table:

Jobs	Machine B		Machine A	
	Time in	Time out	Time in	Time out
J_1	0	3	3	$3 + 2 = 5$
J_2	3	$3 + 6 = 9$	9	$9 + 4 = 13$

Here, the total elapsed time is 13 hours.

The idle time for machine B

$$= \text{Total elapsed time} - \text{Time when the last job is out of machine B}$$

$$= 13 - 9 = 4 \text{ hrs}$$

The idle time for machine A

$$= \text{Time of starting the first job at A}$$

$$+ (\text{Time of starting the second job at A}$$

$$- \text{time of finishing the first job at A})$$

$$= 3 + (9 - 5) = 7 \text{ hrs}$$

For **the third sequence**, the various times can be calculated as in the following table:

Jobs	Machine A		Machine B	
	Time in	Time out	Time in	Time out
J ₂	0	4	4	4 + 6 = 10
J ₁	4	4 + 2 = 6	10	10 + 3 = 13

Here, the total elapsed time is 13 hours.

The idle time for machine A

$$= \text{Total elapsed time} - \text{Time when the last job is out of the machine A}$$

$$= 13 - 6 = 7 \text{ hrs}$$

The idle time for machine B

$$= \text{Time of starting the first job at machine B}$$

$$+ (\text{Time of starting the second job at machine B}$$

$$- \text{time of finishing the first job at machine B})$$

$$= 4 + (10 - 10) = 4 \text{ hrs}$$

For **the fourth sequence**, the various times can be calculated as in the following table:

Jobs	Machine B		Machine A	
	Time in	Time out	Time in	Time out
J ₂	0	6	6	6 + 4 = 10
J ₁	6	6 + 3 = 9	10	10 + 2 = 12

Here, the total elapsed time is 12 hours.

The idle time for machine B

$$= \text{Total elapsed time} - \text{Time when the last job is out of machine B}$$

$$= 12 - 9 = 3 \text{ hrs}$$

The idle time for machine A

$$= \text{Time of starting the first job at machine A}$$

$$+ (\text{Time of starting the second job at machine A}$$

$$- \text{time of finishing the first job at machine A})$$

$$= 6 + (10 - 10) = 6 \text{ hrs}$$

So, in the above example, the technologically feasible sequence for processing the jobs, which gives the minimum total elapsed time for all the jobs is either the first sequence or the fourth sequence. Idle times of the machines are also less for these sequences. The processing, therefore, should be done adopting either the first or the fourth sequence.

7.3 PROCESSING OF n JOBS THROUGH 2 MACHINES

The n -job, 2-machine sequencing problem is described as follows:

Let there be n jobs where each job is to be processed by two machines, say A and B. Each job is processed in a pre-specified order AB, i.e., each job goes to machine A first and then to machine B. In other words, passing is not allowed. The actual or expected processing times for the i^{th} job on machine A and machine B, respectively, are known where $i = 1, 2, 3, \dots, n$.

The problem is now to determine the sequence (order) of jobs which minimises the total elapsed time (T) for processing all the jobs from the start of the first job to the completion of the last job. The procedure suggested by S.M. Johnson and R. Bellman for finding such a sequence consists of the following steps:

Step 1: Select the minimum processing time among A_i and B_i ($i = 1, 2, \dots, n$) where A_i and B_i are the processing times of i^{th} job on machines A and B, respectively. If there are two or more minimum processing times, then select any one of them arbitrarily.

Step 2: a) If the minimum processing time is one of A_i , say A_r , then process the job r first and place it at the beginning of the sequence.
 b) If the minimum processing time is one of B_i , say B_s , then process the job s last and place it at the end of the sequence.
 c) If in step 2(a) and 2(b) above, $A_r = B_s$, then process job r first and job s last.
 d) If there is a tie for minimum among A_i ($i = 1, 2, \dots, n$), then process any one of these jobs first.
 e) If there is a tie for a minimum among B_i ($i = 1, 2, \dots, n$), then process any one of these jobs last.

Step 3: Eliminate the job which has already been assigned from further consideration and repeat Steps 1 and 2, placing the remaining jobs next to the first job or before the last job as the case may be.

Calculations of Various Times

1. The time at which the i^{th} job in a sequence finishes on machine A
 = Time at which the $(i - 1)^{\text{th}}$ job in a sequence finishes on machine A
 + Processing time for the i^{th} job on machine A for $i = 1, 2, \dots, n$
2. The time when the i^{th} job in a sequence starts on machine A
 = Time when the $(i - 1)^{\text{th}}$ job in a sequence finishes on machine A
3. The time the first job in a sequence starts on machine B
 = Time when the first job in a sequence finishes on machine A

4. The time the i^{th} job in a sequence finishes on machine B
 $=$ Time when the i^{th} job in a sequence starts on machine B
 $+ \text{Processing time of the } i^{\text{th}} \text{ job on machine B for } i = 1, 2, \dots, n$
5. The time at which the $(i + 1)^{\text{th}}$ job in a sequence starts on machine B
 $= \text{Max}\{\text{Time when the } (i+1)^{\text{th}} \text{ job in a sequence finishes at machine A; time when the } i^{\text{th}} \text{ job in a sequence finishes at machine B}\}$
6. Total elapsed time
 $=$ Time when the n^{th} job in a sequence finishes on machine B
7. Idle time for Machine A
 $= (\text{Time when the } n^{\text{th}} \text{ job in a sequence finishes on machine B})$
 $- (\text{Time when the } n^{\text{th}} \text{ job in a sequence finishes on machine A})$
OR
 $= (\text{Total elapsed time})$
 $- (\text{Time when the last job in the sequence is out of machine A})$
8. Idle time for Machine B
 $= \text{Time at which the first job in a sequence finishes on machine A}$
 $+ \sum_{i=2}^n [(\text{Time when the } i^{\text{th}} \text{ job in a sequence starts on machine B})$
 $- (\text{Time when the } (i - 1)^{\text{th}} \text{ job in a sequence finishes on machine B})]$

Let us consider an example to explain this sequencing process.

Example 1: There are 5 jobs, each of which has to go through the machines A and B in the order AB. The processing times (in hours) are given as

Job	:	J_1	J_2	J_3	J_4	J_5
Machine A	:	2	4	5	7	1
Machine B	:	3	6	1	4	8

Determine a sequence of these jobs that will minimise the total elapsed time T . Also obtain:

- i) the minimum elapsed time; and
- ii) the idle time for each of the machines.

Solution: From the given information, we notice that the minimum processing time is 1 hour, which occurs for two jobs J_3 and J_5 . For J_3 , it is on machine B and for J_5 , it is on machine A. Thus, job J_5 is placed first in the sequence and job J_3 , the last. The partial allocation for the jobs will appear as follows:

J_5				J_3
-------	--	--	--	-------

The two jobs J_3 and J_5 are now deleted from further consideration. The reduced set of processing times will be as follows:

Job	:	J_1	J_2	J_4
Machine A	:	2	4	7
Machine B	:	3	6	4

In the above set, the minimum processing time is 2 hours for job J_1 on machine A. Hence, this job is placed next to the first job (i.e., next to J_5) in the sequence. The allocation of jobs till this stage would be as follows:

J_5	J_1			J_3
-------	-------	--	--	-------

Proceeding as above, the optimal sequence will be as follows:

J_5	J_1	J_2	J_4	J_3
-------	-------	-------	-------	-------

The various times can now be calculated as in the following table:

Jobs	Machine A		Machine B	
	Time in	Time out	Time in	Time out
J_5	0	1	1	9
J_1	1	3	9	12
J_2	3	7	12	18
J_4	7	14	18	22
J_3	14	19	22	23

- a) Thus, the total elapsed time = 23 hours
- b) The idle time for machine A
 = (Total elapsed time) – (Time when the last job is out of machine A)
 = 23 – 19 = 4 hours
- c) The idle time for machine B
 = Time of starting of first job at B
 + Total of time of starting of 2nd job onwards
 – Total of time of completion of 1st job to $(n - 1)^{\text{th}}$ job
 = 1 + (9 + 12 + 18 + 22) – (9 + 12 + 18 + 22) = 1 hour

You may now like to solve the following exercises to check your understanding.

- E1)** A ready-made garments manufacturer has to process five items through 2 stages of production, viz. cutting and sewing. The time taken for each of these items at the different stages is given below (in hours):

	Item :	1	2	3	4	5
Processing time (hours) {	Cutting:	5	7	3	4	6
	Sewing :	2	6	7	5	9

Find an order in which these items should be processed so as to minimise the total processing time. Also calculate the various idle times.

- E2)** There are 7 jobs, each of which has to go through the machines A and B in the order AB. Processing times (in hours) are given as follows:

Job :	1	2	3	4	5	6	7
Machine A :	3	12	15	6	10	11	9
Machine B :	8	10	10	6	12	1	3

Determine a sequence of these jobs that will minimise the total elapsed time. Also calculate the various idle times.

7.4 PROCESSING OF 2 JOBS THROUGH m MACHINES

For these types of problems, there are two jobs J_1, J_2 each of which has to be processed on m machines $M_1, M_2, \dots, M_m$ with different sequences of machines for processing each of the jobs. Each machine can perform only one job at a time. The exact or expected processing times on all the given machines are known. The objective is to find a sequence of processing the jobs, which minimises the total elapsed time from the start of the first job to the completion of the last job.

A problem of this type can be solved with the help of graphical method. We have to use the following steps in this method:

Step 1: Represent the processing times of job 1 on different machines along the x-axis and processing times of job 2 on different machines along the y-axis.

Step 2: Draw the vertical/horizontal lines through processing times as given in Step 1 and shade the common area for processing of two jobs on the same machine at different times since each machine can perform only one job at a time.

Step 3: Starting from the origin, we move diagonally, horizontally or vertically through various stages of completion of the processing of both the jobs through different machines at the same time until all the processing times are finished. If both the jobs are being processed simultaneously at different machines at the same time, we move diagonally. If Job 2 is under process at any machine and Job 1 is idle, i.e., Job 1 is waiting to be processed at the same machine through which Job 2 is being processed, we move vertically; and we move horizontally when Job 1 is being processed and Job 2 is idle.

Step 4: An optimal sequence for the jobs is the shortest line consisting of combinations of horizontal, vertical and diagonal (45°) lines from the origin. Since both the jobs are being processed simultaneously on the diagonal line, effort should be made to select a path on which diagonal movement is maximum.

So, we first try to move diagonally but keep in mind that diagonal movement through the blocked out (shaded) areas is not allowed since both the jobs cannot be processed simultaneously on a machine. Then horizontal or vertical movement is chosen keeping in view that total elapsed time should be minimum. The elapsed time can be obtained by adding the idle time for either job to the processing time for that job.

Let us explain this method with the help of an example.

Example 2: A machine shop has five machines A, B, C, D and E. Two jobs must be processed through each of these machines. The time (in hours) taken on each of these machines and the necessary sequence of jobs through the shops are given below:

Job 1	Sequence	:	A	B	C	D	E
	Time	:	2	4	3	6	6
Job 2	Sequence	:	C	A	D	E	B
	Time	:	4	6	3	3	6

Use the graphical method to obtain the total minimum elapsed time.

Solution: We are given the job sequences and processing time of 2 jobs at 5 machines. We follow the graphical method to find the minimum total elapsed time from starting the first job at the first machine to completion of the second job at the last machine.

Refer to Fig. 7.1. We first represent the processing time of job 1 on different machines, i.e., 2, 4, 3, 6, 6 along the x-axis and the processing time of job 2, i.e., 4, 6, 3, 3, 6 along the y-axis.

Then we draw the vertical lines through processing times of Job 1. We draw the first vertical line at 2 hrs, the second at $2 + 4 = 6$ hrs, the third at $6 + 3 = 9$ hrs, and so on. Similarly, we draw the horizontal lines through processing times of Job 2. We draw the horizontal lines at 4, 10, 13, 16 and 22 hrs in this case. We now shade the common area for processing of jobs 1 and 2 on one machine at the same time as each machine can perform only one job at a time. In this case we find the common areas to be:

- i) between 0 to 2 hours on the x-axis and 4 to 10 hours on the y-axis for machine A,
- ii) between 2 to 6 hours on the x-axis and 16 to 22 hours on the y-axis for machine B,
- iii) between 6 to 9 hours on the x-axis and 0 to 4 hours on the y-axis for machine C,
- iv) between 9 to 15 hours on the x-axis and 10 to 13 hours on the y-axis for machine D,
- v) between 15 to 21 hours in the x-axis and 13 to 16 hours on the y-axis for machine E.

All these areas are shaded in Fig. 7.1.

We now have to find an optimal sequence for processing of the jobs 1 and 2, i.e., the shortest line consisting of combinations of horizontal, vertical and diagonal (45°) lines from the origin. For this, we start from the origin and move diagonally first up to the point (9, 9) since both the jobs 1 and 2 are being processed simultaneously.

After being processed through machines A, B and C, job 1 becomes idle since job 2 is being processed in machine A and it is then being processed in machine D. Therefore, we move vertically which means Job 2 is under process and Job 1 is idle. The idle time for job 1 is 4 hours in this case.

Again we move diagonally up to the point (18, 22) since both the jobs 1 and 2 are being processed simultaneously. At this stage, job 2 is completed but job 1 is being processed through machine E. Therefore, we move horizontally, which means Job 1 is being processed and Job 2 is idle. The idle time for job 2 is 3 hours since job 1 is processed through machine E between 15 to 21 hours.

In this case, the idle time for job 1 is found to be 4 hours. Therefore, the total elapsed time for Job 1 is $21 + 4 = 25$ hours.

Similarly, the idle time for job 2 is seen to be 3 hours when job 1 was under processing between 18 to 21 hrs and the processing for job 2 had been completed. Therefore, the total elapsed time for job 2 is $22 + 3 = 25$ hours, which is shown in Fig. 7.1.

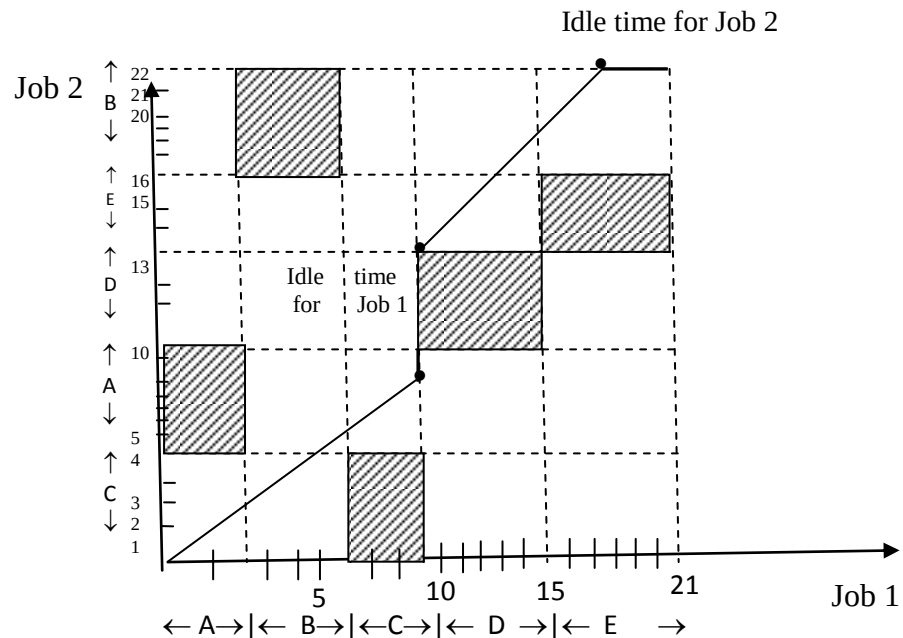


Fig. 7.1

You may now like to solve the following problems to check your understanding.

- E3)** Using the graphical method, calculate the minimum time needed to process job 1 and job 2 on five machines A, B, C, D and E, that is, for each machine find the job which should be done first. Also calculate the total time needed to complete both jobs.

Job 1	Sequence	:	A	B	C	D	E
	Time (in hrs)	:	1	2	3	5	1
Job 2	Sequence	:	C	A	D	E	B
	Time (in hrs)	:	3	4	2	1	5

- E4)** Use the graphical method to minimise the time required to process the following jobs on the machines, that is, for each machine specify the job that should be done first. Also calculate the total elapsed time for completing both jobs.

Job 1	Sequence	:	A	B	C	D	E
	Time (in hrs)	:	6	8	4	12	3
Job 2	Sequence	:	B	C	A	D	E
	Time (in hrs)	:	10	8	6	4	12

Let us now summarise the main points which have been covered in this unit.

7.5 SUMMARY

- The selection of the appropriate order in which waiting customers are to be served is called **sequencing**. In sequencing problems, there are two or more jobs to be done on one or more machines available for this purpose. We are interested in finding when each job is to begin and what its due date or time is. We are also interested in determining the sequence (order) of jobs which minimises the total elapsed time (T) for processing all the jobs from the start of the first job to the completion of the last job.

2. In case of **processing of n-jobs through two machines**, the procedure suggested by S.M Johnson and R. Bellman for finding the sequence of jobs which minimises the total elapsed time is used. **Total elapsed time** in this case is the time when the n^{th} job in a sequence finishes on second machine. **Idle time for the first machine** is the time when the last job in a sequence finishes on the first machine subtracted from the time when the last job in a sequence finishes on the second machine. **Idle time for second machine** is = the time at which the first job in a sequence finishes on the first machine + $\sum [(\text{Time when the } i^{\text{th}} \text{ job in a sequence starts on the second machine}) - (\text{Time when the } (i - 1)^{\text{th}} \text{ job in a sequence finishes on the second machine})]$
3. In case of **processing of 2 jobs through m machines**, each of the two jobs is to be processed on m machines $M_1, M_2, \dots, M_m$ in different sequence of machines. The exact/ expected processing times on all the given machines are known. The objective is to find a sequence of processing the jobs, which minimises the total elapsed time from the start of the first job to the completion of the last job. A problem of this type can be solved with the help of the graphical method.

7.6 SOLUTIONS/ANSWERS

- E1)** We first make an optimal sequence for processing on cutting and sewing machines. If minimum processing time is that of the cutting machine then we place it in the beginning of the sequence and if it is that of sewing machine, then it will be placed at the end of sequence. In this way, we obtain the following optimal sequence:

3	4	5	2	1
---	---	---	---	---

The various times can be calculated as follows:

Item	Cutting		Sewing	
	Time in	Time Out	Time in	Time Out
3	0	3	3	10
4	3	7	10	15
5	7	13	15	24
2	13	20	24	30
1	20	25	30	32

Total elapsed time = 32 hours.

Idle time for cutting machine = $32 - 25 = 7$ hours.

Idle time for sewing machine = 3 hours.

- E2)** The optimal sequence is

1	4	5	3	2	7	6
---	---	---	---	---	---	---

OR

1	5	3	2	4	7	6
---	---	---	---	---	---	---

Here, either of the above 2 sequences can be taken as an optimal solution. We get the same answer using either of these sequences. Let us take the first sequence to obtain various elapsed times, which have been calculated in the following table :

Job	Machine A		Machine B	
	Time in	Time Out	Time in	Time Out
1	0	3	3	11
4	3	9	11	17
5	9	19	19	31
3	19	34	34	44
2	34	46	46	56
7	46	55	56	59
6	55	66	66	67

Total elapsed time = 67 hours

Idle time of Machine A = $67 - 66 = 1$ hour

Idle time of Machine B = $3 + (11 + 19 + 34 + 46 + 56 + 66)$
 $- (11 + 17 + 31 + 44 + 56 + 59)$
 $= 3 + 232 - 218 = 17$ hours

Let us take the second sequence to obtain various elapsed times, which have been calculated in the following table:

Job	Machine A		Machine B	
	Time in	Time Out	Time in	Time Out
1	0	3	3	11
4	3	13	11	23
5	13	28	28	38
3	28	40	40	50
2	40	46	50	56
7	46	55	56	59
6	55	66	66	67

Total elapsed time = 67 hours

Idle time of Machine A = $67 - 66 = 1$ hour

Idle time of Machine B = $3 + (11 + 28 + 40 + 50 + 56 + 66)$
 $- (11 + 23 + 38 + 50 + 56 + 59)$
 $= 3 + 251 - 237 = 17$ hours

- E3)** Let us draw a set of axes at right angle to each other where the x-axis represents the time of job 1 on different machines while job 2 remains idle and the y-axis represents the processing time of job 2 while job 1 remains idle. We draw vertical/horizontal lines through processing times as shown in the Fig. 7.2 and shade the common area for each machine.

Starting from the origin, we move diagonally, horizontally or vertically through various stages of completion until all the processing times are finished. Elapsed time obtained by adding the idle time for job 1 to its processing time is $(12 + 3) = 15$ hours.

Elapsed time obtained by adding the idle time for job 2 to its processing time is $(15 + 0) = 15$ hours.

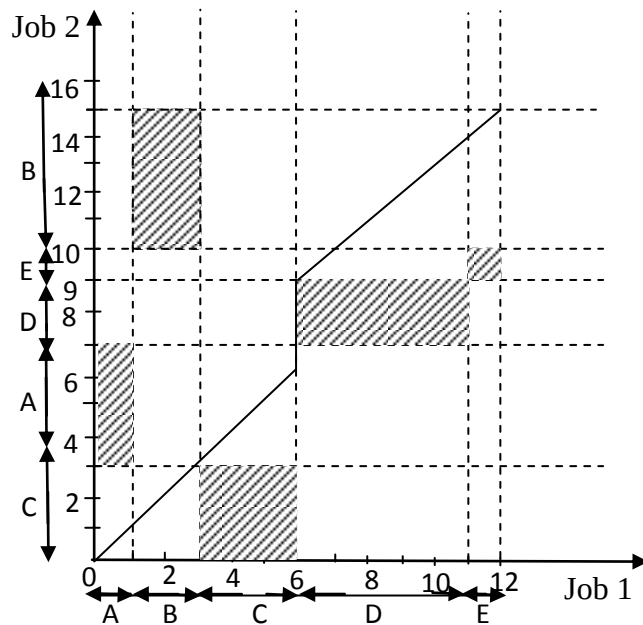


Fig. 7.2

- E4)** Let us draw a set of axes at right angle to each other where the x-axis represents the processing time of job 1 on different machines while job 2 remains idle and the y-axis represents the processing time of job 2 while job 1 remains idle. We draw vertical/horizontal lines through processing times as shown in Fig. 7.3 and shade the common area for each machine.

Starting from the origin, we move diagonally, horizontally or vertically through various stages of completion until all the processing times are finished.

Idle time of job 1 in hours = $4 + 6 = 10$ hours

Idle time of job 2 in hours = 3 hours

Elapsed time = Processing time of Job 1 + its idle time

$$= 33 + 10 = 43 \text{ hours}$$

OR

= Processing time of job 2 + its idle time

$$= 40 + 3 = 43 \text{ hours}$$

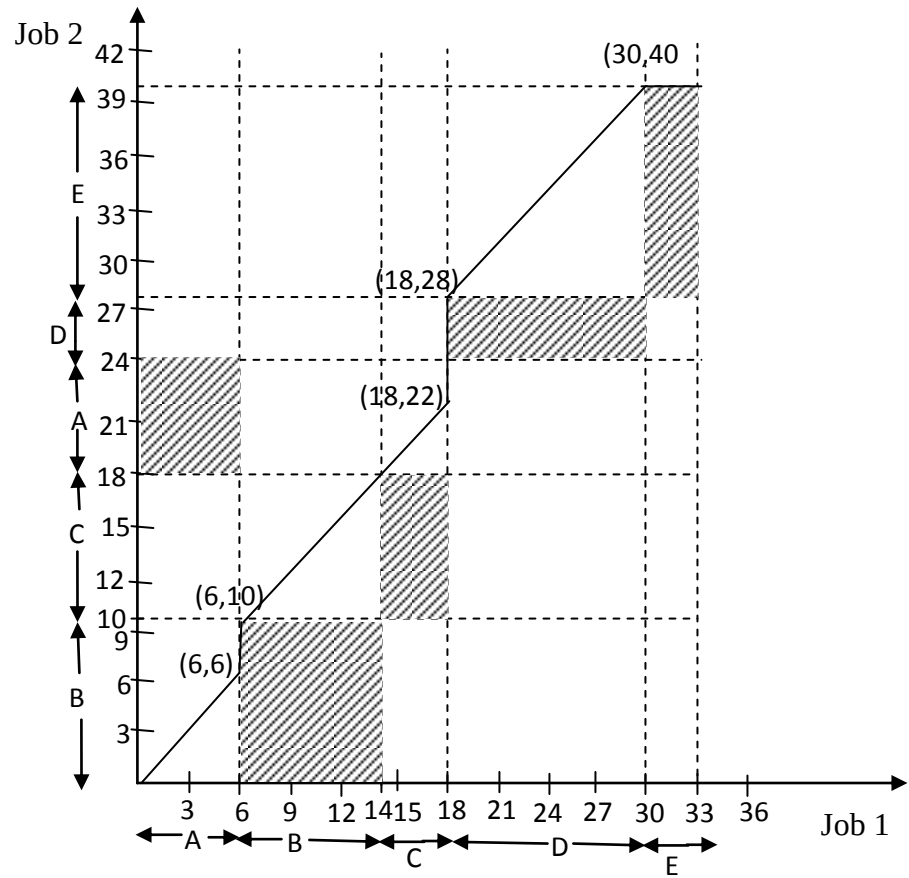


Fig. 7.3

UNIT 8 INVENTORY MODELS

Structure

- 8.1 Introduction
 - Objectives
- 8.2 Inventory Control
- 8.3 Economic Order Quantity (EOQ) Model with Uniform Demand
- 8.4 EOQ Model with Different Rates of Demand in Different Cycles
- 8.5 EOQ Model when Shortages are Allowed
- 8.6 EOQ Model with Uniform Replenishment
- 8.7 EOQ Model with Price (or Quantity) Discounts
- 8.8 Summary
- 8.9 Solutions/Answers

8.1 INTRODUCTION

In Unit 7, you have studied the sequencing problem, which involves determining the optimum order or sequence of jobs for a process to optimise the total time. We have discussed two types of sequencing problems: the ones with n number of jobs to be completed through 2 machines and those with 2 jobs to be completed through m number of machines, in some pre-assigned order.

In this unit, we discuss various inventory models. Inventory refers to a stock of goods, materials, human resources or financial resources or any other idle resource having some economic value, which is stocked in order to meet the demand expected in future. Almost every business must maintain an inventory for running its operations efficiently and smoothly.

Although inventories are essential for business, maintenance of inventories also costs money by way of expenses on stores, equipment, personnel, insurance, etc. Thus, excess inventories are undesirable. This calls for controlling the inventories in the most profitable way. In the present unit, we discuss the models for inventory control known as **economic order quantity** models. These models help in deciding as to how much quantity should be kept in stock in order to balance the costs of holding too much stock vis-à-vis the costs of ordering in small quantities.

In this unit, we discuss inventory control and various factors involved in inventory analysis in Sec. 8.2. In Secs. 8.3 to 8.7, we describe five models for determining the economic order quantity: i) when demand is uniform; ii) when rates of demand are different in different cycles; iii) when shortages are allowed; iv) when replenishment is uniform; and v) when price (or quantity) discounts are given.

Objectives

After studying this unit, you should be able to:

- explain the concept of inventory control;
- determine the economic order quantity when demand is uniform;
- determine the economic order quantity when rates of demand are different in different cycles;

- determine the economic order quantity when shortages are allowed;
- determine the economic order quantity when replenishment is uniform; and
- determine the economic order quantity when there are price discounts.

8.2 INVENTORY CONTROL

An inventory means a physical stock of idle resources of any kind having some economic value kept for the purpose of meeting future demand. It indicates the raw material required before production, the finished goods after production ready for delivery to consumers, human resources, financial resources, etc., which are stocked in order to meet an expected demand in the future. Almost every business must maintain an inventory for running its operations efficiently and smoothly. If an enterprise does not maintain an inventory, it may suddenly find at some point in its operations that it has no materials or goods to supply to its customers. Then on receiving a sales order, it will first have to place order for purchase of raw materials, wait for their receipt and then start production. The customer will, thus, have to wait for a long time for the delivery of the goods and may turn to other suppliers, resulting in loss of business/goodwill for the enterprise.

Maintaining an inventory is necessary because of the following reasons:

- It helps in smooth and efficient running of an enterprise.
- It provides service to the customer at short notice. Timely delivery can fetch more goodwill and orders.
- In the absence of the inventory, an enterprise may have to pay high prices because of piecemeal purchasing. Maintaining an inventory may earn price discounts because of bulk purchasing. Such purchases entail less orders and, therefore, less clerical costs.
- It also takes advantage of favourable market.
- It acts as a buffer stock when raw materials are received late and shop rejections are too many.
- Process and movement inventories (also called pipeline stocks) are quite necessary in big enterprises wherein a significant amount of time is required to ship items from one location to another.

Though inventories are essential, their maintenance also costs money by way of expenses on stores, equipment, personnel, insurance, etc. Thus, excess inventories are undesirable. So, only that quantity should be kept in stock, which balances the costs of holding too much stock vis-à-vis the costs of ordering in small quantities. This calls for controlling the inventories in the most profitable way and that is why we need inventory analysis. We now discuss various factors involved in inventory analysis.

1. Inventory related costs

Various costs associated with inventory control are often classified as follows:

- Set-up cost:** This is the cost associated with the setting up of machinery before starting production. The set-up cost is generally assumed to be independent of the quantity ordered for.
- Ordering cost:** This is the cost incurred each time an order is placed. This cost includes the administrative costs (paper work, telephone calls, postage), transportation, receiving and inspection of goods, etc.

- iii) **Purchase (or production) cost:** It is the actual price at which an item is purchased (or produced). It may be constant or variable. It becomes variable when quantity discounts are allowed for purchases above a certain quantity.
- iv) **Carrying (or holding) cost:** The cost includes the following costs for maintaining the inventory: i) Rent for the space; ii) cost of equipment or any other special arrangement for storage; iii) interest of the money blocked; iv) the expenses on stationery; v) wages of the staff required for the purpose; vi) insurance and depreciation; and vii) deterioration and obsolescence, etc.
- v) **Shortage (or Stock-out) cost:** This is the penalty cost for running out of stock, i.e., when an item cannot be supplied on the customer's demand. These costs include the loss of potential profit through sales of items demanded and loss of goodwill in terms of permanent loss of the customer.

2. Demand

Demand is the number of units required per period and may either be known exactly or known in terms of probabilities. Problems in which demand is known and fixed are called **deterministic problems** whereas problems in which demand is known in terms of probabilities are called **probabilistic problems**.

3. Selling Price

The amount which one gets on selling an item is called its selling price. The unit selling price may be constant or variable, depending upon whether quantity discount is allowed or not.

4. Order Cycle

The period between placement of two successive orders is referred to as an order cycle. The order may be placed on the basis of either of the following two types of inventory review systems:

- a) The record of the inventory level is checked continuously until a specified point is reached where a new order is placed. This is called continuous review.
- b) The inventory levels are reviewed at equal intervals of time and orders are placed accordingly at such levels. This is called periodic review.

5. Time Horizon

The period over which the time cost will be minimised and inventory level will be controlled is termed as time horizon. This can be finite or infinite depending on the nature of demand.

6. Stock Replenishment

The rate at which items are added to the inventory is called the rate of replenishment. The actual replenishment of items may occur at a uniform rate or be instantaneous over time. Usually uniform replacement occurs in cases when the item is manufactured within the factory while instantaneous replacement occurs in cases when the items are purchased from outside sources.

7. Lead Time

The time gap between placing an order for an item and actually receiving the item into the inventory is referred to as lead time.

8. Reorder Level

The lower limit for the stock is fixed at which the purchasing activities must be started for replenishment. With this replenishment, the stock reached at a level is known as maximum stock. The level between maximum and minimum stock is known as the reorder level.

9. Economic Order Quantity (EOQ)

The order in quantity that balances the costs of holding too much stock vis-à-vis the costs of ordering in small quantities too frequently is called Economic Order Quantity (or Economic lot size).

10. Reorder Quantity

The quantity ordered at the level of minimum stock is known as the reorder quantity. In certain cases it is the 'Economic Order Quantity'.

In Secs. 8.3 to 8.7, we shall discuss the following inventory models for obtaining economic order quantity:

- i) EOQ Model with Uniform Demand
- ii) EOQ Model with Different Rates of Demand in Different Cycles
- iii) EOQ Model when Shortages are Allowed
- iv) EOQ Model with Uniform Replenishment
- v) EOQ Model with Price (or Quantity) Discounts

However, before discussing these models, we give the notations that we shall use in the development of the models.

The notation used in the Models

Q = Number of units ordered (supplied) per order

D = Demand in units of inventory per year

N = Number of orders placed per year

TC = Total Inventory cost

C_o = Ordering cost per order

C = Purchase or manufacturing price per unit inventory

C_h = Carrying or holding cost per unit per period of time the inventory is kept

C_s = Shortage cost per unit of inventory

t = The elapsed time between placement of two successive orders

r_p = Replenishment rate at which lot size Q is added to inventory.

8.3 ECONOMIC ORDER QUANTITY (EOQ) MODEL WITH UNIFORM DEMAND

The objective of the **EOQ model with uniform demand** is to determine an optimum economic order quantity such that the total inventory cost is minimised. We make the following assumptions for this model:

1. Demand rate (D) is constant and known;
2. Replenishment rate (r_p) is instantaneous;
3. Lead time is constant and zero;

4. Purchase price is constant, i.e., discounts are not allowed;
5. Carrying cost and ordering cost are known and constant; and
6. Shortages are not allowed.

The situation can be graphically represented as shown in Fig. 8.1.

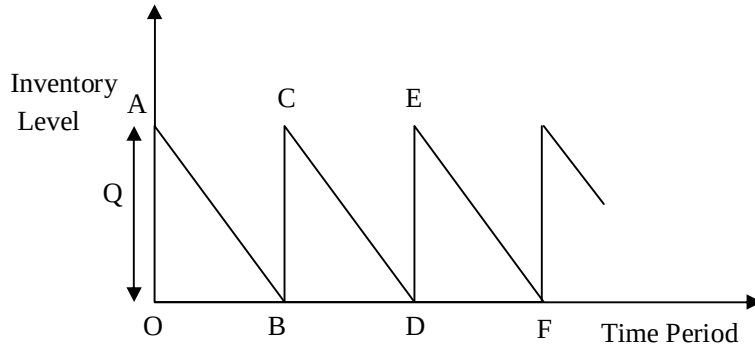


Fig. 8.1

The graph in Fig. 8.1 shows that initially there were Q units in the stock. The number of units goes on decreasing with respect to time to meet the demand and this is represented by the line AB in the graph. When the stock vanishes, i.e., the point B is reached, the stock level rises to Q instantaneously as the replenishment is instantaneous. Since the demand is uniform, the rate of decrease of the quantity remains the same as earlier. Therefore, this is represented by the line CD in the graph, which is parallel to AB . Similarly, EF is parallel to AB and CD due to uniform demand, and so on.

Since the demand is uniform, the average inventory is simply the arithmetic mean of the maximum and the minimum levels of inventory. Let Q be the quantity ordered (or replenished) when the minimum level, i.e., zero is reached.

$$\text{Therefore, the average inventory} = \frac{Q+0}{2} = \frac{Q}{2}$$

The ordering cost = Number of orders per year $\times$ ordering cost per order

$$= N \times C_o = \frac{D}{Q} \times C_o$$

The carrying cost = Average units in inventory $\times$ carrying cost per unit

$$= \frac{Q}{2} \times C_h$$

The total inventory cost is the ordering cost plus the carrying cost. Therefore,

$$TC = \frac{D}{Q} C_o + \frac{Q}{2} C_h$$

The total inventory cost (TC) is minimum at that value of Q where the derivative of TC with respect to Q is zero. Differentiating TC with respect to Q and then equating it to zero, we get

$$-\frac{D}{Q^2} C_o + \frac{1}{2} C_h = 0$$

$$\Rightarrow \frac{D}{Q} C_o = \frac{Q}{2} C_h \quad (\text{i.e., ordering cost} = \text{carrying cost})$$

$$\Rightarrow Q^2 = \frac{2D C_o}{C_h} \Rightarrow Q = \sqrt{\frac{2D C_o}{C_h}}$$

This value of Q minimises the total inventory cost (TC) and hence it is the economic order quantity. Let us denote it by Q*. Thus, the EOQ for this model is

$$Q^* = \sqrt{\frac{2D C_o}{C_h}}$$

Hence, the optimum number of orders placed per year (N^*) = $\frac{D}{Q^*}$

The minimum total yearly inventory cost

$$\begin{aligned} (TC^*) &= \frac{D}{Q^*} C_o + \frac{Q^*}{2} C_h = \frac{2D C_o + (Q^*)^2 C_h}{2Q^*} \\ &= \sqrt{2D C_o C_h} \quad (\text{on simplification}) \end{aligned}$$

and the total minimum cost = (TC*) + cost of material.

Let us consider an example to obtain the total minimum cost in this model.

Example 1: An enterprise requires 1000 units per month. The ordering cost is estimated to be ₹ 50 per order. In addition to ₹ 1, the carrying costs are 5% per unit of average inventory per year. The purchase price is ₹ 20 per unit. Find the economic lot size to be ordered and the total minimum cost.

Solution: We are given that

$$\begin{aligned} D &= \text{Monthly demand} \times 12 \\ &= 1000 \times 12 = 12000 \text{ units per year} \\ C_o &= ₹ 50 \text{ per order}; C = ₹ 20 \text{ per unit} \\ C_h &= 1 + 5\% \text{ of } ₹ 20 \\ &= 1 + 1 = ₹ 2 \text{ per unit of average inventory} \end{aligned}$$

The economic lot size, therefore, is given by

$$Q^* = \sqrt{\frac{2D C_o}{C_h}} = \sqrt{\frac{2 \times 12000 \times 50}{2}} = 775 \text{ units}$$

Total minimum cost = TC* + Cost of material

$$\begin{aligned} &= \sqrt{2D C_o C_h} + (12000 \times 20) \\ &= \sqrt{2 \times 12000 \times 50 \times 2} + 240000 \\ &= \sqrt{2400000} + 240000 = \sqrt{240} \times 100 + 240000 \\ &= 15.5 \times 100 + 240000 = ₹ 241550 \end{aligned}$$

You may now like to solve the following problems to assess your understanding.

- E1)** The XYZ manufacturing company uses 12000 units of raw material annually, which costs ₹1.25 per unit. Placing each order costs ₹15 and the carrying costs are 15% per year per unit of the average inventory. Find the economic order quantity.
- E2)** The XYZ manufacturing company has determined from an analysis of its accounting and production data for a part, that its cost to purchase is ₹36 per order and ₹2 per part. Its inventory carrying charge is 18% of the average inventory. The demand of this part is 10000 units per annum.
- What should the economic order quantity be?
 - What is the optimum number of days supply per optimum order?

8.4 EOQ MODEL WITH DIFFERENT RATES OF DEMANDS IN DIFFERENT CYCLES

In this EOQ model, the same quantity is ordered each time when the stock vanishes with the assumption that the replenishment is instantaneous. The demand is not uniform and the stock vanishes in different periods of time. The situation can be graphically represented as shown in Fig. 8.2.

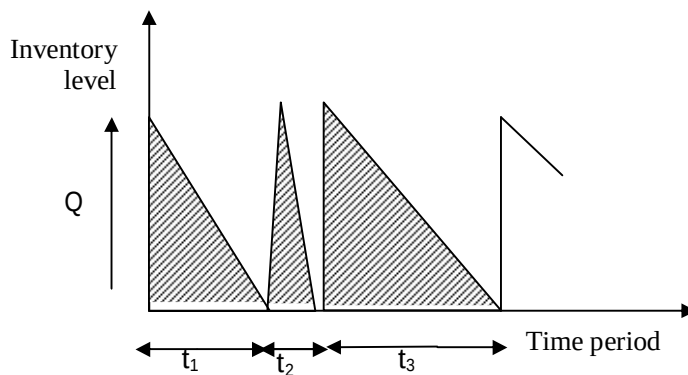


Fig. 8.2

Let the demand in different periods of time $t_1, t_2, \dots, t_n$ be $D_1, D_2, \dots, D_n$, respectively, so that the total demand in time T is given by

$$D = D_1 + D_2 + \dots + D_n \quad \text{where } T = t_1 + t_2 + \dots + t_n$$

The cost of ordering in time T

$$\begin{aligned}
 &= \text{No. of orders per year} \times \text{ordering cost per order} \\
 &= NC_o \\
 &= \frac{\text{Total Demand in time } T \times \text{Ordering cost per order}}{\text{No. of units ordered per order}} \\
 &= \frac{D}{Q} \times C_o
 \end{aligned}$$

The carrying cost for time T

$$= \text{Average unit in inventory} \times \text{carrying cost per unit}$$

$$= \frac{Q \times t_1}{2} C_h + \frac{Q \times t_2}{2} C_h + \dots + \frac{Q \times t_n}{2} C_h$$

$$= \frac{Q}{2} C_h [t_1 + t_2 + \dots + t_n] = \frac{Q C_h T}{2}$$

∴ The total inventory cost (TC) = ordering cost + carrying cost

$$TC = \frac{D C_o}{Q} + \frac{Q C_h T}{2}$$

This cost is minimum when its derivative with respect to Q is zero, i.e., if

$$-\frac{D}{Q^2} C_o + \frac{1}{2} T C_h = 0$$

$$\Rightarrow \frac{D}{Q} C_o = \frac{Q}{2} T C_h \quad (\text{i.e., ordering cost} = \text{carrying cost})$$

$$\Rightarrow Q^2 = \frac{2D C_o}{C_h T} \quad \text{p} \quad Q^* = \sqrt{\frac{2D C_o}{C_h T}}$$

This result is similar to that of the model discussed in Sec. 8.3. The only difference is that the uniform demand (D) is replaced by the average demand (D/T).

$$(TC^*) = \frac{D C_o}{Q^*} + \frac{Q^* T C_h}{2} = \frac{2D C_o + (Q^*)^2 T C_h}{2Q^*}$$

$$= \sqrt{2D C_o C_h T} \quad (\text{on simplification});$$

The total minimum cost = (TC*) + Cost of material

If T = 1 year, the result of this model is exactly the same as the EOQ model with uniform demand.

8.5 EOQ MODEL WHEN SHORTAGES ARE ALLOWED

The assumptions made for this model are the same as those for the EOQ model with uniform demand discussed in Sec. 8.3, except that shortages are allowed, which may occur frequently. The cost of shortages is assumed to be directly proportional to the number of units short. The assumptions for the model are:

- i) Demand is constant and known;
- ii) Replenishment is instantaneous;
- iii) Lead time is zero;
- iv) Purchasing cost is constant and known;
- v) Carrying cost and ordering cost are constant and known; and
- vi) Shortage is allowed and the shortage cost is directly proportional to the number of units short.

The situation can be represented graphically as shown in Fig. 8.3.

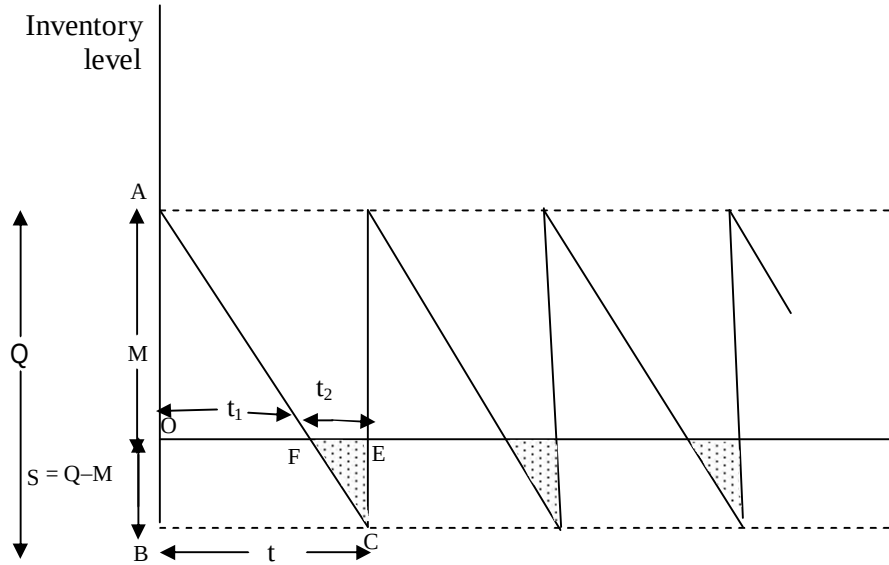


Fig. 8.3

Here,

Q = No. of units ordered per order

D = No. of units required per year

t_1 = Time when there are no shortages

t_2 = Time period during which there is shortage

t = Total cycle time, i.e., $t = t_1 + t_2$

S = Maximum shortage

The same quantity Q is ordered and received every time. The maximum shortage is equal to S . The remaining quantity M is placed in the inventory as surplus to satisfy the demand in the next cycle. Note that S units out of Q are always in the shortage list, i.e., these are never placed in the inventory. Thus, it results in a saving on the inventory carrying cost.

$$\text{No. of orders placed per year} = N = \frac{D}{Q}$$

Ordering cost = No. of orders placed per year $\times$ ordering cost per order

$$= NC_o = \frac{D}{Q} C_o$$

$$\text{Average inventory} = \frac{M}{2}$$

$$\text{Carrying cost per cycle} = C_h \left(\frac{M}{2} \times t_1 \right)$$

Note from Fig. 8.3 that $\triangle ABC$ is similar to $\triangle AOF$. Therefore,

$$\frac{t_1}{t} = \frac{M}{Q} \Rightarrow t_1 = \frac{M}{Q} t$$

$$\text{Therefore, carrying cost} = \left(\frac{M^2 \times t}{2Q} \right) C_h$$

$$\text{Carrying cost per unit time} = \frac{M^2}{2Q} C_h$$

Shortage cost = Average no. of units short $\times$ time period of shortage
 $\times$ shortage cost per unit per time period

$$= \frac{S}{2} \times t_2 \times C_s = \left(\frac{S}{2} t_2 \right) C_s$$

Now, $\triangle ABC$ is similar to $\triangle CEF$ in Fig. 8.3. Therefore,

$$\frac{t_2}{t} = \frac{S}{Q} \Rightarrow t_2 = \frac{S}{Q} t$$

$$\text{Hence, shortage cost} = \left(\frac{S^2 \times t}{2Q} \right) C_s$$

$$\text{Shortage cost per unit time} = \frac{S^2}{2Q} C_s$$

Therefore, the total inventory cost is given as:

$$TC = \text{ordering cost} + \text{carrying cost} + \text{shortage cost}$$

$$\text{or } TC = \frac{D}{Q} C_o + \frac{M^2}{2Q} C_h + \frac{S^2}{2Q} C_s$$

$$\text{or } TC = \frac{D}{Q} C_o + \frac{M^2 C_h}{2Q} + \frac{(Q - M)^2 C_s}{2Q} \quad \dots (1)$$

Now, TC is a function of two variables Q and M. Therefore, to find the minimum inventory cost, we differentiate TC with respect to Q and M separately and equate them to zero. Thus, we get

$$0 = \frac{d}{dQ}(TC) = -\frac{DC_o}{Q^2} - \frac{M^2 C_h}{2Q^2} + \frac{1}{2} \left[\frac{2Q(Q - M)^2 - (Q - M)^2}{Q^2} \right] C_s$$

$$0 = -\frac{DC_o}{Q^2} - \frac{M^2 C_h}{2Q^2} + \left(\frac{Q^2 - M^2}{2Q^2} \right) C_s$$

$$\Rightarrow -\frac{DC_o}{Q^2} - \frac{M^2}{2Q^2} (C_h + C_s) + \frac{C_s}{2} = 0$$

$$\Rightarrow \frac{C_s Q^2}{2} = \frac{M^2}{2} (C_h + C_s) + DC_o$$

$$\Rightarrow Q^2 = M^2 \left(\frac{C_h + C_s}{C_s} \right) + \frac{2DC_o}{C_s} \quad \dots (2)$$

and

$$\frac{d}{dM}(TC) = \frac{MC_h}{Q} - \frac{(Q - M)}{Q} C_s = 0$$

$$\Rightarrow \frac{M}{Q} (C_h + C_s) - C_s = 0$$

$$\Rightarrow M = \left(\frac{C_s}{C_h + C_s} \right) Q \quad \dots(3)$$

Substituting the value of M in equation (2), we get

$$Q^2 = \left(\frac{C_s}{C_h + C_s} \right)^2 \times \left(\frac{C_h + C_s}{C_s} \right) Q^2 + \frac{2DC_0}{C_s}$$

$$Q^2 \left[1 - \frac{C_s}{C_h + C_s} \right] = \frac{2DC_0}{C_s}$$

$$\Rightarrow Q^2 \left[1 - \frac{C_s}{C_h + C_s} \right] = \frac{2DC_0}{C_s}$$

$$\Rightarrow Q^2 \left(\frac{C_h}{C_h + C_s} \right) = \frac{2DC_0}{C_s}$$

$$\Rightarrow Q^2 = \frac{2DC_0}{C_h} \left(\frac{C_s + C_h}{C_s} \right)$$

Thus, the optimum value of Q is given by

$$Q^* = \sqrt{\frac{2DC_0}{C_h} \left(\frac{C_s + C_h}{C_s} \right)}$$

The optimum value of M is given by

$$M^* = \left(\frac{C_s}{C_h + C_s} \right) Q^* = \sqrt{\frac{2DC_0}{C_h} \left(\frac{C_s}{C_h + C_s} \right)}$$

Therefore, the minimum inventory cost

$$TC^* = \sqrt{2DC_0 C_h \left(\frac{C_s}{C_h + C_s} \right)}$$

$$\text{Total Cycle time} = t^* = \frac{Q^*}{D} = \sqrt{\frac{2C_0}{DC_h} \left(\frac{C_s + C_h}{C_s} \right)}$$

Let us explain this model with the help of an example.

Example 2: A contractor has to supply engines to a truck manufacturer at the rate of 20 per day. The penalty in the contract is ₹100 per engine per day for missing the scheduled delivery date. The cost of holding an engine in stock for one month is ₹150. His production process is such that each month (30 days) he starts producing a batch of engines through the agencies and all are available for supply after the end of the month. Determine the maximum inventory level at the beginning of each month.

Solution: Here we have

$$C_h = \frac{150}{30} \text{ per engine per day} = ₹5 \text{ per engine per day;}$$

$$C_s = ₹100 \text{ per day per engine; } D = 20 \text{ engines per day and } t^* = 30 \text{ days.}$$

The maximum inventory level M^* at the beginning of each month will be

$$M^* = \frac{C_s}{C_h + C_s} Q^* = \frac{C_s}{C_h + C_s} (D \times t^*)$$

$$= \frac{100}{5 + 100} \times 20 \times 30 = 571 \text{ engines}$$

You may now like to solve the following problem to assess your understanding.

- E3)** A manufacturer has to supply his customer with 24000 units of his product every year. The demand is fixed and known. Since the unit is used by the customer in an assembly operation and the customer has no storage space for units, if the manufacturer fails to supply the required units, the shortage cost is `2 per unit per month. The inventory cost is `1 per unit per month and the set-up cost per run is `3500. Determine
- the optimal run size (Q);
 - the optimal level of inventory (M) at the beginning of any period;
 - the optimal scheduling period; and
 - the minimum inventory cost.

8.6 EOQ MODEL WITH UNIFORM REPLENISHMENT

For this model, it is assumed that the production run may take significant time to complete, that is, the replenishment is not instantaneous. The stock is added at a constant rate, i.e., the replenishment is uniform. Other assumptions are the same as the ones made for the EOQ model with uniform demand as explained in Sec. 8.3. The order size here is taken as the production size.

The graph representing this situation is shown in Fig. 8.4

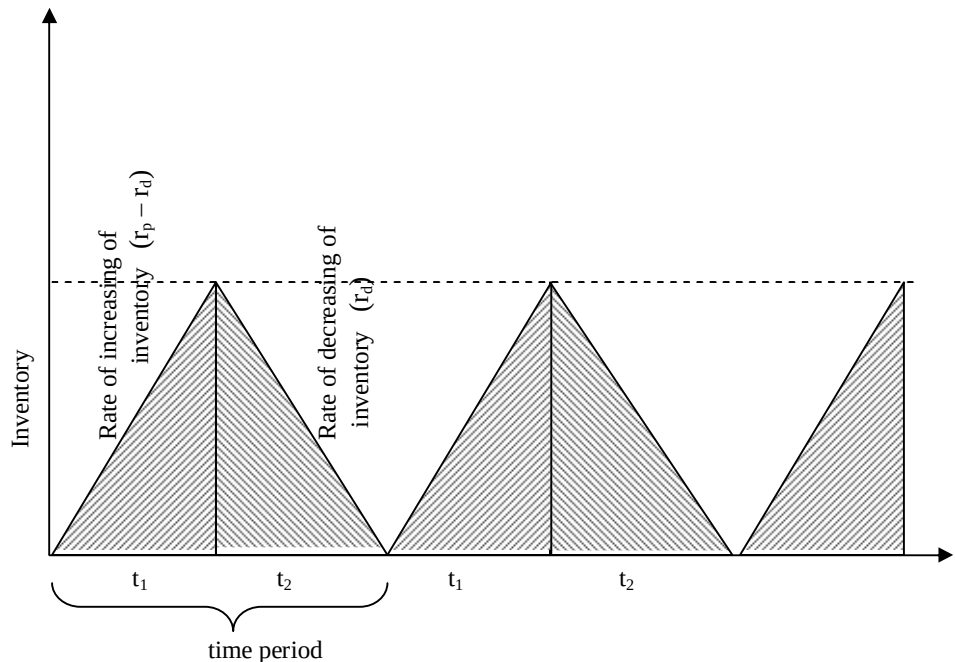


Fig. 8.4

Let r_d be the demand rate in units per time period, r_p , the replenishment rate per unit of time, t_1 , the time of production run or time of replenishment, and t_2 , the time required for the inventory to be exhausted. Then $t_1 + t_2 = t$ (say) is the time for each order cycle.

Let Q be the number of units produced per order cycle. Then

$$t_1 = \frac{Q}{r_p} \quad (\because \text{Time for producing 1 unit} = 1/\text{Rate of replenishment} = 1/r_p)$$

Inventory is building up at the rate of $r_p - r_d$.

$$\therefore \text{Maximum inventory level} = (r_p - r_d) t_1$$

$$\text{Average inventory} = \frac{(r_p - r_d) t_1}{2} = \frac{(r_p - r_d) Q}{2 r_p} = \frac{Q}{2} \left(1 - \frac{r_d}{r_p} \right)$$

$$\text{Now, ordering (or set-up) cost} = \frac{D}{Q} C_o \text{ and carrying cost} = \frac{Q}{2} \left(1 - \frac{r_d}{r_p} \right) C_h$$

$\therefore$ the total annual inventory cost is given by

$$TC = \frac{D}{Q} C_o + \frac{Q}{2} \left(1 - \frac{r_d}{r_p} \right) C_h$$

$$\text{The cost will be minimum if } \frac{D}{Q} C_o = \frac{Q}{2} \left(1 - \frac{r_d}{r_p} \right) C_h$$

$$\Rightarrow Q^2 = \frac{2DC_o}{C_h} \left(\frac{r_p}{r_p - r_d} \right) \Rightarrow Q^* = \sqrt{\frac{2DC_o}{C_h} \left(\frac{r_p}{r_p - r_d} \right)}$$

Thus, the characteristics of the model are as follows:

1. Optimum number of production runs per year:

$$N^* = \frac{D}{Q^*} = \sqrt{\frac{DC_h (r_p - r_d)}{2C_o r_p}}$$

2. Length of each lot size production run:

$$t_1 = \frac{Q^*}{r_p} = \sqrt{\frac{2DC_o}{C_h r_p (r_p - r_d)}}$$

3. The total minimum inventory cost:

$$TC^* = \frac{D}{Q^*} C_o + \frac{Q^*}{2} \left(1 - \frac{r_d}{r_p} \right) C_h = \sqrt{2DC_o C_h \left(1 - \frac{r_d}{r_p} \right)}$$

We take up an example to apply this model.

Example 3: A tyre producer makes 1600 tyres per day and sells them at approximately half that rate. Accounting figures show that the production

set-up cost is ₹1000 and carrying cost per unit is ₹5. If the annual demand is 160000 tyres, what is the optimal lot size and how many production runs should be scheduled per year?

Solution: We are given that the annual demand (D) = 160000 tyres, C_h = ₹5, C_o = ₹1000, Production rate (r_p) = 1600 tyres per day and Demand rate (r_d) = 800 tyres per day.

$$\therefore \text{The optimal lot size } Q^* = \sqrt{\frac{2DC_o}{C_h} \times \frac{r_p}{r_p - r_d}}$$

$$Q^* = \sqrt{\frac{2 \times 160000 \times 1000}{5} \times \frac{1600}{1600 - 800}} = 11314 \text{ tyres}$$

$$\text{Optimal production runs per year } N^* = \frac{D}{Q^*} = \frac{160000}{11314} = 14 \text{ runs/year approx.}$$

You may now like to solve the following problems to assess your understanding.

-
- E4)** A product is manufactured at the rate of 50 items per day. The demand occurs at the rate of 30 items per day. It is given that the set-up cost per order is ₹1000 and holding cost per unit time is ₹0.05. Find the economic lot size and the associated total cost per cycle assuming that no shortage is allowed.
- E5)** A company uses 100000 units of a particular item per year. Each item costs ₹2. The production engineering department estimates the holding cost as 12.5% of the value of the inventory per day. The replenishment rate is uniform at 500 units per day. Assuming 250 working days (for replenishment purpose), calculate the
- optimal set-up quantity;
 - total cost on the basis of optimal policy; and
 - optimal number of set-ups.
-

8.7 EOQ MODEL WITH PRICE (OR QUANTITY) DISCOUNTS

If the items are purchased in bulk, some discount in price is usually offered by the supplier. The discount is called **all units discount** if it is applicable for all the units purchased. It is said to be **incremental discount** if discounts are offered only for the items which are in excess of the specified amount. In the incremental discount, the prices offered in different slabs are applicable in finding the total cost while in 'all units discount', only one price at any one slab is applicable for finding the total cost. Buying in large quantities may result in the following advantages:

Less unit price, less ordering costs, less transportation cost, fewer stock-outs and hence less chances of loss of goodwill, mass display by retailers and preferential treatment by the sellers.

At the same time, large quantity buying may involve the following disadvantages:

Higher carrying costs, older stocks, lower stock turnover, more capital required, blocked money, less flexibility, heavier deterioration and depreciation.

If the items are purchased with quantity discounts, the price C may vary according to the following scheme:

$$\begin{aligned}
 C &= p_0 \text{ if purchased quantity } Q = Q_0 < b_1 \\
 &= p_1 \text{ if purchased quantity } b_1 \leq Q_1 < b_2 \\
 &\dots\dots\dots \\
 &\dots\dots\dots \\
 &\dots\dots\dots \\
 &= p_n \text{ if purchased quantity } b_n < Q_n
 \end{aligned}$$

where p_{j-1} is greater than p_j , for $j = 1, 2, \dots, n$. Here p_j is the price per unit for the j^{th} lot size. If the shortages are not allowed and supply is instantaneous, the total cost per year is given by the set of relations:

$$TC(Q_j) = Dp_j + \frac{C_0 D}{Q_j} + \frac{1}{2} C_h Q_j, \text{ where } b_j \leq Q_j < b_{j+1}$$

and $C_h = ip_j$, where i is the percentage change for $j = 0, 1, 2, \dots, n$.

We follow the procedure given below to find the overall optimum lot size:

Step 1: We find EOQ for the lowest price, i.e., we calculate

$$Q_n^* = \sqrt{\frac{2C_0 D}{ip_n}}$$

If $Q_n^* \geq b_n$, the optimum order quantity is Q_n^* . If $Q_n^* < b_n$, we go to Step 2.

Step 2: We compute $Q_{n-1}^* = \sqrt{\frac{2C_0 D}{ip_{n-1}}}$ for the next lowest price.

If $Q_{n-1}^* \geq b_{n-1}$, then we compare the total cost TC_{n-1} for purchasing Q_{n-1}^* with the total cost TC_n for purchasing quantity b_n and select the one that gives the least cost.

If $Q_{n-1}^* < b_{n-1}$, we go to Step 3.

Step 3: We compute $Q_{n-2}^* = \sqrt{\frac{2C_0 D}{ip_{n-2}}}$. If $Q_{n-2}^* \geq b_{n-2}$, we compare the total

cost TC_{n-2} , TC_{n-1} and TC_n for purchase of quantities Q_{n-2}^* , b_{n-1} and b_n , respectively, and select the optimum purchase quantity.

If $Q_{n-2}^* < b_{n-2}$, then we go to Step 4.

Step 4: We continue in this fashion until $Q_{n-j}^* \geq b_{n-j}$. Then we compare total

costs TC_{n-j} with $TC_{n-j+1}, \dots, TC_{n-1}, TC_n$ for purchase quantities Q_{n-j}^* , $b_{n-j+1}, \dots, b_{n-1}, b_n$, respectively, and select the optimum purchase quantity.

We now apply this method to an example.

Example 5: The annual demand for an item is 2400 units. The inventory carrying charge is 24% of the purchase price per year. Purchase prices are:

$$p_1 = ₹ 10 \quad \text{for purchasing } Q_1 < 500$$

$$p_2 = ₹ 9.25 \quad \text{for purchasing } 500 \leq Q_2 < 750$$

$$p_3 = ₹ 8.75 \quad \text{for purchasing } 750 \leq Q_3$$

Determine the optimum purchase quantity taking the ordering cost as

$$\text{i) } ₹ 437.5$$

$$\text{ii) } ₹ 87.5$$

Solution: To determine the optimum purchase quantity, we follow the procedure given below:

$$\text{i) As per Step 1, } Q_3^* = \sqrt{\frac{2 \times ₹ 437.5 \times 2400}{(0.24)(8.75)}} = 1000 \text{ units}$$

Since 1000 is greater than 750, the optimum purchase quantity is 1000 units.

$$\text{ii) Here } Q_3^* = \sqrt{\frac{2 \times ₹ 87.5 \times 2400}{(0.24)(8.75)}} = 447 \text{ units}$$

Since $447 < 750 = b_3$, we next compute

$$Q_2^* = \sqrt{\frac{2 \times ₹ 87.5 \times 2400}{(0.24)(9.25)}} = 435 \text{ units}$$

Since $435 < 500 = b_2$, we next compute

$$Q_1^* = \sqrt{\frac{2 \times ₹ 87.5 \times 2400}{(0.24)(10)}} = 418 \text{ units} > 0 \text{ and } < 500$$

We now compare the total cost for purchasing $Q_1^* = 418$, $b_2 = 500$ and $b_3 = 750$ units, respectively.

$$\begin{aligned} TC_1(\text{for purchasing } 418) &= ₹ 10 \times 2400 + \frac{₹ 87.5 \times 2400}{418} + \frac{1}{2}(0.24) \times (₹ 10) \times 418 \\ &= ₹ 24504 \end{aligned}$$

$$\begin{aligned} TC_2(\text{for } Q_2 = 500) &= ₹ 9.25 \times 2400 + \frac{₹ 87.5 \times 2400}{500} + \frac{1}{2}(0.24) \times (₹ 9.25) \times (500) \\ &= ₹ 23175 \end{aligned}$$

$$\begin{aligned} TC_3(\text{for } Q_3 = 750) &= ₹ 8.75 \times 2400 + \frac{₹ 87.5 \times 2400}{750} + \frac{1}{2}(0.24) \times (₹ 8.75) \times (750) \\ &= ₹ 22067 \end{aligned}$$

As the total inventory cost is minimum for $Q_3 = 750$, therefore, the optimum purchase quantity is 750.

You may now like to solve the following problems to assess your understanding.

E6) Consider Example 5 and determine the optimum purchase quantity with the ordering cost as ₹ 175.

8.8 SUMMARY

1. An inventory is a physical stock of idle resources of any kind having some economic value kept for the purpose of meeting future demand. It indicates the raw material required before production, the finished goods after production ready for delivery to consumers, human resources, financial resources, etc. stocked in order to meet an expected demand in the future.
2. Inventories are essential for almost all businesses for efficient and smooth operations. Hence, inventories need to be controlled in the most profitable way.
3. The quantity that is kept in stock in order to balance the costs of holding too much stock vis-à-vis the costs of ordering in small quantities is called the economic order quantity.
4. Five models have been discussed for obtaining the economic order quantity for the following situations:
 - i) when demand is uniform,
 - ii) when rates of demand are different in different cycles,
 - iii) when shortages are allowed,
 - iv) when replenishment is uniform, and
 - v) when price discounts are given.

8.9 SOLUTIONS/ANSWERS

E1) We have $D = \frac{12000}{1.25} = \frac{12000}{125} \times 100 = 9600$; $C = 1.25$ per unit;

$$C_o = 15 \text{ and } C_h = 15\% \text{ of } 1.25 \Rightarrow \frac{15}{100} \times \frac{125}{100}$$

$$Q^* = \sqrt{\frac{2DC_o}{C_h}} = \sqrt{\frac{2 \times 9600 \times 15 \times 100}{15}} = \sqrt{1920000} = 1385 \text{ units}$$

E2) Here $C_o = 36$, $C = 2$, $D = 10000$, and $C_h = 18\% \text{ of } 2 = \frac{18 \times 2}{100} = \frac{9}{25}$

$$\text{Therefore, a) } Q^* = \sqrt{\frac{2 \times 10000 \times 36 \times 25}{9}} = \sqrt{2000000} = 1414 \text{ units}$$

$$\text{and b) } \frac{D}{Q^*} = \frac{10000}{1414} = 7.072$$

$$\text{Thus, the optimum no. of days} = \frac{365}{7} = 52 \text{ days}$$

E3) We have $D = \text{number of units supplied per year} = 24000$

Ordering cost per unit = ₹ 3500

Shortage cost = ₹ 2 per unit per month

= ₹ 24 per unit per year

Carrying cost = ₹ 1 per unit per month

= ₹ 12 per unit per year

i) The optimal value of Q is

$$\begin{aligned} Q^* &= \sqrt{\frac{2DC_o}{C_h} \left(\frac{C_s + C_h}{C_s} \right)} \\ &= \sqrt{\frac{2 \times 24000 \times 3500}{12} \left(\frac{24 + 12}{24} \right)} \\ &= \sqrt{2 \times 1000 \times 3500 \times 3} = \sqrt{210 \times 100000} \\ &= \sqrt{21} \times 1000 = 4.583 \times 1000 = 4583 \text{ units} \end{aligned}$$

$$\begin{aligned} \text{ii) } M^* &= \left(\frac{C_s}{C_h + C_s} \right) Q^* = \left(\frac{24}{12 + 24} \right) \times 4583 \\ &= \frac{2}{3} \times 4583 = 3055 \text{ units} \end{aligned}$$

$$\text{iii) } t^* = \frac{Q^*}{D} = \frac{4583}{24000} = \frac{4.583}{2} = 2.29 \text{ months}$$

iv) The minimum inventory cost = TC*

$$\begin{aligned} &= \sqrt{2DC_o C_h \left(\frac{C_s}{C_h + C_s} \right)} \\ &= \sqrt{2 \times 24000 \times 12 \times \left(\frac{24}{12 + 24} \right) \times 3500} \\ &= 576000 \times \frac{24}{36} = 37200 \end{aligned}$$

E4) We have D = 30 items per day, $r_p = 50$ items per day,
 $C_o = 1000$, $C_h = 0.05$, $r_d = 30$ items per day

$$\begin{aligned} \text{Therefore, } Q^* &= \sqrt{\frac{2DC_o}{C_h} \left(\frac{r_p}{r_p - r_d} \right)} = \sqrt{\frac{2 \times 30 \times 1000}{0.05} \left(\frac{50}{20} \right)} \\ &= \sqrt{3000000} = 1732.05081 = 1732 \text{ units} \end{aligned}$$

Total minimum inventory cost

$$\begin{aligned} TC^* &= \sqrt{2DC_o C_h \left(1 - \frac{r_d}{r_p} \right)} \\ &= \sqrt{2 \times 30 \times 1000 \times 0.05 \left(1 - \frac{30}{50} \right)} \\ &= \sqrt{2 \times 30 \times 1000 \times \frac{5}{100} \times \frac{20}{50}} = \sqrt{1200} = 34.64 \end{aligned}$$

E5) We are given that

$$r_p = 500 \times 250 \text{ units per year, } r_d = 100000 \text{ units per year, } C_o = 25$$

$$C_h = 12.5\% \text{ of } 2 = \frac{12.5 \times 2}{100} = \frac{1}{4} = 0.25$$

$$\begin{aligned} \text{a)} \quad Q^* &= \sqrt{\frac{2DC_o}{C_h} \left(\frac{r_p}{r_p - r_d} \right)} \\ &= \sqrt{\frac{2 \times 100000 \times 25}{0.25} \left(\frac{125000}{25000} \right)} \\ &= \sqrt{4000000 \times 25} = 10000 \end{aligned}$$

$$\begin{aligned} \text{b)} \quad TC^* &= \sqrt{2DC_o C_h \left(1 - \frac{r_d}{r_p} \right)} \\ &= \sqrt{2 \times 100000 \times 0.25 \times 25 \left(1 - \frac{100000}{125000} \right)} \\ &= \sqrt{250000} = 500 \end{aligned}$$

$$\begin{aligned} \text{Total cost on the basis of optimal policy} &= TC^* + 2 \times 100000 \\ &= 200500 \end{aligned}$$

$$\text{c)} \quad \text{Optimum number of set ups} = \frac{D}{Q^*} = \frac{100000}{10000} = 10 \text{ set-ups}$$

$$\text{E6)} \quad \text{Here} \quad Q_3^* = \sqrt{\frac{2 \times 175 \times 2400}{(.24)(8.75)}} = 632 \text{ units}$$

Since $632 < 750 = b_3$, we next compute

$$Q_2^* = \left(\frac{2 \times 175 \times 2400}{(.24)(9.25)} \right)^{1/2} = 615 \text{ units}$$

Since $615 > 500 = b_2$ we compare the total costs for purchasing

$Q_2^* = 615$ and $b_3 = 750$ units respectively.

$$\begin{aligned} TC_2(\text{for } 615 \text{ units}) &= 9.25 \times 2400 + \frac{100 \times 2400}{615} + \frac{1}{2} \times (0.24) \times (9.25) \times (615) \\ &= ₹ 23,273 \end{aligned}$$

$$\begin{aligned} TC_3(\text{for } Q_3 = 750) &= 8.75 \times 2400 + \frac{100 \times 2400}{750} + \frac{1}{2} \times (0.24) \times (8.75) \times (750) \\ &= ₹ 22,119.50 \end{aligned}$$

Therefore, the economic purchase quantity for this problem is

$$Q_3^* = 750 \text{ units}$$

Block

3

REGRESSION MODELLING

UNIT 9

Simple Linear Regression	5
---------------------------------	----------

UNIT 10

Statistical Inference in Simple Linear Regression	27
--	-----------

UNIT 11

Multiple Linear Regression	45
-----------------------------------	-----------

UNIT 12

Selection of Variables and Testing Model Assumptions	71
---	-----------

Curriculum and Course Design Committee

Prof. K. R. Srivathsan
Former Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Parvin Sinclair
Former Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Geeta Kaicker
Former Director, School of Sciences
IGNOU, New Delhi

Prof. Jagdish Prasad
Department of Statistics
University of Rajasthan, Jaipur

Prof. R. M. Pandey
Department of Bio-Statistics
AIIMS, New Delhi

Dr. Manish Trivedi
School of Sciences, IGNOU

Dr. Nitin Gupta
School of Sciences, IGNOU

Dr. Neha Garg
School of Sciences, IGNOU

Mr. Rajesh Kaliraman
School of Sciences, IGNOU

Prof. Rahul Roy
Theoretical Statistics and Mathematics Unit
Indian Statistical Institute, New Delhi

Prof. Diwakar Shukla
Department of Mathematics and Statistics
Dr. Hari Singh Gaur University, Sagar

Prof. Rakesh Srivastava
Department of Statistics
M.S. University of Baroda, Vadodara

Prof. G. N. Singh
Department of Applied Mathematics
I.S.M., Dhanbad

Prof. Gulshan Lal Taneja
Department of Mathematics
M.D. University, Rohtak

Prof. Poornima Mital
School of Sciences, IGNOU

Prof. Sujatha Varma
School of Sciences, IGNOU

Dr. S. Venkataraman
School of Sciences, IGNOU

Dr. Deepika
School of Sciences, IGNOU

Block Preparation Team

Prof. Rakesh Srivastava (Editor)
Department of Statistics
M.S. University of Baroda,
Vadodara (GUJ)

Prof. Vijayshri (Language Editor)
School of Sciences, IGNOU

Prof. G. K. Shukla (Units 10 to 12)
(Retd.), Department of Applied Mathematics,
Indian Institute of Technology,
Kanpur (UP)

Dr. Manish Trivedi (Units 9 to 12)
School of Sciences, IGNOU

Programme and Course Coordinator: Dr. Manish Trivedi

Block Production

Mr. Sunil Kumar, AR (P)
School of Sciences, IGNOU

September, 2014

© Indira Gandhi National Open University, 2014

ISBN-978-81-266-

All rights reserved. No part of this work may be reproduced in any form, by mimeograph or any other means, without permission in writing from the Indira Gandhi National Open University

Further information on the Indira Gandhi National Open University may be obtained from the University's Office at Maidan Garhi, New Delhi-110068 or visit University's website <http://www.ignou.ac.in>

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by the Director, School of Sciences.

Laser Typeset by: Tessa Media & Computers, C-206, A.F.E.-II, Okhla, New Delhi

Printed at:

BLOCK 3 REGRESSION MODELLING

In Blocks 1 and 2 of this course, you have learnt some basic methods of solving linear programming problems, transportation problems and assignment problems. In these blocks, you have also learnt various techniques to optimise the processing/service time or total cost while dealing with the queueing systems, sequential problems or inventory problems, respectively.

In this block, we discuss the concepts of regression modelling, which is one of the most widely used techniques for analysing multivariable data. Regression analysis is the process for investigating and analysing the average relationship between the dependent or response variable and a set of independent or regressor variables. The purpose of regression analysis is to establish a relationship between a dependent variable and one or more than one regressor variables and to predict the optimum value of dependent variable using this relationship. Regression analysis helps us model practical problems dealing with real world data. Therefore, a thorough knowledge of regression modelling is essential for a statistician. We present the concepts of regression modelling in 4 units of this block.

In Unit 9 entitled **Simple Linear Regression**, we discuss how to establish the linear relationship between a dependent variable Y and an independent variable X using an equation $Y = a + bX$. The coefficients a and b are unknown and for given data on Y and X , we obtain the least squares estimates of these coefficients. Residual analysis is also performed for detecting the deficiencies in this regression model.

In Unit 10 entitled **Statistical Inference in Simple Linear Regression**, we discuss the inferential aspect of simple linear regression. It is of interest to examine whether a significant linear relationship exists between Y and X . The coefficient of determination, R^2 , which gives a measure of goodness of fit of the model, is explained in this unit. We also discuss the confidence interval for the predicted value of $E(Y)$ corresponding to a given value of X .

Unit 11 entitled **Multiple Linear Regression** deals with multiple regression modelling. We examine the adequacy and overall fit of the model with the help of the coefficient of determination R^2 . We also discuss a method for calculating R^2 and adjusted R^2 .

In Unit 12 entitled **Selection of Variables and Testing Model Assumptions**, we discuss how to build a regression model for satisfactory explanation of variability in the dependent (response) variable Y . The forward selection, backward elimination and stepwise selection procedures for selection of variables in multiple regression modelling are also described in this unit. The other important aspects covered in this unit are verification of assumptions such as normality and the homogeneity of variances.

Notations and Symbols

Y_i	:	i^{th} observation of the dependent variable Y
X_i	:	i^{th} observation of the independent variable X
$\hat{Y}_i$	:	i^{th} observation of the fitted value
r_i	:	i^{th} residual value, i.e., $Y_i - \hat{Y}_i$
d_i	:	i^{th} standardised residual value
e_i	:	i^{th} error term
p_i	:	Cumulative probability of i^{th} ranked residual value
P_i	:	Percentile cumulative probability of i^{th} ranked residual value
E	:	Sum of squares of the error terms
$\text{Var}(\bar{Y})$	:	Variance of $\bar{Y}$
SS_{Res}	:	Sum of squares of residuals
$\hat{a}$ and $\hat{b}$	:	Unbiased estimates of regression parameters a and b
R^2	:	Coefficient of determination
F_X	:	Lowest partial F-value
SS_{Reg}	:	Sum of squares due to regressor variables
SS_T	:	Total sum of squares
B_j	:	Amount of change in Y for a unit change in X_j
B_0	:	Intercept and coefficient of dummy variable X_0
I_n	:	Identity matrix of order n
$X'X$	:	A Non-singular Matrix
$V(\hat{B})$	:	Variance-Covariance matrix of $\hat{B}$
Secs.	:	Sections/Sub-sections
Fig.	:	Figure

UNIT 9 SIMPLE LINEAR REGRESSION

Structure

- 9.1 Introduction
 - Objectives
- 9.2 Simple Linear Regression
- 9.3 Least Squares Estimation of Parameters
- 9.4 Fitting of Regression Line
- 9.5 Residual Analysis
 - Scaling of Residuals
 - Residual Plots
 - Normal Probability Plot
- 9.6 Summary
- 9.7 Solutions/Answers

9.1 INTRODUCTION

In Blocks 1 and 2, you have learnt some basic methods of optimisation of various problems such as LPP, transportation problem, assignment problem, queueing problem, scheduling and sequencing problem and inventory problem. In this unit, we discuss the concepts of regression modelling. Regression analysis is a statistical tool for investigating and analysing the average relationship between two or more variables. In regression analysis, one variable is referred to as the dependent variable or response variable, whereas the other variable is referred to as independent variable, predictor variable or regressor variable. The regression analysis with only one predictor variable is known as **simple regression analysis** and with two or more predictor variables is known as **multiple regression analysis**.

The term simple linear regression refers to a regression equation with only one predictor variable and the equation is linear. This means that if Y is the dependent variable and X, the independent variable, the regression equation is of the form $Y = a + bX$. Such an analysis is useful in many situations, e.g., in analysing the relationship between the number of customers and monthly sales of a product. In regression analysis, we use the method of least squares to fit the regression equation to the data. We discuss the concept of simple linear regression, the method of least squares and fitting of regression line in Secs. 9.2 to 9.4. In Sec. 9.5, we present residual analysis.

In the next unit, we shall discuss the test of significance of regression coefficients, the coefficient of determination and confidence intervals of regression coefficients.

Objectives

After studying this unit, you should be able to:

- describe the simple linear regression model;
- estimate the model parameters using the method of least squares;
- describe the properties of the estimates;
- fit the simple linear regression line; and
- check the model assumptions with the help of residual analysis.

9.2 SIMPLE LINEAR REGRESSION

Regression analysis may broadly be defined as the analysis of relationships among variables. It is popular because it explores the power of statistics as a tool for establishing average relationships among variables. This relationship is given as an equation that helps to predict the dependent variable Y through one or more independent variables. In regression analysis, the variable whose values vary with the variations in the values of the other variable(s) is called the **dependent variable** or **response variable**. The other variables which are independent in nature and influence the response variable are called **independent variables**, **predictor variables** or **regressor variables**. A regression equation with a single independent variable is called a **simple regression equation** and it is linear when the model parameters have a linear relation.

The equation $Y = a + b X^2$ may also be classified as simple linear regression because it has only one independent variable X and by change of variable, e.g., $X^2 = T$, we can get a straight line of the form $Y = a + b T$ even though Y bears a quadratic relationship with X . Its conceptual usefulness exists in analysing the relationship between a variable of interest and a set of related predictor variables using an equation.

Let us take an example where regression technique may be useful. Suppose a statistician employed by a cold drink bottler is analysing the product delivery and service operation for vending machines. He would like to find how the delivery time taken by the delivery man to load and service a machine is related to the volume of delivery cases. The statistician visits 50 randomly chosen retailer shops having vending machines and observes the delivery time (in minutes) and the volume of delivery cases for each shop. He plots those 50 observations on a graph, which shows that an approximate linear relationship exists between the delivery time and delivery volume. If Y represents the delivery time and X , the delivery volume, the equation of a straight line relating these two variables may be given as

$$Y = a + bX \quad \dots (1)$$

where a is the intercept and b , the slope.

In such cases, we draw a straight line in the form of equation (1) so that the data points generally fall near the straight line. Now, suppose the points do not fall exactly on the straight line. Then we should modify equation (1) to minimise the difference between the observed value of Y and that given by the straight line ($a + b X$). This is known as error.

The error e , which is the difference between the observed value and the predicted value of the variable of interest Y , may be conveniently assumed as a statistical error. This error term accounts for the variability in Y that cannot be explained by the linear relationship between X and Y . It may arise due to the effects of other factors. Thus, a more plausible model for the variable of interest (Y) may be given as

$$Y = a + b X + e \quad \dots (2)$$

where the intercept a and the slope b are unknown constants and e is a random error component.

Equation (2) is called a **linear regression model**. Since this model involves only one regressor or independent variable, it is called a **simple linear regression model**. In case of two or more than two independent variables, it is known as a **multiple linear regression model**.

The following assumptions are made about the error term e in the regression model given in equation (2):

- i) the error term e is a random variable with mean zero; i.e., $E(e) = 0$;
- ii) the variance of the error term e , denoted by σ^2 , is the same for all values of X ;
- iii) all the error terms, $e_1, e_2, \dots, e_n$, are independent. This means that the value of e for a particular value of X is not related to the value of e for any other value of X . Thus, the value of Y for a particular X is independent of any other value of X ; and
- iv) the error term e is a normally distributed random variable.

These assumptions provide the theoretical basis for the t-test and F-test used to determine whether the relationship between X and Y is significant and for the estimation of confidence and prediction interval.

You may now like to answer the following questions, which will help you assess your understanding of simple linear regression.

-
- E1)** What is simple linear regression ? Write the equation for simple linear regression model.
- E2)** Answer the following questions for the error term e in the regression model $Y = a + bX + e$:
- i) What are the mean and variance of the random error e ?
 - ii) Which distribution does it follow?
 - iii) Are the errors $e_1, e_2, \dots$ independent?
-

9.3 LEAST SQUARES ESTIMATION OF PARAMETERS

In the previous section, we have discussed the simple regression equation with only one regressor variable X and the variable of interest Y . We have also discussed the simple linear regression model with a single regressor variable X . The simple linear regression model has two unknown parameters a and b , which are known as **intercept** and **regression coefficient**, respectively. Their values are unknown. Therefore, they must be estimated using sample data. The estimation of the parameters a and b is done by minimising the error term e .

Let $(Y_1, X_1), (Y_2, X_2), \dots, (Y_n, X_n)$ be n pairs of values in the data. The equation of the simple linear regression model may be written as

$$Y = a + bX + e \quad \dots (3)$$

where e represents the error term which arises due to the difference of the observed Y and the fitted line $\hat{Y} = \hat{a} + \hat{b}X$. We use the method of least squares to minimise the error term e . From equation (3), we may write a simple regression model as

$$Y_i = a + bX_i + e_i \quad i = 1, 2, \dots, n \quad \dots (4)$$

for a sample data of n pairs of values given in terms of (Y_i, X_i) , $(i = 1, 2, \dots, n)$.

We estimate a and b so that the sum of the squares of the differences between the observed values (Y_i) and the points lying on the straight line is minimum, i.e., the sum of squares of the error terms given by

$$E = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - a - bX_i)^2 \quad \dots (5)$$

is minimum. To find the values of a and b for which the sum of squares of the error terms, i.e., E is minimum, we differentiate it with respect to the parameters a and b and equate the results to zero:

$$\frac{\partial E}{\partial a} = -2 \sum_{i=1}^n (Y_i - a - bX_i) = 0 \quad \dots (6)$$

and

$$\frac{\partial E}{\partial b} = -2 \sum_{i=1}^n (Y_i - a - bX_i) X_i = 0 \quad \dots (7)$$

Simplifying equations (6) and (7), we get

$$n a + b \sum_{i=1}^n X_i = \sum_{i=1}^n Y_i \quad \dots (8)$$

$$a \sum_{i=1}^n X_i + b \sum_{i=1}^n X_i^2 = \sum_{i=1}^n Y_i X_i \quad \dots (9)$$

Equations (8) and (9) are called the **least-squares normal equations**. The solution to these normal equations is

$$\hat{a} = \bar{Y} - \hat{b}\bar{X} \quad \dots (10)$$

where $\bar{Y}$ and $\bar{X}$ are the averages of Y_i and X_i , respectively. On putting the value of $\hat{a}$ from equation (10) in equation (9), we get

$$\hat{b} = \frac{\sum_{i=1}^n Y_i X_i - \left(\sum Y_i \right) \left(\sum X_i \right) / n}{\sum_{i=1}^n X_i^2 - \left(\sum X_i \right)^2 / n} \quad \dots (11)$$

Since the denominator of equation (11) is the corrected sum of squares of X_i , we may rewrite it as

$$SS_X = \sum_{i=1}^n X_i^2 - \frac{\left(\sum X_i \right)^2}{n} = \sum_{i=1}^n (X_i - \bar{X})^2$$

Similarly, the numerator is the corrected sum of the cross product of X_i and Y_i and may be rewritten as:

$$SS_{XY} = \sum_{i=1}^n Y_i X_i - \frac{\left(\sum Y_i \right) \left(\sum X_i \right)}{n} = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Therefore, the expression for $\hat{b}$ may be rewritten as

$$\hat{b} = \frac{SS_{XY}}{SS_X} \quad \dots (12)$$

Thus, $\hat{a}$ and $\hat{b}$ are the least squares estimates of the intercept a and slope b , respectively. Therefore, the fitted simple linear regression model is given by

$$\hat{Y} = \hat{a} + \hat{b} X \quad \dots (13)$$

Equation (13) gives a point estimate of the mean of Y for a particular X . The difference between the fitted value $\hat{Y}_i$ and Y_i is known as the **residual** and is denoted by r_i :

$$r_i = Y_i - \hat{Y}_i, \quad i = 1, 2, \dots, n \quad \dots(14)$$

The role of the residuals and its analysis is very important in regression modelling. We shall discuss it in detail in Sec. 9.5.

Properties of Estimates of Parameters

So far you have learnt that the estimates of parameters of a simple linear regression model are obtained with the help of the method of least squares. On putting the values of the estimates of the parameters in the equation, we get a fitted simple linear regression model. The least squares estimates $\hat{a}$ and $\hat{b}$ have several important properties, which are given below:

- a) The means of $\hat{a}$ and $\hat{b}$ (the estimates of the parameters a and b) of a simple linear regression model

$$E(Y_i) = a + b X_i$$

are given by

$$E(\hat{a}) = a \text{ and } E(\hat{b}) = b,$$

respectively. Thus, if we assume that the model is correct, then $\hat{a}$ and $\hat{b}$ are unbiased estimators of a and b .

The variance of $\hat{b}$ is given as

$$\text{Var}(\hat{b}) = \frac{\sigma^2}{SS_x} \quad \dots(15)$$

The variance of $\hat{a}$ is given as

$$\begin{aligned} \text{Var}(\hat{a}) &= \text{Var}(\bar{Y} - \hat{b} \bar{X}) \\ &= \text{Var}(\hat{\bar{Y}}) + \bar{X}^2 \text{Var}(\hat{b}) - 2\bar{X} \text{Cov}(\bar{Y}, \hat{b}) \end{aligned}$$

Now the variance of $\hat{\bar{Y}}$ is just $\text{Var}(\bar{Y}) = \frac{\sigma^2}{n}$ and the covariance between $\hat{\bar{Y}}$ and $\hat{b}$ would be zero. Thus, we get

$$\text{Var}(\hat{a}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_x} \right) \quad \dots(16)$$

- b) The sum of the residuals in any regression model that contains an intercept a is always zero, i.e.,

$$\begin{aligned} \sum_{i=1}^n r_i &= \sum_{i=1}^n (Y_i - \hat{Y}_i) = \sum_{i=1}^n (Y_i - \hat{a} - \hat{b} X_i) \\ &= \sum_{i=1}^n [Y_i - \bar{Y} - \hat{b} (X_i - \bar{X})] \quad \because \hat{a} = \bar{Y} - \hat{b} \bar{X} \\ &= \sum_{i=1}^n (Y_i - \bar{Y}) - \hat{b} \sum_{i=1}^n (X_i - \bar{X}) = 0 \end{aligned} \quad \dots(17)$$

- c) The sum of the observed values (Y_i) is equal to the sum of the fitted values ($\hat{Y}_i$):

$$\sum_{i=1}^n Y_i = \sum_{i=1}^n \hat{Y}_i$$

- d) The least square regression lines always pass through the centroid $(\bar{X}, \bar{Y})$ of the data since we know that the two lines of regression intersect at this point.
- e) The sum of the residuals weighted by the corresponding value of the regressor variables is always zero, i.e., $\sum_{i=1}^n X_i r_i = 0$
- f) The sum of the residuals weighted by the corresponding fitted values is always equal to zero, i.e., $\sum_{i=1}^n \hat{Y}_i r_i = 0$

Now, you should answer the following questions to assess your understanding.

E3) Describe the method of obtaining the least squares estimates of the parameters a and b for the regression model $Y = a + bX + e$.

E4) Describe the properties (in brief) of the estimates of the parameters a and b for the regression model $Y = a + bX + e$.

9.4 FITTING OF REGRESSION LINE

In Sec. 9.3, you have learnt the method of least squares for estimating the parameters a and b in the simple linear regression model $Y = a + bX + e$. You have also learnt the properties of the least squares estimates. In this section, we discuss the method of fitting the simple linear regression equation for the given data of n pairs of observations on (X, Y) .

Let the given data of n pairs of observations on X and Y be as follows:

X:	X_1	X_2	X_3	$\dots$	$\dots$	X_i	$\dots$	X_n
Y:	Y_1	Y_2	Y_3	$\dots$	$\dots$	Y_i	$\dots$	Y_n

where Y is the dependent variable and X , the independent variable. Suppose, we wish to fit the following simple regression equation to the data:

$$Y = a + bX \quad \dots(18)$$

where a is the intercept and b is the slope of the equation. For fitting equation (18) to the data on (X, Y) , we follow the steps given below:

Step 1: We draw a scatter diagram by plotting the (X, Y) points given in data.

Step 2: We construct a table as given below and take the sum of the values of X_i , $X_i Y_i$, and X_i^2 . We write the values of $\sum X$, $\sum Y$, $\sum XY$ and $\sum X^2$ in the last row:

X	Y	XY	X ²
X ₁	Y ₁	X ₁ Y ₁	X ₁ ²
X ₂	Y ₂	X ₂ Y ₂	X ₂ ²
X ₃	Y ₃	X ₃ Y ₃	X ₃ ²
⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮
X _i	Y _i	X _i Y _i	X _i ²
⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮
X _n	Y _n	X _n Y _n	X _n ²
∑X	∑Y	∑XY	∑X ²

Step 3: We express of $\hat{a}$ given in equation (10) as follows:

$$\hat{a} = \bar{Y} - b\bar{X} = \frac{1}{n}[\sum Y - b\sum X] \quad \dots (19)$$

Step 4: We substitute the values of $\sum X$, $\sum Y$, $\sum XY$ and $\sum X^2$ in equation (11) and calculate the optimum value of $\hat{b}$ as

$$\hat{b} = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} \quad \dots (20)$$

Step 5: Now for obtaining the value of $\hat{a}$, we substitute the value of $\sum X$, $\sum Y$ and slope $\hat{b}$ in equation (19) and obtain the optimum value of intercept $\hat{a}$ as

$$\hat{a} = \frac{1}{n}[\sum Y - \hat{b}\sum X]$$

Step 6: We put these values of $\hat{a}$ and $\hat{b}$ in the regression equation and get

$$\hat{Y} = \hat{a} + \hat{b}X$$

Let us obtain the regression line for a given set of data.

Example 1: Sales data of 10 months for a coffee house situated near a prime location of a city comprising the number of customers (in hundreds) and monthly sales (in Thousand Rupees) are given below:

S. No.	No. of Customers (in hundreds)	Monthly Sales (in thousand Rs.)
1	6.0	01
2	6.1	06
3	6.2	08
4	6.3	10
5	6.5	11
6	7.1	20
7	7.6	21
8	7.8	22
9	8.0	23
10	8.1	25

Find the simple linear regression equation that fits the given data.

Solution: It is given that $n = 10$. Let us assume an equation of simple linear regression as follows:

$$Y = a + bX$$

where X is the independent variable and Y , the dependent variable. Now for finding the fitted regression equation, we follow the procedure described in Steps 1 to 6.

We first draw a scatter diagram by plotting the (X, Y) points as shown in Fig. 9.1.

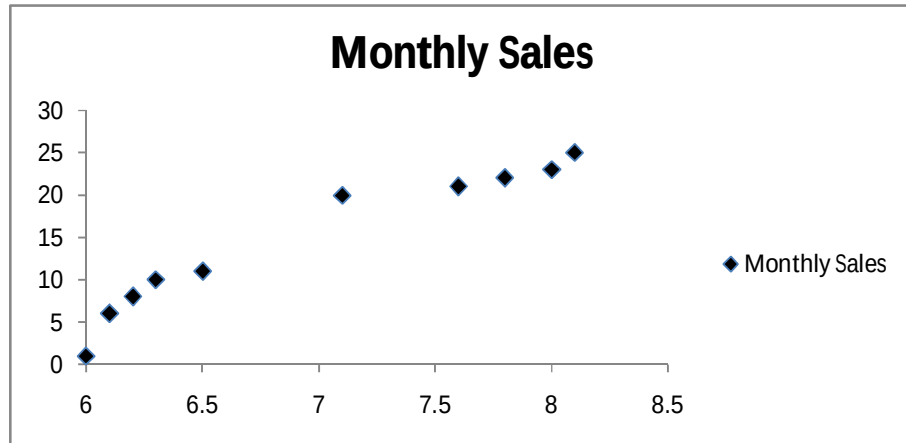


Fig. 9.1: Scatter diagram for the (X, Y) points given in the data of monthly sales.

We construct a table for finding the values of ΣX , ΣY , ΣXY and ΣX^2 as follows:

X	Y	XY	X^2
6.0	01	6.0	36.00
6.1	06	36.6	37.21
6.2	08	49.6	38.44
6.3	10	63.0	39.69
6.5	11	71.5	42.25
7.1	20	142.0	50.41
7.6	21	159.6	57.76
7.8	22	171.6	60.84
8.0	23	184.0	64.00
8.1	25	202.5	65.61
$\Sigma X = 69.7$	$\Sigma Y = 147$	$\Sigma XY = 1086.4$	$\Sigma X^2 = 492.21$

From the table,

$$\Sigma X = 69.7 \quad \Sigma Y = 147 \quad \Sigma XY = 1086.4 \quad \text{and} \quad \Sigma X^2 = 492.21$$

We obtain the intercept a and slope b from equations (19) and (20) by substituting the values of ΣX , ΣY , ΣXY and ΣX^2 in them. Thus,

$$\hat{b} = \frac{10 \cdot 1086.4 - (147)(69.7)}{10 \cdot 492.21 - (69.7)(69.7)}$$

$$= \frac{1086.40 - 10245.9}{4922.10 - 4858.09} = \frac{618.10}{64.01} = 9.656$$

and

sg

$$= \frac{1}{10} [147 - 9.656 \times 69.7] = \frac{1}{10} [147 - 673.02] = -52.6$$

The best fitted regression equation for the given data using the method of least squares is given as

$$\hat{Y} = -52.6 + 9.656 X$$

We may also show the fitted regression line for the given data as in Fig. 9.2.

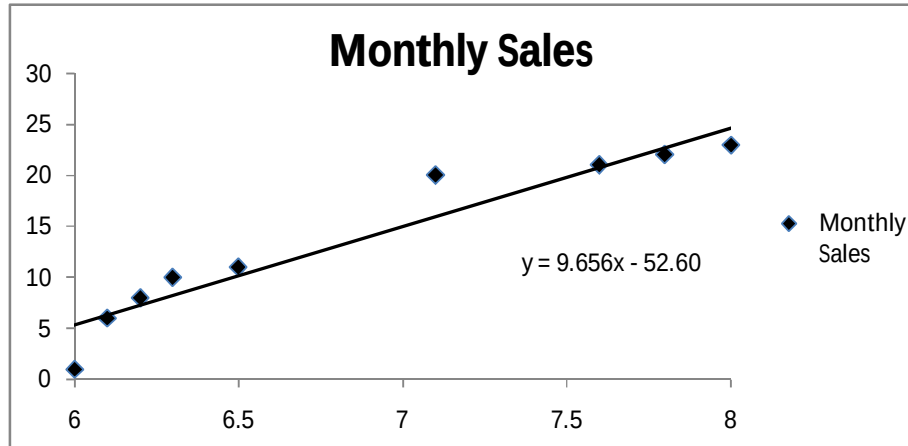


Fig. 9.2: Fitted regression equation for Example 1.

You may now like to solve the following problem to check your understanding:

- E5)** A statistician collected the sales data of a restaurant near a metro station for 12 months. The data comprising the number of customers (in hundreds) and monthly sales (in thousand Rupees) is given as follows:

Month	No. of Customers (in hundreds)	Monthly Sales (in thousand Rs.)
1	04	1.8
2	06	3.5
3	06	5.8
4	08	7.8
5	10	8.7
6	14	9.8
7	18	10.7
8	20	11.5
9	22	12.9
10	26	13.6
11	28	14.2
12	30	15.0

Find the fitted simple linear regression equation for the given data.

9.5 RESIDUAL ANALYSIS

In the previous section, we have discussed how to obtain the estimated values of the parameters a and b for a simple linear regression model given by equation (1):

$$Y = a + bX$$

We have also explained how to get the following fitted model after minimising the error term e :

$$\hat{Y} = \hat{a} + \hat{b}X$$

We are now interested in knowing: How well does this equation fit the data? Is the model likely to be useful as a predictor? Can the fitted model be used to predict the value of the dependent variable?

All these issues must be investigated before adopting the model for use. A simple but efficient method of detecting model deficiencies is to examine the residuals. Residuals are the differences between the observed values and the corresponding fitted values. Mathematically, if Y_i is the i^{th} observation of the dependent variable Y and $\hat{Y}_i$, the corresponding fitted value, then

$$r_i = Y_i - \hat{Y}_i, \quad i = 1, 2, \dots, n \quad \dots (21)$$

From equation (21), note that the residual r_i is the deviation between the value given in the data and the corresponding fitted value. Upon substituting $\hat{Y}_i = \hat{a} + \hat{b}X_i$ in equation (21), we get

$$r_i = Y_i - (\hat{a} + \hat{b}X_i), \quad i = 1, 2, \dots, n \quad \dots (22)$$

where $\hat{a}$ and $\hat{b}$ are the estimated values of the parameters a and b . As we have discussed earlier, residuals play a leading role in detecting the adequacy of a model. In other words, residuals may be defined as realised or observed values of the model errors.

9.5.1 Scaling of Residuals

Analysis and plotting of residuals is an effective way of discovering the model inadequacies, investigating how well the model fits the data and checking the assumptions listed below:

- i) the error term e_i has zero mean;
- ii) the error term e_i has constant variance σ^2 ;
- iii) the errors are uncorrelated;
- iv) the errors are normally distributed.

In this section, we discuss the standardised method for scaling residuals.

Some observations in the data are separated or very far away from the rest of the data. These are called **outliers** or **extreme values**. The scaled residuals are helpful in finding the outliers or the extreme values.

Residuals have zero mean and their approximate average variance is estimated by

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (r_i - \bar{r})^2}{n - k} = \frac{\sum_{i=1}^n r_i^2}{n - k} = SS_{\text{Res}} \quad (\because \bar{r} \text{ is zero}) \quad \dots (23)$$

where $n - k$ is the degree of freedom. Alternatively $\sum r_i^2$ can be written as

$$\begin{aligned}\hat{\sigma}_{r_i}^2 &= \hat{\sigma}^2 (Y_i - \hat{Y}_i)^2 = \hat{\sigma}^2 \left(\hat{\epsilon} Y_i - \bar{Y} - \hat{b}(X_i - \bar{X}) \right)^2 \\ &= \hat{\sigma}^2 (Y_i - \bar{Y})^2 - 2\hat{b}\hat{\sigma}^2 (Y_i - \bar{Y})(X_i - \bar{X}) + \hat{b}^2 \hat{\sigma}^2 (X_i - \bar{X})^2\end{aligned}$$

Thus,

$$\begin{aligned}SS_{Res} &= \sum_{i=1}^n \hat{\sigma}_{r_i}^2 = \sum_{i=1}^n \left(\hat{\sigma}^2 (Y_i - \bar{Y})^2 - 2\hat{b}\hat{\sigma}^2 (Y_i - \bar{Y})(X_i - \bar{X}) + \hat{b}^2 \hat{\sigma}^2 (X_i - \bar{X})^2 \right) / (n - k) \\ &= SS_Y - 2SS_{XY} / SS_X + SS_X^2 / SS_X \\ &= SS_Y - SS_{XY}^2 / SS_X\end{aligned}$$

Residuals are not independent because estimates of k parameters impose k constraints on them.

A logical method of scaling the residuals is to divide the i^{th} error term by its standard deviation that will provide the i^{th} standardised residual d_i , i.e.,

$$d_i = \frac{r_i}{\sqrt{SS_{Res}}}, \quad \forall i=1, 2, 3, \dots, n \quad \dots (24)$$

This method is known as the **standardised method of scaling the residuals** and d_i is called the i^{th} **standardised residual**. The standardised residuals have zero mean and their variance is approximately 1, i.e.,

$$E(d_i) = \frac{E(r_i)}{\sqrt{SS_{Res}}} = 0$$

$$\text{and} \quad \text{Var}(d_i) = \left(\frac{1}{\sqrt{SS_{Res}}} \right)^2 \text{Var}(r_i) \approx 1 \quad \dots (25)$$

$\forall$ is the symbol used to denote the phrase “for every”.

The values of the standardised residuals indicate the positions of the observed data. A large value of a standardised residual indicates its distance from the fitted value of the model. Consequently a very large value of a standardised residual, i.e., $d_i \geq 3$ indicates an extreme observation or an outlying observation.

9.5.2 Residual Plots

A very effective way to investigate the adequacy of the fit of a regression model and to check the underlying assumptions is the graphical analysis of residuals. The basic residual plots are generated by plotting the residual values against the predicted Y values. These should be examined in all regression modelling problems.

A satisfactory residual plot should be more or less a horizontal band of points shown in Fig. 9.3.

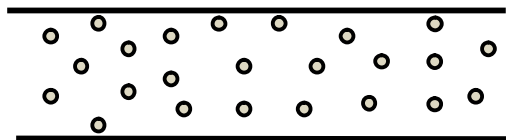


Fig. 9.3

A Heteroscedastic data (Non-constant variance) will have a residual plot as shown in Fig. 9.4.

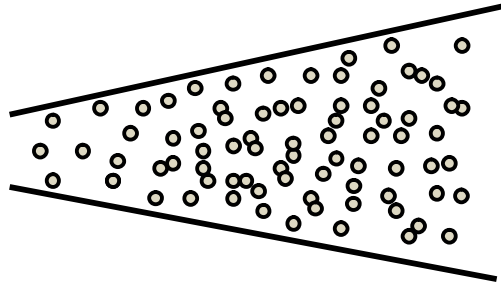


Fig. 9.4

The trend shown in Fig. 9.5 may be exhibited for data for which there exists an error in the regression calculation or some additional regression is needed in the models.

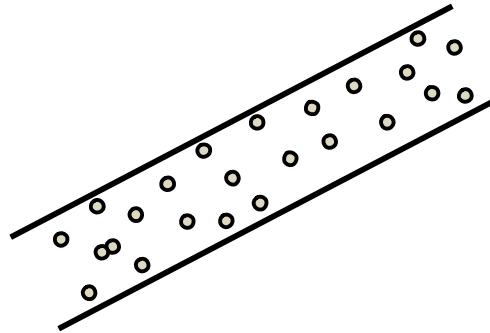


Fig. 9.5

If the relationship between X and Y is nonlinear, the pattern of residuals shown in Fig. 9.6 will be observed, i.e., a curvilinear relationship is suggested.

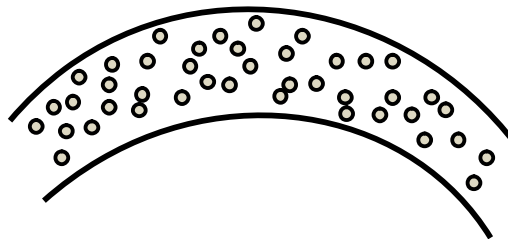


Fig. 9.6

9.5.3 Normal Probability Plot

A small departure from the normality assumptions does not affect the model extensively. Since the confidence intervals depend on the normality assumptions, the gross normality is potentially more serious. Construction of the normal probability plot of the residuals is a very simple method to check the normality assumptions. The method of constructing a normal probability plot is as follows:

1. We arrange the standardised residuals in increasing order.
2. We rank the residuals, which are arranged in increasing order.
Let $r_1, r_2, r_3, \dots, r_n$ be the calculated standardised residuals ranked in increasing order.
3. After ranking the residuals, we calculate the cumulative probability (p_i) as

$$p_i = \frac{\left(i - \frac{1}{2}\right)}{n} \quad i = 1, 2, \dots, n. \quad \dots (26)$$

for the corresponding ranked residual values.

4. Then we plot the cumulative probability by multiplying p_i by 100. It is known as percentile cumulative probability (P_i), i.e.,

$$P_i = \frac{\left(i - \frac{1}{2}\right)}{n} \times 100; \quad i = 1, 2, \dots, n. \quad \dots (27)$$

We plot P_i versus the standardised residuals to obtain the normal probability plot. The resulting points would lie approximately on a straight line.

In the normal probability plot, we usually determine the straight line visually with emphasis on the central values of all the points rather than the extreme points at both ends. You should remember the following points about the normal probability plot for the fitted model:

1. A normal probability plot is called **ideal** if all points lie approximately along a straight line as shown in Fig. 9.7.

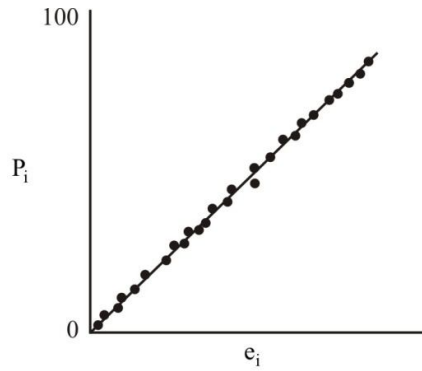


Fig. 9.7

2. A sharp upward and downward curve at both ends indicates that the end points of this distribution are too light for it to be considered normal. These do not constitute the normal probability plot for light-tailed distribution as shown in Fig. 9.8a. A heavy-tailed distribution shows flattening at the extremes, which is a pattern of samples from a distribution with heavier tails than normal as shown in Fig. 9.8b.

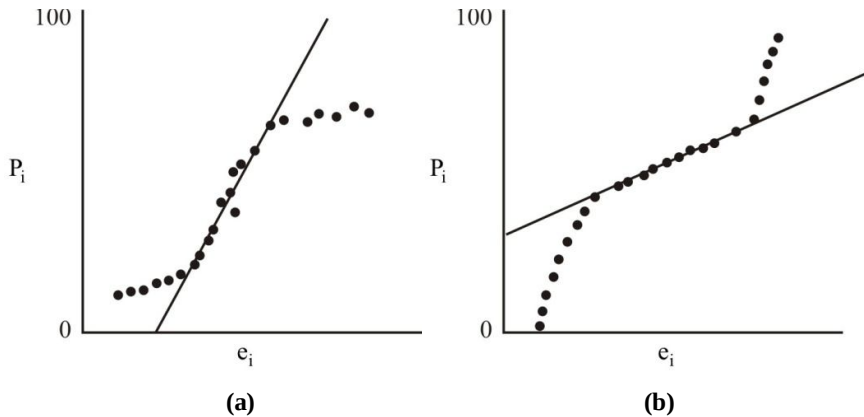


Fig. 9.8

3. The positive and negative skewed normal probability plot shows the pattern of an upward trend (Fig. 9.9a) and downward trend (Fig. 9.9b) of the distribution, respectively, at the ends of the distribution.

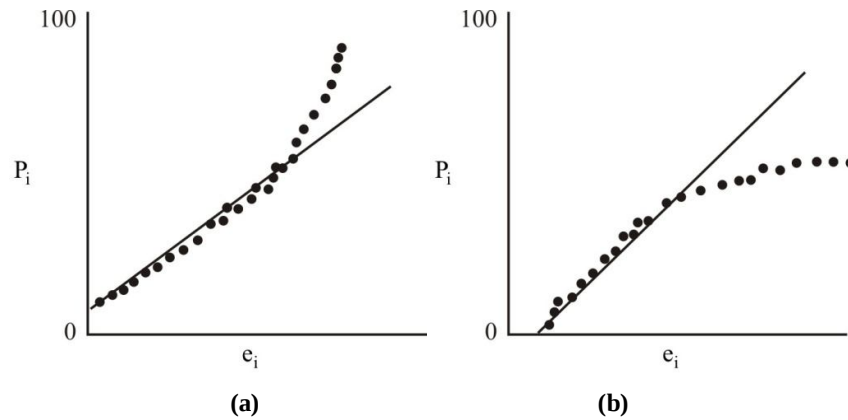


Fig. 9.9

Let us take up an example to explain the concepts of residual analysis.

Example 2: For the data given in Example 1, calculate the residuals and determine the standardised residuals for the model. Draw the corresponding residual plot and normal probability plot.

Solution: In Example 1, we have fitted the regression equation $Y = a + bX$ for the given set of data. We have obtained the best fitted regression equation as

$$\hat{Y} = -52.6 + 9.656X$$

We now calculate the fitted value of the dependent variable from the best fitted regression line given above. We also calculate the values of residuals and their squares for all values of Y given in the data by substituting the values of X , and arrange them in a table as follows:

Y	$\hat{Y}$	$r_i = Y_i - \hat{Y}$	r_i^2
1	5.34	- 4.34	$(-4.34)^2 = 18.83$
6	6.30	- 0.30	$(-0.30)^2 = 0.09$
8	7.26	0.74	$(+0.74)^2 = 0.55$
10	8.23	1.77	$(+1.77)^2 = 3.13$
11	10.16	+0.84	$(+0.84)^2 = 0.70$
20	15.95	+4.05	$(+4.05)^2 = 16.40$
21	20.78	+0.22	$(+0.22)^2 = 0.48$
22	22.72	- 0.72	$(-0.72)^2 = 0.52$
23	24.65	-1.65	$(-1.65)^2 = 2.72$
25	25.61	- 0.61	$(-0.61)^2 = 0.37$
Total	147.00	0.00	$\sum (r_i - \bar{r})^2 = 43.79$

Let us check how well the regression line fits the given data.

We calculate the standard deviation of residuals from equation (23) to calculate the standardised residuals for the given data as follows:

$$SS_{Res}^{1/2} = \sqrt{\frac{\sum r_i^2}{n-k}} = \sqrt{\frac{43.79}{8}} = \sqrt{5.47} = 2.34$$

Here $n - k = 10 - 2 = 8$.

We now construct a table which contains the values of residuals and the standardised residuals as follows:

S. No.	Residuals r_i	$d_i = r_i / \sqrt{SS_{Res}}$
1	-4.34	$-4.34/2.34 = -1.855$
2	-0.30	$-0.30/2.34 = -0.128$
3	-0.74	$+0.74/2.34 = +0.316$
4	+1.77	$+1.77/2.34 = +0.756$
5	+0.84	$+0.84/2.34 = +1.036$
6	+4.05	$+4.05/2.34 = +1.730$
7	+0.22	$+0.22/2.34 = +0.094$
8	-0.72	$-0.72/2.34 = -0.307$
9	-1.65	$-1.65/2.34 = -0.705$
10	-0.61	$-0.61/2.34 = -0.260$
Total	$\sum r_i = 0.00$	$\sum d_i = 0.000$

Note that the sum of the residuals is zero. We now plot the standardised residuals against the fitted values of Y and obtain a residual plot as shown in Fig. 9.10.

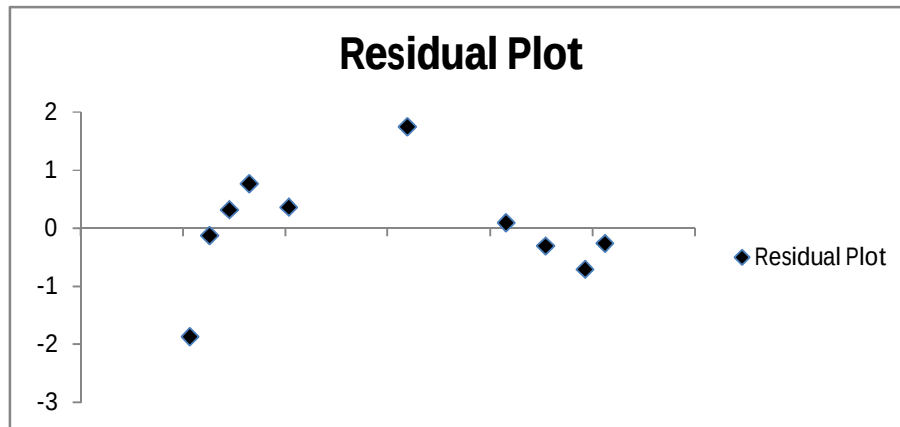


Fig. 9.10: Residual plot for the given data.

We follow the method explained in Sec.9.5.3 to construct the normal probability plot. We arrange the standardised residuals in increasing order and rank the residuals, which are also arranged in increasing order. After ranking the residuals, we calculate the percentile cumulative probability using equation (27) for the corresponding ranked residual values.

Ordered Residuals	Ranks (i)	Cumulative Probability $p_i = (i - 1/2)/n$	Percentile Cumulative Probability $P_i = \{(i - 1/2)/n\} \times 100$
-1.88807	1	0.041667	04.1667
-1.27982	2	0.125	12.5
-0.53461	3	0.20833	20.833
-0.48012	4	0.291667	29.1667
-0.27835	5	0.375	37.5
0.106038	6	0.45833	45.833
0.16053	7	0.541667	54.1667
0.413844	8	0.625	62.5
0.493373	9	0.70833	70.833
0.784978	10	0.791667	79.1667
1.243004	11	0.875	87.5
1.26215	12	0.95833	95.833

Then we plot the percentile cumulative probability against the standardised residuals to obtain the normal probability plot. The resulting points would lie approximately on a straight line as shown in Fig. 9.11.

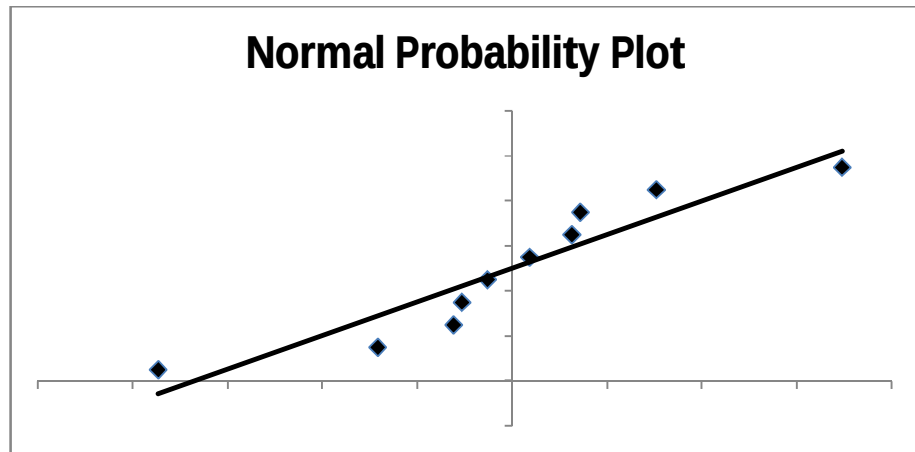


Fig. 9.11: Normal probability plot for the given data.

From Fig. 9.11, note that some points of the distribution deviate slightly from the straight line, but do not lie very far from the central points.

On the basis of the above residual analysis for the given data, the best fitted regression equation is obtained as shown in Fig. 9.12.

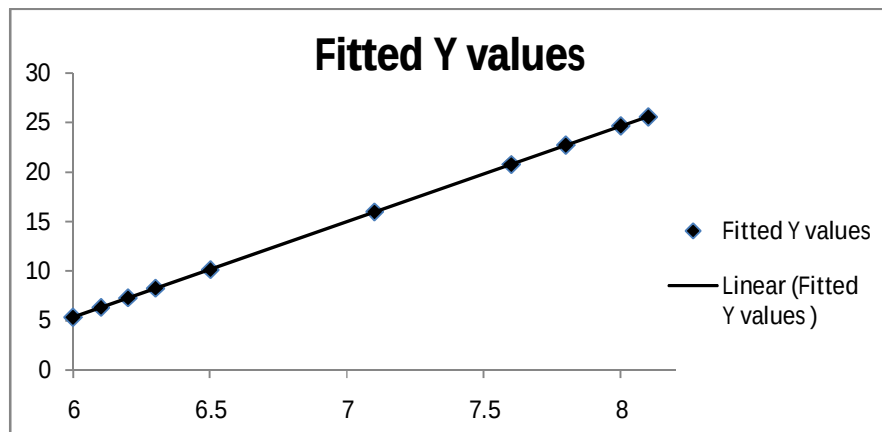


Fig. 9.12: Best fitted regression equation after residual analysis for the given data.

You may now like to solve the following questions to assess your understanding.

-
- E6)** Describe the method of scaling the residuals for the simple linear regression model $Y = a + bX + e$.
- E7)** Calculate the residuals and determine the standardised residuals for the simple linear regression model obtained for the data given in E2 of this unit. Generate the corresponding residual plot and the normal probability plot.
-

We now summarise the concepts that we have discussed in this unit.

9.6 SUMMARY

1. Simple linear regression fits a straight line through the set of n points in such a way that makes the sum of squared residuals of the model (that is, vertical distances between the points of the data set and the fitted line) as small as possible.
2. The simple linear regression model has two unknown parameters a and b , which are known as **intercept** and **regression coefficient**, respectively. Their values are unknown. Therefore, they must be estimated using the sample data. The estimation of the parameters a and b is done by minimising the error term e .
3. An essential part of regression analysis includes a careful examination of the residuals to guarantee that the assumptions in the least squares theory are met. Some observations in the data are separated or very far away from the rest of the data. These are called **outliers** or **extreme values**. The scaled residuals are helpful in finding the outliers or the extreme values.
4. A very effective way to investigate the adequacy of the fit of a regression model and to check the underlying assumptions is the graphical analysis of the residuals. The basic residual plots are generated by plotting the residual values against the predicted Y values. These should be examined in all regression modelling problems.
5. The values of the standardised residuals indicate the positions of the observed data. Large values of standardised residuals indicate their distance from the fitted values of the model. Consequently a very large value of standardised residuals, i.e., $d_i \geq 3$, indicates an extreme observation or an outlying observation.
6. A small departure from the normality assumptions does not affect the model extensively. Since the confidence intervals depend on the normality assumptions, the gross normality is potentially more serious. Construction of the normal probability plot of the residuals is a very simple method to check the normality assumptions.
7. To construct the normal probability plot, we arrange the standardised residuals in increasing order and rank the residuals, which are also arranged in increasing order. After ranking the residuals, we calculate the percentile cumulative probability for the corresponding ranked residual values.

9.7 SOLUTIONS/ANSWERS

- E1)** Refer to Sec. 9.2.
E2) Refer to Sec. 9.2.
E3) Refer to Sec. 9.3.
E4) Refer to Sec. 9.3.
E5) Refer to Sec. 9.4.
E6) We are given $n = 12$. Let us assume an equation of a simple linear regression model as

$$Y = a + bX$$

For finding the fitted regression equation, we first draw a scatter diagram by plotting the (X, Y) points as shown Fig. 9.13.

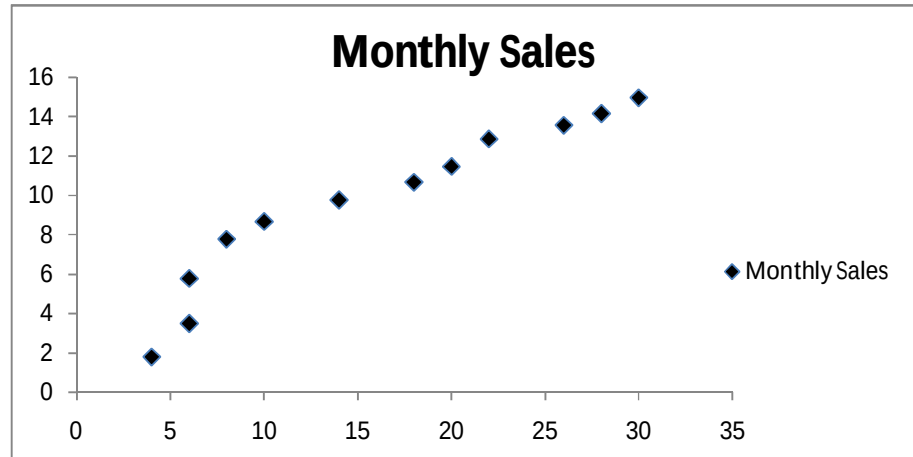


Fig. 9.13: Scatter diagram for the (X, Y) points given in the data.

We construct the following table for calculating the values of $\hat{a}$ and $\hat{b}$:

S. No.	X	Y	XY	X ²
1	4	1.8	7.2	16
2	6	3.5	21.0	36
3	6	5.8	34.8	36
4	8	7.8	62.4	64
5	10	8.7	87.0	100
6	14	9.8	137.2	196
7	18	10.7	192.6	324
8	20	11.5	230.0	400
9	22	12.9	283.8	444
10	26	13.6	353.6	676
11	28	14.2	397.6	784
12	30	15.0	450.0	900
	$\sum X = 192$	$\sum Y = 115.3$	$\sum XY = 2257.2$	$\sum X^2 = 4016$

Substituting the values of $\sum X$, $\sum Y$, $\sum XY$ and $\sum X^2$ in equations (19) and (20), we get

$$\begin{aligned}
 \hat{b} &= \frac{12 \times 2257.2 - (192)(115.3)}{12 \times 4016 - (192)(192)} \\
 &= \frac{27086.4 - 22137.6}{48192 - 36864} = 0.4369
 \end{aligned}$$

$$\begin{aligned}\text{and } \hat{a} &= \frac{1}{12} [115.3 - (0.437)(192)] \\ &= 2.6185\end{aligned}$$

Therefore, the best fitted regression equation (Fig. 9.14) is

$$\hat{Y} = 2.6185 + 0.4369X$$

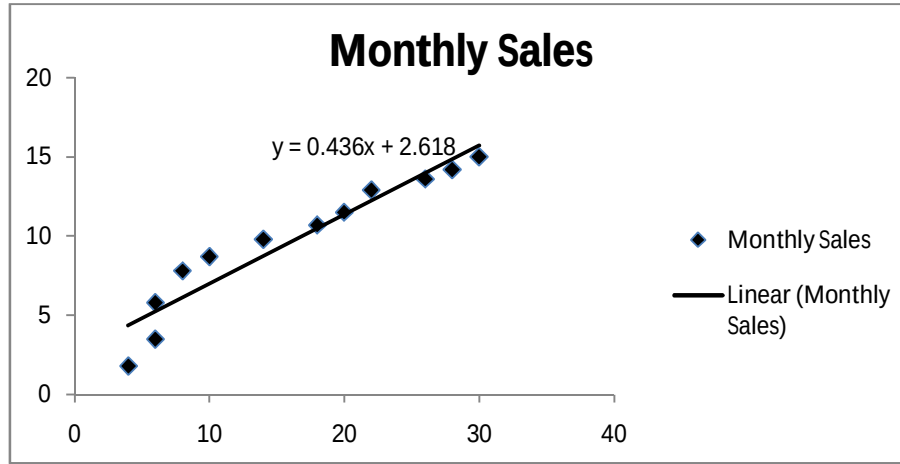


Fig. 9.14: Best fitted regression equation for the given set of (X, Y) values.

E7) Refer to Sec. 9.5.

E8) In Example 2, we have fitted the simple linear regression equation $Y = a + bX + e$ for the given set of data and obtained the best fitted regression equation as

$$\hat{Y} = 2.6185 + 0.4369X$$

From the above equation of regression, we calculate the fitted values of the dependent variable Y, i.e., $\hat{Y}_i$ for all i, and obtain the values of residuals, i.e., r_i for all i, as given in the following table:

S. No.	Y Values	$\hat{Y}$ Values	r_i	r_i^2
1	1.8	4.36	-2.56	$(-2.56)^2 = 6.5536$
2	3.5	5.24	-1.74	$(-1.74)^2 = 3.0276$
3	5.8	5.24	+0.56	$(+0.56)^2 = 0.3136$
4	7.8	6.11	+1.69	$(+1.69)^2 = 2.8561$
5	8.7	6.99	+1.71	$(+1.71)^2 = 2.9241$
6	9.8	8.73	+1.07	$(+1.07)^2 = 1.1449$
7	10.7	10.48	+0.22	$(+0.22)^2 = 0.0484$
8	11.5	11.36	+0.14	$(+0.14)^2 = 0.0196$
9	12.9	12.23	+0.67	$(+0.67)^2 = 0.4489$
10	13.6	13.98	-0.38	$(-0.38)^2 = 0.1444$
11	14.2	14.85	-0.65	$(-0.65)^2 = 0.4225$
12	15.0	15.73	-0.73	$(-0.73)^2 = 0.5329$
Total	$\sum Y_i = 115.30$	$\sum \hat{Y}_i = 115.30$	$\sum r_i = 0.00$	$\sum r_i^2 = 18.4366$

To understand how well the regression line fits the given data, we first calculate the standard deviation of the residuals:

$$\begin{aligned}
 SS_{\text{Res}}^{1/2} &= \sqrt{\frac{\sum r_i^2}{n-k}} \\
 &= \sqrt{\frac{18.4366}{10}} \\
 &= \sqrt{1.84366} = 1.3578
 \end{aligned}$$

where $n - k = 10$

We now construct the following table for obtaining the values of standardised residuals for all values of the residuals:

S. No.	Residuals r_i	Standardised Residuals $d_i = r_i / \sqrt{SS_{\text{Res}}}$
1	-2.56	-1.8854
2	-1.74	-1.28148
3	+0.56	+ 0.41243
4	+1.69	+ 1.24466
5	+1.71	+ 1.25939
6	+1.07	+ 0.78803
7	+0.22	+ 0.16202
8	+0.14	+ 0.10310
9	+0.67	+ 0.49344
10	-0.38	- 0.27986
11	-0.65	-0.47871
12	-0.73	-0.53763
Total	$\sum r_i = 0.00$	$\sum d_i = 0.00$

The standardised residuals are obtained as above and it is observed that their total is zero. We plot these residuals against the fitted values of Y as shown in Fig. 9.15.

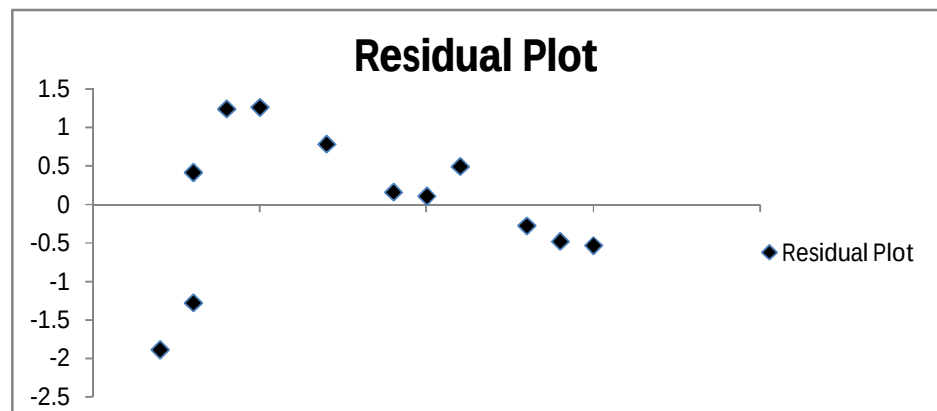


Fig. 9.15: Residual plot for the given data.

Then we follow the method of constructing a normal probability plot as explained in Sec 9.5.3. We arrange the standardised residuals in increasing order and rank the residuals, which are arranged in increasing order. After ranking the residuals, we calculate the cumulative probability using equation (26) and percentile cumulative probability using equation (27) for the corresponding ranked residual values.

Ordered Residual	Ranks (i)	Cumulative Probability $p_i = (i - 1/2)/n$	Percentile Cumulative Probability $P_i = \{(i - 1/2)/n\} \times 100$
-1.86495	1	0.05	05
-0.70882	2	0.15	15
-0.3083	3	0.25	25
-0.26391	4	0.35	35
-0.12972	5	0.45	45
0.092215	6	0.55	55
0.315183	7	0.65	65
0.35957	8	0.75	75
0.760086	9	0.85	85
1.738667	10	0.95	95

Then we plot the percentile cumulative probability versus the standardised residuals to obtain the normal probability plot. The resulting points would lie approximately on a straight line as shown in Fig. 9.16.

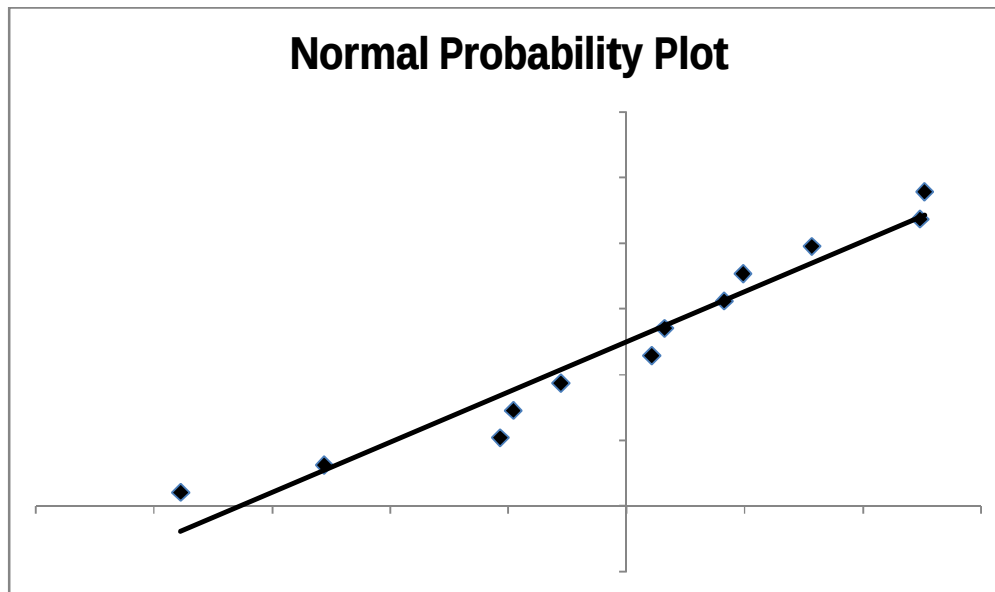


Fig. 9.16: Normal probability plot for the given data.

From the normality probability plot (Fig. 9.16), you may note that some points of the distribution are deviating slightly from the straight line, but do not lie very far from the central points.

On the basis of the above residual analysis for the given data, we obtain the best fitted regression equation shown in Fig. 9.17.

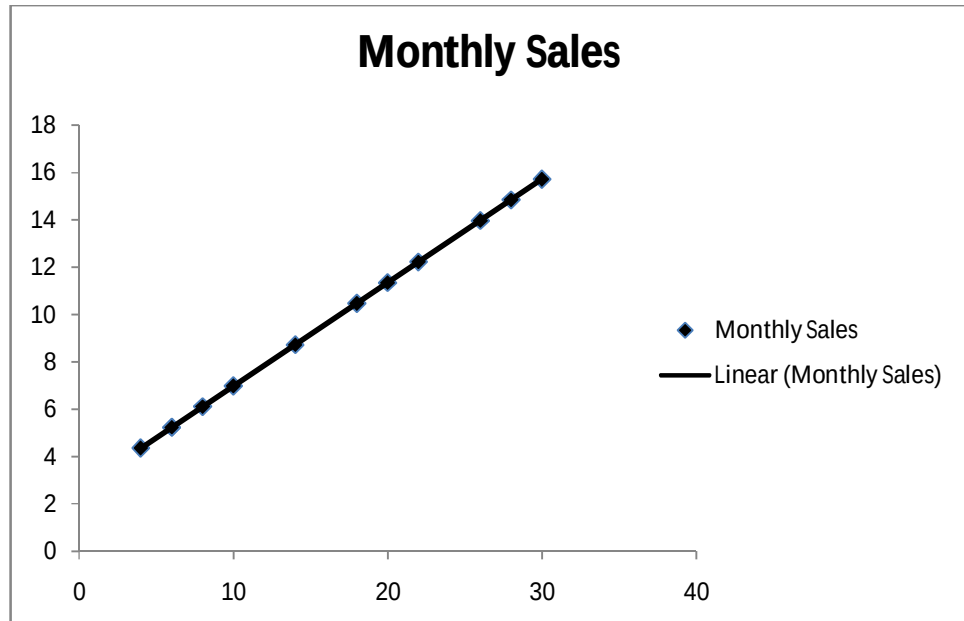


Fig. 9.17: Best fitted regression equation after residual analysis for the given data.

UNIT 10 STATISTICAL INFERENCE IN SIMPLE LINEAR REGRESSION

Structure

- 10.1 Introduction
 - Objectives
- 10.2 Test of Significance of Regression Coefficients
 - Distribution of Regression Coefficients
 - Hypothesis Testing of Regression Coefficients
- 10.3 Coefficient of Determination
- 10.4 Confidence Interval
 - Confidence Interval for b
 - Confidence Interval for Predicted Value at $X=X_0$
- 10.5 Summary
- 10.6 Solutions/Answers

10.1 INTRODUCTION

In Unit 9, you have studied the simple linear regression model for one dependent variable Y and one independent variable X . In linear regression analysis, the parameters a and b are assumed to be unknown and for the given data on Y and X , we obtain the least squares estimates of these parameters. The estimates of parameters a and b are linear functions of observations Y . Since Y is a random variable, the least squares estimates $\hat{a}$ and $\hat{b}$ are also random variables. In Unit 9, we have also described the method of detecting model deficiencies using residual analysis.

In this unit, we discuss the inferential aspect of simple linear regression. We examine whether there is a significant linear relationship between Y and X —we test whether regression coefficient b is zero or different from zero in Sec. 10.2. If it is zero, then there is no significant dependence of Y on X . Hence, the application of regression will not be useful. If b is not zero, it may be of interest to explore how much variability in Y is accounted for by this linear relationship, in other words, how good the fit is.

The coefficient of determination, R^2 , gives a measure of goodness of fit of the model and is discussed in Sec. 10.3. It is simply the square of the correlation coefficient between X and Y and lies between 0 and 1. When the linear relationship between Y and X is good, most observations lie very close to the fitted line $\hat{Y} = \hat{a} + \hat{b}X$. If the relationship between Y and X is poor, the value of R^2 would be close to zero, and most values of Y lie far away from the fitted line. We also obtain the confidence interval of b and the predicted value of $E(Y)$ corresponding to a given value of X in Sec. 10.4.

In the next unit, we shall discuss the multiple linear regression model and the estimation of model parameters. We shall also discuss a method for calculating the coefficient of determination (R^2) and Adjusted R^2 .

Objectives

After studying this unit, you should be able to:

- test the significance of regression coefficients;

- describe and determine the coefficient of determination, R^2 ; and
- form the confidence interval of regression coefficient b and calculate the predicted value of $E(Y)$ corresponding to a given value of X .

10.2 TEST OF SIGNIFICANCE OF REGRESSION COEFFICIENTS

For testing the significance of regression coefficients, we need to follow the assumptions, which have been discussed in Sec. 9.2.1 in Unit 9. For convenience, we state them again:

- We always assume that the independent variable or predictor variable X is not a random variable and it takes fixed values. In some cases, these may be random variables. But here we make inferences by putting the condition on them that these are fixed (not random) variables.
- We assume that for a given (fixed) X , Y has the following mean and variance:

$$\text{Mean} = E(Y|X) = a + bX$$

$$\text{and} \quad \text{Variance} = V(Y|X) = \sigma^2$$

Further, the error e is distributed normally with mean zero and variance σ^2 , i.e.,

$$E(e) = 0; \quad V(e) = \sigma^2$$

- We further assume that Y_i , $i = 1, 2, \dots, n$, are independent. Thus, they are normally distributed random variables with mean $(a+bX)$ and variance σ^2 .

10.2.1 Distribution of Regression Coefficients

In Sec. 9.3.1 of Unit 9, we have discussed the estimation of regression coefficients a and b using the method of least squares. We have denoted these least squares estimates by $\hat{a}$ and $\hat{b}$. These are given by

$$\hat{a} = \bar{Y} - \hat{b}\bar{X} \quad \text{and} \quad \hat{b} = \frac{\sum (Y_i - \bar{Y})(X_i - \bar{X})}{\sum (X_i - \bar{X})^2} \quad \dots (1a)$$

The parameters $\hat{a}$ and $\hat{b}$ are unbiased estimates of regression parameters a and b , respectively, and their variances are given as

$$V(\hat{a}) = s^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_X} \right) \quad \text{and} \quad V(\hat{b}) = \frac{s^2}{SS_X} \quad \dots (1b)$$

$$\text{where } SS_X = \sum (X_i - \bar{X})^2$$

Note that $\hat{a}$ and $\hat{b}$ are linear functions of Y_i 's. Here we use a simple property of normally distributed variables that a new variable, which is a linear combination of normally distributed random variables, also follows the normal distribution with mean and variance of the linear combination. Thus, $\hat{a}$ and $\hat{b}$ are normally distributed with mean and variance as given above. Therefore, we write

$$\hat{a} \sim N\left(\bar{Y} - \hat{b}\bar{X}, s^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_X} \right)\right) \quad \hat{b} \sim N\left(\hat{b}, \frac{s^2}{SS_X}\right) \quad \dots (2)$$

We cannot determine variances of $\hat{a}$ and $\hat{b}$ unless we know the value of σ^2 . Generally, it is not known and has to be estimated from the data. Its estimation has already been discussed in Unit 9 and an unbiased estimate of σ^2 is given by

$$\hat{\sigma}^2 = \sum (r_i - \bar{r})^2 / (n - k) = SS_{\text{Res}}$$

In the case of simple linear regression model, we have two parameters a and b . Thus, $k = 2$, and an unbiased estimate of σ^2 is given by

$$SS_{\text{Res}} = \sum (r_i - \bar{r})^2 / (n - 2) = \sum r_i^2 / (n - 2) \quad \dots(3)$$

Let us understand these concepts with the help of an example.

Example 1: Using the data of Example 1 of Unit 9, calculate SS_{Res} as an estimate of σ^2 . Use this estimate for calculating the variances of $\hat{a}$ and $\hat{b}$.

Solution: We have the following values (from the solution of Example 1, Sec. 9.4 of Unit 9):

$$\bar{X} = 6.97, SS_X = \sum (X_i - \bar{X})^2 = 6.40$$

$$SS_{XY} = \sum (X_i - \bar{X})(Y_i - \bar{Y}) = 61.81$$

$$SS_Y = \sum (Y_i - \bar{Y})^2 = \sum Y_i^2 - n \bar{Y}^2 = 2801 - 10 \times (14)^2 = 640.1$$

$$\begin{aligned} \sum r_i^2 &= \left(SS_Y - \frac{(SS_{XY})^2}{SS_X} \right) = \left(640.1 - \frac{(61.81)^2}{6.40} \right) \\ &= 640.10 - 596.95 = 43.51 \end{aligned}$$

Putting the above values in equation (3), we get

$$\hat{\sigma}^2 = \sum \frac{r_i^2}{8} = 5.42$$

Using equation (1a) for $\hat{a}$ and $\hat{b}$ and substituting $\hat{\sigma}^2$ for σ^2 in equation (1b), we get

$$\begin{aligned} V(\hat{a}) &= \hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_X} \right) = 5.42 \left\{ \frac{1}{10} + \frac{(6.97)^2}{6.40} \right\} \\ &= 5.42 (0.10 + 7.59) = 5.42 \times 7.69 = 41.68 \end{aligned}$$

$$V(\hat{b}) = \frac{\hat{\sigma}^2}{SS_X} = \frac{5.42}{6.40} = 0.85$$

10.2.2 Hypothesis Testing of Regression Coefficients

The regression coefficients, which we have estimated in Unit 9, are subject to random variation as they depend on Y . It means that while working with different samples, we obtain different estimates. Hence, the sample estimates

of a and b may differ from the population values. As population values of a and b are not known, one may hypothesise that the sample has come from a population whose true values are $a = a_0$ and $b = b_0$. On the basis of a sample (Y_i, X_i) , $i = 1, 2, \dots, n$, one may like to test whether these hypotheses are acceptable or not.

This is particularly important when one is interested in knowing whether Y and X are linearly related or not. If Y and X are not linearly related, $b = 0$ and the population relationship is given as:

$$Y = a + e$$

rather than $Y = a + bX + e$

Hence, we may be interested in testing a null hypothesis

$$H_0: b = 0$$

against the alternative hypothesis

$$H_1: b \neq 0$$

Or, in general, we may like to test two hypotheses

i) $H_0: a = a_0$ against $H_1: a \neq a_0$

ii) $H_0: b = b_0$ against $H_1: b \neq b_0$

These are two hypotheses and we shall derive the tests for these hypotheses separately.

i) Test of $H_0: a = a_0$ against $H_1: a \neq a_0$

We have seen that $\hat{a}$ is an unbiased estimate of a . From equation (2), it is normally distributed as follows:

$$\hat{a} \sim N \left\{ a, \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_X} \right) \right\}$$

Thus, if we consider deviation of $\hat{a}$ from a and divide it by its standard error, we obtain a variable which follows standard normal $N(0, 1)$.

If the null hypothesis $H_0: a = a_0$ is correct, then we write the z-statistic as

$$Z = \frac{(\hat{a} - a_0)}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_X} \right)}} \sim N(0, 1)$$

As we do not know σ^2 , we cannot use this as a test statistic. Therefore, we replace σ^2 by $\hat{\sigma}^2$ and get a t-statistic as

$$t = \frac{(\hat{a} - a_0)}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_X} \right)}} \quad \dots (4)$$

which is distributed as a Student's 't'-statistic with $(n - 2)$ d.f.

This is an appropriate test statistic to test the null hypothesis:

$$H_0: a = a_0 \quad \text{against} \quad H_1: a \neq a_0$$

If we want to test at α level of significance, this will be a two-sided test and one can compare the calculated value of $|t|$ with tabulated value at α level of significance, i.e., $t_{\alpha/2}$. We accept H_0 , if $|t| < t_{\alpha/2}$. Otherwise, we reject the null hypothesis. One-sided test can also be constructed in a similar way.

ii) Test of $H_0: b = b_0$ against $H_1: b \neq b_0$

From equation (2), $\hat{b}$ is distributed as

$$\hat{b} \sim N\left\{b, \frac{\sigma^2}{SS_x}\right\}$$

If the null hypothesis $H_0: b = b_0$ is correct, then we write the z-statistic as

$$Z = \frac{(\hat{b} - b)}{\sqrt{\sigma^2 / SS_x}} \sim N(0,1)$$

Since σ^2 is not known, we replace σ^2 by $\hat{\sigma}^2$ and get a t-statistic as

$$t = \frac{(\hat{b} - b)}{\sqrt{\hat{\sigma}^2 / SS_x}} \quad \dots (5)$$

where the statistic 't' follows Student's t-distribution with $(n - 2)$ d.f., under H_0 . This is an appropriate test for the null hypothesis. The value $|t|$ is calculated from equation (5) and is compared with the tabulated value at α level of significance, i.e., $t_{\alpha/2}$ (for two-sided hypothesis) and if $|t| < t_{\alpha/2}$, we accept the hypothesis. Otherwise, we reject H_0 . One-sided null hypothesis can also be tested in a similar way. Let us explain this further taking an example.

Example 2: Using the data of Example 1 given in Sec. 9.4 of Unit 9, test the following hypotheses, taking $\alpha = 0.05$:

- i) $H_0: b = 8$ against $H_1: b \neq 8$
- ii) $H_0: b = 8$ against $H_1: b > 8$
- iii) $H_0: b = 8$ against $H_1: b < 8$

Solution: From the solution of Example 1, we have

$$\bar{X} = 6.97, \bar{Y} = 14.7, V(\hat{a}) = 41.68, V(\hat{b}) = 0.85 \text{ and } \hat{\sigma}^2 = 5.48;$$

From equation (12) of Unit 9, we get

$$\hat{b} = SS_{xy} / SS_x = 61.81 / 6.40 = 9.66$$

$$\hat{a} = \bar{Y} - \hat{b}\bar{X} = 14.7 - 9.66 \times 6.97 = -52.63$$

i) $H_0: b = 8$ against $H_1: b \neq 8$

Here $b_0 = 8$ and from equation (5), we have

You have learnt about testing of hypotheses in the course MST-004 entitled Statistical Inference.

$$t = \frac{(\hat{b} - b)}{\sqrt{\hat{\sigma}^2 / SS_x}} = \frac{(9.66 - 8)}{\sqrt{5.48 / 6.40}}$$

$$= \frac{1.66}{0.925} = 1.79 \quad \dots (i)$$

The Statistics tables are provided in the last pages in most books on Statistics. However, these are also given in the Appendix in Block 3 of the course MST-004.

From the Statistics table, the value $t_{\alpha/2}$ at 8 d.f. is 2.31. Since the calculated value $|t| = 1.79 < 2.31$, we accept the null hypothesis, i.e., there is no evidence against the null hypothesis H_0 .

ii) $H_0: b = 8$ against $H_1: b > 8$

We have calculated in (i) that $t = 1.79$. It is a right tail test and we reject the hypothesis H_0 if $t > t_\alpha$ at 8 d.f.

In this case, $t_{0.05}$ at 8 d.f. = 1.86 (from Statistics table). Since $t < t_\alpha$, i.e., $1.79 < 1.86$, we accept the null hypothesis and conclude that there is no evidence against the null hypothesis.

iii) $H_0: b = 8$; against $H_1: b < 8$

It is a left tail test and we reject the hypothesis when $t \leq t_{(1-\alpha)}$ or $t \leq -t_\alpha$.

From the Statistics table, $t_{0.95} = -t_{0.05} = -1.86$ at 8 d.f. and $t = 1.79$. Since $t > -t_{0.05}$, we accept H_0 and conclude that there is no evidence against the null hypothesis.

From all the three tests we conclude that the null hypothesis $H_0: b = 8$ cannot be rejected.

10.3 COEFFICIENT OF DETERMINATION

In the previous section, we have tested for the difference of coefficient b from b_0 . If b is zero, it means that Y is not influenced by X at least in a linear fashion. If b is not zero, it may be of interest to explore how much variability in Y is accounted for by this linear relationship. In other words, how good the fit is. When the fit is good, most points must lie on the fitted line

$$\hat{Y} = \hat{a} + \hat{b} X$$

or very close to it. This means that the deviation of the fitted values from Y must be small for all the observations, i.e., the absolute value of the **residual**

$$r_i = Y_i - \hat{Y}_i$$

must be small for all observations. If this is so, then

$$\sum_{i=1}^n r_i^2$$

should be small. Since the values of r_i are affected by the unit in which observations are measured, we should have a measure of goodness of fit, which is not affected by the measurement unit. This is what we shall determine now.

The sum of squares of residuals denoted by SS_{Res} is obtained as follows:

$$\begin{aligned} SS_{Res} &= \sum_{i=1}^n r_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n [Y_i - \bar{Y} - \hat{b}(X_i - \bar{X})]^2 \\ &= \sum_{i=1}^n (Y_i - \bar{Y})^2 - 2\hat{b} \sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X}) + \hat{b}^2 \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= SS_Y - 2(SS_{XY})^2 / SS_X + (SS_{XY})^2 / SS_X \end{aligned}$$

or $SS_{Res} = \left(SS_Y - (SS_{XY})^2 / SS_X \right) \quad \dots (6)$

This is the sum of squares which involves variability due to random errors. It does not arise due to the regressor variable. However, if we subtract it from the total variability of Y around its mean, i.e., SS_Y , the sum of squares will have the contribution of the regressor variable in the variability, i.e.,

$$SS_Y - (SS_Y - \frac{(SS_{XY})^2}{SS_X}) = \frac{(SS_{XY})^2}{SS_X}$$

This is always positive and is known as **variability accounted for by X**. Now, this is not unit free. To make it unit free, we divide it by total variability in Y around mean SS_Y . The ratio of variability in Y accounted for by X to the total variability in Y around the mean is called the **coefficient of determination** and is denoted by R^2

$$R^2 = \frac{(SS_{XY})^2}{SS_X \cdot SS_Y} \quad \dots (7)$$

You may recognise that it is the square of the correlation coefficient between X and Y, i.e., r^2 . Since r lies between -1 and 1, R^2 lies between 0 and 1. It is closer to 1 when the linear relationship between Y and X is very good. In this case most observations lie very close to the fitted line of $Y = \hat{a} + \hat{b}X$. When R^2 is closer to zero, the relationship between Y and X is very poor, and most observations lie far away from the fitted line. Hence, the value of R^2 gives a good measure of the fitness of the line. Let us take up an example to illustrate this concept.

Example 3: Using the data of Example 1, calculate the coefficient of determination, R^2 , and comment on the goodness of fit of the regression line. Also calculate the correlation coefficient r.

Solution: For the data of Example 1, we have

$$SS_X = 6.40, \quad SS_{XY} = 61.81, \quad SS_Y = 640.1$$

On putting these values in equation (7), we get

$$R^2 = \frac{(SS_{XY})^2}{SS_X \cdot SS_Y} = \frac{(61.81)^2}{6.40 \cdot 640.1} = 0.933$$

Since R^2 is close to 1 and 93.3% variability in variable Y is due to X, the regression line is a good fit.

Now $r = \sqrt{R^2}$ and we have

$$r = \sqrt{0.933} = 0.966$$

Alternatively,

$$r = \frac{SS_{XY}}{\sqrt{SS_X \times SS_Y}} = \frac{61.81}{\sqrt{6.40 \times 640.1}} = 0.966$$

Adjusted R^2

The coefficient of determination, R^2 , has a weakness. In equation (7), the denominator is fixed and the numerator increases if additional independent variables are included in the analysis. This is due to the contribution of the regressor variables to the variability. This results in a higher R^2 , even when the new variable causes the equation to become less efficient (worse).

The **Adjusted R^2** value is an effort to correct this drawback by adjusting both the numerator and the denominator by their respective degrees of freedom. It is given by

$$\tilde{R}^2 = 1 - (1 - R^2) \left(\frac{n-1}{n-k} \right) \quad \dots (8)$$

where R^2 is the coefficient of determination, n , the number of observations and k , the number of independent variables.

Note that unlike R^2 , the Adjusted R^2 contains the d.f. term. So even if the contribution of the additional variable is less, the value of $\tilde{R}^2$ may reduce. This implies that the equation with the smallest standard error of the estimate may have the highest Adjusted R^2 .

Example 4: Determine Adjusted R^2 for Example 3.

Solution: From the solution of Example 3, we have

$$R^2 = 0.933, n = 10 \text{ and } k = 2$$

$$\text{Therefore, } \tilde{R}^2 = 1 - (1 - (0.933)^2) \left(\frac{9}{8} \right)$$

$$\tilde{R}^2 = 1 - (0.13)(1.125) = 0.925$$

You may now like to solve the following exercises to check your understanding.

-
- E1)** What is the statistic t for testing a hypothesis? How is it used?
- E2)** Define the Coefficient of Determination.
- E3)** A study was conducted on the effect of temperature (X) on the yield of a chemical process (Y). The following data (in coded form) were collected:

X	-5	-4	-3	-2	-1	0	1	2	3	4	5
Y	-1	5	4	7	10	8	9	13	14	13	18

- Assuming a model $Y = a + bX + e$, what are the least squares estimates of a and b ?
- Calculate the variances of the estimated regression coefficients.

- iii) Test the hypothesis that the temperature (X) has no effect on the yield (Y) of the chemical process, ie., $H_0: b = 0$; $H_1: b \neq 0$ at $\alpha = 0.05$.

10.4 CONFIDENCE INTERVAL

In Sec. 10.2, we have derived the variances of the estimated coefficients. We have also discussed how to test their significance. If the regression line is a good fit, which is indicated by a high value of R^2 , it may be of interest to predict the mean value of Y for some given value of $X = X_0$. It may be of interest to calculate its variance and confidence interval to get an idea of its accuracy.

Suppose a regression line $Y = a + bX + e$ is fitted and the fitted line is given by

$$\hat{Y} = \hat{a} + \hat{b}X$$

Since $E(\hat{a}) = a$, and $E(\hat{b}) = b$, we have

$$E(\hat{Y}) = a + bX = E(Y)$$

Thus, if we want to estimate $E(Y)$ at a given value of X, say X_0 , an unbiased estimate is given by

$$\begin{aligned}\hat{Y} &= \hat{a} + \hat{b}X_0 = \bar{Y} - \hat{b}\bar{X} + \hat{b}X_0 \\ &= \bar{Y} + \hat{b}(X_0 - \bar{X}) \\ V(\hat{Y}) &= V[\bar{Y} + \hat{b}(X_0 - \bar{X})] \\ &= V(\bar{Y}) + V(\hat{b})(X_0 - \bar{X})^2 + 2\text{Cov}(\bar{Y}, \hat{b})(X_0 - \bar{X}) \\ &= \left[\frac{\sigma^2}{n} + \frac{\sigma^2(X_0 - \bar{X})^2}{SS_x} \right] \quad \dots (9)\end{aligned}$$

since $\text{Cov}(\bar{Y}, \hat{b}) = 0$. We note that $V(\hat{Y})$ is minimum when $X_0 = \bar{X}$ and is given by $\frac{\sigma^2}{n}$. It increases rapidly in proportion to $(X_0 - \bar{X})^2$. Equation (9) provides the variance of predicted value of Y, which depends on σ^2 .

10.4.1 Confidence Interval for b

We have discussed the tests of hypotheses about regression coefficients a and b in Sec. 10.2.2. Once we have tested the t-statistic, we can construct the confidence intervals for b. From equation (5), we have

$$t = \frac{(\hat{b} - b)}{\sqrt{\hat{\sigma}^2 / SS_x}} \sim \text{Student's } t - \text{distribution on } (n - 2) \text{ d.f.}$$

Since the variable t follows the Student's t-distribution, we may write

$$P\{|t| \leq t_{\alpha/2}\} = 1 - \alpha$$

$$\text{or } P\{-t_{\alpha/2} \leq t \leq t_{\alpha/2}\} = 1 - \alpha$$

$$\text{or } P\left\{-t_{\alpha/2} \leq \frac{(\hat{b} - b)}{\sqrt{\hat{\sigma}^2/SS_X}} \leq t_{\alpha/2}\right\} = 1 - \alpha$$

$$\text{or } P\left\{\hat{b} - t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_X}} \leq b \leq \hat{b} + t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_X}}\right\} = 1 - \alpha \quad \dots (10)$$

Let us denote the lower confidence limit and upper confidence limit by b_L and b_U . Then, we have

$$b_L = \hat{b} - t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_X}} \quad \text{and} \quad b_U = \hat{b} + t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_X}} \quad \dots (11)$$

Hence, the probability that b lies between b_L and b_U is $(1 - \alpha)$ and the interval (b_L, b_U) is known as $(1 - \alpha)\%$ confidence interval. You must note that the parameter b is not a random variable, but the interval (b_L, b_U) is random. This means that in repeated sampling, the intervals will be different in each case and $100(1 - \alpha)\%$ of interval will contain parameter b and $100\alpha\%$ intervals will not contain parameter b . In one sample case, the interval (b_L, b_U) may contain b or may not contain b . However, it certainly tells us that if the interval is big, the estimate is not very accurate. If it is small, the estimate is better.

10.4.2 Confidence Interval for Predicted Value at $X = X_0$

We have to find the confidence interval for the mean value of Y at $X = X_0$, i.e.,

$$E(Y | X_0) = a + bX_0$$

We have seen that an unbiased estimate of $E(Y | X_0)$ is

$$\hat{Y} = \hat{a} + \hat{b}X_0$$

We have already seen that the variance of estimate of $E(Y | X_0)$ is

$$V(\hat{Y}) = \sigma^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_X} \right)$$

As we do not know σ^2 , we substitute $\hat{\sigma}^2$ for it. Thus, we get

$$V(\hat{Y}) = \hat{\sigma}^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_X} \right)$$

We have discussed in Block-3 of MST-004 that

$$t = \frac{\hat{Y} - E(Y)}{\sqrt{V(\hat{Y})}} \sim \text{Student's } t\text{-distribution on } (n - 2)\text{d.f.}$$

Thus, we have

$$P\{|t| \leq t_{\alpha/2}\} = 1 - \alpha$$

or
$$P\{-t_{\alpha/2} \leq t \leq t_{\alpha/2}\} = 1 - \alpha$$

or
$$P\{\hat{Y} - t_{\alpha/2} \sqrt{V(\hat{Y})} \leq E(Y) \leq \hat{Y} + t_{\alpha/2} \sqrt{V(\hat{Y})}\} = 1 - \alpha$$

Let us denote the limits by $\hat{Y}_L$ and $\hat{Y}_U$, respectively, such that

$$\begin{aligned}\hat{Y}_L &= \hat{Y} - t_{\alpha/2} \sqrt{V(\hat{Y})} = \hat{Y} - t_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_X} \right)} \\ \hat{Y}_U &= \hat{Y} + t_{\alpha/2} \sqrt{V(\hat{Y})} = \hat{Y} + t_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_X} \right)}\end{aligned} \quad \dots (12)$$

Then the confidence interval for

$$E(Y | X = X_0) = a + bX_0$$

is given by $(\hat{Y}_L, \hat{Y}_U)$. Note that the interval is of minimum length at $X_0 = \bar{X}$ and increases rapidly at the rate of $(X_0 - \bar{X})^2$.

An important application of the regression analysis is to predict the new optimum value Y_0 corresponding to a specific value of the regressor variable X , i.e., $X = X_0$. The confidence interval on the mean response at $X = X_0$ given above is inappropriate for this prediction because it is an interval estimate based on the mean of Y . Therefore, the variance of the estimate of Y_0 becomes

$$V(\hat{Y}_0) = V(Y_0 - \hat{Y}_0) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_X} \right)$$

as the variance of the error term is also to be included.

Then the confidence interval for a new predicted value

$$E(Y | X = X_0) = a + bX_0$$

is given by $(\hat{Y}_L, \hat{Y}_U)$ as follows:

$$\begin{aligned}\hat{Y}_L &= \hat{Y}_0 - t_{\alpha/2} \sqrt{V(\hat{Y}_0)} = \hat{Y}_0 - t_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_X} \right)} \\ \hat{Y}_U &= \hat{Y}_0 + t_{\alpha/2} \sqrt{V(\hat{Y}_0)} = \hat{Y}_0 + t_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_X} \right)}\end{aligned}$$

Fig. 10.1 shows the predicted values of $E(Y | X_0)$ and their confidence intervals.

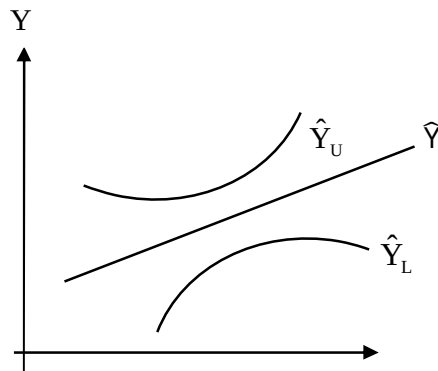


Fig. 10.1: Predicted values of $E(Y | X_0)$ and their confidence intervals.

This shows that the confidence interval is of minimum length when $X_0 = \bar{X}$ and it increases rapidly at a rate in proportion to $(X_0 - \bar{X})^2$. This shows that the prediction of $E(Y)$ is most accurate at $X_0 = \bar{X}$ and the accuracy reduces rapidly as X_0 departs from $\bar{X}$.

Example 4: For the data given in E3, find the 95% confidence intervals for b and $E(Y)$ at $X_0 = -5, -3, -1, 0, 1, 3, 5$.

Solution: From the solution of Example 3, we have

$$\hat{a} = 9.27, \quad \hat{b} = 1.44, \quad \hat{\sigma}^2 = 2.36, \quad \bar{X} = 0, \quad SS_X = 110$$

$$V(\hat{a}) = 0.236, \quad V(\hat{b}) = 0.021, \quad t_{0.025,9} = 2.262$$

Hence from equation (11), the lower confidence limit and upper confidence limit for b are given by

$$\hat{b}_L = \hat{b} - t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_X}} = 1.44 - 2.262 \times \sqrt{0.021} = 1.112$$

$$\hat{b}_U = \hat{b} + t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_X}} = 1.44 + 2.262 \times \sqrt{0.021} = 1.768$$

Therefore, the fitted straight line equation is given by

$$\hat{Y} = 9.27 + 1.44X_0$$

Thus, from equation (12), the upper and lower confidence limits for $E(Y)$ are given by

$$\hat{Y}_L = 9.27 + 1.44X_0 - 2.262 \sqrt{2.36 \left\{ \frac{1}{11} + \frac{(X_0 - \bar{X})^2}{SS_X} \right\}}$$

$$\hat{Y}_U = 9.27 + 1.44X_0 + 2.262 \sqrt{2.36 \left\{ \frac{1}{11} + \frac{(X_0 - \bar{X})^2}{SS_X} \right\}}$$

The following table gives the predicted values of $E(Y | X_0)$ and their confidence intervals:

S. No.	X_0	$\hat{Y}$	$\hat{Y}_L$	$\hat{Y}_U$
1	-5	2.07	0.41	3.73
2	-3	5.07	3.63	6.51
3	-1	7.83	6.74	8.92
4	0	9.27	8.22	10.32
5	1	10.71	9.62	11.80
6	3	13.59	12.15	15.03

7	5	16.47	14.81	18.13
---	---	-------	-------	-------

You may now like to solve the following problems to assess your understanding.

- E4)** A firm wants to know whether there is any linear relationship between the size of its sales force (X) and its yearly sales revenue (Y). The records of 10 years were examined and the data are listed in the following table:

Year	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989
X	15	18	24	22	25	29	30	32	35	38
Y	1.35	1.62	2.33	2.41	2.63	2.93	3.41	3.26	3.63	4.15

where X and Y are the number of salespersons and sales in hundred thousand rupees, respectively. The following results are given:

$$\Sigma X = 265, \Sigma Y = 27.73, SS_X = 485.6, SS_Y = 6.978, SS_{XY} = 57.456$$

- Fit a regression line taking Y as the dependent variable and X as the independent variable.
 - Test whether the sales force has any effect on Y, by testing the significance of regression coefficient b, at 5% significance level.
 - Find the coefficient of determination R^2 and comment on the goodness of fit of the regression line.
 - Estimate the sales value (Y) for the year 1990 for which $X = 40$.
 - Find the 95% confidence interval for b and also for the estimated value of $E(Y|1990)$.
- E5)** Derive $(1-\alpha)\%$ confidence interval for the regression coefficient a.
- E6)** The following table gives the data concerning the monthly usage of electricity Y (in KWA) and the size of the home X (in square feet) for a sample of ten homes:

Y:	11820	11720	12640	14930	15710	17110	18040	18400	19560	19540
X:	12900	13500	14700	16000	17100	18400	19800	22300	24000	29300

- Fit the regression line.
 - Test whether the monthly usage of electricity is influenced by the size of the home.
 - Find out what percentage of variability in Y is accounted for by variation in X, by calculating the coefficient of determination R^2 .
- E7)** An economist suspects that a firm's profitability, as measured by the percentage rate of return on investment, is largely a function of its production efficiency. She has gathered information on the returns and the production efficiency index (expressed in percentage; the higher its value, the more efficient the firm) for 8 local companies. The sample data are shown in the following table:

Firm	1	2	3	4	5	6	7	8
Profitability (Y)	12.3	25.8	23.3	16.9	15.4	6.9	8.9	10.2
Efficiency Index (X)	40.8	53.2	60.1	38.4	33.7	20.8	22.9	20.7

- Calculate a and b and find the regression equation for the data.

- ii) Comment on the goodness of fit of the regression line after calculating the coefficient of determination R^2 .
- iii) Does the regression line seem to provide a reasonable fit?

We now summarise the concepts that we have discussed in this unit.

10.5 SUMMARY

1. The parameters $\hat{a}$ and $\hat{b}$ are unbiased estimates of regression parameters a and b , respectively, and their variances are given as

$$V(\hat{a}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_x} \right), \quad V(\hat{b}) = \frac{s^2}{SS_x}$$

where $\hat{a}$ and $\hat{b}$ are linear functions of Y_i .

2. A variable, which is a linear combination of normally distributed random variables, also follows the normal distribution with mean and variance of the linear combination. Thus, $\hat{a}$ and $\hat{b}$ are normally distributed with mean and variance as given above.
3. In simple linear regression model, we have two parameters a and b . Thus, $k = 2$, and an unbiased estimate of σ^2 is given by

$$SS_{\text{Res}} = \sum (r_i - \bar{r})^2 / (n - 2) = \sum r_i^2 / (n - 2)$$

4. Since population values of a and b are not known, we may hypothesise that the sample has come from a population whose true values are $a = a_0$ and $b = b_0$. On the basis of a sample (Y_i, X_i) , $i = 1, 2, \dots, n$, we test whether these hypotheses are acceptable or not. This is particularly important when we are interested in knowing whether Y and X are linearly related or not.
5. The ratio of variability in Y accounted for by X to the total variability in Y around the mean is called the **coefficient of determination** and is denoted by R^2

$$R^2 = \frac{(SS_{XY})^2}{SS_x \cdot SS_y}$$

The coefficient of determination, R^2 , gives a measure of the goodness of fit of the regression line.

6. Confidence interval for regression coefficient b is

$$b_L = \hat{b} - t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_x}}$$

$$\text{and} \quad b_U = \hat{b} + t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_x}}$$

and of predicted value of $E(Y|X=X_0)$ is

$$\hat{Y}_L = \hat{Y} - t_{\alpha/2} \sqrt{V(\hat{Y})} = \hat{Y} - t_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_X} \right)}$$

$$\hat{Y}_u = \hat{Y} + t_{\alpha/2} \sqrt{V(\hat{Y})} = \hat{Y} + t_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_X} \right)}$$

7. The parameter b is not a random variable, but the interval (b_L, b_U) is random. This means that in repeated sampling, the intervals will be different in each case and $100(1 - \alpha)\%$ of interval will contain parameter b and $100 \alpha\%$ intervals will not contain parameter b . In one sample case, the interval (b_L, b_U) may contain b or may not contain b .

10.6 SOLUTIONS/ANSWERS

E1) Refer to Section 10.2.

E2) Refer to Section 10.3.

E3) From the given data, we have the following:

$$\begin{aligned} \sum X &= 0, \quad \sum X^2 = 110 & \sum Y &= 102, \quad \sum Y^2 = 1194 \\ \sum XY &= 158 \quad SS_X = 110, \quad SS_Y = 248.18, \quad SS_{XY} = 158 \end{aligned}$$

i) Therefore, from equation (1a) for $\hat{a}$ and $\hat{b}$, we get

$$\hat{b} = SS_{XY} / SS_X = 1.44, \text{ and } \hat{a} = \bar{Y} - \hat{b}\bar{X} = 9.27 - 0 = 9.27$$

ii) On putting the values of SS_X, SS_Y and SS_{XY} in equation (1b), we get

$$\begin{aligned} SS_{Res} &= (SS_Y - (SS_{XY})^2 / SS_X) / (n - 2) \\ &= \{248.18 - (158)^2 / 110\} / 9 = 21.23 / 9 = 2.36 \end{aligned}$$

$$\text{Thus } V(\hat{a}) = \hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_X} \right) = 2.36 \left[\frac{1}{11} + 0 \right] = 0.236$$

$$\text{and } V(\hat{b}) = \frac{\hat{\sigma}^2}{SS_X} = 2.36 / 110 = 0.021$$

iii) To test the null hypothesis $H_0: b = 0$; against $H_1: b \neq 0$

$$\text{we have } t = (\hat{b} - 0) / \sqrt{\frac{\hat{\sigma}^2}{SS_X}} = 1.44 / \sqrt{0.021} = 9.93$$

It is a two-tailed test, hence from table $t_{0.025} = 2.262$ at 9 d.f. Therefore, the calculated value is $|t| > 2.262$. Thus, we reject the null hypothesis and conclude that the temperature affects the yield of chemical process.

E4) From the given data, we get the following results:

$$\begin{aligned} \hat{a} \quad X &= 268, \hat{a} \quad Y = 27.73, SS_X = 485.6, SS_Y = 6.978, SS_{XY} = 57.456 \text{ and} \\ n &= 10 \end{aligned}$$

- i) On putting these values in equation (1a) we get

$$\hat{b} = SS_{XY} / SS_X = 0.118,$$

$$SS_{Res} = (SS_Y - (SS_{XY})^2 / SS_X) / (n - 2) = 0.022$$

and $\hat{a} = \bar{Y} - \hat{b}\bar{X} = 2.773 - 0.118 \times 26.8 = -0.389$

Thus, the fitted line is

$$\hat{Y} = -0.389 + 0.118X$$

- ii) We have to test

$$H_0: b = 0 \text{ against } H_1: b \neq 0$$

From equation (5)

$$\begin{aligned} t &= \frac{\hat{b} - 0}{\sqrt{\hat{\sigma}^2 / SS_X}} = \frac{0.118}{\sqrt{0.022 / 485.6}} \\ &= \frac{0.118}{\sqrt{0.00045}} = \frac{0.118}{0.021} = 5.62 \end{aligned}$$

where the tabulated value of $t_{0.025}$ on 8 d.f. is 2.306.

Since $|t| > 2.306$, the null hypothesis is rejected. This means that the sales force X has a significant effect on sales revenue Y.

- iii) Correlation Coefficient r

$$\begin{aligned} r &= \frac{SS_{XY}}{\sqrt{SS_X \times SS_Y}} = \frac{57.456}{\sqrt{485.6 \times 6.978}} \\ &= \frac{57.456}{58.210} = 0.987 \end{aligned}$$

Hence the coefficient of determination, $R^2 = r^2 = 0.974$. Thus 97% of the variability is accounted for by X. Hence, the regression model gives a very good fit.

- iv) The value of estimated sales for the year 1990 is obtained by substituting $X = 40$ in the formula for $\hat{Y}$

$$\hat{Y} = \hat{a} + \hat{b}X = -0.389 + 0.118 \times 40 = 4.33$$

- v) On putting the values in the equations

$$\hat{b}_L = \hat{b} - t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_X}} \quad \text{and} \quad \hat{b}_U = \hat{b} + t_{\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{SS_X}}$$

We have

$$\hat{b}_U = 0.118 + 2.306 \times 0.0067 = 0.133$$

and $\hat{b}_L = 0.118 - 2.306 \times 0.0067 = 0.102$

where tabulated value of $t_{0.025}$ on 8 d.f. is 2.306.

E5) Test statistic for testing hypothesis

$$H_0: a = a_0 \text{ against } H_1: a \neq a_0$$

given in equation (3) as

$$t = \frac{(\hat{a} - a_0)}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{SS_X} \right)}}$$

has Students-t distribution on $(n-2)$ d.f. We replace a_0 by a and construct confidence intervals as before, using

$$P\{|t| \leq t_{\alpha/2}\} = 1 - \alpha$$

giving $(1-\alpha)\%$ confidence limits. Thus,

$$\hat{a}_L = \hat{a} - t_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(\bar{X})^2}{SS_X} \right)}$$

$$\text{and } \hat{a}_U = \hat{a} + t_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(\bar{X})^2}{SS_X} \right)}$$

E6) We have calculated the following values from the given data:

$$\Sigma X = 1,88,000;$$

$$\Sigma Y = 1,59,470;$$

$$SS_X = 2,41,14,000; \quad SS_Y = 84,64,020; \quad SS_{XY} = 1,30,28,900$$

i) On putting the above values in equation (1a), we get

$$\hat{b} = SS_{XY} / SS_X = 0.5403, \text{ and } \hat{a} = \bar{Y} - \hat{b}\bar{X} = 5789.36$$

The fitted line is

$$\hat{Y} = 5789.4 + 0.5403X$$

$$SS_{Res} = (SS_Y - (SS_{XY})^2 / SS_X) / 8 = 178056.02$$

ii) Here we have to test the hypothesis

$$H_0: b = 0 \text{ against } H_1: b \neq 0$$

We consider the test statistic

$$t = \frac{\hat{b} - 0}{\sqrt{\hat{\sigma}^2 / SS_X}} = \frac{0.5403}{0.0859} = 6.289$$

where $|t| > t_{0.025}$ at 8 d.f. = 2.306.

Hence, the size of the house affects the electricity use considerably.

iii) Similarly, from equation (4), we have

$$r = \frac{SS_{XY}}{\sqrt{SS_X \times SS_Y}} = \frac{13028900}{14286405.3} = 0.9119$$

Therefore, the coefficient of determination, $R^2 = r^2 = 0.83156$.

Hence, 83.156% of variability in Y is accounted for by X.

- E7)** We have calculated the following values from the given data:
 $\Sigma Y = 119.7$; $\Sigma X = 290.6$; $\Sigma X^2 = 12102.68$; $\Sigma Y^2 = 2113.45$,
 $\Sigma XY = 5001.14$; $SS_X = 1546.64$; $SS_Y = 322.439$, $SS_{XY} = 653.037$,
 $\hat{\sigma}^2 = 7.785$.

Therefore, we have

$$\text{i) } \hat{b} = SS_{XY}/SS_X = 0.422 \text{ and } \hat{a} = \bar{Y} - \hat{b}\bar{X} = -0.367$$

$$\therefore \hat{Y} = -0.367 + 0.422X$$

- ii) Similarly, from equation (4), we have

$$r = \frac{SS_{XY}}{\sqrt{SS_X SS_Y}} = \frac{653.037}{706.185} = 0.925$$

- iii) Coefficient of determination, $R^2 = r^2 = 0.855$

Thus 85.5% of variability in Y is accounted for by X. Hence, the regression line is a good fit.

UNIT 11 MULTIPLE LINEAR REGRESSION

Structure

- 11.1 Introduction release relies
 - Objectives
- 11.2 Multiple Linear Regression Model
- 11.3 Estimation of Model Parameters
 - Use of Matrix Notation
 - Properties of Least Squares Estimates
- 11.4 Test of Significance in Multiple Regression
- 11.5 Coefficient of Determination (R^2) and Adjusted R^2
- 11.6 Regression with Dummy Variables
- 11.7 Summary
- 11.8 Solutions/Answers

11.1 INTRODUCTION

In previous units, we have discussed the linear relationship between the dependent variable Y and an independent variable X . The coefficients a and b were unknown and for the given data on Y and X , we have obtained least squares estimates of parameters, i.e., $\hat{a}$ and $\hat{b}$. We have also gone through the inferential study to examine whether there exists a significant linear relationship between Y and X or not. We have discussed the simple linear regression model and estimation of model parameters, and determined standard errors.

In this unit, we discuss the multiple linear regression model along with the estimation of parameters in Secs. 11.2 and 11.3. In multiple linear regression, the basic concept is the same as that of simple regression. However, instead of one independent variable, there are several independent variables, say, $X_1, X_2, X_3, \dots, X_p$. For example, the number of units sold by a car manufacturing company per year may not depend on only one independent variable such as price, but also on mileage per unit of fuel, appearance of the car, comfort level, durability and money spent on advertising, etc. Here we may like to identify the important independent variables, which contribute more to the variation in the dependent variable(s). For this purpose, a mathematical relationship between the dependent and independent variables is established and this relation is further used for prediction purposes. We also discuss the inferential study in multiple linear regression in Sec. 11.4.

Since the model may involve several independent variables affecting the dependent variable because of their relationship via regression, it may be of interest to estimate their importance by estimating regression coefficients along with their standard errors. The adequacy of model fit may be examined by overall fit of the model with the help of coefficient of determination (R^2). In this unit, we also discuss a method for calculating R^2 and adjusted R^2 in Sec. 11.5. The regression analysis with dummy variables is also discussed in Sec. 11.6.

In the next unit, we shall discuss how to calculate the extra sum of squares explained by the regressor variables on the response variable. We shall also discuss the methods of selection of important regressor variables which play an important role in selection of the best fitted models.

Objectives

After studying this unit, you should be able to:

- explain the concept of multiple linear regression;
- formulate a multiple linear regression model;
- estimate the regression coefficients and their standard errors;
- calculate the coefficient of determination (R^2) and adjusted R^2 ; and
- predict the dependent variable for given values of independent variables.

11.2 MULTIPLE LINEAR REGRESSION MODEL

In this section, we generalise the simple regression model considered in Unit 9. We have assumed in Unit 9 that $(Y_1, X_1), (Y_2, X_2), \dots, (Y_n, X_n)$ are n pairs of values. The equation of the simple linear regression model may be written as

$$Y = a + bX + e$$

where e represents the error term, which arises from the difference of the observed Y and the straight line $Y = a + bX$. To minimise the term e , we use the method of least squares. From the above equation, we may write a simple regression model as

$$Y_i = a + bX_i + e_i \quad i = 1, 2, \dots, n$$

for the sample data of n pairs given in terms of (Y_i, X_i) ($i = 1, 2, \dots, n$).

In agriculture, the crop yield depends on more than one variable such as fertility of the soil, amount of rainfall, amount of fertilisers, etc. A multiple regression model that might describe this relationship is

$$Y = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + e$$

where Y denotes the yield, X_1 denotes the fertility of soil, X_2 denotes the rainfall and X_3 denotes the amount of fertilisers used. This is called the **multiple linear regression model** with three independent/regressor variables. The term linear is used because the dependent/response variable Y is a linear function of the unknown parameters B_0, B_1, B_2 and B_3 .

In general, the response variable may be related to p regressors or independent variables. Let Y be the dependent variable and $X_1, X_2, \dots, X_p$ be p independent variables. Then the multiple regression model can be written as:

$$Y = B_0 + B_1X_1 + B_2X_2 + \dots + B_pX_p + e \quad \dots (1)$$

The parameters $B_0, B_1, \dots, B_p$ are called the **regression coefficients**. The parameters B_i ($i = 0, 1, 2, \dots, p$) represent the expected change in the response variable Y per unit change in X_i when the remaining regressor variables are treated as constant.

For the sake of simplicity, we shall attach a dummy variable X_0 with the intercept B_0 ; X_0 takes value 1 for all n observations. Now the model in equation (1) can be written as:

$$Y = B_0X_0 + B_1X_1 + B_2X_2 + \dots + B_pX_p + e \quad \dots (2)$$

The simple regression model considered in Unit 9 becomes a particular case of this model with $X_0 = 1$, $B_0 = a$, $B_1 = b$ and $B_i = 0$, ($i \geq 2$). The interpretation of coefficients B_j ($j = 1, 2, \dots, p$) is that B_j represents the amount of change in Y for a unit change in X_j , keeping the other independent variables X_k ($k \neq j$) fixed. These coefficients are known as **partial regression coefficients** as the effect of one independent variable is studied on the dependent variable while the other variables are held fixed or constant. We use the term multiple linear regression for this model because two or more than two variables are included in the regression analysis and the parameters $B_0, B_1, \dots, B_p$ appear in a linear form. Moreover, the effect of these variables can be studied jointly. Here X_i can be any continuous function such as $\log X, X^2, X^3, X^{-1}$, etc. However, it is necessary that the equation is linear. Let us consider a polynomial model

$$Y = B_0 + B_1X + B_2X^2 + \dots + B_pX^p + e$$

If we let $X_1 = X$, $X_2 = X^2$, $X_3 = X^3$ and so on, the above model can be written in a linear form as given in equation (2).

As in the case of simple linear regression (Unit 9), here too we make the assumptions that e is normally and independently distributed with mean zero and constant variance σ^2 .

11.3 ESTIMATION OF MODEL PARAMETERS

Recall that in Unit 9, we have estimated the parameters a and b of a simple linear regression equation using the method of least squares. In this method, we minimise the total error term, so that the sum of the squares of the differences between the observed values and their expected values is minimum, i.e., the sum of squares of the error terms is minimum. We also use the method of least squares to estimate the regression coefficients given in equation (2).

Let the number of observations be n ($> p$). Let y_i denote the i^{th} observed value and x_{ij} denote the j^{th} observation of the regressor variable X_i . The data is represented as given in the table below:

Response Variable Y	Regressor Variables			
	X_1	X_2	$\dots$	X_p
y_1	x_{11}	x_{21}	$\dots$	x_{p1}
y_2	x_{12}	x_{22}	$\dots$	x_{p2}
y_3	x_{13}	x_{23}	$\dots$	x_{p3}
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
y_n	x_{1n}	x_{2n}	$\dots$	x_{pn}

Then the multiple regression model for the i^{th} observation Y_i can be written as:

$$Y_i = B_0 + B_1X_{1i} + B_2X_{2i} + \dots + B_pX_{pi} + e_i, \quad i = 1, 2, \dots, n$$

where $X_{1i}, X_{2i}, \dots, X_{pi}$ are the corresponding values of p independent variables, B_0 is the intercept, $B_1, B_2, \dots, B_p$ are p regression coefficients corresponding to independent variables $X_1, X_2, \dots, X_p$, respectively.

We now minimise $\sum e_i^2$, the sum of squares of errors in the model given in equation (2):

$$E = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n \left(Y_i - B_0 X_{0i} - B_1 X_{1i} - \dots - B_p X_{pi} \right)^2 \quad \dots (3)$$

with respect to $B_0, B_1, \dots, B_p$ to obtain their least squares estimates. For estimating the model parameters $B_0, B_1, B_2, \dots, B_p$, we differentiate E with respect to $B_0, B_1, B_2, \dots, B_p$, respectively, and equate the result to zero. If we differentiate E with respect to B_j , we obtain the j^{th} ($j = 0, 1, \dots, p$) normal equations as follows:

$$\frac{\partial E}{\partial B_i} = -2 \sum_{j=1}^n (Y_j - B_0 X_{0j} - B_1 X_{1j} - \dots - B_p X_{pj}) X_{ji} = 0, \quad j = 0, 1, 2, \dots, p \dots (4)$$

Simplifying equation (4), we obtain the least squares normal equations:

$$\begin{aligned}
& nB_0 + B_1 \sum_{i=1}^n X_{li} + B_2 \sum_{i=1}^n X_{2i} + \dots + B_p \sum_{i=1}^n X_{pi} = \sum_{i=1}^n Y_i \\
& B_0 \sum_{i=1}^n X_{li} + B_1 \sum_{i=1}^n X_{li}^2 + B_2 \sum_{i=1}^n X_{li} X_{2i} + \dots + B_p \sum_{i=1}^n X_{li} X_{pi} = \sum_{i=1}^n X_{li} Y_i \\
& B_0 \sum_{i=1}^n X_{2i} + B_1 \sum_{i=1}^n X_{2i} X_{li} + B_2 \sum_{i=1}^n X_{2i}^2 + \dots + B_p \sum_{i=1}^n X_{2i} X_{pi} = \sum_{i=1}^n X_{2i} Y_i \\
& \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \quad \quad \quad \dots (5) \\
& \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \\
& \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \quad \quad \quad \cdot \\
& B_0 \sum_{i=1}^n X_{pi} + B_1 \sum_{i=1}^n X_{pi} X_{li} + B_2 \sum_{i=1}^n X_{pi} X_{2i} + \dots + B_p \sum_{i=1}^n X_{pi}^2 = \sum_{i=1}^n X_{pi} Y_i
\end{aligned}$$

These are $p + 1$ normal equations and can be solved using the methods of solving simultaneous linear equations.

The solutions of the above normal equations called the **least squares estimates** are $\hat{B}_0, \hat{B}_1, \hat{B}_2, \dots, \hat{B}_p$, respectively.

For simplicity, we shall rewrite the model in equation (1) by centralising the independent variables $X_1, X_2, \dots, X_p$, i.e., by taking differences from their means:

$$Y = B_0 + B_1 \bar{X}_1 + \dots + B_p \bar{X}_p + B_1 (X_1 - \bar{X}_1) + \dots + B_p (X_p - \bar{X}_p) + e_i$$

$$= B'_0 X_0 + B_1 (X_1 - \bar{X}_1) + \dots + B_p (X_p - \bar{X}_p) + e$$

where $B'_0 = B_0 + B_1\bar{X}_1 + \dots + B_p\bar{X}_p$. Here $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_p$ are the means of p independent/ regressor variables. With this, the normal equation becomes

$$\frac{\partial E}{\partial B_j} = -2 \sum_{i=1}^n \left(Y_i - B'_0 X_{0i} - B_1 (X_{1i} - \bar{X}_1) - \dots - B_p (X_{pi} - \bar{X}_p) \right) (X_{ji} - \bar{X}_j) = 0, \quad \dots (6)$$

Note that $X_{0i} = 1$ for all i . The coefficients $B_1, B_2, \dots, B_p$ remain the same, but the intercept changes from B_0 to B'_0 . Once we have obtained the estimates of $B'_0, B_1, B_2, \dots, B_p$, we can obtain $\hat{B}_0$ from the following equation:

$$\hat{B}_0 = \hat{B}_0' - \hat{B}_1 \bar{X}_1 - \dots - \hat{B}_p \bar{X}_p \quad \dots (7)$$

Let us consider an application of these results.

Example 1: A statistical analyst is analysing the vending machine routes in the distribution system. He/she is interested in predicting the amount of time required by the route driver to service the vending machines in an outlet. The company manager responsible for the study has suggested that the two most important variables affecting the delivery time Y (in minutes), are (i) the number of cases (X_1) and (ii) the distance travelled (in m) by the route driver (X_2). The delivery time data collected by the statistical analyst is given below:

Time (Y)	No. of Cases (X_1)	Distance (X_2)
20	10	50
10	5	20
10	5	30
15	5	10
15	10	10
20	10	30
10	5	10
25	15	40
30	10	80
15	10	20
20	10	10
10	5	40

Check whether there is a linear relationship between Y (Time) and the two independent variables X_1 (number of cases) and X_2 (distance). Calculate the values of the regression coefficients and fit the regression equation.

Solution: To find the values of regression coefficients and fit the regression equation for the given data, we form the following table:

Time (Y)	No. of Cases (X_1)	Distance (X_2)	Y^2	$(X_1)^2$	$(X_2)^2$	X_1Y	X_2Y	X_1X_2
20	10	50	400	100	2500	200	1000	500
10	5	20	100	25	400	50	200	100
10	5	30	100	25	900	50	300	150
15	5	10	225	25	100	75	150	50
15	10	10	225	100	100	150	150	100
20	10	30	400	100	900	200	600	300
10	5	10	100	25	100	50	100	50
25	15	40	625	225	1600	375	1000	600
30	10	80	900	100	6400	300	2400	800
15	10	20	225	100	400	150	300	200
20	10	10	400	100	100	200	200	100
10	5	40	100	25	1600	50	400	200
$\sum Y_i$ =200	$\sum X_{1i}$ =100	$\sum X_{2i}$ =350	$\sum Y_i^2$ =3800	$\sum X_{1i}^2$ =950	$\sum X_{2i}^2$ =15100	$\sum X_{1i}Y_i$ =1850	$\sum X_{2i}Y_i$ =6800	$\sum X_{1i}X_{2i}$ =3150

On putting the values from the above table in the normal equations (5) for $p = 2$, and noting that $X_0 = 1$, we get

$$n \hat{B}_0 + \hat{B}_1 \sum X_{1i} + \hat{B}_2 \sum X_{2i} = \sum Y_i$$

$$\hat{B}_0 \sum X_{1i} + \hat{B}_1 \sum X_{1i}^2 + \hat{B}_2 \sum X_{1i} X_{2i} = \sum Y_i X_{1i}$$

$$\hat{B}_0 \sum X_{2i} + \hat{B}_1 \sum X_{1i} X_{2i} + \hat{B}_2 \sum X_{2i}^2 = \sum Y_i X_{2i}$$

On putting the values calculated in the table in the above equations, we get

$$12 \hat{B}_0 + 100 \hat{B}_1 + 350 \hat{B}_2 = 200 \quad \dots (i)$$

$$100 \hat{B}_0 + 950 \hat{B}_1 + 3150 \hat{B}_2 = 1850 \quad \dots (ii)$$

$$350 \hat{B}_0 + 3150 \hat{B}_1 + 15100 \hat{B}_2 = 6800 \quad \dots (iii)$$

From equation (i), we have

$$\hat{B}_0 = \frac{(200 - 100 \hat{B}_1 - 350 \hat{B}_2)}{12} \quad \dots (iv)$$

On putting the value of $\hat{B}_0$ in equations (ii) and (iii) and simplifying, we get

$$1400 \hat{B}_1 + 2800 \hat{B}_2 = 2200 \quad \dots (v)$$

$$2800 \hat{B}_1 + 58700 \hat{B}_2 = 11600 \quad \dots (vi)$$

On solving equations (v) and (vi), we get

$$\hat{B}_1 = 1.3002 \quad \hat{B}_2 = 0.1356$$

and
$$\hat{B}_0 = \frac{(200 - 100(1.3002) - 350(0.1356))}{12} = 1.8765$$

Hence, the fitted equation is

$$Y = 1.8765 + 1.3002 X_1 + 0.1356 X_2$$

So we can conclude that there is a linear relationship between Y (time in seconds) and the two independent variables X_1 (number of cases) and X_2 (distance). As the regression coefficients for both variables are positive, these affect the delivery time. The numerical value of the regression coefficient $\hat{B}_1$ associated with X_1 is higher than the value of $\hat{B}_2$ associated with X_2 . It shows that the number of cases affects the delivery time more than the distance travelled.

11.3.1 Use of Matrix Notation

When p is greater than 2, it is more convenient to write the normal equations in matrix form. The regression equations in matrix notation can be written as:

$$Y = X B + e \quad \dots (8)$$

where

$$Y_{n \times 1} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, X_{n \times (p+1)} = \begin{pmatrix} 1 & X_{11} & \dots & X_{p1} \\ 1 & X_{12} & \dots & X_{p2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \dots & X_{pn} \end{pmatrix}, B_{(p+1) \times 1} = \begin{pmatrix} B_0 \\ B_1 \\ \vdots \\ B_p \end{pmatrix} \text{ and } e_{n \times 1} = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix}$$

In general, Y is an $n \times 1$ vector of the observed values of the response variable Y , X is a $(p+1) \times n$ matrix of the values of regressor variables, B is a $(p+1) \times 1$ vector of regression coefficients and e is an $n \times 1$ vector of random errors.

Here, we shall use the following notation:

In matrix notation, the $(p+1)$ normal equations can be written as follows:

$$X'X\hat{B} = X'Y \quad \dots (9a)$$

Equation (9a) represents the normal least squares equations. For the sake of simplicity, we may write them as

$$\begin{pmatrix} n & \sum x_{1i} & \dots & \sum x_{pi} \\ \sum x_{1i} & \sum x_{1i}^2 & \dots & \sum x_{1i}x_{pi} \\ \vdots & \vdots & \ddots & \vdots \\ \sum x_{pi} & \sum x_{pi}x_{1i} & \dots & \sum x_{pi}^2 \end{pmatrix} \begin{pmatrix} B_0 \\ B_1 \\ \vdots \\ B_p \end{pmatrix} = \begin{pmatrix} \sum y_i \\ \sum x_{1i}y_i \\ \vdots \\ \sum x_{pi}y_i \end{pmatrix}$$

$$Y_{n \times 1} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, X_{n \times (p+1)} = \begin{pmatrix} X_{01} & X_{11} - \bar{X}_1 & \dots & X_{p1} - \bar{X}_p \\ \vdots & \vdots & \ddots & \vdots \\ X_{0n} & X_{n1} - \bar{X}_1 & \dots & X_{pn} - \bar{X}_p \end{pmatrix}$$

Note that X_{0i} 's are all unity and other variables are centralised (deviations from mean). Then $(k+1)$ normal equations can be written as

$$\dots (9b) \quad X'X\hat{B} = X'Y$$

To solve the normal least squares equations given in equation (9a), we multiply both sides by the inverse of $X'X$. Thus, the estimates of the regression coefficients are given by

$$\hat{B} = (X'X)^{-1} X'Y \quad \dots (10)$$

On putting the values of the estimates in equation (2), we get the fitted regression model corresponding to the observations of the regressor variables $X_1, X_2, \dots, X_p$ as

$$\hat{Y} = \hat{B}_0 + \hat{B}_1 X_1 + \hat{B}_2 X_2 + \dots + \hat{B}_p X_p + e \quad \dots (11)$$

The matrix representation of the fitted values corresponding to the observed values are similar to the equation (9a) and are given as

$$\hat{Y} = X\hat{B} = X(X'X)^{-1} X'Y \quad \dots (12)$$

The difference between the observed value y_i and the corresponding estimated value $\hat{y}_i$ is called the i^{th} residual r_i , i.e.,

$$r_i = y_i - \hat{y}_i \quad i = 1, 2, 3, \dots, n.$$

The residuals may be written in matrix notation as

$$r = Y - \hat{Y} = Y - X\hat{B} \quad \dots (13)$$

11.3.2 Properties of Least Squares Estimates

We now describe the statistical properties of least squares estimates. When X_j , ($j = 0, 1, \dots, p$) are linearly related, $(X'X)$ is not invertible. In this case we cannot obtain unique estimates of B . We shall not consider this case any more.

It is to be noted that $\hat{B}$ is an unbiased estimate of B because

$$\begin{aligned} E(\hat{B}) &= (X'X)^{-1} X' E(Y) \\ &= (X'X)^{-1} X' E(XB + e) = (X'X)^{-1} (X'X)B \\ &= B \quad \text{since } E(e) = 0 \text{ and } (X'X)^{-1} (X'X) = I \end{aligned}$$

This shows that $\hat{B}$ is unbiased.

The variance of Y (which is actually the variance-covariance matrix as Y is a vector) is given as

$$V(Y) = s^2 I_n$$

where I_n is an identity matrix of order n . The variance-covariance matrix of $\hat{B}$ is given by

$$\begin{aligned} V(\hat{B}) &= (X'X)^{-1} X' V(Y) X (X'X)^{-1} \\ &= (X'X)^{-1} X' \sigma^2 I_n X (X'X)^{-1} \\ &= \sigma^2 (X'X)^{-1} \end{aligned} \quad \dots (14)$$

where

$$X'X = \begin{bmatrix} n & \sum X_{1i} & \sum X_{2i} & \dots & \sum X_{pi} \\ \sum X_{1i} & \sum X_{1i}^2 & \sum X_{1i}X_{2i} & \dots & \sum X_{1i}X_{pi} \\ \sum X_{2i} & \sum X_{1i}X_{2i} & \sum X_{2i}^2 & \dots & \sum X_{2i}X_{pi} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum X_{pi} & \sum X_{1i}X_{pi} & \sum X_{2i}X_{pi} & \dots & \sum X_{pi}^2 \end{bmatrix}$$

$$\text{and } s_{jk} = \sum_{i=1}^n X_{ji} X_{ki}.$$

Here $V(\hat{B}) = \sigma^2 (X'X)^{-1}$ is a $(p+1) \times (p+1)$ matrix and its diagonal elements give the variances of coefficients and off diagonal elements give the covariances. If we use the notation

$$V(\hat{B}) = s^2 (X'X)^{-1} = (s_{jk}), \quad j, k = 0, 1, \dots, p$$

we can write

$$V(\hat{B}_j) = s_{jj}, \quad \text{and} \quad \text{Cov}(\hat{B}_j, \hat{B}_k) = s_{jk} \quad \dots (15)$$

The standard error of $\hat{B}_j$ is given by

$$\text{S. E. } (\hat{B}_j) = \sqrt{s_{jj}} \quad \dots (16)$$

The residual sum of squares SS_{Res} is obtained by substituting the least squares estimates of $B_0, B_1, \dots, B_p$ in equation (3):

$$SS_{Res} = \sum_{i=1}^n \left(Y_i - \hat{B}_0 X_{0i} - \hat{B}_1 X_{1i} - \dots - \hat{B}_p X_{pi} \right)^2$$

This is the sum of squares not accounted for by the regression model. In matrix notation, this can be written as

$$\begin{aligned} SS_{Res} &= Y'Y - Y'X\hat{B} \\ &= \sum Y_i^2 - \hat{B}_0(Y'X_0) - \hat{B}_1(Y'X_1) - \hat{B}_2(Y'X_2) \dots - \hat{B}_p(Y'X_p) \dots (17) \end{aligned}$$

Note that $X_1, X_2, \dots, X_p$ are deviations from respective means. As we have fitted $(p+1)$ parameters, the degree of freedom of residual sum of squares is $(n-p-1)$. An unbiased estimate of σ^2 is obtained by dividing the residual sum of squares, i.e., SS_{Res} , by its degree of freedom $(n-p-1)$. Thus

$$\hat{\sigma}^2 = SS_{Res} / (n-p-1) \quad \dots (18)$$

If we are interested in predicting the mean value of Y for a given set of independent variables $X_1, \dots, X_p$, then we use the fitted model. The predicted mean value of Y for given $X_{10}, \dots, X_{p0}$ is given by

$$\hat{Y}_0 = \hat{B}_0 + \hat{B}_1 X_{10} + \dots + \hat{B}_p X_{p0}$$

Let us explain the matrix method with the help of an example.

Example 2: Using the data of Example 1, find the estimate of regression coefficients and SS_{Res} by using the matrix method. Also predict the expected time Y at $X_1 = 7, X_2 = 20$.

Solution: Using the matrix notation we have from the data:

$$Y'_{12} = [20, 10, 10, 15, 15, 20, 10, 25, 30, 15, 20, 10]$$

$$X'X = \begin{pmatrix} 12 & 100 & 350 \\ 100 & 950 & 3150 \\ 350 & 3150 & 15100 \end{pmatrix}, \quad X'Y = \begin{pmatrix} 200 \\ 1850 \\ 6800 \end{pmatrix}$$

$$(X'X)^{-1} = \begin{pmatrix} 0.7139 & -0.0658 & -0.0028 \\ -0.0658 & 0.0095 & -0.0005 \\ -0.0028 & -0.0005 & 0.0002 \end{pmatrix}$$

and

$$\begin{aligned} \begin{pmatrix} \hat{B}_0 \\ \hat{B}_1 \\ \hat{B}_2 \end{pmatrix} &= (X'X)^{-1} X'Y \\ &= \begin{pmatrix} 0.7139 & -0.0658 & -0.0028 \\ -0.0658 & 0.0095 & -0.0005 \\ -0.0028 & -0.0005 & 0.0002 \end{pmatrix} \times \begin{pmatrix} 200 \\ 1850 \\ 6800 \end{pmatrix} = \begin{pmatrix} 1.8765 \\ 1.3002 \\ 0.1356 \end{pmatrix} \end{aligned}$$

Hence the fitted equation is

$$Y = 1.8765 + 1.3002 X_1 + 0.1356 X_2$$

Now, we calculate the value of residual sum of squares to obtain an estimate of $\hat{\sigma}^2$ as follows:

$$\begin{aligned} SS_{\text{Res}} &= YY' - Y'XB_0 \\ &= 3800 - 200 \times (1.8765) - 1850 \times (1.3002) - 6800 \times (0.1356) \\ &= 3800 - 375.3 - 2405.37 - 922.08 = 97.25 \end{aligned}$$

Therefore, on putting the value of SS_{Res} in equation (18), we get

$$\hat{\sigma}^2 = 97.25 / (12 - 3) = 10.8017$$

Using the above results and putting the values $X_1 = 7$ and $X_2 = 20$ in the fitted equation for multiple regression, we get

$$\hat{Y} = 1.8765 + 1.3002 X_1 + 0.1356 X_2$$

$$\hat{Y} = 1.8765 + 1.3002 \times 7 + 0.1356 \times 20 = 13.6899$$

As far as the interpretation of coefficients is concerned, there is an increase of 1.3002 seconds in time for one unit increase in X_1 . Similarly, for one unit increase in X_2 there is an increase of 0.1356 seconds in time.

You may like to pause here and solve the following exercises to check your understanding.

E1) In a study of 10 firms, the dependent variable was the total delivery time (Y) and the independent variables were the distance covered (X_1) and the packaging time (X_2). The delivery time data collected by the statistical analyst is given below:

Time (Y)	Distance (X_1)	Packaging Time (X_2)
18	61	30
14	95	25
17	72	30
14	84	25
13	98	10
24	53	35
13	68	15
22	54	40
12	89	30
19	73	20
$\sum Y_i = 166$	$\sum X_{1i} = 747$	$\sum X_{2i} = 260$

Estimate the parameters B_0 , B_1 , and B_2 by solving normal equations and find the estimated multiple linear regression equation.

E2) Use the matrix method to estimate parameters from the data given in E1).

11.4 TEST OF SIGNIFICANCE IN MULTIPLE REGRESSION

So far you have learnt how to estimate the parameters and fit the multiple regression model. You may now like to test the adequacy of the fitted model and examine whether the independent variables contribute significantly in explaining the variability in Y or not. For this purpose, we use the test of significance of equality of variances of the regressor variables.

If there is a linear relationship between the response variable Y and any of the independent variables $X_1, X_2, \dots, X_p$, we use the **test of significance of regression**. The test of significance of regression is a test to determine the linear relationship between the response variable and regressor variables and is often used to examine the adequacy of the model.

In order to test whether the contribution of independent variables $X_1, \dots, X_p$ is significant or not, we test whether $B_1, B_2, \dots, B_p$ are all zero in the model or at least one of them is not zero. This hypothesis can be written as:

$$H_0 : B_1 = B_2 = \dots = B_p = 0$$

$$H_1 : \text{At least one of the regression coefficients is not zero}$$

It can be tested by considering the following F-ratio:

$$F = \frac{SS_{\text{Reg}}/p}{SS_{\text{Res}}/(n-p-1)} \quad \dots (19)$$

In this test, the total sum of squares SS_T is partitioned into a sum of squares due to the contribution of regressor variables (SS_{Reg}) and a residual sum of squares (SS_{Res}). From equation (17), the residual sum of squares (SS_{Res}) is:

$$SS_{\text{Res}} = \sum Y_i^2 - \hat{B}_0(Y'X_0) - \hat{B}_1(Y'X_1) - \dots - \hat{B}_p(Y'X_p)$$

$$\text{or } SS_{\text{Res}} = Y'Y - Y'X\hat{B} \quad \dots (20)$$

If $B_1, B_2, \dots, B_k$ are all zeros, i.e., independent variables do not contribute to the variability in Y , then the total sum of squares, denoted by SS_T , is given as:

$$SS_T = \sum Y_i^2 - n\bar{Y}^2 = Y'Y - \frac{\left(\sum Y_i\right)^2}{n} \quad \dots (21)$$

This is the total variability present in Y around the mean $\bar{Y}$. We can rewrite equation (20) as

$$SS_{\text{Res}} = \sum Y_i^2 - \frac{\left(\sum Y_i\right)^2}{n} - \left\{ \sum B_j Y'X_j - \frac{\left(\sum Y_i\right)^2}{n} \right\}$$

that is, $SS_{\text{Res}} = SS_T - SS_{\text{Reg}}$

Hence, the difference of $SS_T - SS_{\text{Res}}$ gives the contribution of independent variables $X_1, X_2, \dots, X_p$, in explaining the variability in Y , i.e.,

$$SS_T - SS_{Res} = \hat{B}_0(Y'X_0) + \hat{B}_1(Y'X_1) + \dots + \hat{B}_p(Y'X_p) - n\bar{Y}^2$$

$$\Rightarrow SS_{Reg} = \left\{ Y'X\hat{B} - \frac{(\sum Y_i)^2}{n} \right\}$$

$$\text{or } SS_{Reg} = \sum \hat{B}_j Y'X_j - n\bar{Y}^2 \quad \dots (22)$$

We now summarise these results in the following ANOVA Table:

ANOVA TABLE

Sources of Variation	Degree of Freedom (d.f.)	Sum of Squares (S.S.)	Mean Sum of Squares	Variance Ratio
Independent Variables ($X_1, X_2, \dots, X_p$)	p	$SS_{Reg} = \sum_{j=0}^p \hat{B}_j Y'X_j - n\bar{Y}^2$	$\frac{SS_{Reg}}{p}$	$F = \frac{SS_{Reg}/p}{SS_{Res}/(n-p-1)}$
Residuals (SS_{Res})	n-p-1	$SS_{Res} = Y'Y - \sum_{j=0}^p \hat{B}_j Y'X_j$	$\frac{SS_{Res}}{(n-p-1)}$	
Total	n-1	$Y'Y - n\bar{Y}^2$		

Under the null hypothesis, i.e., when $B_1 = B_2 = \dots = B_p = 0$, F is distributed as Fisher's F-distribution with p and (n-p-1) degree of freedom, i.e.,

$$F \sim F_{p, (n-p-1)} \quad \dots (23)$$

If the calculated F is less than the tabulated $F_{p, (n-p-1)}$ at α level of significance, then we conclude that the contribution of $X_1, X_2, \dots, X_p$ to the variability in Y is not significant. Thus, they have no contribution in prediction. It may be of further interest to examine whether any one coefficient (say B_j) corresponding to the independent variable X_j is different from zero, after accounting for other variables X_k (all $k \neq j$). This can be tested by considering the statistic t:

$$t = \frac{\hat{B}_j}{S.E.(\hat{B}_j)} \quad \dots (24)$$

where $S.E.(\hat{B}_j)$ uses the estimated value of $\hat{\sigma}^2$ given in equation (18). Under the null hypothesis, i.e., $B_j = 0$, the proposed statistic t follows the Student's t-distribution with (n-p-1) d.f. Thus, if

$$|t| \leq t_{\alpha/2, n-p-1} \quad \dots (25)$$

we accept H_0 . Otherwise, we reject it. If B_j is significantly different from zero, it contributes significantly to the variability in Y after taking into account the contribution of other variables. If B_j is not significantly different from zero, its contribution is not significant after accounting for other variables in the model.

Example 4: Using the data of Example 1 and the results of Example 2, construct the ANOVA table, apply a relevant test of hypothesis and interpret the results.

Solution: As per the data given in Example 1 and the results of Example 2, we have

$$SS_{Res} = 97.2155 \text{ and } \sum_{j=1}^p \hat{B}_j Y'X_j = 3702.7845$$

Using these values, we construct the ANOVA table as follows:

ANOVA TABLE

Sources of Variation	Degree of Freedom (d.f.)	Sum of Squares (S.S.)	Mean Sum of Squares	Variance Ratio
Independent Variables (X_1, X_2)	2	$SS_{Reg} = \sum_{j=0}^p \hat{B}_j Y'X_j - n\bar{Y}^2$ $= 369.4512$	$\frac{SS_{Reg}}{2} = 184.7256$	$F = \frac{SS_{Reg}/p}{SS_{Res}/(n-p-1)}$ $= 17.1015$
Residuals (SS_{Res})	9	$SS_{Res} = Y'Y - \sum_{j=0}^p \hat{B}_j Y'X_j$ $= 97.2155$	$\frac{SS_{Res}}{(n-p-1)} = 10.8017$	
Total	11	$Y'Y - n\bar{Y}^2 = 466.6667$		

We have obtained the Variance Ratio $F = 17.1012$, whereas the tabulated value of $F_{2,9}$ at $\alpha = 0.05$ is 4.26. Hence, we reject H_0 and conclude that X_1 and X_2 contribute significantly to the variability. It may be of further interest to examine whether the coefficient B_j corresponding to independent variable X_j is different from zero, after accounting for other variables X_k (all $k \neq j$). This can be tested by considering the statistic t:

$$t = \frac{\hat{B}_j}{S.E.(\hat{B}_j)}$$

From the result of Example 2, we also have

$$\hat{B}_1 = 1.3002 \text{ and } \hat{B}_2 = 0.1356$$

The Variance-Covariance matrix is

$$V(\hat{B}) = \hat{\sigma}^2 (X'X)^{-1}$$

$$= 10.8017 \begin{pmatrix} 0.7139 & -0.0658 & -0.0028 \\ -0.0658 & 0.0095 & -0.0005 \\ -0.0028 & -0.0005 & 0.0002 \end{pmatrix}$$

Thus
$$V(\hat{B}) = \begin{pmatrix} 7.7112 & -0.7105 & -0.0305 \\ -0.7105 & 0.1024 & -0.0049 \\ -0.0305 & -0.0049 & 0.0024 \end{pmatrix}$$

Using equation (15), we obtain

$$V(\hat{B}_0) = 7.7112, \quad V(\hat{B}_1) = 0.1024 \quad \text{and} \quad V(\hat{B}_2) = 0.0024$$

and therefore,

$$S.E.(\hat{B}_0) = \sqrt{7.7112} = 2.7769$$

$$\text{S.E.}(\hat{B}_1) = \sqrt{0.1024} = 0.3199$$

$$\text{S.E.}(\hat{B}_2) = \sqrt{0.0024} = 0.0494$$

Therefore, the statistic t is given as:

$$t_0 = \frac{\hat{B}_0}{\text{S.E.}(\hat{B}_0)} = \frac{1.8765}{2.7769} = 0.6758$$

$$t_1 = \frac{\hat{B}_1}{\text{S.E.}(\hat{B}_1)} = \frac{1.3002}{0.3199} = 4.0642$$

$$t_2 = \frac{\hat{B}_2}{\text{S.E.}(\hat{B}_2)} = \frac{0.1356}{0.0494} = 2.7444$$

But the tabulated value of t -statistic for $\alpha = 0.05$ is

$$t_{0.025, 22} = 2.262$$

Hence, both variables contribute significantly to the variability in Y .

You may now like to solve the following exercise.

E3) Make the ANOVA table, calculate standard errors of estimates and test their significance using the data in E1. Interpret the results.

11.5 COEFFICIENT OF DETERMINATION (R^2) AND ADJUSTED R^2

We define the coefficient of determination, R^2 , in the same way as for simple regression. It gives a measure of adequacy of model fit.

We define R^2 as follows:

$R^2 = \text{Variability accounted by independent variables} / \text{Total variability around the mean}$

$$= \frac{\sum_{j=0}^p \hat{B}_j Y' X_j - n \bar{Y}^2}{Y' Y - n \bar{Y}^2} \quad \dots (26)$$

Its value always lies between 0 and 1. When the fit is good, $R^2 \sim 1$. Otherwise, $R^2 \sim 0$.

The value of R^2 always increases with p . The increase may be negligible, but R^2 never decreases. When we compare two models with different values of p , the model with larger p is preferable if R^2 corresponding to it is significantly larger than R^2 with smaller p . A model with smaller p with large R^2 is always preferable as it is a simple model. Hence, you should choose a model with small p if its R^2 is not much smaller than R^2 for a model with a larger p .

For this, we define an adjusted R^2 , viz., R^2_{Adj} , which penalises R^2 when p increases but R^2 does not increase significantly. We know that

$$R^2 = \frac{SS_{\text{Reg}}}{SS_T} \quad \dots(27)$$

$$1 - R^2 = \frac{SS_T - SS_{\text{Reg}}}{SS_T} = \frac{SS_{\text{Res}}}{SS_T}$$

Then we define R^2_{Adj} as

$$1 - R^2_{\text{Adj}} = \frac{SS_{\text{Res}} / (n - p - 1)}{SS_T / (n - 1)} = \frac{(n - 1)(1 - R^2)}{(n - p - 1)} \quad \dots (28)$$

Here, we have divided the numerator and denominator by their degree of freedom.

$SS_{\text{Res}} / (n - p - 1)$ may decrease with increase in p even when there is no appreciable decrease in R^2 . Hence,

$$R^2_{\text{Adj}} = 1 - \frac{(n - 1)(1 - R^2)}{(n - p - 1)} \quad \dots (29)$$

Therefore, we should stop including the terms in the model if R^2_{Adj} starts decreasing. We prefer a model with larger R^2_{Adj} and smaller p than a model with smaller R^2_{Adj} but larger p .

Example 5: Using the data of Example 1 and the results of Examples 2 and 3, calculate R^2 , R^2_{Adj} and interpret the results.

Solution: Using the data of Example 1 and the results of Examples 2 and 3, and on putting the values in equation (27), we get

$$R^2 = \frac{SS_{\text{Reg}}}{SS_T} = \frac{369.4512}{466.6667} = 0.7917$$

Therefore, the adjusted R^2 is obtained as follows:

$$R^2_{\text{Adj}} = 1 - \frac{(12 - 1)(1 - 0.7917)}{(12 - 2 - 1)} = 0.7454$$

From the coefficient of determination, R^2 , we see that 79% variability in Y is due to X . This is quite a good fit. Adjusted R^2 is 0.7454, which is quite large. Hence we conclude that both X_1 and X_2 contribute adequately to the model fit.

You may now like to calculate R^2 and adjusted R^2 yourself. Try the following exercise.

E4) Calculate R^2 and adjusted R^2 and comment on the goodness of fit of the model, for the data given in E1.

11.6 REGRESSION WITH DUMMY VARIABLES

In previous sections, we have dealt with multiple linear regression when the independent / regressor variables are quantitative. The quantitative variables such as height, distance, temperature, time, income, pressure, etc. have a well

defined scale of measurement. However, sometimes independent variables include qualitative variables such as sex (male/female), regions (north, south, east, west, etc.), religion such as Hindu, Muslim, Christian, etc. Such variables called **categorical variables** cannot be measured and hence no quantitative number can be assigned to them. We define dummy variables to account for the effect that the qualitative variables may have on the response variable. Dummy variables are also known as **indicator variables**. Suppose, k is the number of levels a categorical variable takes. Then we define $(k - 1)$ dummy variables. For example, if we have two categories of male or female in the data, i.e., $k = 2$ and we define one dummy variable.

Suppose that a statistical analyst is analysing the vending machine's efficiency in the distribution of a product. She/he is interested in relating the time required to service the consumer with the distance travelled by the product in the vending machine for machines of two types, A and B.

The second regressor variable, machine type is qualitative, and has two levels: Type A and Type B. It allows us to code the types of machines used. Therefore, we define a dummy variable X_2 which takes on the values 0 and 1 to identify the types of machines as follows:

$$X_2 = \begin{cases} 1, & \text{if distribution is done by machine A} \\ 0, & \text{if distribution is done by machine B} \end{cases}$$

The variable X_2 is called an indicator variable because it is used to indicate the presence or absence of Machine A or B.

For such situations, we have a multiple linear regression model given by

$$Y = B_0 + B_1X_1 + B_2X_2 + e \quad \dots (30)$$

To determine the regression coefficients in this model, we first consider machine type A for which X_2 takes value 0. Then the regression model is given by:

$$\begin{aligned} Y &= B_0 + B_1X_1 + B_2(0) + e \\ \Rightarrow Y &= B_0 + B_1X_1 + e \quad \dots (31) \end{aligned}$$

The relationship between the response variable Y and regressor variable X_1 , i.e., distance travelled by the product in the machine is a straight line with intercept B_0 and slope B_1 .

For machine of type B, we have $X_2 = 1$. Then the regression model becomes

$$\begin{aligned} Y &= B_0 + B_1X_1 + B_2(1) + e \\ \Rightarrow Y &= (B_0 + B_2) + B_1X_1 + e \quad \dots (32) \end{aligned}$$

which shows that the relationship between Y and X_1 is also a straight line with slope B_1 but intercept $(B_0 + B_2)$.

Note that these models are linear with the same slope B_1 but different intercepts. Hence, these two models describe two parallel regression lines, i.e., two lines with a common slope and different intercepts. The vertical distance between these two lines is the difference in the intercepts, i.e., B_2 . The two parallel regression lines formed by the above models given in equations (31) and (32) are shown Fig. 11.1.

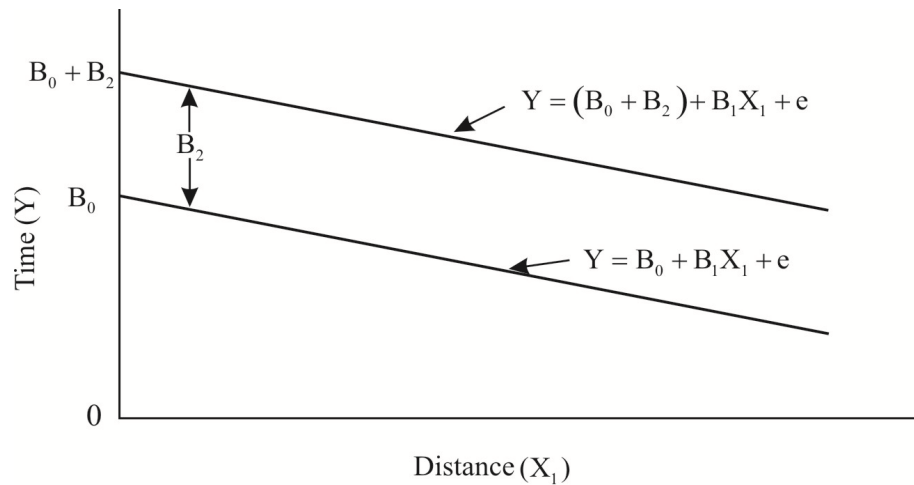


Fig. 11.1

For three Machine types A, B and C, two dummy variables X_2 and X_3 are used. The model becomes

$$Y = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + e \quad \dots (33)$$

The levels of dummy variable would be:

$X_2=0$	$X_3=0$	For Machine Type A
$X_2=1$	$X_3=0$	For Machine Type B
$X_2=0$	$X_3=1$	For Machine Type C

In general, a categorical variable with k categories is denoted by $(k - 1)$ dummy variables.

Let us try to understand regression analysis using dummy variables with the help of an example.

Example 6: A statistical analyst is analysing the performance of washing machines in the distribution system. He/she is interested in predicting the amount of time required by the driver to service washing machines of two types: i) Type A and ii) Type B. The data on the required time collected by the statistical analyst is given below:

Time (Y)	Distance (X_1)	Machine Type (X_2)
20	50	1
10	20	1
10	30	1
15	10	0
15	10	0
20	30	0
10	10	0
25	40	0
30	80	1
15	20	0
20	10	0
10	40	1

Check whether there is a linear relationship between Y (time) and the two independent variables X_1 (distance) and X_2 (type). Calculate the values of the coefficients and fit the regression equation.

Solution: Since two types of washing machines A and B have been used, $k = 2$. Here we have to define one dummy variable X_2 , which takes two values:

$$X_2 = 0 \quad \text{if the observation is from machine A}$$

$$= 1 \quad \text{if the observation is from machine B}$$

We form the following table from the given data to fit the regression equation:

Time (Y)	Distance (X_1)	Machine Type (X_2)	Y^2	$(X_1)^2$	$(X_2)^2$	X_1Y	X_2Y	X_1X_2
20	50	1	400	2500	1	1000	20	50
10	20	1	100	400	1	200	10	20
10	30	1	100	900	1	300	10	30
15	10	0	225	100	0	150	0	0
15	10	0	225	100	0	150	0	0
20	30	0	400	900	0	600	0	0
10	10	0	100	100	0	100	0	0
25	40	0	625	1600	0	1000	0	0
30	80	1	900	6400	1	2400	30	80
15	20	0	225	400	0	300	0	0
20	10	0	400	100	0	200	0	0
10	40	1	100	1600	1	400	10	40
$\sum Y_i = 200$	$\sum X_{1i} = 350$	$\sum X_{2i} = 05$	$\sum Y_i^2 = 3800$	$\sum X_{1i}^2 = 15100$	$\sum X_{2i}^2 = 5$	$\sum X_{1i}Y_i = 6800$	$\sum X_{2i}Y_i = 80$	$\sum X_{1i}X_{2i} = 220$

The normal equations (5) for $p = 2$ and $X_{0i} = 1$ are:

$$n \hat{B}_0 + \hat{B}_1 \sum X_{1i} + \hat{B}_2 \sum X_{2i} = \sum Y_i$$

$$\hat{B}_0 \sum X_{1i} + \hat{B}_1 \sum X_{1i}^2 + \hat{B}_2 \sum X_{1i}X_{2i} = \sum Y_i X_{1i}$$

$$\hat{B}_0 \sum X_{2i} + \hat{B}_1 \sum X_{1i}X_{2i} + \hat{B}_2 \sum X_{2i}^2 = \sum Y_i X_{2i}$$

On putting the values of the sums calculated in the above table, we get

$$12 \hat{B}_0 + 350 \hat{B}_1 + 05 \hat{B}_2 = 200 \quad \dots (i)$$

$$350 \hat{B}_0 + 15100 \hat{B}_1 + 220 \hat{B}_2 = 6800 \quad \dots (ii)$$

$$05 \hat{B}_0 + 220 \hat{B}_1 + 05 \hat{B}_2 = 80 \quad \dots (iii)$$

From equation (iii), we have

$$\hat{B}_0 = 16 - 44 \hat{B}_1 - \hat{B}_2 \quad \dots (iv)$$

On putting the value of $\hat{B}_0$ in equations (i) and (ii) and simplifying, we get

$$-178\hat{B}_1 - 7\hat{B}_2 = 8 \quad \dots (v)$$

$$-300\hat{B}_1 - 130\hat{B}_2 = 1200 \quad \dots (vi)$$

On solving equations (v) and (vi), we get

$$\hat{B}_1 = 0.3498, \quad \hat{B}_2 = -10.038$$

and $\hat{B}_0 = 16 - 44\hat{B}_1 - \hat{B}_2 = 10.646$

Hence, the fitted regression equation is

$$Y = 10.646 + 0.3498 X_1 - 10.038 X_2 \quad \dots (vii)$$

We conclude that there is a linear relationship between Y (time in seconds) and the two independent variables X_1 (distance) and X_2 (type of machine). Since the regression coefficient for the variable X_2 is negative, it affects the delivery time. The numerical value of the regression coefficient associated with X_2 is higher than that of the other regressor variable. It shows that distance travelled (in m) affects the delivery time less than the type of machines.

To determine the regression coefficients in this model for each type of machine, we first consider machine A for which X_2 takes value 0. We put the values of regression coefficients in equation (28). Then the regression model becomes

$$Y = 10.646 + 0.3498 X_1 \quad \dots (viii)$$

For machine B, we put the value of the regression coefficient and $X_2 = 1$. Then the regression model becomes

$$Y = (0.606) + 0.3498 X_1 \quad \dots (ix)$$

Note that as discussed in Sec 11.5, these estimated regression lines have the same slope, i.e., 0.3498, but have different intercepts, i.e., 10.646 and 0.606.

You may now like to solve the following problem to check your understanding:

E5) Using the data given in the following table, find the regression coefficients and obtain the estimated regression equations for the model given in equations (27), (28) and (29) :

Time (hour) Y	Distance (feet) X_1	Machine Type	X_2
18	61	A	0
14	95	A	0
17	72	A	0
14	84	A	0
13	98	A	0
24	53	B	1
13	68	B	1
22	54	B	1
12	89	B	1
19	73	B	1

Check whether there is a linear relationship between Y (time) and the two independent variables X_1 (distance) and X_2 (machine type). Calculate the values of the coefficients and fit the regression equation.

We now summarise the concepts that we have discussed in this unit.

11.7 SUMMARY

1. The basic concept of multiple linear regression is the same as that of simple regression. However, instead of one independent variable, there are several independent variables, say, $X_1, X_2, X_3, \dots, X_p$.
2. A **multiple regression model** is given by

$$Y = B_0 + B_1X_1 + B_2X_2 + \dots + B_pX_p + e$$

where Y is the dependent variable and $X_1, X_2, \dots, X_p$ are p independent variables. This is called the multiple linear regression model with p independent/regressor variables. The term linear is used because the dependent/response variable Y is a linear function of the unknown parameters $B_0, B_1, B_2, \dots, B_p$.

3. The simple regression model considered in Unit 9 becomes a particular case of this model with $X_0 = 1, B_0 = a, B_1 = b$ and $B_i = 0$ ($i \geq 2$). The interpretation of coefficients B_j ($j = 1, 2, \dots, p$) is that B_j represents the amount of change in Y for a unit change in X_j , keeping the other independent variables X_k ($k \neq j$) fixed. These coefficients are known as **partial regression coefficients** as the effect of one independent variable is studied on the dependent variable while the other variables are held fixed or constant. We use the term multiple linear regression for this model because several variables are included in the regression and the parameters $B_0, B_1, \dots, B_p$ appear in a linear form.
4. We estimate the parameters of a multiple linear regression equation using the method of least squares. In this method, we minimise the total error term, so that the sum of the squares of the differences between the observed values Y_i and its expected values is minimum, i.e., the sum of squares of the error terms is minimum.

When p is greater than 2, it is more convenient to write the normal equations in matrix form. The regression equations in matrix notation can be written as $Y = X B + e$, where Y is an $n \times 1$ vector of the observed values of the response variable Y, X is a $(p + 1) \times n$ matrix of the values of regressor variables, B is a $(p + 1) \times 1$ vector of regression coefficients and e is an $n \times 1$ vector of random errors. In matrix notation, the $(p + 1)$ normal equations can be written as

$$X'X\hat{B} = X'Y$$

5. The variance-covariance matrix of $\hat{B}$ is given by

$$V(\hat{B}) = \sigma^2 (X'X)^{-1}$$

where $V(\hat{B}) = \sigma^2 (X'X)^{-1}$ is a $(p + 1) \times (p + 1)$ matrix and its diagonal elements give the variances of coefficients and off diagonal elements give the covariances. If we use the notation

$$V(\hat{B}) = \sigma^2 (X'X)^{-1} = (\sigma_{jk}), \quad j, k = 0, 1, \dots, p.$$

we can write

$$V(\hat{B}_j) = \sigma_{jj}, \quad \text{and} \quad \text{Cov}(\hat{B}_j, \hat{B}_k) = \sigma_{jk}$$

The standard error of $\hat{B}_j$ is given by

$$\text{S.E.}(\hat{B}_j) = \sqrt{\sigma_{jj}}$$

6. If there is a linear relationship between the response variable Y and any of the independent variables $X_1, X_2, \dots, X_p$, we use the **test of significance of regression**. The test of significance of regression is a test to determine the linear relationship between the response variable and regressor variables and is often used to examine the adequacy of the model.
7. The **coefficient of determination**, R^2 and adjusted R^2 are measures of goodness of fit of the multiple regression model. The value of R^2 always increases with p . The increase may be negligible, but R^2 never decreases. When we compare two models with different values of p , the model with larger p is preferable if R^2 corresponding to it is significantly larger than R^2 with smaller p . A model with smaller p with large R^2 is always preferable as it is a simple model. Hence, one should choose a model with small p if its R^2 is not much smaller than R^2 for a model with a larger p .
8. We define dummy variables to account for the effect that qualitative variables may have on the response variable. Dummy variables are also known as **categorical** as **indicator variables**. Suppose, k represents the number of levels a categorical variable takes, then we define $(k - 1)$ dummy variables. For example, if we have two categories, male or female, in the data, $k = 2$ and we define one dummy variable.

11.8 SOLUTIONS/ANSWERS

E1) We do the following calculations for the given data:

Time (Y)	Distance (X ₁)	Vending Time (X ₂)	Y ²	(X ₁) ²	(X ₂) ²	X ₁ Y	X ₂ Y	X ₁ X ₂
18	61	30	324	3721	900	1098	540	1830
14	95	25	196	9025	625	1330	350	2375
17	72	30	289	5184	900	1224	510	2160
14	84	25	196	7056	625	1176	350	2100
13	98	10	169	9604	100	1274	130	980
24	53	35	576	2809	1225	1272	840	1855
13	68	15	169	4624	225	884	195	1020
22	54	40	484	2916	1600	1188	880	2160
12	89	30	144	7921	900	1068	360	2670
19	73	20	361	5329	400	1387	380	1460
$\sum Y_i$ = 166	$\sum X_{1i}$ = 747	$\sum X_{2i}$ = 260	$\sum Y_i^2$ = 2908	$\sum X_{1i}^2$ = 58189	$\sum X_{2i}^2$ = 7500	$\sum X_{1i} Y_i$ = 11901	$\sum X_{2i} Y_i$ = 4535	$\sum X_{1i} X_{2i}$ = 18610

From the above table, putting the values of $\sum Y_i$, $\sum X_{1i}$, $\sum X_{2i}$, $\sum X_{1i}^2$, $\sum X_{2i}^2$, $\sum X_{2i} Y_i$, $\sum X_{1i} Y_i$ and $\sum X_{1i} X_{2i}$ in normal equations, we get

$$10 \hat{B}_0 + 747 \hat{B}_1 + 260 \hat{B}_2 = 166 \quad \dots (i)$$

$$747 \hat{B}_0 + 58189 \hat{B}_1 + 18610 \hat{B}_2 = 11901 \quad \dots (ii)$$

$$260 \hat{B}_0 + 18610 \hat{B}_1 + 7500 \hat{B}_2 = 4535 \quad \dots (iii)$$

Solving these equations, we get

$$\hat{B}_0 = 26.7569, \hat{B}_1 = -0.1729 \text{ and } \hat{B}_2 = 0.1062$$

The fitted regression equation is:

$$Y = 26.7569 - 0.1729 X_1 + 0.1062 X_2$$

E2) Using the matrix notation, we have from the data:

$$Y'_{1 \times 10} = [18, 14, 17, 14, 13, 24, 13, 22, 12, 19]$$

$$X'X = \begin{pmatrix} 10 & 747 & 260 \\ 747 & 58189 & 18610 \\ 260 & 18610 & 7500 \end{pmatrix}, \quad X'Y = \begin{pmatrix} 166 \\ 11901 \\ 4535 \end{pmatrix}$$

$$(X'X)^{-1} = \begin{pmatrix} 8.1316 & -0.0690 & -0.1108 \\ -0.0690 & 0.0007 & 0.0007 \\ -0.1108 & 0.0007 & 0.0022 \end{pmatrix}$$

$$\text{and } \begin{pmatrix} \hat{B}_0 \\ \hat{B}_1 \\ \hat{B}_2 \end{pmatrix} = (X'X)^{-1} X'Y = \begin{pmatrix} 26.7569 \\ -0.1729 \\ 0.1062 \end{pmatrix}$$

Hence, the fitted equation is

$$Y = 26.7569 - 0.1729 X_1 + 0.1062 X_2$$

We now calculate the value of residual sum of squares to obtain an estimate of $\hat{\sigma}^2$ as follows:

$$\begin{aligned} SS_{\text{Res}} &= Y'Y - Y'X\hat{B} \\ &= 2908 - 166 \times (26.7569) - 11901 \times (-0.1729) - 4535 \times (0.1062) \\ &= 2908 - 4441.6454 + 2057.6829 - 481.617 = 42.4205 \end{aligned}$$

Therefore, on putting the value of SS_{Res} in equation (18), we get

$$\hat{\sigma}^2 = 42.4205 / (10 - 3) = 6.06$$

E3) Using the data of E1 and the results of E2, we get

$$\hat{B}_0 = 26.7569, \hat{B}_1 = -0.1729 \text{ and } \hat{B}_2 = 0.1062$$

As per the data given in Example 1 and the result of Example 2, we have

$$SS_{Res} = 42.4205 \text{ and } \sum_{j=1}^p \hat{B}_j Y' X_j = 2865.5795$$

Using these values we construct the ANOVA table as follows:

ANOVA TABLE

Sources of Variation	Degree of Freedom (d.f.)	Sum of Squares (S.S.)	Mean Sum of Squares	Variance Ratio
Independent Variables (X_1, X_2)	2	$SS_{Reg} = \sum_{j=1}^p \hat{B}_j Y' X_j - n \bar{Y}^2$ $= 109.97956$	$\frac{SS_{Reg}}{2} = 54.9897$	$F = \frac{SS_{Reg}/p}{SS_{Res}/(n-p-1)}$ $= 9.074$
Residuals (SS_{Res})	7	$SS_{Res} = Y'Y - \sum_{j=1}^p \hat{B}_j Y' X_j$ $= 42.4205$	$\frac{SS_{Res}}{(n-p-1)} = 6.06$	
Total	9	$Y'Y - n \bar{Y}^2 = 152.4$		

The calculated value of Variance Ratio $F = 9.074$, whereas the tabulated value of $F_{2,22}$ at $\alpha = 0.05$ is 3.44. Hence, we reject H_0 and conclude that X_1 and X_2 contribute significantly in explaining the variability.

It may be of further interest to examine whether the coefficient B_j , corresponding to independent variable X_j , is different from zero, after accounting for other variables X_k (all $k \neq j$). This can be tested by considering statistic t :

$$t = \frac{\hat{B}_j}{S.E.(\hat{B}_j)}$$

From the result of Example 2, we have

$$\hat{B}_1 = -0.1729 \text{ and } \hat{B}_2 = 0.1062$$

The Variance-Covariance matrix is

$$V(\hat{B}) = \hat{\sigma}^2 (X'X)^{-1}$$

$$\text{Thus } V(\hat{B}) = \begin{pmatrix} 49.7393 & -0.4218 & -0.6777 \\ -0.4218 & 0.0041 & 0.0045 \\ -0.6777 & 0.0045 & 0.0132 \end{pmatrix}$$

Using equation (15), we obtain

$$V(\hat{B}_0) = 49.7393, \quad V(\hat{B}_1) = 0.0041 \quad \text{and} \quad V(\hat{B}_2) = 0.0132$$

and therefore,

$$\text{S.E.}(\hat{B}_0) = \sqrt{49.7393} = 7.0526$$

$$\text{S.E.}(\hat{B}_1) = \sqrt{0.0041} = 0.064$$

$$\text{S.E.}(\hat{B}_2) = \sqrt{0.0132} = 0.11489$$

Therefore, the statistic t is given as:

$$t_0 = \frac{\hat{B}_0}{\text{S.E.}(\hat{B}_0)} = \frac{26.7569}{7.0526} = 3.6521$$

$$t_1 = \frac{\hat{B}_1}{\text{S.E.}(\hat{B}_1)} = \frac{-0.1729}{0.064} = -2.7014$$

$$t_2 = \frac{\hat{B}_2}{\text{S.E.}(\hat{B}_2)} = \frac{0.1062}{0.11489} = 0.924$$

But the tabulated value of t-statistic for $\alpha = 0.05$ is

$$t_{0.025,7} = 2.37$$

Hence, variable X_1 contributes significantly in explaining the variability in Y but the variable X_2 does not.

As far as the interpretation of coefficients is concerned, there is an increase of 0.1062 seconds in time for one unit change in cases (X_1). Similarly, for one unit increase in X_2 , there is a 0.1729 seconds decrease in time.

E4) Using the data of E1) and the results of E2) and E3) , we get

$$\begin{aligned} R^2 &= \text{Sum of Squares due to } X_1, X_2 / \text{Total Sum of Squares} \\ &= 109.58/152.4 = 0.719 \end{aligned}$$

$$\begin{aligned} R^2_{\text{Adj}} &= 1 - \frac{(n-1)(1-R^2)}{(n-p-1)} \\ &= 1 - \frac{9(1-0.719)}{7} = 0.6387 \end{aligned}$$

R^2 indicates that only 72% of variability in Y is explained by X_1 and X_2 .

E5) Two types of washing machines A and B have been used. Hence, $k = 2$. Here we have to define one dummy variable X_2 , which takes two values:

$$\begin{aligned} X_2 &= 0 \quad \text{if the observation is from machine type 'A'} \\ &= 1 \quad \text{if the observation is from machine type 'B'} \end{aligned}$$

From the given data, we form the following table to find and fit the regression equation:

Time Y	Distance (X ₁)	(X ₂)	Y ²	(X ₁) ²	(X ₂) ²	X ₁ Y	X ₂ Y	X ₁ X ₂
18	61	0	324	3721	0	1098	0	0
14	95	0	196	9025	0	1330	0	0
17	72	0	289	5184	0	1224	0	0
14	84	0	196	7056	0	1176	0	0
13	98	0	169	9604	0	1274	0	0
24	53	1	576	2809	1	1272	24	53
13	68	1	169	4624	1	884	13	68
22	54	1	484	2916	1	1188	22	54
12	89	1	144	7921	1	1068	12	89
19	73	1	361	5329	1	1387	19	73
$\sum Y_i$ =166	$\sum X_{1i}$ =747	$\sum X_{2i}$ = 05	$\sum Y_i^2 =$ 2908	$\sum X_{1i}^2$ = 58189	$\sum X_{2i}^2$ =05	$\sum X_{1i}Y_i$ = 11901	$\sum X_{2i}Y_i$ = 90	$\sum X_{1i}X_{2i}$ =337

From the above table, putting the values in the normal equations (5) for $p = 2$ and noting that $X_0 = 1$, we get

$$10 \hat{B}_0 + 747 \hat{B}_1 + 05\hat{B}_2 = 166 \quad \dots (i)$$

$$747 \hat{B}_0 + 58189\hat{B}_1 + 337 \hat{B}_2 = 11901 \quad \dots (ii)$$

$$05 \hat{B}_0 + 337 \hat{B}_1 + 05 \hat{B}_2 = 90 \quad \dots (iii)$$

From equation (iii), we have

$$\hat{B}_0 = \frac{90 - 337 \hat{B}_1 - 05\hat{B}_2}{05} \quad \dots (iv)$$

On putting the value of $\hat{B}_0$ in equations (i) and (ii) and simplifying, we get

$$365 \hat{B}_1 - 25 \hat{B}_2 = -70 \quad \dots (v)$$

$$39206 \hat{B}_1 - 2050 \hat{B}_2 = -7135 \quad \dots (vi)$$

On solving equations (v) and (vi), we get

$$\hat{B}_1 = -0.214, \quad \hat{B}_2 = -0.3244$$

and
$$\hat{B}_0 = \frac{90 - 337(-0.214) - 05(-0.3244)}{05} = 32.748$$

Hence the fitted equation for the model given in equation (27) is

$$Y = 32.748 - 0.214 X_1 - 0.3244 X_2 \quad \dots (vii)$$

Now we can conclude that there is a linear relationship between Y (time) and the two independent variables X_1 (distance) and X_2 (type of machines). As the regression coefficient for the variable X_1 is negative, it affects the delivery time.

To determine whether the regression coefficients in this model are correct, we first consider machine A for which X_2 takes value 0. We put the values of regression coefficients in equation (28). Then the regression model becomes

$$Y = 32.748 - 0.214X_1 \quad \dots \text{(viii)}$$

For machine B, we put the value of regression coefficients and $X_2 = 1$. Then the regression model becomes

$$Y = 31.126 - 0.214X_1 \quad \dots \text{(ix)}$$

Note that as discussed in Sec 11.5, these estimated regression lines have the same slope, i.e., -0.214 , but different intercepts, i.e., 32.748 and 31.126 .

UNIT 12 SELECTION OF VARIABLES AND TESTING MODEL ASSUMPTIONS

Structure

- 12.1 Introduction
 - Objectives
- 12.2 Extra Sum of Squares
- 12.3 Selection of Independent Variables
 - Forward Selection Method
 - Backward Elimination Method
 - Stepwise Selection Method
- 12.4 Testing Model Assumptions
 - Multicollinearity
 - Outlying Observations
 - Fitting the Correct Model
 - Homogeneity of Variances
 - Normality of Data
- 12.5 Summary
- 12.6 Solutions/Answers

12.1 INTRODUCTION

In Unit 11, we have discussed the multiple linear regression model and estimation of model parameters. You have learnt how to identify the important independent variables which contribute to the variation in the dependent variables. You have estimated their importance by estimating regression coefficients and their standard errors. You have also learnt how to examine the overall fit of the model by determining the coefficient of determination R^2 and Adjusted R^2 .

In this unit, we discuss how to build a regression model that explains the variability in the dependent (response) variable Y . We assume that p independent variables $X_1, X_2, \dots, X_p$ are available for inclusion in the model and we search for a set of variables that explains the variability in Y . Since we have to select a subset of these variables, we take care that the procedure of choosing it is such that the model is a good fit and the numbers of parameters is as small as possible. In Sec. 12.2, we discuss the test of significance for the extra sum of squares. In Sec. 12.3, we describe three selection procedures, namely, forward selection, backward elimination and stepwise selection procedures for selecting variables. We discuss various aspects of testing the model assumptions to check for multicollinearity, outlying observations, homogeneity of variances and normality of data in Sec.12.4.

Objectives

After studying this unit, you should be able to:

- calculate the extra sum of squares of the additional variable(s) in the model;
- test the usefulness of including additional variable(s) in the model;

- describe the forward selection, backward elimination and stepwise selection procedures for selecting variable(s);
- select an adequate model by using the methods of selection of independent variables; and
- test the model assumptions for multicollinearity, normality, outlying variables (outliers), homogeneity.

12.2 EXTRA SUM OF SQUARES

In the previous unit, you have learnt that in a regression model, the independent variables are responsible for satisfactory explanation of variability in the dependent (response) variable Y . If p independent variables $X_1, X_2, \dots, X_p$ are available for inclusion in the model, each and every variable contributes to explanation of variability in the response variable Y in the model.

In this section, we discuss how to determine the variability accounted for by the variable which enters in the model at a later stage when one or more variables have already been included. This additional variability accounted for by the variable entered at a later stage over the variability accounted for by the variables already included in the model is called the **extra sum of squares**.

Suppose we have three independent variables X_1, X_2 , and X_3 . We can fit the following four types of models:

Model 1: $Y = B_0 + e$ which has only an intercept.

Model 2: $Y = B_0 + B_1X_1 + e$ which has one independent variable X_1 .

Model 3: $Y = B_0 + B_1X_1 + B_2X_2 + e$ which has two independent variables X_1 and X_2 .

Model 4: $Y = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + e$ which has three independent variables X_1, X_2 and X_3 .

We first fit Model 1 with only the intercept:

$$Y = B_0 + e \quad \dots(1)$$

The sum of squares due to the intercept B_0 is $n\bar{Y}^2$ and the residual sum of squares SS_{Res} is given by

$$SS_{\text{Res}}(B_0) = \sum (Y_i - \bar{Y})^2 = \sum Y_i^2 - n\bar{Y}^2 = \sum Y_i^2 - SS(B_0) \quad \dots (2)$$

Next we fit Model 2. Let us denote the sum of squares due to B_0 and B_1 by $SS(B_0, B_1)$. The residual sum of squares is given by

$$SS_{\text{Res}}(B_0, B_1) = \sum Y_i^2 - SS(B_0, B_1) \quad \dots (3)$$

Hence, the additional (extra) sum of squares accounted for by B_1 is

$$SS(B_1|B_0) = SS_{\text{Res}}(B_0) - SS_{\text{Res}}(B_0, B_1) = SS(B_0, B_1) - SS(B_0) \quad \dots (4)$$

If we have an estimate of σ^2 , i.e., $\hat{\sigma}^2$, we can test whether it is worthwhile to include X_1 in the model. In other words, we test whether $B_1 = 0$ or not, i.e.,

$$H_0: B_1 = 0 \quad \text{against} \quad H_1: B_1 \neq 0$$

If H_0 is accepted, $SS(B_1|B_0)$ is not significant. Here $SS(B_1|B_0)$ is the **extra sum of squares** accounted for by B_1 (or X_1).

We proceed in this way and fit Model (3). We find the sum of squares accounted for by B_0, B_1 and B_2 , i.e., $SS(B_0, B_1, B_2)$. We also calculate the extra sum of squares due to B_2 after accounting for B_0 and B_1 , i.e., $SS(B_2|B_0, B_1)$ by calculating $SS_{Res}(B_0, B_1, B_2)$ as follows:

$$SS_{Res}(B_0, B_1, B_2) = \sum Y_i^2 - SS(B_0, B_1, B_2) \quad \dots (5)$$

and

$$\begin{aligned} SS(B_2|B_0, B_1) &= SS_{Res}(B_0, B_1) - SS_{Res}(B_0, B_1, B_2) \\ &= SS(B_0, B_1, B_2) - SS(B_0, B_1) \end{aligned} \quad \dots (6)$$

and so on. At each stage we can test for the significance of the extra sum of squares by comparing it with $\hat{\sigma}^2$.

Note from equation (6) that the extra sum of squares depends on the variables, which have been accounted for. For example, in general, $SS(B_2|B_0, B_1)$ is not the same as $SS(B_1|B_0, B_2)$.

The test of significance of the extra sum of squares helps us in deciding whether it is worthwhile to include the additional variable and thereby an additional parameter or not. Let us try to understand this concept with the help of an example.

Example 1: A statistician collected data of 25 values with two independent variables X_1 and X_2 . The following four models were considered one at a time:

1. $Y = B_0 + e$
2. $Y = B_0 + B_1X_1 + e$
3. $Y = B_0 + B_2X_2 + e$
4. $Y = B_0 + B_1X_1 + B_2X_2 + e$

The results obtained were:

$$\begin{aligned} \hat{B}_0 &= 22.38, \hat{B}_1 = 1.6161, \hat{B}_2 = 0.0144, SS(B_0) = 12526.08, \\ SS(B_0, B_1) &= 17908.47, SS(B_0, B_2) = 17125.23, SS(B_0, B_1, B_2) = 18079.0, \\ \hat{\sigma}^2 &= 10.53, SE(\hat{B}_1) = 0.17 \text{ and } SE(\hat{B}_2) = 0.0035. \end{aligned}$$

Find the additional contribution of (i) X_2 over X_1 and (ii) X_1 over X_2 . Test whether their inclusion in the model is justified or not.

Solution: Using the results given above, we calculate the extra sum of squares due to X_1 after accounting for X_0 :

$$\begin{aligned} SS(B_1|B_0) &= SS(B_0, B_1) - SS(B_0) \\ &= 17908.47 - 12526.08 = 5382.39 \end{aligned}$$

The extra sum of squares due to X_2 after accounting for X_0 is

$$\begin{aligned} SS(B_2|B_0) &= SS(B_0, B_2) - SS(B_0) \\ &= 17125.23 - 12526.08 = 4599.15 \end{aligned}$$

The extra sum of squares due to X_2 after accounting for X_0 and X_1 is:

$$\begin{aligned} SS(B_2 | B_0, B_1) &= SS(B_0, B_1, B_2) - SS(B_0, B_1) \\ &= 18079.00 - 17908.47 = 170.53 \end{aligned}$$

The extra sum of squares due to X_1 and X_2 after accounting for X_0 is

$$\begin{aligned} SS(B_1, B_2 | B_0) &= SS(B_0, B_1, B_2) - SS(B_0) \\ &= 18079.00 - 12526.08 = 5552.92 \end{aligned}$$

The extra sum of squares due to X_1 after accounting for X_0 and X_2 is

$$\begin{aligned} SS(B_1 | B_0, B_2) &= SS(B_0, B_1, B_2) - SS(B_0, B_2) \\ &= 18079.00 - 17125.23 = 953.77 \end{aligned}$$

With only one term B_0 in the model, i.e., for Model 1, we see that the sum of squares due to X_0 is given by

$$SS(B_0) = 12526.08$$

If we add one more variable (X_1) to the model, (Model 2), the sum of squares due to X_0 and X_1 is

$$SS(B_0, B_1) = 17908.47$$

and the extra sum of squares accounted for by B_1 over B_0 is

$$SS(B_1 | B_0) = 5382.39$$

If we test against $\hat{\sigma}^2 = 10.53$, the calculated value of F-statistic is

$$F = 5382.39 / 10.53 = 511.48$$

We compare this calculated value with the tabulated value $F_{(1,22)}$ for $\alpha = 0.05$, which is 4.30. Hence, the term is highly significant. This means that the contribution of X_1 is significant.

After X_1 has been included, it is of interest to see whether X_2 contributes significantly to the model. The additional or extra sum of squares due to Model 3 over Model 2, i.e., the extra sum of squares due to X_2 after accounting for X_0 and X_1 is given by

$$SS(B_2 | B_0, B_1) = 170.53$$

If we test against $\hat{\sigma}^2 = 10.53$, the calculated value of F-statistic

$$F = 170.53 / 10.53 = 16.19$$

On comparing it with the tabulated value $F_{(1,22)}$ for $(\alpha = 0.05) = 4.30$, we find that the contribution of X_2 after X_1 is significant and it has to be included in the model for it to be worthwhile.

We may also be interested in testing for the contribution of X_1 when X_2 is already included in the model. The contribution in terms of the extra sum of squares due to X_1 after accounting for X_0 and X_2 is given by

$$SS(B_1 | B_0, B_2) = 953.77$$

If we test against $\hat{\sigma}^2 = 10.53$, the calculated value of F-statistic is

$$F = 953.77 / 10.53 = 90.58$$

This is also significant.

Hence, it is worthwhile to include both X_1 and X_2 in the model. Therefore, the final model is

$$Y = B_0 + B_1X_1 + B_2X_2 + e$$

You may now like to solve the following exercises which will help you to assess your understanding of Sec.12.2.

-
- E1)** A statistician collected data of 78 values with two independent variables X_1 and X_2 , the four models considered are the same as in Example 1 and the results are: $SS(B_0) = 652.42$, $SS(B_0, B_1) = 679.34$, $SS(B_0, B_2) = 654.00$, $SS(B_0, B_1, B_2) = 687.79$ and $\hat{s}^2 = 0.91$.

Find the additional contribution of (i) X_2 over X_1 and (ii) X_1 over X_2 .
Test whether their inclusion in the model is justified.

12.3 SELECTION OF INDEPENDENT VARIABLES

In Unit 11, you have learnt that the regressor variables included in the multiple regression model are considered to be important since the variability in the response variable is accounted for only by them. So far we have focused on the techniques to ensure that the functional form of the fitted model is correct and the underlying assumptions are not violated.

Along with the above considerations and assumptions we have set the following main objectives to obtain an adequate multiple regression model:

- i) to cover as much information as possible in the model by including as many regressor variables as possible, and
- ii) to increase the variability accounted for by the regressor variables by including only a few important regressor variables, so that fitted model is simple and adequate.

These two objectives contradict each other since adding a large number of regressor variables increases the accounted variability, and selecting only a few of them may reduce the accounted variability and may cause loss of useful information.

The problem of building an adequate model by striking a balance between both these objectives is not trivial. In multiple linear regression, we determine a group of regressor variables, which are proved to be important significantly, to find an adequate model. In some cases, theoretical considerations and personal experience may be helpful in selecting the important regressor variables to be included in the model. However, in most cases, the appropriate subset of the important variables to be included in the model needs to be examined. The methods of finding an appropriate subset of regressor variables, which adequately explain the variability in the response variable and need to be included in the model are called **variable selection methods**.

When a number of independent variables $X_1, X_2, \dots, X_p$ are available, we may be interested in retaining only a few of them in the model, without sacrificing much of the explanatory power of the model. In this section, we describe some methods that help us in selecting important variables, which adequately explain the variability in the dependent variable Y . The main criterion is that the number of parameters should be as small as possible and all the variables which contribute significantly should be included in the model. When a variable X_i is included in the model, then any other variable, say, X_k , which is

highly correlated with X_i , need not be included as its contribution has already been accounted for by X_i . In the same way if X_k is included in the model first, then X_i may not be included as its contribution has already been taken into account by X_k . Thus, the order in which variables enter in the model is important.

We now describe three methods for selection of variables and model building: The forward selection method, the backward elimination method and the stepwise selection method.

12.3.1 Forward Selection Method

In this method, the most important variable is included first. The most important variable is the one which accounts for a maximum sum of squares (SS). After including this variable, the next most important variable is chosen which accounts for a maximum *additional* sum of squares. The contribution is tested at each stage and the most important variable, which accounts for a maximum additional sum of squares is included in the model. When the maximum contribution is not significant, the process is stopped. Otherwise, the process continues by including the variables with significant contributions.

In the forward selection method, variables are added to the model one at a time until the addition of a variable does not significantly improve the model. Thus, variables are added one at a time until further addition of a variable results in the acceptance of the reduced model.

Let us explain this method for two variables.

Suppose we have two possible regressor variables X_1 and X_2 , which are linearly related with the response variable Y . The steps followed to obtain the adequate final model using forward selection method are explained below:

We start with two one variable models:

$$Y = B_0 + B_1 X_1 \quad \dots (i)$$

$$Y = B_0 + B_2 X_2 \quad \dots (ii)$$

We fit these models to the given data. The value of the coefficient of determination (R^2) is calculated for each model. The one with the highest R^2 is chosen and compared to the reduced model:

$$Y = B_0$$

Let the model $Y = B_0 + B_2 X_2$ have larger R^2 . In this case, we test

$$H_0: B_2 = 0 \text{ (reduced model } Y = B_0 \text{ is appropriate)}$$

$$H_1: B_2 \neq 0 \text{ (model } Y = B_0 + B_2 X_2 \text{ is needed)}$$

If H_0 is rejected, the variable X_2 is included in the model. Otherwise, it is not included.

We now try to include the variable X_1 . For this, we fit the two variable model to the given data:

$$Y = B_0 + B_1 X_1 + B_2 X_2$$

The value of the coefficient of determination (R^2) is calculated for the model $Y = B_0 + B_1 X_1 + B_2 X_2$ and compared to the reduced model $Y = B_0 + B_2 X_2$.

At this stage we again test

$H_0: B_1 = 0$ (reduced model $Y = B_0 + B_2 X_2$ is appropriate)

$H_1: B_1 \neq 0$ (model $Y = B_0 + B_1 X_1 + B_2 X_2$ is needed)

If H_0 is rejected, at this stage the variable X_1 is included in the model. The final model we obtain is the two variable model:

$$Y = B_0 + B_1 X_1 + B_2 X_2$$

12.3.2 Backward Elimination Method

In this method, all p variables are included in the model. Then the least important variable is deleted if its contribution to the sum of squares is not significant. At the second stage, the next least important variable whose contribution is not significant is deleted. This process goes on till a variable cannot be deleted from the model.

In backward elimination method, we begin with the model that includes all potential regressor or independent variables. Variables are deleted from the model one at a time until further deletion of a variable results in the rejection of the reduced model.

It is considered to be an economical procedure as it attempts to examine only the best regression equation containing a certain number of variables.

Suppose we have two potential regressor or independent variables X_1 and X_2 . We obtain a best fitted model using the backward elimination method as follows:

First of all, we fit a full model with two regressor variables X_1 and X_2 :

$$Y = B_0 + B_1 X_1 + B_2 X_2$$

Then we determine the value of R^2 .

Now, we reduce the model by discarding one variable at a time. Suppose the two reduced models obtained on discarding the variables X_1 and X_2 one at a time are, respectively, given by

$$Y = B_0 + B_1 X_1$$

$$Y = B_0 + B_2 X_2$$

The value of R^2 and the extra sum of squares is obtained for both the reduced models and the model with higher extra sum of squares and R^2 is chosen and compared with the full model.

Let the reduced model $Y = B_0 + B_2 X_2$ have a higher value of R^2 . Then, in this case, we test

$H_0: B_1 = 0$ (reduced model $Y = B_0 + B_2 X_2$ is adequate)

$H_1: B_1 \neq 0$ (full model $Y = B_0 + B_1 X_1 + B_2 X_2$ is needed)

If H_0 is accepted, we delete the regressor variable X_1 from the model since it appears to be unnecessary.

Now to test the significance of the variable X_2 in model $Y = B_0 + B_2 X_2$, we fit the model without any regressor variable, i.e.,

$$Y = B_0$$

We calculate R^2 and test it against the model with only variable X_2 , i.e.,

$$H_0: B_2 = 0 \text{ (reduced model } Y = B_0 \text{ is adequate)}$$

$$H_1: B_2 \neq 0 \text{ (model } Y = B_0 + B_2 X_2 \text{ is needed)}$$

If H_0 is rejected, we are left with the model that contains one regressor variable X_2 , i.e.,

$$Y = B_0 + B_2 X_2$$

12.3.3 Stepwise Selection Method

The stepwise selection method begins with selecting an equation containing the single best independent variable X . Then, the second most important variable is included in the model. But at this stage, each of the variables selected earlier in the model are tested for their significance. If they are found to be significant, then we try to build an equation with the subsequent inclusion of other regressor variables one at a time. These additional variables are included as long as their inclusion is significant. A variable is excluded from the model if found insignificant. The selection process continues till no additional variable can be included or deleted in the model.

The order of addition is determined by using a test of significance based on partial F-test values. In order to decide the next variable to be included, the highest partial F-value is compared with the selected F-tabulated value (for the corresponding variable to be included). After a variable has been included, the fitted equation is examined to check whether some variable(s) can be deleted.

Stepwise selection method is a modified forward selection method. In the forward selection method, once a variable enters the model, it stays. Unfortunately, it is possible for a variable entering at a later stage to render a previously selected variable unimportant when these variables are themselves closely related. The forward selection method does not consider this possibility. In stepwise selection method, each time a new variable is entered into the model, all previous variables in the model are checked for continued importance.

It may be noted here that these procedures may not select a unique model. There could be several models which may explain the variability adequately and could be considered equivalent.

Suppose we have two regressor variables X_1 and X_2 and both are closely related. Let X_2 be the best single predictor. We now explain the stepwise selection method for this problem.

First of all, we fit two single variable models to the given data:

$$Y = B_0 + B_1 X_1$$

$$Y = B_0 + B_2 X_2$$

The value of R^2 is computed for each model and the model with larger R^2 and extra sum of squares is chosen and compared with the model with no

variables, i.e., $Y = B_0$. Let the selected model be $Y = B_0 + B_1 X_1$. We test this model with the reduced model as follows:

$$H_0 : B_1 = 0 \text{ (reduced model } Y = B_0 \text{ is adequate)}$$

$$H_1 : B_1 \neq 0 \text{ (selected model } Y = B_0 + B_1 X_1 \text{ is needed)}$$

If H_0 is rejected, we are left with the model that contains one regressor variable X_1 , i.e.,

$$Y = B_0 + B_1 X_1$$

Now we fit the model with two variables X_1 and X_2 , i.e.,

$$Y = B_0 + B_1 X_1 + B_2 X_2$$

and calculate R^2 and the extra sum of squares for this model. We test this model against the previously selected model as follows:

$$H_0 : B_2 = 0 \quad (\text{model } Y = B_0 + B_1 X_1 \text{ is adequate})$$

$$H_1 : B_2 \neq 0 \quad (\text{model } Y = B_0 + B_1 X_1 + B_2 X_2 \text{ is needed})$$

Let H_0 be rejected. This means that X_2 is to be entered in the model.

At this stage we check the importance of X_1 in this model. Therefore, we test the full model against the model with X_2 as the only variable, i.e.,

$$H_0 : B_1 = 0 \quad (\text{model } Y = B_0 + B_2 X_2 \text{ is needed})$$

$$H_1 : B_1 \neq 0 \quad (\text{model } Y = B_0 + B_1 X_1 + B_2 X_2 \text{ is adequate})$$

If H_0 is not rejected, it means that after entering the variable X_2 into the model, the variable X_1 does not have any importance. Therefore, we delete the variable X_1 from the model.

We end this section by applying all three methods to a problem.

Example 2: Consider the results of Example 1. Apply all three selection procedures to choose an appropriate model.

Solution: From the results of Example 1, we have

$$SS_{\text{Res}} = \sum Y_i^2 - \hat{B}' Y' X = 231.63$$

Therefore, on putting the value of SS_{Res} in the expression of $\hat{\sigma}^2$, we get

$$\hat{\sigma}^2 = 231.63 / (25 - 3) = 10.529$$

Using the results of Example 1, we get

$$\hat{B}_0 = 22.38, \quad \hat{B}_1 = 1.6161 \quad \text{and} \quad \hat{B}_2 = 0.0144$$

Using the above results along with those of Example 1 of this unit, we apply variable selection methods, one at a time, to find the adequate model as follows:

1. Forward Selection Method

We start with a model with only the intercept, i.e.,

$$Y = B_0 + e$$

and try to see if other variables can be added. For this, we fit two models with one variable each:

$$Y = B_0 + B_1 X_1 + e$$

and $Y = B_0 + B_2 X_2 + e$

According to the calculations and the models fitted by including the extra variables X_1 and X_2 , separately and one at a time, the extra sum of squares due to X_1 over X_0 , and X_2 over X_0 are:

$$SS(B_1|B_0) = 5382.39$$

$$SS(B_2|B_0) = 4599.15$$

As the extra sum of squares due to X_1 over X_0 , i.e., $SS(B_1|B_0)$ is larger than the extra sum of squares due to X_2 over X_0 , i.e., $SS(B_2|B_0)$, we test the hypothesis

$$H_0: B_1 = 0 \quad \text{against} \quad H_1: B_1 \neq 0$$

and get the t-statistic value

$$t = \frac{\hat{B}_1}{SE(\hat{B}_1)} = \frac{1.6161}{0.17} = 9.15$$

This value is significant in comparison with the t-tabulated value $t_{(0.5, 22)} = 2.074$. Thus, the first variable to enter in the model is X_1 . Hence at this stage, the selected model is

$$Y = B_0 + B_1X_1 + e$$

We now check whether the variable X_2 can be included in the model. The extra residual sum of squares due to X_2 , after including X_1 , is given by

$$SS(B_2|B_0, B_1) = 170.53$$

Then the value of F-statistic is calculated as

$$F = 170.53/10.53 = 16.19$$

which is significant. Hence, it is worthwhile to include X_2 after X_1 has been included in the model. The final model is

$$Y = B_0 + B_1X_1 + B_2X_2 + e$$

2. Backward Elimination Method

Here we consider the full model

$$Y = B_0 + B_1X_1 + B_2X_2 + e$$

and then try to identify and delete the least important variable if its contribution is not significant. For this, we fit two one variable models

$$Y = B_0 + B_1X_1 + e$$

and

$$Y = B_0 + B_2X_2 + e$$

We calculate the sum of squares due to X_1 and X_2 for the respective models. According to the calculation and model fitted by excluding the extra variables X_1 and X_2 , one at a time, from the model, the sum of squares due to X_0 , X_1 and X_2 , the sum of squares due to X_2 over X_0 and X_1 , and the sum of squares due to X_1 over X_0 and X_2 are given as:

$$SS(B_0, B_1, B_2) = 18079.00$$

$$SS(B_2|B_0, B_1) = 170.53$$

$$SS(B_1|B_0, B_2) = 953.77$$

From the above results, we see that the least important variable is X_2 since the extra sum of squares is least for X_2 . We test its significance against $\hat{\sigma}^2 = 10.53$ and get the F-statistic as:

$$F = 170.53 / \hat{\sigma}^2 = 170.53 / 10.53 = 16.50$$

which is significant since the tabulated value $F_{(1, 22)} = 4.33$ at $\alpha = 0.05$.

We also test the hypothesis

$$H_0: B_2 = 0 \text{ against } H_1: B_2 \neq 0$$

and get the t-statistic value as

$$t = \frac{\hat{B}_2}{SE(\hat{B}_2)} = \frac{0.0144}{0.0035} = 4.11$$

which is significant in comparison to the tabulated value $t_{(0.5, 22)} = 2.074$.

Hence, we cannot delete any variable in the model and the final model is

$$Y = B_0 + B_1X_1 + B_2X_2 + e$$

3. Stepwise Selection Method

Here we start with the model with only the intercept, i.e.,

$$Y = B_0 + e$$

and try to see if regressor variables can be added. For this, we fit two models with one variable

$$Y = B_0 + B_1X_1 + e$$

and

$$Y = B_0 + B_2X_2 + e$$

According to the calculations and the models fitted by including the extra variables X_1 over X_0 separately, one at a time, the extra sum of squares due to X_1 over X_0 , and X_2 over X_0 are as follows:

$$SS(B_1|B_0) = 5382.39$$

$$SS(B_2|B_0) = 4599.15$$

Since the extra sum of squares due to X_1 over X_0 , i.e., $SS(B_1|B_0)$ is larger than the extra sum of squares due to X_2 over X_0 , i.e., $SS(B_2|B_0)$, we now test the hypothesis

$$H_0: B_1 = 0 \text{ against } H_1: B_1 \neq 0$$

and get the t-statistic value as follows:

$$t = \frac{\hat{B}_1}{SE(\hat{B}_1)} = \frac{1.6161}{0.17} = 9.15$$

This value is significant in comparison with the tabulated value $t_{(0.5, 22)} = 2.074$. Thus, the first variable to enter in the model is X_1 . Hence, at this stage the selected model is

$$Y = B_0 + B_1X_1 + e$$

We now check whether the variable X_2 can be included in the model. The extra residual sum of squares due to X_2 , after including X_1 , is given by

$$SS(B_2|B_0, B_1) = 170.53$$

Then we calculate the value of F-statistic:

$$F = 170.53/10.53 = 16.19$$

which is significant. We also test the hypothesis

$$H_0: B_2 = 0 \quad \text{against} \quad H_1: B_2 \neq 0$$

and get the t-statistic value:

$$t = \frac{\hat{B}_2}{SE(\hat{B}_2)} = \frac{0.0144}{0.0035} = 4.11$$

which is significant in comparison with the tabulated value $t_{(0.5, 22)} = 2.074$. Hence, it is worthwhile to include X_2 after X_1 has been included in the model.

At this stage we have to check whether X_1 can be excluded from the model. For this we calculate the sum of squares due to X_1 over X_0 and X_2 , i.e., $SS(B_1|B_0, B_2)$ as follows:

$$SS(B_1|B_0, B_2) = 953.77$$

We now have to test whether this is significant or not. In case it is not significant, we can exclude X_1 from the model. We test its significance against $\hat{\sigma}^2 = 10.53$ and get

$$F = 953.77/10.53 = 90.58$$

which is significant. This suggests that we cannot delete any term from the model. Hence, the selected model is:

$$Y = B_0 + B_1X_1 + B_2X_2 + e$$

Therefore, all three procedures select the same final model.

You may now like to solve the following exercise which will help you assess your understanding of the three methods.

E2) Using the data and the results of E1), apply all three methods of selection of variables and comment on the selected models.

12.4 TESTING MODEL ASSUMPTIONS

So far we have discussed the linear regression model when certain assumptions are satisfied. Many problems arise when we fit the model to actual data and make inferences. There may be situations when the assumptions are violated due to some problems. Then we may need to test the model assumptions and modify the model. In this section, we consider the major problems and briefly discuss methods of overcoming them. Some of the major problems are associated with:

- Multicollinearity;
- Outlying observations;
- Fitting the correct model;
- Homogeneity of variances; and
- Normality of data.

12.4.1 Multicollinearity

This problem often occurs when the number of independent variables is large and some of them are linearly related. When this happens, it is difficult to solve normal equations. The solution of equations is not very stable and estimates are not very accurate. The variables Y and X may show very high correlation but their contribution, as shown by t values of test of significance, may be very small. Eliminating some variables from the model may make a very large change in the coefficients of other variables in the model. In such a situation, the prediction may not be very reliable. To overcome this problem we have to examine the correlations among independent variables and delete the variables which are highly correlated. However, this may not be very easy. In such cases it may be worthwhile to include only those variables which contribute significantly after accounting for the other variables in the model. To a great extent, this problem can be overcome by using an appropriate model selection procedure.

12.4.2 Outlying Observations

Sometimes when we collect observations, we may commit mistakes in recording them. In some cases the variable (Y) may have a large error component e. Such extreme observations have great influence on the estimates $\hat{B}$. In case such observations are removed, the estimates obtained may differ considerably. Therefore, it may be better to remove them from the data under consideration. For detecting such observations, we calculate the residuals $Y - \hat{Y}$ for each observation and standardise them by dividing them by $\hat{\sigma}$. If the absolute value of a standardised residual is larger than 3, we call it an outlying observation. Thus, if

$$\left| \frac{Y - \hat{Y}}{\hat{\sigma}} \right| \geq 3$$

we consider it as an outlying observation. We examine such observations carefully and in some cases we may decide to analyse the data after neglecting such observations. However, in some cases, such observations may suggest that the model itself is not correct and we may revise the model.

12.4.3 Fitting the Correct Model

When we fit a model given by

$$Y = B_0 + B_1 X_1 + \dots + B_p X_p + e \quad \dots (1)$$

we assume that e is a random error with mean zero. However, very often the correct form of the model is not known and the error component e may not behave as random error with zero mean. This may create problem in the estimation of parameters and they may not give unbiased estimates. In such cases, inference obtained from the data may not be correct. Let us consider an example. Suppose a set of data comes from a quadratic model:

$$Y = B_0 + B_1X + B_2X^2 + e \quad \dots (2)$$

However, a wrong model is chosen and a straight line is fitted:

$$Y = B_0 + B_1X + e \quad \dots (3)$$

This means that now the error component e' is

$$e' = B_2X^2 + e \quad \dots (4)$$

It not only contains the error term e but also the term B_2X^2 . Hence, its expected value is given by

$$E(e') = B_2X^2$$

This is not zero but depends upon X . If we estimate B_0 and B_1 using the wrong model (3), then the estimates of B_0 and B_1 may not be unbiased and the fitted model may not be correct.

In case the correct model is used, we expect that the residuals r given by

$$r = Y - \hat{Y}$$

should behave like random error. In case they do not, it suggests that the correct form of the model has not been used. For this, we calculate the residuals r from the proposed model and then plot them against all regressor variables. If we find a random pattern of residuals, we may accept the model. Otherwise, we may modify it by including appropriate terms.

12.4.4 Homogeneity of Variances

One of our assumptions is that the variances of Y are constant. Very often this assumption is violated. Consequently, estimates of parameters are not very efficient. Moreover, the distribution of the test statistic may not follow the desired distribution. Very often the variances of Y may depend upon $E(Y)$ and when $E(Y)$ increases, variances may also increase. For checking this, we plot the residuals against the estimated means $\hat{Y}$ (Fig. 12.1). If the spread of residuals increases or decreases, it indicates that variances may depend on means.

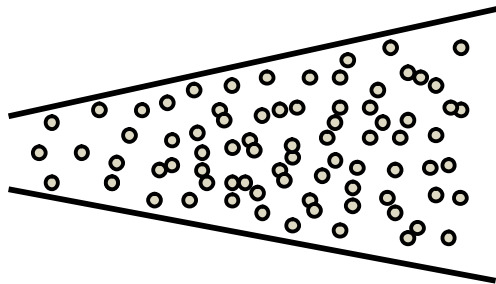


Fig.12.1: Residual plot indicating the increase in variability.

Fig. 12.1 shows that variances are increasing with estimated values of the dependent variable.

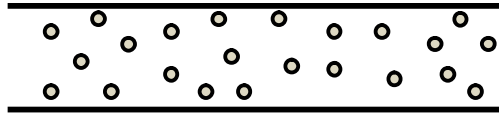


Fig.12.2: Residual plot of constant variability.

Fig. 12.2 shows that variances may be constant (homogeneous). In case variances are not constant, we may have to consider suitable transformation of Y . Sometimes heterogeneity of variances gives an indication that the model may not be a good fit.

12.4.5 Normality of Data

We have also assumed in our models that the observations (Y) are coming from a normal population. If this assumption is violated, the estimation of parameters may not be affected but the inferences regarding the test of hypotheses and confidence intervals are affected. The level of significance (α) and the probability level of confidence interval are affected. Moreover, the level of departure may not be known. For diagnosing their departure, we usually calculate the standardised residuals:

$$d_i = \frac{Y_i - \hat{Y}_i}{\hat{s}}, \quad i = 1, 2, \dots, n$$

These are plotted on a normal probability plot. In case of normality, these residuals lie on a straight line of slope 1. If there is a significant departure from a straight line, it is an indication of departure from normality. Sometimes, the transformation of data with a change of variable using the log, inverse, square root functions may help them to behave like normally distributed variables.

We stop here and summarise the concepts we have discussed in this unit.

12.5 SUMMARY

1. The variability accounted for by the variable which enters in the model at a later stage over the variability accounted for by the variables already present in the model is called the **extra sum of squares**. Three methods are used for choosing important variables from a large number of independent variables: **forward selection method**, **backward elimination method** and **stepwise selection method**.
2. In a regression model, the independent variables are responsible for the satisfactory explanation of variability in the dependent (response) variable Y . If p independent variables $X_1, X_2, \dots, X_p$ are available for inclusion in the model, each and every variable contributes for explanation of variability in the response variable Y in the model. In multiple linear regression, a group of significant regressor variables needs to be selected to find an adequate model.
3. In the forward selection method, variables are added to the model one at a time until the addition of a variable does not significantly improve the model. Thus, variables are added one at a time until further addition of a variable results in the acceptance of the reduced model.

4. In backward elimination method, we begin with the model that includes all potential regression or independent variables. Variables are deleted from the model one at a time until further deletion of a variable results in rejection of the reduced model.
5. Stepwise selection is a modified forward selection method. In forward selection, once a variable enters the model it stays. It is possible for a variable entering at a later stage to render a previously selected variable unimportant. Forward selection method does not consider this possibility. In stepwise regression, each time a new variable is entered into the model, all the variables in the previous model are checked for continued importance.
6. Very often the correct form of the model is not known and the error component may not behave as random error with mean value zero. This may create problems with the estimation of parameters and they may not give unbiased estimates. In such cases inference obtained from the data may not be correct.
7. The residuals plot is used for identifying the departure from assumptions, such as normality and homogeneity of variances.

12.6 SOLUTIONS/ANSWERS

E1) The following extra sums of squares for the data are given as:

$$SS = (B_1, B_2/B_0) = 35.37, \quad SS = (B_1/B_0) = 26.82, \quad SS = (B_2/B_0) = 1.58$$

Therefore, the extra sum of squares due to X_2 over X_0 and X_1 and the extra sum of squares due to X_1 over X_0 and X_2 are

$$SS(B_2/B_0, B_1) = SS(B_1, B_2/B_0) - SS(B_1/B_0) = 8.55$$

$$SS(B_1/B_0, B_2) = SS(B_1, B_2/B_0) - SS(B_2/B_0) = 33.79$$

and the estimated value of σ^2 is $\hat{\sigma}^2 = 0.91$ at 76 d.f.

Testing for X_2 (after X_1):

$$F = SS(B_2/B_0, B_1) / \hat{\sigma}^2 = 9.39$$

where $F_{1,86}$ at $\alpha = 0.05 = 3.95$. Hence, the additional contribution of X_2 over X_1 is significant.

Testing for X_1 (after X_2):

$$F = SS(B_1/B_0, B_2) / \hat{\sigma}^2 = 33.79 / 0.91 = 37.13$$

where $F_{1,86}$ at $\alpha = 0.05 = 3.95$. Hence, the additional contribution of X_1 over X_2 is also significant.

Hence their inclusion in the model is justified.

E2) Forward Selection Method

First we include X_1 as the extra sum of squares due to X_1 over X_0 is 26.82 and is much larger than the extra sum of squares due to X_2 over X_0 , which is 1.58.

After X_1 has been included, the extra sum of squares due to X_2 is

$$SS(B_2|B_0, B_1) = 8.55$$

We now test for X_2 by comparing it with the estimate of σ^2 :

$$F = 8.55 / 0.91 = 9.39$$

It is significant in comparison with $F_{(1, 86)}$ at $\alpha = 0.05$, which is 3.95.

Hence, the selected model should contain both X_1 and X_2 .

Backward Deletion Method

In this method, we consider the full model with X_1 and X_2 :

$$Y = B_0 + B_1X_1 + B_2X_2 + e$$

We now try to delete the variable which accounts for the least sum of squares. We fit the two one variable models by deleting one variable, one at a time

$$Y = B_0 + B_1X_1 + e$$

$$Y = B_0 + B_2X_2 + e$$

We calculate the extra sum of squares due to X_1 and X_2 over X_0 . From E1, we have

$$SS(B_2|B_0, B_1) = 8.55, \quad SS(B_1|B_0, B_2) = 33.79$$

We can observe from the above results that X_2 is the least important variable in the model. Hence, we test for X_2 and calculate the F-statistic:

$$F = 8.55 / 0.91 = 9.39$$

It is significant in comparison with $F_{(1, 86)}$ at $\alpha = 0.05$, which is 3.95.

Hence, we cannot delete X_2 from the model. Thus, both variables X_1 and X_2 are important.

Stepwise Selection Method

In this method, we start with the forward selection method and include X_1 in the model. Then we add X_2 at the second stage and examine whether X_1 can be excluded from the model. For this, we calculate the extra sum of squares due to X_1 after X_2 has been included in the model, i.e., $SS(B_1|B_0, B_2) = 33.79$.

We test for X_1 by comparing it to the estimate of σ^2 and get the F-statistic

$$F = 33.79 / 0.91 = 37.13$$

This is also significant in comparison to $F_{(1, 86)}$ at $\alpha = 0.05$, which is 3.95. Hence, we cannot delete X_1 after X_2 has been selected. Thus, the final model contains both variables X_1 and X_2 .

Hence, all three methods select the model with both variables X_1 and X_2 , i.e.,

$$Y = B_0 + B_1X_1 + B_2X_2 + e$$

Block

4

TIME SERIES MODELLING

UNIT 13

Trend Component Analysis	5
---------------------------------	----------

UNIT 14

Seasonal Component Analysis	33
------------------------------------	-----------

UNIT 15

Stationary Processes	55
-----------------------------	-----------

UNIT 16

Time Series Models	73
---------------------------	-----------

Curriculum and Course Design Committee

Prof. K. R. Srivathsan
Former Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Parvin Sinclair
Former Pro-Vice Chancellor
IGNOU, New Delhi

Prof. Geeta Kaicker
Former Director, School of Sciences
IGNOU, New Delhi

Prof. Jagdish Prasad
Department of Statistics
University of Rajasthan, Jaipur

Prof. R. M. Pandey
Department of Bio-Statistics
AIIMS, New Delhi

Dr. Manish Trivedi
School of Sciences, IGNOU

Dr. Nitin Gupta
School of Sciences, IGNOU

Dr. Neha Garg
School of Sciences, IGNOU

Mr. Rajesh Kaliraman
School of Sciences, IGNOU

Prof. Rahul Roy
Theoretical Statistics and Mathematics Unit
Indian Statistical Institute, New Delhi

Prof. Diwakar Shukla
Department of Mathematics and Statistics
Dr. Hari Singh Gaur University, Sagar

Prof. Rakesh Srivastava
Department of Statistics
M.S. University of Baroda, Vadodara

Prof. G. N. Singh
Department of Applied Mathematics
I.S.M., Dhanbad

Prof. Gulshan Lal Taneja
Department of Mathematics
M.D. University, Rohtak

Prof. Poornima Mital
School of Sciences, IGNOU

Prof. Sujatha Varma
School of Sciences, IGNOU

Dr. S. Venkataraman
School of Sciences, IGNOU

Dr. Deepika
School of Sciences, IGNOU

Block Preparation Team

Prof. Rakesh Srivastava (Editor)
Department of Statistics
M.S. University of Baroda
Vadodara (GUJ)

Prof. Vijayshri (Language Editor)
School of Sciences, IGNOU

Prof. G. K. Shukla (Units 13 to 16)
(Retd.), Department of Applied Mathematics
Indian Institute of Technology
Kanpur (UP)

Dr. Manish Trivedi (Units 13 to 16)
School of Sciences, IGNOU

Programme and Course Coordinator: Dr. Manish Trivedi

Production

Mr. Sunil Kumar, AR (P)
School of Sciences, IGNOU

September, 2014

© Indira Gandhi National Open University, 2014

ISBN-978-81-266-

All rights reserved. No part of this work may be reproduced in any form, by mimeograph or any other means, without permission in writing from the Indira Gandhi National Open University

Further information on the Indira Gandhi National Open University may be obtained from the University's Office at Maidan Garhi, New Delhi-110068 or visit University's website <http://www.ignou.ac.in>

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by the Director, School of Sciences.

Laser Typeset by: Tessa Media & Computers, C-206, A.F.E.-II, Okhla, New Delhi

Printed at:

BLOCK 4 TIME SERIES MODELLING

In the previous block, we have discussed the fundamental concepts of regression modelling. You have learnt that regression analysis is a statistical tool for investigating and analysing the average relationship between two or more variables. We have also discussed the applications of regression analysis in decision making. In this block, we discuss time series modelling, which helps us in analysing past data to forecast and plan for future. For example, suppose the Government wishes to bring in a policy to ensure low cost housing for economically weaker sections (EWS). Then it would like to know: What would the number of people in this category be in the next five years? In general, institutions, organisations, governments and even individuals take most decisions based on the situations expected in future. Planning for future is an essential aspect of decision making and managing any organisation. This requires that one should be able to forecast the future demands or trends.

For this, it is natural to use historical data of the last few years. Based on the features observed from the past data, we try to understand their role in causing variability and use them for forecasting. This is done with the help of time series modelling, which is the analysis of a collection of observations made sequentially through time to forecast future trends. We discuss basic concepts of time series modelling in the last unit of this block. It comprises four units.

In Unit 13 entitled **Trend Component Analysis**, we introduce the concept of time series. We discuss different types of time series and different components of time series along with examples. We also explain the methods of smoothing and filtering the time series data and estimation of trend by curve fitting method and curvilinear method.

When time series data do not exhibit any trend and cyclic components but reflect seasonal variations, we have to estimate the seasonal component by removing irregular components. In Unit 14 entitled **Seasonal Component Analysis**, we discuss some methods for estimating the seasonal components, namely, the simple average method, ratio to moving average and ratio to trend methods.

In Unit 15 entitled **Stationary Processes**, we describe a very important class of time series called the stationary time series. If a time series shows a particular type of non-stationarity then some simple transformation makes it stationary and we can model it. We also explain the concepts of autocovariance, autocorrelation and correlogram for a stationary time series.

In Unit 16 entitled **Time Series Models**, we discuss the Autoregressive (AR), Moving Average (MA), mixed Autoregressive Moving Average (ARMA) and Autoregressive Integrated Moving Average (ARIMA) processes. We also analyse their properties in the form of autocorrelations and discuss the fitting of suitable models to the given data.

Notations and Symbols

Y_t	:	Time series value at time t of variable of interest Y
X_t	:	t^{th} observation of the independent variable X
$\hat{Y}_t$	:	t^{th} observation of the fitted value
e_t	:	t^{th} forecast error
T_t	:	Trend component or trend effect
S_t	:	Seasonal component or seasonal effect
C_t	:	Cyclic component or cyclic effect
I_t	:	Irregular component or irregular effect
S_i	:	i^{th} seasonal index
$\bar{y}_i$	:	Average for i^{th} year, quarter or month
μ	:	Mean
a_t	:	t^{th} white noise with mean zero, variance σ_a^2
ψ_i	:	i^{th} weight
β_i	:	i^{th} constant
γ_k	:	Autocovariance at lag k
c_k	:	Autocovariance coefficient at lag k
ρ_k	:	Autocorrelation at lag k
r_k	:	Autocorrelation coefficient at lag k
B	:	Backward shift operator
$\rho = \text{a.c.f.}$	:	Autocorrelation function
$\alpha = \text{p.a.c.f.}$	:	Partial autocorrelation function
$\text{MA}(q)$	:	Moving Average process of order q
$\text{AR}(p)$	:	Autoregressive process of order p
$\text{ARMA}(p, q)$	:	Autoregressive Moving Average process of order (p, q)
$\text{ARIMA}(p, d, q)$	:	Autoregressive Integrated Moving Average process of order (p, q) with d differences
$\{Y(t)\}=Y_t$	:	Discrete time random process
$V\{Y(t)\}$	:	Variance of time dependent random variable Y
Sec.	:	Section/subsection
Secs.	:	Sections/subsections
Fig.	:	Figure
Figs.	:	Figures

UNIT 13 TREND COMPONENT ANALYSIS

Structure

- 13.1 Introduction
 - Objectives
- 13.2 Introduction to Time Series
 - Time Series with Trend Effect
 - Time Series with Seasonal Effect
 - Time Series with Cyclic Effect
- 13.3 Components of Time Series
 - Trend Component (T)
 - Seasonal Component (S)
 - Cyclic Component (C)
 - Irregular Component (I)
- 13.4 Basic Models of Time Series
 - Additive Model
 - Multiplicative Model
- 13.5 Smoothing or Filtering Time Series
 - Equal Weight (Simple) Moving Average (MA) Method
 - Weighted (Unequal) Moving Average Method
 - Exponential Smoothing
- 13.6 Estimation of Trends by Curve Fitting
 - Fitting a Linear Trend Equation
 - Fitting a Quadratic Trend Equation
 - Fitting the Exponential Trend Equation
- 13.7 Measurement of Trend Effect Using Centred Moving Average Method
- 13.8 Summary
- 13.9 Solutions/Answers

13.1 INTRODUCTION

In the previous block, we have discussed simple and multiple linear regression analysis where we have dealt with bivariate as well as multivariate data. While studying that block, you have learnt how useful regression analysis is in decision making. If we carefully analyse most decisions and actions of the Government, an institution, an industrial organisation or an individual, we find that, to a large extent, these depend on the situations expected to arise in future. For example, suppose the Delhi Government wishes to frame a housing development policy for providing houses to all families of the central government employees in Delhi over the next five years. Then the Government would like to know: What would the number of families of government employees in Delhi be in the next five years? A similar assessment is required while formulating the employment policy, and so on.

Planning for future is an essential aspect of managing an organisation. This requires that we should be able to **forecast** the future requirements of that organisation. For example, suppose we are asked to provide quarterly forecasts of the sales volume for a particular product during the coming one

year period. Our forecasts for sales would also affect production schedules, raw material requirements, inventory policies, and sales quota. A good forecast of the future requirements will result in good planning. A poor forecast results in poor planning and may result in increased cost. In order to provide such forecasts, we use historical data of the past few years to assess the average requirement, trend (if any) over the years and seasonal variations. Based on these features observed from the past data, we try to understand their role in causing variability and use them for forecasting requirements.

This exercise is done with the help of **time series analysis** which is a collection of observations made sequentially over a period of time. **The main objectives of time series analysis are description, explanation and forecasting.** It has applications in many fields including economics, engineering, meteorology, etc.

In this unit, we discuss the concept of time series and explain different types of time series in Sec. 13.2. In Secs. 13.3 and 13.4, we describe different components and basic models of time series. We explain the methods of smoothing and filtering the time series data along with the estimation of trend by the curve fitting and curvilinear methods in Secs. 13.5 and 13.6. Finally, in Sec. 13.7, we describe the methods of measurement of trend and cyclic effect in time series data.

In the next unit, we shall discuss some methods for estimating the seasonal component (S). We shall also discuss the method of estimating the trend component from deseasonalised time series data.

Objectives

After studying this unit, you should be able to:

- explain the concept of time series;
- describe the components of time series;
- explain the basic models of time series;
- decompose a time series into its different components for further analysis;
- describe the trend component of the time series;
- describe different types of trends;
- explain various methods for smoothing time series and estimation of trends; and
- describe the centred moving average method of measuring the trend effect.

13.2 INTRODUCTION TO TIME SERIES

Generally it is seen that forecasting involves studying the behaviour of a characteristic over time and examining data for any pattern. The forecasts are made by assuming that, in future, the characteristic will continue to behave according to the same pattern. The data gathered could be of sales per day, units of productions per week, the running cost of a machine per month, etc.

A time series (TS) is a collection of observations made sequentially over a period of time. In other words, the data on any characteristic collected

with respect to time over a span of time is called a time series. Normally, we assume that observations are available at equal intervals of time, e.g., on an hourly, daily, monthly or yearly basis. Some time series cover a period of several years.

The methods of analysing time series constitute an important area of study in statistics. But before we discuss time series analysis, we would like to show you the plots of some time series from different fields. In the next three sub-sections, we look at the plots of **three time series**, namely, **time series with trend effect**, **time series with seasonal effect** and **time series with cyclic effect**. These plots are called **time plots**.

13.2.1 Time Series with Trend Effect

A **trend** is a long term smooth variation (increase or decrease) in the time series. When values in a time series are plotted in a graph and, on an average, these values show an increasing or decreasing trend over a long period of time, the time series is called the **time series with trend effect**. You should note that all time series do not show an increasing or decreasing trend. In some cases, the values of the time series fluctuate around a constant reading and do not show any trend with respect to time. You should also remember that an increase or decrease may not necessarily be *in the same direction* throughout the given period. Time series may show an upward trend, a downward trend or have no trend at all. Let us explain all three cases with the help of examples.

Time Series with Upward Trend

When values in a time series are plotted in a graph and these show an upward trend with respect to time, we call it a **time series with upward trend**. For example, the profit of a company plotted for the time period 1981 to 2012 in Fig. 13.1 shows an upward trend.

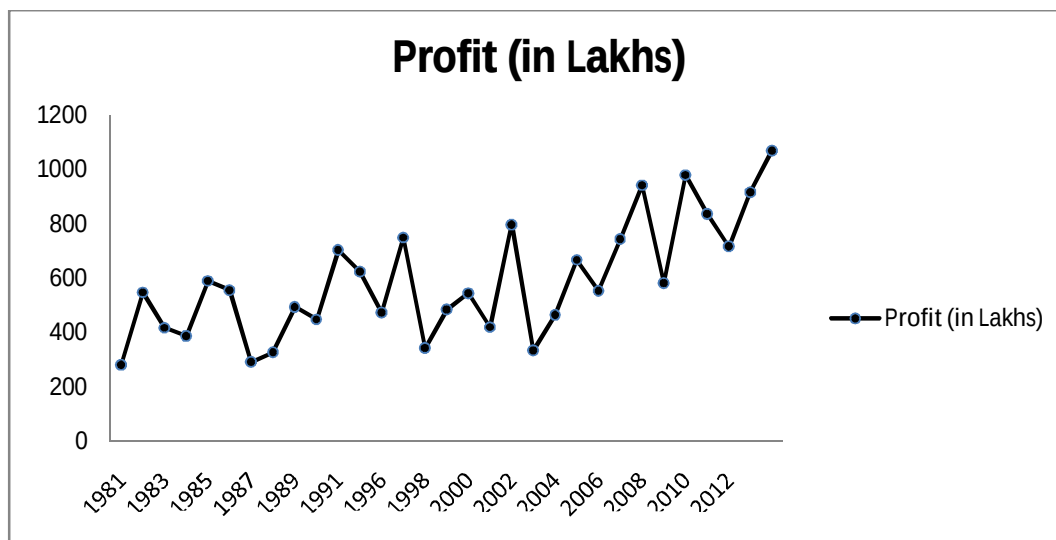


Fig. 13.1: Profit of a company from 1981 to 2012.

Time Series with Downward Trend

If data in a time series plotted in a graph show a downward trend with respect to time, it is called the **time series with downward trend**. For example, the values of mortality rate of a developing country from 1991 to 2000 show a downward trend in the time plot given in Fig. 13.2.

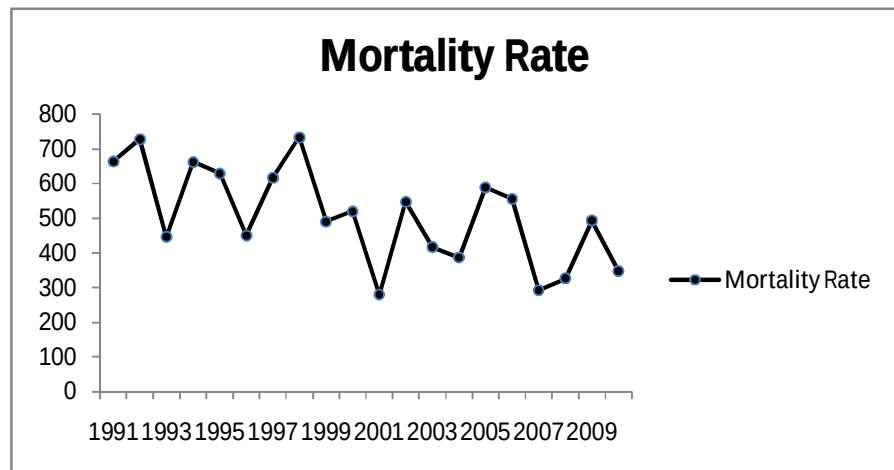


Fig. 13.2: Mortality rates of a developing country from 1991 to 2009.

Time Series with No Trend

If data of a time series is plotted on the graph paper and does not show any trend, that is, neither an upward nor a downward trend is reflected in the time plot, the time series is called the **time series with no trend**. For example, Fig. 13.3 shows the time plot of the production of a commodity in a factory from 1988 to 2012. Notice that the time series shows no trend.

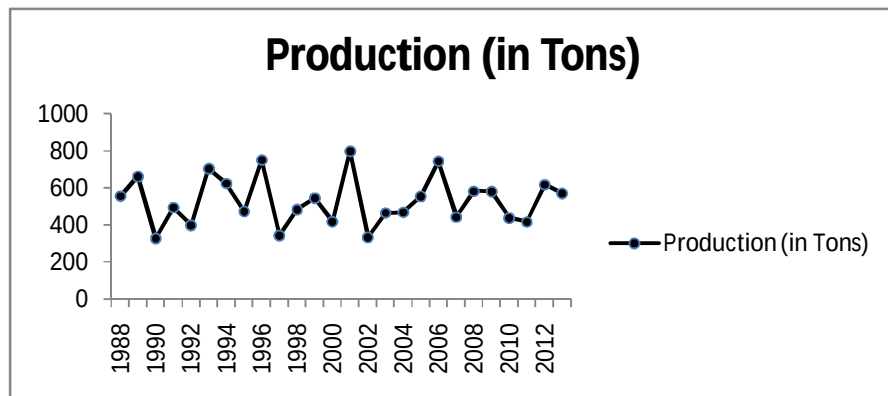
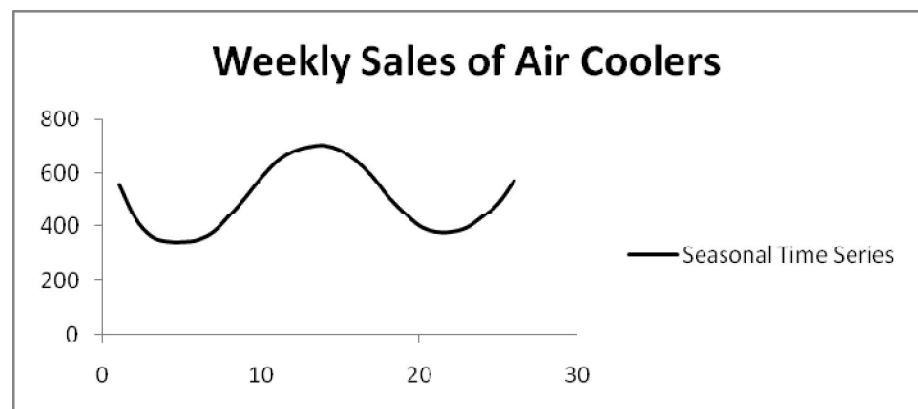


Fig. 13.3: Production of a commodity from 1988 to 2012.

13.2.2 Time Series with Seasonal Effect

If values in a time series reflect seasonal variation with respect to a given period of time such as a quarter, a month or a year, the time series is called a **time series with seasonal effect**. For example, the time plot of the data of weekly sales of air coolers shows a seasonal effect (Fig. 13.4).



13.2.3 Time Series with Cyclic Effect

If the time plot of data in a time series exhibits a cyclic trend, the time series is called a **time series with cyclic effect**. For example, time series data of the number of employees in software industry in different phases, i.e., phases of prosperity, recession, depression and recovery shows a cyclic pattern, that is, the pattern repeats itself over an almost fixed period of time (Fig. 13.5).

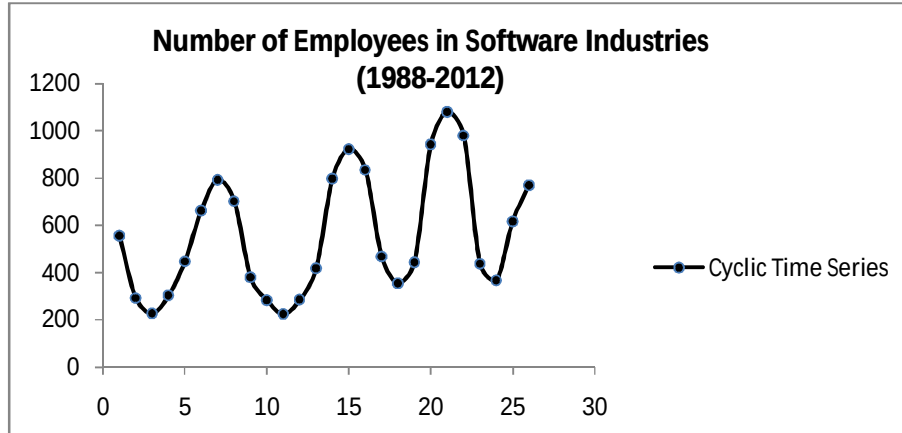


Fig. 13.5: Number of employees in software industries for the last 25 years.

So far, you have learnt about different types of time series plots which exhibit different trends in data. These trends arise due to the effect of various factors on the variations in data. The variations in the values or data are also described in terms of components of time series. Let us learn about them.

13.3 COMPONENTS OF TIME SERIES

In Sec. 13.2, you have learnt that the variations in the time series values are of different types and arise due to a variety of factors. These different types of variations in the values of the data in a time series are also called **components** of the time series.

In the past, time series analysis was mainly concerned with decomposing the variations in a time series into **components** representing **(i) Trend (ii) Seasonal (iii) Cyclic** and **(iv) Remaining variations** attributed to **Irregular fluctuations (sometimes referred to as the Random component)**. This approach is not necessarily the best one and we shall discuss the modern approach in later units of this block. Usually, some or all components may be present in a time series in varying amounts and can be classified in the above-mentioned four categories. For the sake of completeness, we discuss these components in some detail.

13.3.1 Trend Component (T)

Usually time series data show random variation, but over a long period of time, there may be a gradual shift in the mean level to a higher or a lower level. This gradual shift in the level of time series is known as the **trend**. In other words, **the general tendency of values of the data to increase or decrease during a long period of time is called the trend**. Some time series show an upward trend while some show a downward trend as you have learnt in the previous section. For example, upward trends are seen in the data of population growth, currency in circulation, etc., while data of

that some time series may not show any trend at all. The shifting in level is usually the result of changes in the population, demographic characteristic of the population, technology, consumer preferences, purchasing power of the population, and so on.

You should clearly understand that **a trend is a general, smooth, long term and average tendency of a time series data**. The increase or decrease may not necessarily be in the same direction throughout the given period. A time series may show a linear or a nonlinear (curvilinear) trend. If the time series data are plotted on a graph and the points on the graph cluster more or less around a straight line, the tendency shown by the data is called **linear trend in time series**. But if the points plotted on the graph do not cluster more or less around a straight line, the tendency shown by the data is called **nonlinear or curvilinear trend**. Trend need not always be a straight line. It can be quadratic, exponential or may not be present at all.

Trend is also known as long term variation. However, do understand that long term or long period of time is a relative term which cannot be defined uniformly. In some situations, a period of one week may be fairly long while in others, a period of 2 years may not be long enough.

13.3.2 Seasonal Component (S)

In a time series, variations which occur due to rhythmic or natural forces/factors and operate in a regular and periodic manner over a span of **less than or equal to one year** are termed as **seasonal variations**. Although we generally think of seasonal movement in time series as occurring over one year, it can also represent any regularly repeating pattern that is **less than one year** in duration. For example, daily traffic volume data show seasonal behaviour within the same day, with peak level occurring during rush hours, moderate flow during the rest of the day, and light flow from midnight to early morning. Thus, in a time series, seasonal variations may exist if data are recorded on a quarterly, monthly, daily or hourly basis.

However, do remember this point: **Although the data may be recorded over a span of three months, one month, a week or a day, the total period should be one year to assess seasonal variation properly**. Note that the amplitudes of the seasonal variation are different for different spans of time over which data is recorded (quarterly, monthly, daily, etc.). Most of the time series data in the fields of economics or business show a seasonal pattern.

The seasonal pattern in a time series may be either due to natural forces or manmade conventions. Variations in time series that arise due to changes in seasons or weather conditions and climatic changes are known as **seasonal variations due to natural forces**. For example, the sales of umbrellas, rain coats and gumboots pick up very fast in the rainy season; the demand for air conditioners goes up in summers; the sale of woollens goes up in winter. All these arise due to natural factors. Variations in time series that arise due to changes in habits, fashions, customs and conventions of people in any society are termed as **seasonal variations from manmade conventions**. For example, in our country, the sale of gold and silver ornaments goes up in Diwali, Dussehra (Durga Puja) or the marriage season.

13.3.3 Cyclic Component (C)

Apart from seasonal effects, some time series exhibit variation in a fixed period of time due to other physical causes. For example, economic data are sometimes thought to be affected by business cycles with a period varying

from three to ten years (see Fig. 13.5). The cycles could be caused by period of moderate inflation followed by a period of high inflation. However, the existence of such business cycles leads to some confusion about cyclic, trend and seasonal effects. To avoid this confusion, **we shall term a pattern in the time series as cyclic component only when its duration is longer than one year.**

The cyclic variations in a time series are usually called “business cycle” and comprise four phases of a business, namely, prosperity (boom), recession, depression and recovery. These are normally over a span of seven to eleven years. Thus, **the oscillatory variations with a period of oscillation of more than one year are called cyclic variations or the cyclic component in a time series. One oscillation period is called one cycle.**

13.3.4 Irregular Component (I)

The long term variations, i.e., the trend component and short term variations, i.e., the seasonal and cyclic component are known as **regular variations**. Apart from these regular variations, **random or irregular variations**, which are not accounted for by trend, seasonal or cyclic components, exist in almost all time series. In most cases, these irregular variations are random, irregular and unpredictable and are caused by short-term, unanticipated and non-recurring factors that affect time series in some cases. In Unit 16, we shall model the irregular component using probability models such as auto-regressive (AR) and moving average (MA) models.

So far, you have learnt about four components of time series. These components may be present individually or jointly in any time series.

We now take up an example of a time series, which has both trend and seasonal components. Consider the quarterly sales data of washing machines for the period 2001-2007 given in Table 1.

Table 1: Quarterly sales of washing machines for the period 2001-2007

Year	Quarter 1	Quarter 2	Quarter 3	Quarter 4
2001	556	662	327	494
2002	398	704	624	473
2003	750	343	484	545
2004	419	798	334	465
2005	468	554	744	443
2006	582	581	437	417
2007	618	571	517	754

Fig. 13.6 shows the plot of this quarterly data. Note that there are 28 quarters from the year 2001 to 2007 and so 28 values are plotted in Fig. 13.6. These have been numbered from 1 onwards on the horizontal axis. We have connected the data points by a dotted curve to obtain a time series plot of the data. From the plot, we note that the values exhibit an upward linear trend over the long term. We show this trend by a thin straight line in Fig. 13.6. So the thin straight line in Fig. 13.6 reflects the presence of a long-term linear trend. We also notice a seasonal variation in the data, which

we show by a smooth thick free-hand curve. This thick curve shows the approximate movement around the straight trend line.

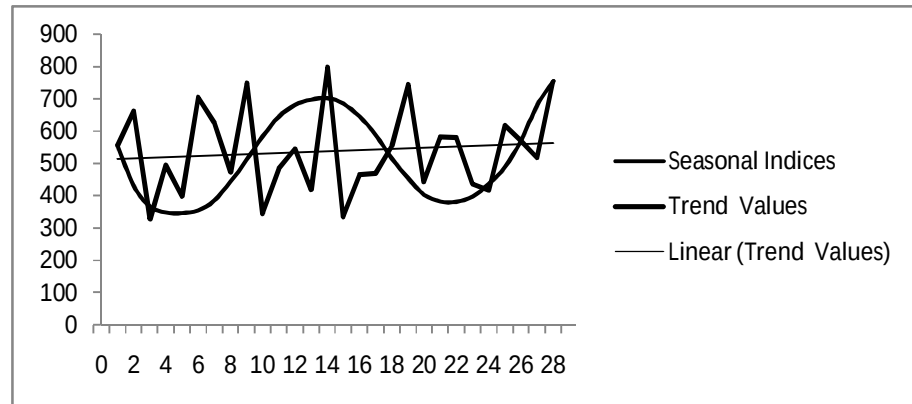


Fig. 13.6: Trend and seasonal variation in quarterly sales of washing machine.

In Fig. 13.6, in most years, the first quarter is a low point and then there is a rise in the second quarter, a decline in the third quarter and a rise in the fourth quarter. This could be due to changes in seasons and festival offers.

13.4 BASIC MODELS OF TIME SERIES

In Sec. 13.3, we have discussed different components of time series with examples. You have learnt of many factors (natural and manmade) that affect a time series. We would now like to describe the effect of these factors and the components of a time series mathematically. In this section, we discuss two commonly used mathematical models, which explain time series data reasonably well. While discussing these models, we shall use the notation y_t for the value of the time series at time t . We shall use serial numbering for time t since all time series are in chronological order. We now describe the two basic time series models, namely, the additive model and the multiplicative model.

13.4.1 Additive Model

The **additive model** is one of the most widely used models. It is based on the assumption that at any time t , the time series value Y_t is the sum of all the components. According to the additive model, a time series can be expressed as

$$Y_t = T_t + C_t + S_t + I_t$$

where T_t , C_t , S_t and I_t are the trend, cyclic, seasonal and irregular variations, respectively, at time t . In this model, we make the following assumptions:

- cyclic effects remain constant for all cycles;
- seasonal effects remain constant during any year or the corresponding period; and
- I_t is an i.i.d. normal variable with mean 0, i.e., the effect of irregular variation remains constant throughout. This implies that the term S_t does not appear in a time series of annual data.

The additive model implies that seasonal variations in different years, cyclic variations in different cycles and irregular variations in different trends show equal absolute effects irrespective of the trend value. In the previous sections, you have learnt that all four components need not necessarily be exhibited in every time series. For example, the time series of annual

production data of a yield does not have seasonal variations. Similarly, a time series for the annual rainfall does not contain cyclic variations.

In additive model, we have assumed that the time series is the sum of the trend, cyclic, seasonal and irregular components. Generally, the additive model is appropriate when seasonal variations do not depend on the trend of the time series. However, there are a number of situations where the seasonal variations exhibit an increasing or decreasing trend over time. In such cases, we use the multiplicative model.

13.4.2 Multiplicative Model

When seasonal variations exhibit any change over time in terms of an increasing or decreasing trend, we can use the **multiplicative model** to describe the time series data. The multiplicative model is appropriate if various components in a time series operate proportionately to the general level of the series. The multiplicative model is based on the assumption that the time series value Y_t at time t is the product of the trend, cyclic, seasonal and irregular component of the series:

$$Y_t = T_t \times C_t \times S_t \times I_t$$

where T_t , C_t , S_t and I_t denote the trend, cyclic, seasonal and irregular variations, respectively. The multiplicative model is found to be appropriate for many business and economic data. Some examples are the time series for production of electricity, time series for number of passengers opting for air travel, time series for sales of soft drinks.

For estimation of any trend in a time series, smoothing (or filtering) the effect of irregular fluctuations present in it becomes important so that trend and seasonal effects may be easily estimated. We now discuss the smoothing or filtering of time series.

13.5 SMOOTHING OR FILTERING TIME SERIES

In this section, we discuss the methods of moving averages and exponential smoothing for smoothing or filtering the time series data. There are two methods of moving averages: the **equal weight** or **simple moving average method** and the **weighted (unequal) moving average method**. We discuss these two moving average methods in the next two sub-sections.

13.5.1 Equal Weight (Simple) Moving Average (MA) Method

In this method, we find the simple moving averages of time series data over m periods of time, called **m-period moving averages**. You can calculate them in the following way:

1. Calculate the average of the first m values of the time series.
2. Then discard the first value and take the average of the next m values again.
3. Repeat this process till all data are exhausted.

These steps yield a new time series of **m-period moving averages**.

Let us explain this method with the help of an example.

Example 1: Compute the 3-year simple moving averages for the time series of annual output of a factory for the period 2006-2011 given in Table 2.

Table 2: Annual output of a factory from 1976 to 1981

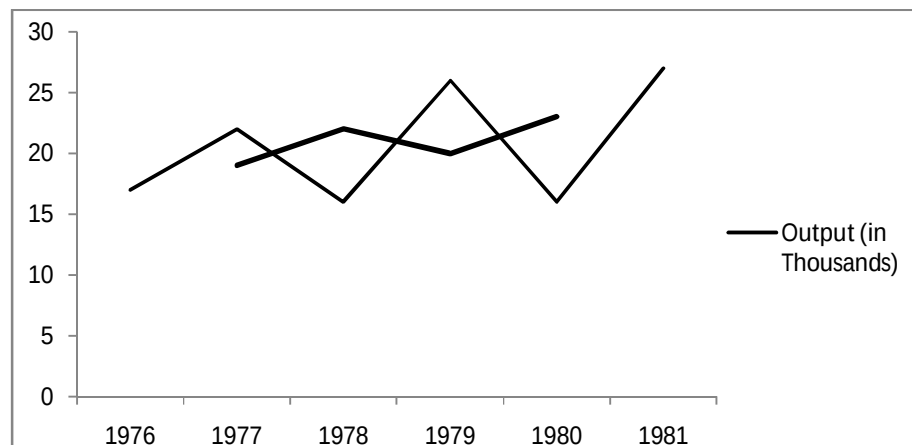
Year	1976	1977	1978	1979	1980	1981
Output (in thousands)	17	22	18	26	16	27

Solution: In this case, $m = 3$ years. The first value of the moving averages for $m = 3$ years is the average of 17, 22 and 18, which is 19. The second value of moving averages is obtained by discarding the first value, i.e., 17 and taking the average of the next 3 values in the time series, i.e., 22, 18 and 26. So we take the average of 22, 18 and 26, which is 22. Again, we discard the first value, i.e., 22 and take the average of the next 3 values in the time series, i.e., 18, 26 and 16. It is 20. We repeat this procedure for calculating the remaining 3-year moving averages. Table 3 gives the 3-year simple moving averages for the data given in Table 2. Note that each moving average is tabulated at the average (centre) of the time period for which it is computed. This method is, therefore, called the **centred moving averages**.

Table 3: Centred simple moving averages for time series data of Table 2

Year	1976	1977	1978	1979	1980	1981
Output (in thousands)	17	22	18	26	16	27
Moving Averages	-	19	22	20	23	-

Remember that moving averages vary less than the data values from which they are calculated as they smooth (or filter) out the effect of irregular component. This helps us in appreciating the effect of trend more clearly. For the given data the original time series varies between 16 and 27 whereas the moving averages vary between 19 and 23, which is much smoother than the original series. Fig.13.7 shows the output of the original time series and the 3-year moving averages series. In this figure, you can clearly see the smoothing property of moving averages.

**Fig. 13.7: Three-year centred moving average of the time series.**

You may ask: What should the value of m be? If m is increased, the series becomes much smoother and it may also smooth out the effect of cyclical and seasonal components, which are our main interest of study. Sometimes 3-year, 5-year or 7-year moving averages are used to expose the combined trend and cyclical movement of time series. But as we shall see in Sec. 13.6, four quarter or 12-months moving averages are more useful for estimating

13.5.2 Weighted (Unequal) Moving Average Method

The simple moving average method described in Sec.13.5.1 is not generally recommended for measuring trend although it can be useful for removing seasonal variation. It may also not lie close to the latest values. Therefore, the weighted (unequal) moving averages method is used. In this method, instead of giving equal weights to all values, unequal weights are given in such a way that all the weights are positive and their sum is equal to 1. If w_i denotes the weight of the i^{th} observation, the weighted moving average value y_t is given by

$$y_t = \sum_{i=-q}^q w_i x_{t+i} \quad w_i \geq 0, \quad \sum_{i=-q}^q w_i = 1 \quad \dots (1)$$

where x_t is the original series.

Simple moving average becomes a particular case of weighted moving average for

$$m = 2q+1 \text{ and } w_i = \frac{1}{(2q+1)}$$

In Example 5, $q = 1$, $m = 2q+1 = 3$ and $w_i = 1/(2q+1) = 1/3$.

Let us consider an example to illustrate this method.

Example 2: Compute the 3-years weighted moving averages for the time series given in Example 1.

Solution: Generally, the most recent observation receives the largest weight and the weights decrease for older data values. For the data given in Example 1, suppose we give 3 times more weight to the most recent observation than the first observation and 2 times more weight to the second observation. Then the weights for each year in every 3-year period are:

$$w_1 = 1/6, \quad w_2 = 2/6, \quad w_3 = 3/6$$

After assigning the weights as above, we get the smoothened trend values as

$$y_t = (x_{t-1} + 2x_t + 3x_{t+1}) / 6 \quad \dots (2)$$

The procedure for getting the smoothened trend values by weighted moving average method is as follows:

1. Calculate the weighted average of the first m values given in time series as explained above.
2. Now discard the first value and include the next one and take the average of the next m values by following the weight structure given in equation (2).
3. Repeat this process till all values are exhausted.

These steps yield a new time series of m -period weighted moving averages and the weighted moving average values are given in the following table:

Table 4: Weighted moving averages for time series data of Table 2

Year	1976	1977	1978	1979	1980	1981
Output (in thousands)	17	22	18	26	16	27
Weighted Moving Averages	-	19.16	22.66	19.66	23.16	-

The reason for giving larger weight to the latest observation rather than the earlier observations is that the latest observation is a better predictor of the trend than the earlier one. However, this may not be true for the cases in which the data contains a very large irregular component.

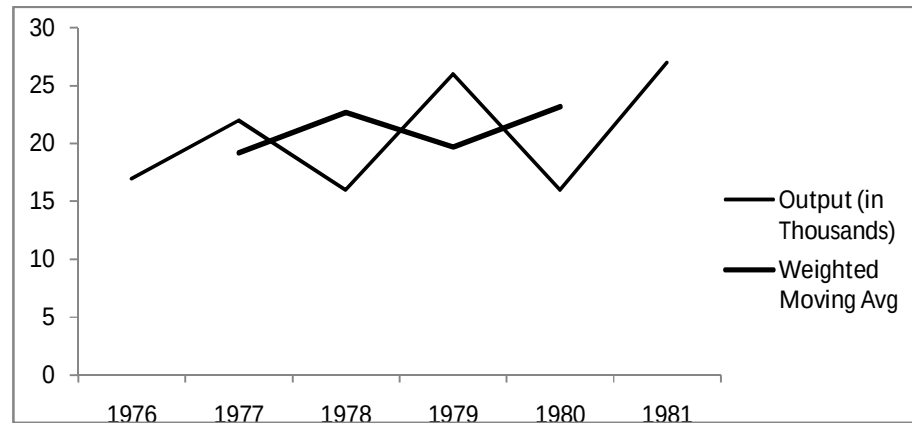


Fig. 13.8: Three-year weighted moving average of the time series.

13.5.3 Exponential Smoothing

We now discuss another technique of smoothing the time series, namely, the **exponential smoothing** technique, which is very popular. In Secs. 13.5.1 and 13.5.2, you have learnt that in the method of moving averages, weights (equal or unequal) are attached to each time series value that is considered. But in this method, the weights assigned to current as well as past time series values are different fixed positive numbers.

In this smoothing technique, weights decrease exponentially, except the last one, and the trend value is given by

$$y_t^C = w y_t + (1 - w) y_{t-1}^C \quad \text{for } t = 1, 2, \dots \quad \dots(3)$$

where $0 < w < 1$. The value of 'w' is chosen as per the requirements. Equation (3) gives weighted average of $y_1, y_2, \dots, y_t$ for calculating moving average series y_t^C . You can see that the latest observation y_{t-1}^C gets the maximum weight and then weights decrease exponentially. Note that the sum of weights is equal to one. This is a very popular technique for forecasting purposes. Let us take up an example to explain this method.

Example 3: Find the forecast for the time series given in Example 1.

Solution: On plotting the given data, it appears that there is no trend in the time series.

This is the same as saying that if $T = a + bt$ is the trend then $b = 0$. Therefore, if we fit a straight line to the given data with $b = 0$, the least squares estimate of a would be the simple average of the time series values, i.e.,

$$\hat{a} = \frac{\sum y_t}{6} = 20.66$$

In this way, the forecast for output for all t would be $\hat{a}$ as the model is

$$T = a + (\text{forecast error})$$

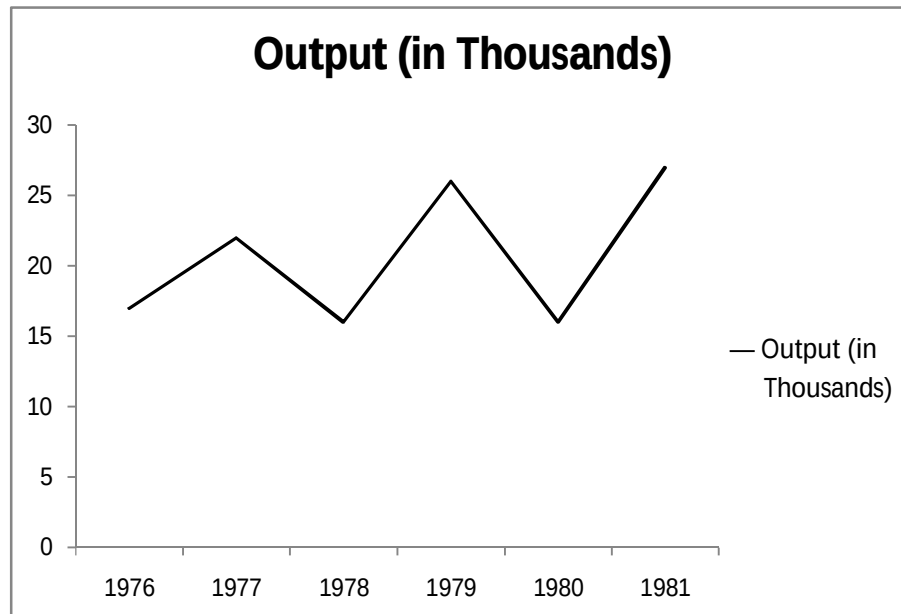


Fig. 13.9: Original time series values output (in thousands).

The result seems somewhat unreasonable because even though there is no trend, the time series values are not the same at all times t . The values of $\hat{a}$ seem to be higher in the year 2007, 09 and 2011 than the values in the year 2006, 08 and 2010. Therefore, it is logical to assume that the value of $\hat{a}$ is gradually changing over time. Therefore, the value of $\hat{a}$ should be denoted by $\hat{a}_t$ rather than ' $\hat{a}$ '.

According to this smoothing technique, we select a single weight w which is called exponential smoothing constant, where w lies between 0 and 1 and there is a method of choosing this constant. We can compute an exponential smoothing series y'_t as follows:

$$y'_1 = w y_1 + (1-w) y_1 = y_1$$

$$y'_2 = w y_2 + (1-w) y'_1$$

$$y'_3 = w y_3 + (1-w) y'_2$$

$$\cdot \quad \cdot$$

$$\cdot \quad \cdot$$

$$\cdot \quad \cdot$$

$$y'_t = w y_t + (1-w) y'_{t-1}$$

We should know the value of y_1 for the computation of $y'_1, y'_2, y'_3, \dots$

According to this technique, we also select a single weight w , which lies between 0 and 1. To start with, let us take $w = 0.01$. Then from equation (3), we get

$$y'_1 = 0.01 y_1 + 0.99 y_1 = 0.01(17) + (0.99)(17) = 17$$

Our forecast for y_1 at $t = 0$ is y'_1 . Therefore, the forecast error is

$$e_1 = y_1 - y'_1 = 17 - 17 = 0.00$$

We get the second forecast value as follows:

$$y'_2 = 0.01 y_2 + 0.99 y'_1 = 0.01(22) + (0.99)(17) = 17.05$$

The forecast error at $t = 2$ is

$$e_2 = y_2 - y'_2 = 22 - 17.05 = 4.95$$

Proceeding in this way, we calculate the forecast value y'_t and forecast error e_t for all t . These are calculated and given in the following table:

Table 5: Forecast values and forecast errors for time series given in Table 2

Year	Output (in thousand) y_t	Forecasts y'_t	Forecast errors e_t
1976	17	17.00	+0.00
1977	22	17.05	+4.95
1978	18	17.06	+0.94
1979	26	17.15	+8.85
1980	16	17.14	-1.14
1981	27	17.24	+9.76

After calculating the forecasts y'_t and errors e_t for all t , we plot the forecast values with the original time series values (Fig. 13.10). The graph of time series shows that there are no peaks in the time series after smoothing.

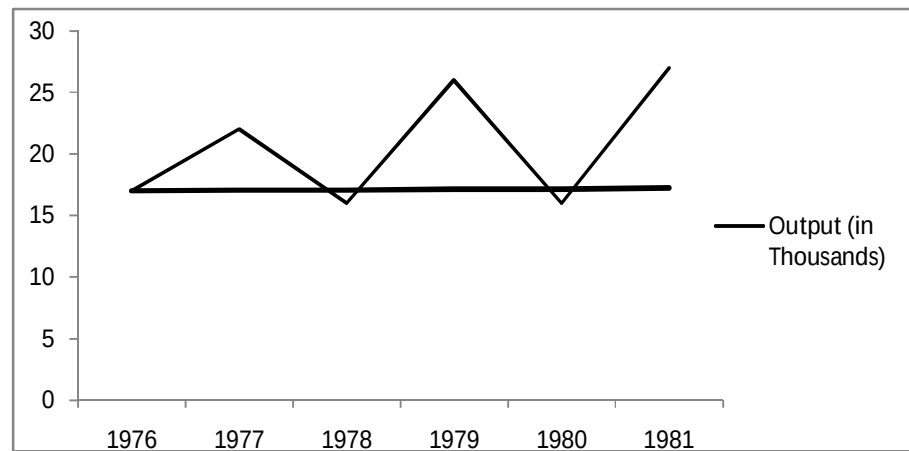


Fig. 13.10: Original and smoothed time series values of output (in thousands).

You should try to solve a few problems to check your understanding of the concepts discussed so far.

-
- E1)** Calculate the last two values of 3-year moving average for the data given in Example 1.
- E2)** Use exponential smoothing to obtain filtered values for the data given in Example 1 taking $w = 0.5$ and compare them with simple moving average values obtained in the same example.
- E3)** Obtain filtered values for the following data using exponential smoothing:

Year	Rainfall (in cm)	Year	Rainfall (in cm)	Year	Rainfall (in cm)	Year	Rainfall (in cm)
1970	664	1981	548	1991	624	2001	468
1971	728	1982	417	1992	473	2002	554
1972	447	1983	387	1993	750	2003	744
1973	663	1984	590	1994	343	2004	943
1974	630	1985	556	1995	484	2005	582
1975	451	1986	292	1996	545	2006	581
1976	617	1987	327	1997	419	2007	437
1977	734	1988	494	1998	798	2008	417
1978	491	1989	448	1999	334	2009	617
1979	520	1990	704	2000	465	2010	571
1980	280						

13.6 ESTIMATION OF TRENDS BY CURVE FITTING

An alternative approach to smoothing is to fit a polynomial to the data. This treats smoothing as a regression problem in which y_t is the trend value and integral powers of time t are the explanatory variables. The resulting smooth function is a polynomial

$$y_t = \sum_{j=0}^p b_j t^j \quad \dots(4)$$

where b_j is the j^{th} coefficient of the polynomial of degree p . The coefficients of the polynomial are estimated by the method of least squares by minimising the quantity

$$E = \sum_{i=1}^n \left(y_i - \sum_{j=0}^p b_j t_i^j \right)^2 \quad \dots(5)$$

We shall not discuss it here in detail as you have already studied the method of least squares and curve fitting in Unit 5 of MST-002. In the next section, we shall discuss the case when $p = 1$, i.e., the case of a linear trend equation.

13.6.1 Fitting a Linear Trend Equation

The trend equation given in equation (4) would be the special case of linear equation for $p = 1$ and we get the equation of the straight line as

$$y_t = b_0 + b_1 t \quad \dots(6)$$

We can write it as

$$y_t = b_0 + b_1 \bar{t} + b_1 (t - \bar{t}) \quad \dots(7)$$

To simplify the calculations, we take $x_t = t - \bar{t}$, where $\bar{t}$ is the mean of all times t .

Equation (6) then becomes

$$y_t = a_1 + b_1 x_t \quad \text{where } a_1 = b_0 + b_1 \bar{t} \quad \dots(8)$$

Using the method of least squares for minimising the error term, we obtain the following normal equations:

$$\begin{aligned} \sum_{t=1}^n y_t &= n a_1 + b_1 \sum_{t=1}^n x_t \\ \sum_{t=1}^n y_t x_t &= a_1 \sum_{t=1}^n x_t + b_1 \sum_{t=1}^n x_t^2 \end{aligned} \quad \dots(9a)$$

On solving the above equations, the least squares estimates of a_1 and b_1 are given by $\hat{a}_1$ and $\hat{b}_1$ as:

$$\hat{a}_1 = \bar{y} \quad \text{and} \quad \hat{b}_1 = \frac{\sum_{t=1}^n y_t x_t}{\sum_{t=1}^n x_t^2} \quad \dots(9b)$$

The estimate of b_0 is given by $\hat{b}_0$:

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{t} \quad \dots(10)$$

The fitted trend line is given by

$$\hat{y}_t = \hat{b}_0 + \hat{b}_1 \bar{t} + \hat{b}_1(t - \bar{t}) = \bar{y} + \hat{b}_1(t - \bar{t}) \quad \dots(11)$$

Let us explain this concept further with the help of an example.

Example 6: Fit a straight line trend for the data of annual profit of a company given below:

Year	2003	2004	2005	2006	2007	2008	2009	2010	2011
Profit (in crores)	93	102.8	126.7	103.5	105.7	133.2	156.7	175.7	161.6

Solution: From the given data, we obtain linear trend values for the annual profit as follows:

Table 6: Trend values for the given time series

Year	Profit (in crores)	$x_t = t - \bar{t}$	$x_t y_t$	x_t^2	Trend Values
2003	93.0	-4	-372.0	16	89.915
2004	102.8	-3	-308.4	9	99.828
2005	126.7	-2	-253.4	4	109.351
2006	103.5	-1	-103.5	1	119.054
2007	105.7	0	0	0	128.767
2008	133.2	1	133.2	1	138.480
2009	156.7	2	313.4	4	148.193
2010	175.7	3	527.1	9	157.906
2011	161.6	4	646.4	16	167.619
	$\Sigma y_t = 1158.9$		$\Sigma x_t y_t = 582.8$	$\Sigma x_t^2 = 60$	

From the above table, we calculate the least squares estimates as follows:

$$\hat{a}_1 = \bar{y} = \frac{\sum y}{9} = 128.76 \quad \hat{b}_1 = \frac{\sum_{t=1}^n y_t x_t}{\sum_{t=1}^n x_t^2} = \frac{582.8}{60} = 9.71$$

The equation of the fitted trend line is given by equation (11) as

$$\begin{aligned} \hat{y}_t &= \hat{b}_0 + \hat{b}_1 \bar{t} = \bar{y} + \hat{b}_1(t - \bar{t}) \\ &= 128.76 + 9.71(t - 2007) \end{aligned}$$

The trend projection for 2013 is

$$Y_{2013} = 128.76 + 9.71(2013 - 1977) = 187.02$$

These points are used to plot the trend line along with the data and projected value for 2013 (Fig. 13.11).

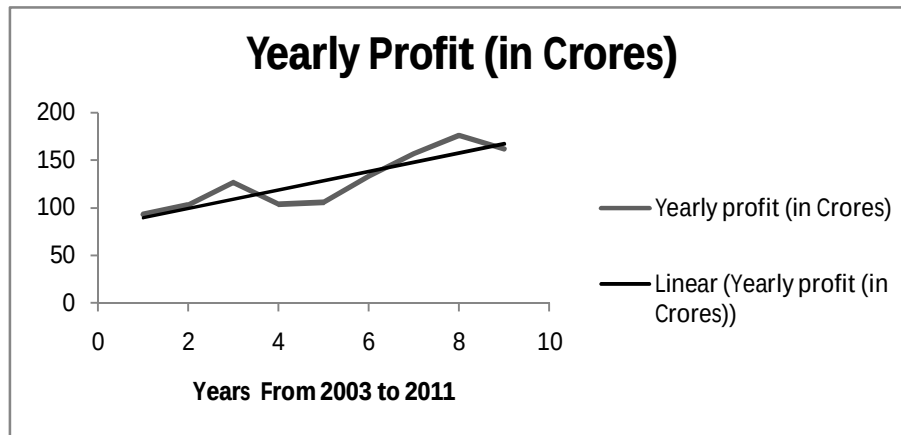


Fig. 13.11: Computed trend line for the data of profit of a company.

The projections are forecasts of future trend values but they do not take into account the cyclical effect. Sometimes cyclical effects are confused with trend curves which are of higher degree of polynomials.

13.6.2 Fitting a Quadratic Trend Equation

Sometimes trend is not linear and shows some curvature. The simplest curvilinear form is a second degree polynomial which can be obtained by taking $p = 2$ in equation (4).

$$y_t = b_0 + b_1t + b_2t^2 \quad \dots(12)$$

Proceeding in the same way as in Sec. 13.6.1 for linear trend equation, the equations for estimating b_0 , b_1 and b_2 are given by

$$\begin{aligned} \sum y_t &= nb_0 + b_1 \sum x_t + b_2 \sum x_t^2 \\ \sum x_t y_t &= b_0 \sum x_t + b_1 \sum x_t^2 + b_2 \sum x_t^3 \\ \sum x_t^2 y_t &= b_0 \sum x_t^2 + b_1 \sum x_t^3 + b_2 \sum x_t^4 \end{aligned} \quad \dots(13)$$

The values of $\sum y_t$, $\sum x_t$, $\sum x_t y_t$, $\sum x_t^2 y_t$, $\sum x_t^2$, $\sum x_t^3$ and $\sum x_t^4$ are obtained from the given data and the normal equations given in equation (13) can now be solved for the optimum values $\hat{b}_0$, $\hat{b}_1$ and $\hat{b}_2$.

With these values, equation (12) gives the desired quadratic trend.

To illustrate this, let us take an example of fitting a quadratic trend for the data of gross revenue of a company.

Example 7: Fit a quadratic trend equation for the data given below:

Year	1991	1992	1993	1994	1995	1996	1997	1998	1999
Gross Revenue (in Lakhs)	240	167	140	120	124	128	142	176	207
Year	2000	2001	2002	2003	2004	2005	2006	2007	
Gross Revenue (in Lakhs)	304	338	397	439	481	577	711	778	

Solution: Let us take the simplest curvilinear form of a second degree polynomial given as

$$y_t = b_0 + b_1t + b_2t^2 \quad \dots (i)$$

Proceeding in the same way as in Sec. 13.6.1, the normal equations for estimating b_0 , b_1 and b_2 are given by

$$\begin{aligned}\sum y_t &= nb_0 + b_1 \sum x_t + b_2 \sum x_t^2 \\ \sum x_t y_t &= b_0 \sum x_t + b_1 \sum x_t^2 + b_2 \sum x_t^3 \\ \sum x_t^2 y_t &= b_0 \sum x_t^2 + b_1 \sum x_t^3 + b_2 \sum x_t^4\end{aligned}\quad \dots (ii)$$

The values of $\sum y_t$, $\sum x_t$, $\sum x_t y_t$, $\sum x_t^2 y_t$, $\sum x_t^2$, $\sum x_t^3$ and $\sum x_t^4$ are obtained from the given data as follows:

Table 7: Calculations for fitting of quadratic trend

Year	y_t	$x_t = t - 1999$	$x_t y_t$	$x_t^2 y_t$	x_t^2	x_t^3	x_t^4
1991	240	-8	-1920	15360	64	-512	4096
1992	167	-7	-1169	8183	49	-343	2401
1993	140	-6	-840	5040	36	-216	1296
1994	120	-5	-600	3000	25	-125	625
1995	124	-4	-496	1984	16	-64	256
1996	128	-3	-384	1152	9	-27	81
1997	142	-2	-284	568	4	-8	16
1998	176	-1	-176	176	1	-1	1
1999	207	0	0	0	0	0	0
2000	304	1	304	304	1	1	1
2001	338	2	676	1352	4	8	16
2002	397	3	1191	3573	9	27	81
2003	439	4	1756	7024	16	64	256
2004	481	5	2405	12025	25	125	625
2005	577	6	3462	20772	36	216	1296
2006	711	7	4977	34839	49	343	2401
2007	778	8	6224	49792	64	512	4096
Total	5469	0	15126	165144	408	0	17544

Putting the values from the above table in the normal equations given in equation (ii), we get

$$17 b_0 + 0 b_1 + 408 b_2 = 5469$$

$$0 b_0 + 408 b_1 + 0 b_2 = 15126$$

$$408 b_0 + 0 b_1 + 17544 b_2 = 165144$$

On solving the normal equations, we get the optimum values: $\hat{b}_0 = 373.22$, $\hat{b}_1 = -41.614$ and $\hat{b}_2 = 4.3715$.

With these values, we get the desired quadratic trend:

$$y_t = 373.22 - 41.614 (t - \bar{t}) + 4.3715 (t - \bar{t})^2$$

Since we have used $x_t = (t - \bar{t})$ in equation (1)

Fig. 13.12 gives a plot of the above quadratic equation that fits the data (shown by continuous line) along with the actual values of the gross revenue of a company shown by dots. The values of time t are coded year values ($t = 1, 2, \dots, 17$). The fitted model with coded year values is shown in Fig. 13.12.

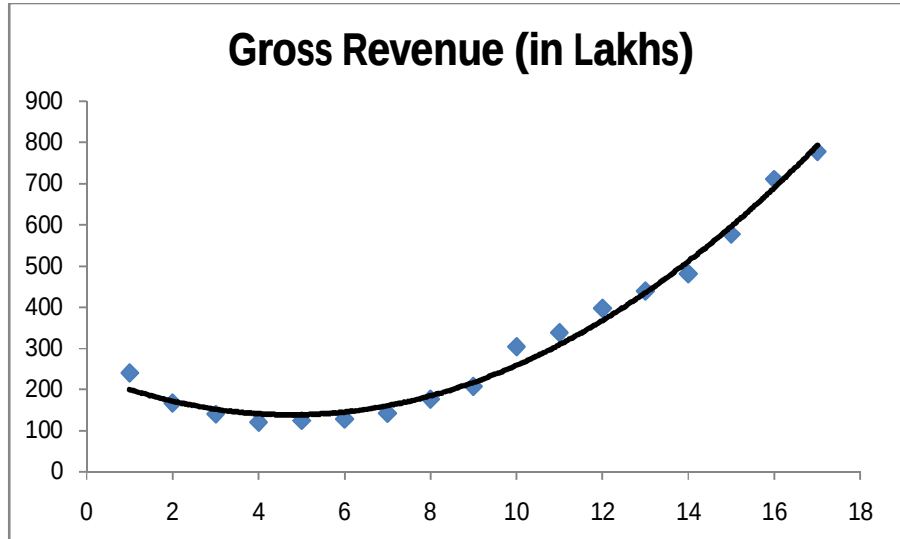


Fig. 13.12: Plot the quadratic trend model to the data of Example 7.

13.6.3 Fitting the Exponential Trend Equation

Sometimes data show that compound annual growth rate is constant over time rather than increasing annually as in the case of linear model. This can be represented by an exponential model

$$y_t = a_0 a_1^t$$

You can see that $(a_1 - 1) \times 100\%$ is the annual compound growth rate (in %), which remains constant. As far as fitting is concerned, we transform this model to a linear trend model by taking natural logarithm of y_t :

$$\log_e y_t = \log_e a_0 + t \log_e a_1$$

$$Y_t = \beta_0 + \beta_1 t$$

where $Y_t = \log_e y_t$, $\beta_0 = \log_e a_0$, $\beta_1 = \log_e a_1$,

Now, we can fit the model with Y_t and t as described in Sec. 13.6.1. Once we know the estimates of β_0 and β_1 , i.e., $\hat{\beta}_0$ and $\hat{\beta}_1$, we can obtain the estimates of a_0 and a_1 , i.e., the values of $\hat{a}_0$ and $\hat{a}_1$ by taking anti-logarithm of $\hat{\beta}_0$ and $\hat{\beta}_1$, respectively. When we fit the exponential model to the data given in Example 7, we get the values $\hat{\beta}_0 = 2.57$ and $\hat{\beta}_1 = 0.0494$, respectively, and the estimated exponential trend equation is obtained as:

$$y_t = 371.535 (1.1205)^t$$

The fitted and raw data are plotted in Fig. 13.13.

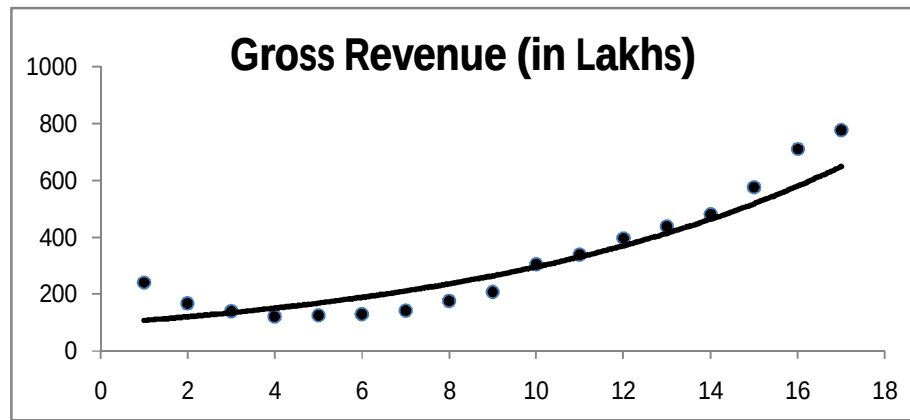


Fig. 13.13: Plot of fitted exponential trend model to the data of Example 7.

An exponential model can also be represented by

$$y_t = \alpha_0 \exp^{\beta_1 t}$$

As far as fitting is concerned, we transform this model to a linear trend model by taking natural logarithm of y_t :

$$\log_e y_t = \log_e \alpha_0 + \beta_1 t$$

$$\text{or } Y_t = \beta_0 + \beta_1 t$$

where $Y_t = \log_e y_t$, $\beta_0 = \log_e \alpha_0$.

On comparing Fig. 13.12 with Fig. 13.13, we find that the quadratic model is a better choice.

You may now like to solve the following exercise.

-
- E4)** Use the data given in Example 1 to fit a linear trend line and obtain the projection for the year 2012.
-

13.7 MEASUREMENT OF TREND EFFECT USING CENTRED MOVING AVERAGE METHOD

In Sec. 13.3, you have learnt that time series data usually have four components: Trend (T), Cyclical (C), Seasonal (S) and Irregular (I). Suppose a set of data are recorded on a monthly basis and there is a seasonal effect of one year. This means that after twelve months the data behaves in the same way as it did twelve months ago. This is called the **period** of seasonal effect. If we take a centred moving average with $m = 12$, then it will smooth out (eliminate) the effect of season. If data have been recorded on a quarterly basis and there is a seasonal effect with a period of twelve months, then a moving average with $m = 4$ will smooth out the seasonal effect. Not only this, it will also smooth out the effect of the irregular component I. Thus, the time series values of centred moving average will be nearly free from the effects of S and I. When the effects of S and I are removed, we are left with the effect of trend (T) and the cyclic effect (C).

In the next unit, we shall discuss the method of estimating seasonal effect (S) by making use of the estimates of trend (T) and cycle (C).

Example 8: Compute the trend values for the given data for quarterly sales of washing machines by an appliance manufacturer for the period 2001-2009.

Year	Quarter 1	Quarter 2	Quarter 3	Quarter 4
2001	935	1215	1045	1455
2002	990	1315	1350	1485
2003	1370	1815	1470	1680
2004	1160	1365	1205	1445
2005	1030	1475	1195	1585
2006	1185	1330	1500	2145
2007	1410	2120	1915	2390
2008	1875	2145	1965	2800
2009	1865	2115	1935	2165

Solution: We shall use the data of four years 2001-2004 from the given data to explain the calculations of moving averages, which are estimates of T and C. The given time series data is shown in Fig. 13.14.

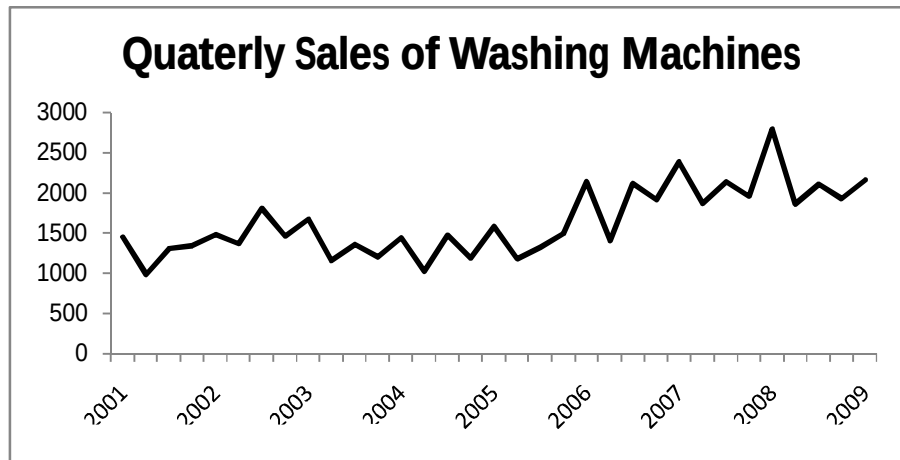


Fig. 13.14: Time series plot for quarterly sales of washing machines from 2001-2009.

We have seen from the plot (Fig. 13.14) that it has a very strong 12 monthly seasonal effect. Hence, to remove the seasonal effect, we have to take moving averages with $m = 4$.

Table 8: Calculation of centred moving averages

Year	Quarter	Sales (in Hundreds)	Centred MA(1)	Centred MA (2)
2001	1	935	-	1169.375
	2	1215		
	3	1045	1162.5	
	4	1455	1176.25	
2002	1	990	1201.25	1188.75
	2	1315	1277.5	1239.375
	3	1350	1285.0	1281.25
	4	1485	1380.0	1332.5

2003	1	1370	1505.0	1520
	2	1815	1535.0	1559.125
	3	1470	1583.25	1619.75
	4	1680	1656.25	1600
2004	1	1660	1543.75	1510.625
	2	1365	1477.5	1448.625
	3	1205	1419.75	-
	4	1445	-	-

The values in MA(1) are the moving averages for $m = 4$ but they do not correspond to any of the given four quarters as the average of 1, 2, 3 and 4 is 2.5. Hence, to make it correspond to a quarter, we usually calculate moving averages with $m = 2$ on MA(1) so that the centred values correspond to one of the four quarters. MA(2) gives the moving average of MA(1) series with $m = 2$ so that the values correspond to quarters 3, 4, 1, 2,... etc.

You should try to solve the following exercises.

-
- E5)** Compute MA(1), the moving average values for $m = 4$ and MA(2), the moving average values for $m = 2$ of MA(1) for the remaining years of the period 2005-2009 for the data given in Example 8.
- E6)** Compute the moving average (MA) values for $m = 3$ for time series for the period 2001-2009 for the data given in Example 8.
-

Let us now summarise the concepts that we have discussed in this unit.

13.8 SUMMARY

1. A good forecast of the future requirements will result in good planning. A poor forecast results in poor planning and may lead to increased cost. In order to provide such forecasts, we use historical data of the past few years to assess the average requirement, trend (if any) over the years and seasonal variations. Based on these features observed from the past data, we try to understand their role in causing variability and use them for forecasting requirements.
2. A **time series** is a collection of observations made sequentially over a period of time. **The main objectives of time series analysis are description, explanation and forecasting.** It has applications in many fields including economics, engineering, meteorology, etc.
3. A **trend** is a long term smooth variation (increase or decrease) in the time series. When values in a time series are plotted in a graph and, on an average, these values show an increasing or decreasing trend over a long period of time, the time series is called the **time series with trend effect**.
4. If values in a time series reflect seasonal variation with respect to a given period of time such as a quarter, a month or a year, the time series is called a **time series with seasonal effect**. If the time plot of data in a time series exhibits a cyclic trend, it is called the **time series with cyclic effect**.

5. The long term variations, i.e., the trend component, and short term variations, i.e., the seasonal and cyclic component, are known as **regular variations**. Apart from these regular variations, **random or irregular variations**, which are not accounted for by trend, seasonal or cyclic variations, exist in almost all time series.
6. The **additive model** is one of the most widely used models. It is based on the assumption that at any time t , the time series value Y_t is the sum of all the components. According to the additive model, a time series can be expressed as

$$Y_t = T_t + C_t + S_t + I_t$$

where T_t , C_t , S_t and I_t are the trend, cyclic, seasonal and irregular variations, respectively, at time t .

9. The multiplicative model is based on the assumption that the time series value Y_t at time t is the product of the trend, cyclic, seasonal and irregular component of the series:

$$Y_t = T_t \times C_t \times S_t \times I_t$$

where T_t , C_t , S_t and I_t denote the trend, cyclic, seasonal and irregular variations, respectively. The multiplicative model is found to be appropriate for many business and economic data.

10. There are two methods of moving averages: the **equal weight** or **simple moving averages method** and the **weighted (unequal) moving average method**. The methods of moving averages and exponential smoothing are used for smoothing or filtering the time series data.
11. In exponential smoothing technique, where weights decrease exponentially, except the last one, the trend value is given by

$$y_t^{\text{C}} = w y_t + (1 - w) y_{t-1}^{\text{C}} \quad \text{for } t = 1, 2, \dots$$

where $0 < w < 1$. The value of 'w' is chosen as per the requirements.

12. An alternative approach to smoothing is to fit a polynomial to the data. This treats smoothing as a regression problem in which y_t is the trend value and integral powers of time t are the explanatory variables. The resulting smooth function is a polynomial

$$y_t = \sum_{j=0}^p b_j t^j$$

where b_j is the j^{th} coefficient of the polynomial of degree p . The coefficients of the polynomial are estimated by the method of least squares.

13.9 SOLUTIONS/ANSWERS

E1)	Year	MA
	1979	$(18 + 26 + 16)/3 = 20$
	1980	$(26 + 16 + 27)/3 = 23$

E2) Using equation (1), we obtain the exponential smoothing values as follows:

$$y_2 = 0.5x_2 + 0.5x_1 = 0.5 \times 22 + 0.5 \times 17 = 19.50$$

$$y_3 = 0.5x_3 + 0.25x_2 + 0.25x_1 \\ = 0.5 \times 18 + 0.25 \times 22 + 0.25 \times 17 = 18.75$$

$$y_4 = 0.5x_4 + 0.25x_3 + 0.125x_2 + 0.125x_1 \\ = 0.5 \times 26 + 0.25 \times 18 + 0.125 \times 22 + 0.125 \times 17 = 22.375$$

$$y_5 = 0.5x_5 + 0.25x_4 + 0.125x_3 + 0.0625x_2 + 0.0625x_1 \\ = 0.5 \times 16 + 0.25 \times 26 + 0.125 \times 18 + 0.0625 \times 22 \\ + 0.0625 \times 17 = 19.1875$$

$$y_6 = 0.5x_6 + 0.25x_5 + 0.125x_4 + 0.0625x_3 + 0.03125x_2 + 0.03125x_1 \\ = 0.5 \times 27 + 0.25 \times 16 + 0.125 \times 26 + 0.0625 \times 18 \\ + 0.03125 \times 22 + 0.03125 \times 17 = 23.09375$$

E3) If we plot the given data in a graph, there seems to be no trend in the time series.

If we fit a linear regression to this data and if $T = a + bt$ is the equation, it seems to have $b = 0$. Therefore, the least squares estimates of all 41 time series values would be the same. Also our forecast for rainfall is a for all t because our model is

$$T = a + (\text{forecast error})$$

This result of forecast seems somewhat unreasonable because a particular place cannot have a constant amount of rainfall every year. From Fig. 13.15, you can see that the value of y_t seems to be higher during the period 1970 to 1980 compared to the value of a for the period 1980 to 1990. Therefore, it is logical to assume that the value of a is gradually changing over time and to denote it by y_t rather than a .

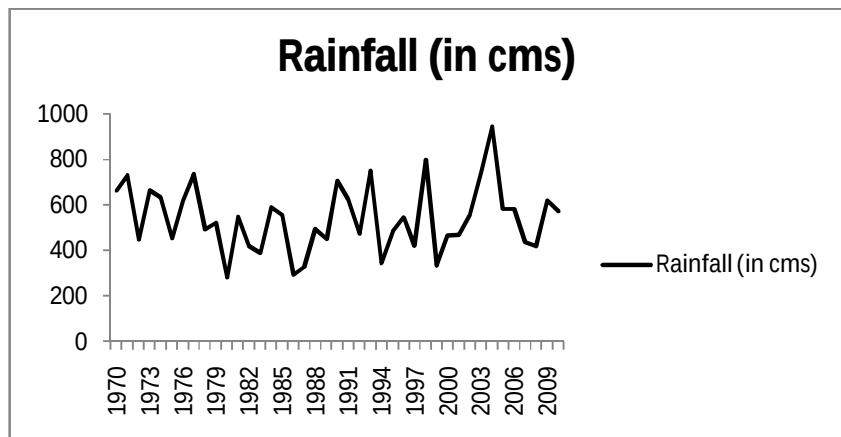


Fig. 13.15: Time series plot for rainfall (in cm) from 1970 to 2009.

According to this smoothing technique, we select a single weight w called the exponential smoothing constant. Here w lies between 0 and 1 and there is a method of choosing this constant. We can compute an exponential smoothing series y'_t as follows:

$$\begin{aligned}
y'_1 &= w y_1 + (1-w) y_1 = y_1 \\
y'_2 &= w y_2 + (1-w) y'_1 \\
y'_3 &= w y_3 + (1-w) y'_2 \\
&\vdots \\
&\vdots \\
&\vdots \\
y'_t &= w y_t + (1-w) y'_{t-1}
\end{aligned}
\quad \dots (i)$$

We should know the value of y_t to initialise the computation of $y_1, y_2, \dots$. For this example we take the initial value of y_t as y_1 , the first and initial value of the given time series. Thus,

$$y'_1 = y_1$$

Now let us take $w = 0.02$ since w can take any value between 0 and 1. From experience it is suggested that the value of w be taken between 0.01 and 0.3. Then from the equation (i) given above, we get

$$y'_2 = 0.02 y_2 + 0.98 y'_1 = 0.02 (728) + (0.98) (664) = 543.71$$

Our forecast for x_1 at time zero is y_1 . Therefore, the forecast error is

$$e_1 = y_1 - y'_1 = 664 - 664 = 0.00$$

Similarly, the second forecast error is

$$e_2 = y_2 - y'_2 = 728 - 665.24 = 62.76$$

Following the same procedure, we get

$$y'_3 = (0.02) y_3 + (0.98) y'_2 = (0.02) 447 + 0.98 \times 665.24 = 660.875$$

and the third forecast error is

$$e_3 = y_3 - y'_3 = 447 - 660.875 = -213.875$$

Proceeding in the same way, we calculate y_t^f and the forecast errors e_t for all t . The calculated forecast values and errors are given in the following table:

Table 9: Forecast values and Forecast errors for given time series

Year	Rainfall y_t	Forecast y_t^f	Error $e_t = y_t - y_t^f$
1970	664	664.00	0.00
1971	728	665.24	+62.76
1972	447	660.875	-213.875
1973	663	660.92	+02.08
1974	630	660.30	-30.30
1975	451	656.11	-05.11
1976	617	655.33	-38.33
1977	734	656.91	+77.09

1978	491	653.58	-162.58
1979	520	650.92	-130.92
1980	280	643.50	-363.50
1981	548	641.59	-93.59
1982	417	637.1	-220.1
1983	387	632.1	-245.1
1984	590	631.25	-41.25
1985	556	629.75	-73.75
1986	292	623.0	-331.0
1987	327	617.07	-290.07
1988	494	614.61	-120.61
1989	448	611.28	-163.28
1990	704	613.13	+90.87
1991	624	613.35	+10.65
1992	473	610.54	-137.54
1993	750	613.33	+136.67
1994	343	607.93	-264.93
1995	484	605.45	-121.45
1996	545	604.24	-59.24
1997	419	600.53	-181.53
1998	798	604.48	+193.52
1999	334	599.07	-265.07
2000	465	596.40	-131.40
2001	468	593.82	-125.82
2002	554	593.03	-39.03
2003	744	596.05	+147.95
2004	943	602.99	+340.01
2005	582	602.56	-20.56
2006	581	602.13	-21.13
2007	437	598.83	-161.83
2008	417	595.20	-178.20
2009	618	595.65	+22.35
2010	571	595.16	-24.16

After calculating the forecast $\hat{a}_t$ and the error e_t for all t , we plot the time series values along with the forecasted values, which are the outcomes after smoothing the time series values and observe the change in the time series values before and after the smoothing (Fig. 13.16). We can also see that the peaks barely exist in the graph of the time series after smoothing.

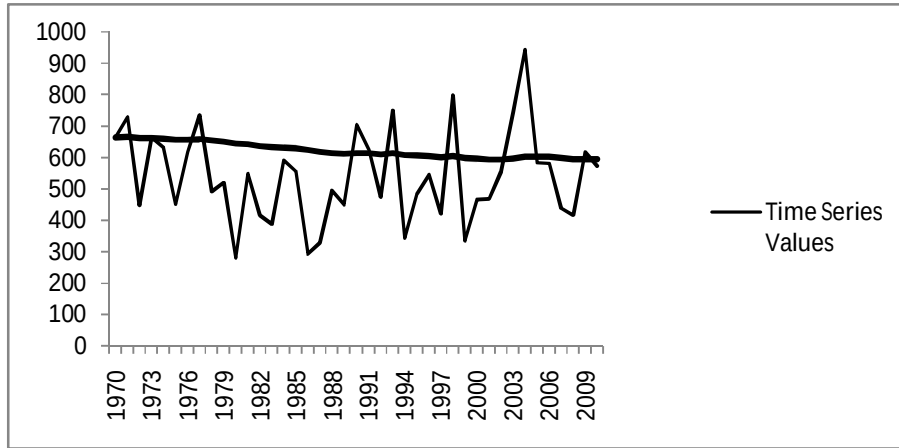


Fig. 13.16: Exponentially smoothed data for rainfall from 1971-2010.

E4) From the given data, we calculate the following values:

$$\bar{t} = 2008.5 \quad \hat{a} = \frac{\sum y_t x_t}{\sum x_t^2} = \frac{21.0}{17.5} = 1.20$$

$$\bar{y} = 20.667$$

On solving the equations, we get

$$\hat{a}_1 = \bar{y} = 20.667$$

$$\hat{b}_1 = \hat{a} = 1.20$$

The trend equation is

$$\hat{y}_t = \bar{y} + \hat{b}_1 (t - \bar{t}) = 20.667 + 1.20(t - 2008.5)$$

The projection for 2012 is:

$$\begin{aligned} \hat{Y}_{2012} &= 20.667 + 1.20(2012 - 2008.5) = 20.667 + 1.20 \times 3.5 \\ &= 24.867 \end{aligned}$$

E5) The complete set of MA(2) values for $m = 2$ are as follows:

Table 10: Centred moving averages for given time series

Quarter Year	1	2	3	4
2001	-	-	1169.375	1188.75
2002	1239.375	1281.25	1332.5	1442.5
2003	1520.0	1559.125	1619.75	1600.0
2004	1510.625	1448.625	1340.5	1275.0
1977	1287.5	1303.75	1340.625	1341.875
2006	1361.875	1470	1568.125	1695.0
2007	1845.625	1928.125	2016.875	2078.125
2008	2087.5	2145.0	2195.0	2190.0
2009	2182.5	2099.375	-	-

E6) The complete set of MA(1) values for $m = 3$ are as follows:

Table 11: Centred moving averages for given time series

Quarter Year	1	2	3	4
2001	-	1065	1238.34	1163.34
2002	1253.34	1218.34	1383.34	1401.67
2003	1556.67	1551.67	1655.0	1436.67
2004	1401.67	1243.34	1338.34	1226.67
2005	1316.67	1233.34	1418.34	1321.67
2006	1366.67	1338.34	1658.34	1685.0
2007	1891.67	1815.0	2141.67	2060.0
2008	2136.67	1995.0	2303.34	2210.0
2009	2260.0	1971.67	2071.67	-

UNIT 15 STATIONARY PROCESSES

Structure

- 15.1 Introduction
 - Objectives
- 15.2 Stationary Processes
 - Stationary Process
 - Strict Stationary Process
 - Weak Stationary Process
- 15.3 Autocovariance and Autocorrelation
 - Autocovariance and Autocorrelation Coefficients
 - Estimation of Autocovariance and Autocorrelation Coefficients
- 15.4 Correlogram of Stationary Processes
 - Interpretation of Correlogram
- 15.5 Summary
- 15.6 Solutions / Answers

15.1 INTRODUCTION

In Units 13 and 14, you have learnt that a time series can be decomposed into four components, i.e., Trend (T), Cyclic (C), Seasonal (S) and Irregular (I) components. We have discussed methods for smoothing or filtering the time series and for estimating Trend, Seasonal and Cyclic components. We have also explained how to use them for forecasting.

In this unit, we describe a very important class of time series, called the **stationary time series**. In Sec. 15.2, we explain the concept of stationary process and define weak and strict stationary processes. We discuss autocovariance, autocorrelation function and correlogram of a stationary process in Secs. 15.3 and 15.4. If a time series is stationary, we can model it and draw further inferences and make forecasts. If a time series is not stationary, we cannot do any further analysis and hence cannot make reliable forecasts. If a time series shows a particular type of non-stationarity and some simple transformations can make it stationary, then we can model it.

In the next unit, we shall discuss certain stationary linear models such as Auto Regressive (AR), Moving Average (MA) and mixed Autoregressive Moving Average (ARMA) processes. We shall also discuss how to deal with models with trend by considering an integrated model called Autoregressive Integrated Moving Average (ARIMA) model.

Objectives

After studying this unit, you should be able to:

- describe stationary processes;
- define weak and strict stationary processes;
- define autocovariance and autocorrelation coefficients;
- estimate autocovariance and autocorrelation coefficients;
- plot the correlogram and interpret it; and
- make proper choice of probability models for further studies.

15.2 STATIONARY PROCESSES

In the course MST-004, you have studied random variables and their properties. Recall that a random variable Y is a function defined on a sample space. A family of random variables defined on the same sample space taking values over time is known as a **random process**. Most physical processes in real life situations involve random components or variables and a random process may be described as a statistical phenomenon that evolves in time. A random process may be defined as a family of random variables defined on a given probability space indexed by the parameter t . Here we denote a stochastic variable by the capital letter Y and assume that it is observable at discrete time points $t_1, t_2, \dots$.

A random process is a statistical phenomenon that evolves in time according to some laws of probability. The length of queue in a system, the number of accidents in a particular city in successive weeks, etc. are examples of a random process. Mathematically, a random process is defined as the family of random variables which are ordered in time, i.e., random process $\{Y(t); t \text{ belongs to } T\}$ is a collection of random variables, where T is a set for which all random variables Y_t are defined on the same sample space. If T takes continuous range of values, the random process is said to be a **continuous parameter process**. On the other hand, if T takes discrete set of values, the process is said to be a **discrete parameter process**. We use the notation Y_t for a random process when we deal with discrete parameter processes. When T represents time, the random process is referred to as a time series.

In Units 13 and 14, we have dealt with one set of observations recorded at different times. Thus, we had only a single outcome of the process and a single observation on the random variable at time t . This sample may be regarded as one time series out of the infinite set of time series, which might have been observed. This infinite set of time series is called an **Ensemble**.

Every member of the ensemble can be taken as a possible realisation of the stochastic process and the observed time series can be considered as one particular realisation.

15.2.1 Stationary Process

Broadly speaking, a time series is said to be **stationary** if there is no systematic change in mean, variance and covariance of the observations over a period of time. This means that the properties of one section of the time series are similar to the properties of the other sections of the time series. In other words, a process is said to be stationary if it is in a state of statistical equilibrium.

A random process is said to be stationary if the joint distribution of $Y_{t_1}, Y_{t_2}, Y_{t_3}, \dots, Y_{t_k}$ is the same as the joint distribution of $Y_{t_1+J}, Y_{t_2+J}, Y_{t_3+J}, \dots, Y_{t_k+J}$, for all $t_1, t_2, \dots, t_k$ and J . In other words, shifting the origin of time by an amount J has no effect on the joint distribution. This means that it depends only on the interval between $t_1, t_2, \dots, t_k$. This definition holds for any value of k .

For $k=1$,

$$E(Y_t) = \mu \quad \text{and} \quad V(Y_t) = \sigma^2$$

This implies that the mean and variance of Y_t are constant and do not depend on time.

For $k=2$, the joint distribution of Y_{t_1} and Y_{t_2} depends only on the time difference $(t_2 - t_1) = J$, say, which is called **lag**.

Thus, the covariance term depends only on lag $J = t_2 - t_1$, i.e.,

$$\begin{aligned}\gamma(t_1, t_2) &= E[(Y_{t_1} - \mu)(Y_{t_2} - \mu)] \\ &= \text{Cov}(Y_{t_1}, Y_{t_2})\end{aligned}$$

The variance function is a special case of the covariance function when $t_1 = t_2$, i.e., $J = 0$.

There are two types of stationary processes: Strict stationary processes and weak stationary processes. Let us discuss them one at a time.

15.2.2 Strict Stationary Process

Strict stationary process imposes the strong condition that the joint probability distribution remains the same and depends only on the time interval. If all the finite dimensional distributions of a random process are invariant under translation of time, then it is called a **Strict Sense Stationary process** or **SSS process**. In other words, if the joint distribution of $Y_{t_1}, Y_{t_2}, \dots, Y_{t_k}$ is the same as the joint distribution of $Y_{t_1+J}, Y_{t_2+J}, \dots, Y_{t_k+J}$, for all $t_1, t_2, \dots, t_k$ and $J (> 0)$ for all $k \geq 1$, the random process Y_t is called a Strict Sense Stationary process.

The strict stationary process requires that for any t_1, t_2 , the distributions of Y_{t_1} and Y_{t_2} for all $i = 1, 2, \dots, n$ must be the same. Also, the bivariate distributions of pairs $[Y_{t_1}, Y_{t_1+J}]$ and $[Y_{t_2}, Y_{t_2+J}]$ are the same for all $i = 1, 2, \dots, n$ and J .

Note that the requirement of strict stationarity is a severe one. Usually it is difficult to establish it mathematically and in most cases, the distributions of $[Y_{t_i}, Y_{t_i+J}]$ for all $i = 1, 2, \dots, n$ and J are not known. That is why, the less restrictive notions of stationarity called **weak stationary processes** have been developed.

15.2.3 Weak Stationary Process

A stationary process is said to have **weak stationarity** of order m , if the moments up to order m depend only on time lag J . If $m = 2$, the stationarity (or weak stationarity) of order 2 implies that moments up to the second order depend only on time lag J . For a weak stationary process:

$$E[y_t] = \mu;$$

and

$$\text{Cov}[Y_t, Y_{t+J}] = \gamma(J)$$

No requirements are placed on the moments of higher order. The definition also implies that the mean and variance remain constant and finite. Thus, a random process Y_t with finite first and second order moments is called a **weak stationary process**, if the means are constant and the covariance depends only on the time lag.

In the subsequent discussion in this unit, we shall assume weak stationarity as many properties of a stationary process depend only on the first and second order moments. One important class of processes is the **normal** process, where joint distributions of $Y_{t_1}, Y_{t_2}, \dots, Y_{t_k}$ are multivariate normal for all $t_1, t_2, \dots, t_k$. This multivariate normal distribution is completely characterised by its first and second moments, i.e., μ_t and $\gamma(t_1, t_2)$.

15.3 AUTOCOVARANCE AND AUTOCORRELATION

In this section, we discuss the autocovariance and autocorrelation function for a stationary process. Of particular interest in the analysis of a time series are the covariance and correlation between Y_{t1} and Y_{t2} . Since these are the covariance and correlation within the same time series, they are called **autocovariance** and **autocorrelation**. Some important properties of time series can be studied with the help of autocovariance and autocorrelation. They measure the linear relationship between observations at different time lags. They provide useful descriptive properties of the time series being studied and are important tools for guessing a suitable model for the time series data. Let us define these parameters.

15.3.1 Autocovariance and Autocorrelation Coefficients

Suppose a weak stationary time series process is denoted by $Y_1, Y_2, \dots, Y_t, Y_{t+1}, \dots, Y_{t+k}, \dots$. We are interested in finding the linear relationship between two consecutive observations, Y_t, Y_{t+1} . We are also interested in the relationship between observations that are apart by a time lag k , e.g., Y_t and Y_{t+k} . We shall study the linear relationship by studying covariances and correlations of observations at different time lags.

In the course MST-002, you have learnt how to calculate the covariance and correlation between two variables for given N pairs of observations on two variables X and Y , say $\{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$. Recall that the formulas for computation of covariance and correlation coefficient are given as:

$$\text{Cov}(X, Y) = E[(X - \mu)(Y - \mu)]$$

$$\rho_{(X,Y)} = \frac{E[(X - \mu)(Y - \mu)]}{\sqrt{E[(X - \mu)^2] E[(Y - \mu)^2]}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Here we apply analogous formulas to the stationary time series data to measure whether successive observations are correlated.

The autocovariance between Y_t and Y_{t+k} , separated by the time interval k , for a stationary process must be the same for all t and is defined as:

$$\gamma_k = \gamma_{-k} = \text{Cov}(Y_t, Y_{t+k}) = E[(Y_{t+k} - \mu)(Y_t - \mu)] \quad \dots (1)$$

Similarly, the autocorrelation at lag k is

$$\rho_k = \frac{E[(Y_{t+k} - \mu)(Y_t - \mu)]}{\sqrt{E[(Y_{t+k} - \mu)^2] E[(Y_t - \mu)^2]}}$$

$$= \frac{\text{Cov}(Y_t, Y_{t+k})}{\sigma_Y^2} \quad \dots (2)$$

From equation (1), we note that

$$\sigma_Y^2 = \gamma_0 \quad \dots (3)$$

Therefore, $\rho_k = \frac{\gamma_k}{\gamma_0}$ and $\rho_0 = 1$... (4)

15.3.2 Estimation of Autocovariance and Autocorrelation Coefficients

So far, we have defined the autocovariance and autocorrelation coefficients for a random process. You would now like to estimate them for a finite time series for which N observations $y_1, y_2, \dots, y_N$ are available. We shall denote a realisation of the random process $Y_1, Y_2, \dots, Y_N$ by small letters $y_1, y_2, \dots, y_N$. The mean μ can be estimated by

$$\bar{y} = \sum_{i=1}^N y_i / N \quad \dots (5)$$

and autocovariance γ_k at lag k can be estimated by the autocovariance coefficient c_k as follows:

$$c_k = \sum_{t=1}^{N-k} (y_t - \bar{y})(y_{t+k} - \bar{y}) / (N - k), \text{ for all } k \quad \dots (6)$$

The sample covariance is a special case of autocovariance when $k = 0$, i.e.,

$$c_0 = \sum_{t=1}^N (y_t - \bar{y})(y_t - \bar{y}) / N = \sum_{t=1}^N (y_t - \bar{y})^2 / N = \sigma_Y^2, \text{ for all } k$$

The autocorrelation coefficients (r_k) are usually calculated by computing the series of autocovariance coefficients (c_k) as follows:

$$r_k = \frac{c_k}{c_0} = \frac{c_k}{\sigma_Y^2} \quad \dots (7)$$

In practice, at least 50 observations are required for the estimation of correlations. It is also advisable that for calculations of r_k , the lag k should not exceed $N/4$.

Let us explain these concepts with the help of an example.

Example 1: A series of 10 consecutive yields from a batch chemical process are given as follows:

47, 64, 23, 71, 38, 64, 55, 41, 59, 48

Calculate the mean, autocovariance c_1 and autocorrelation coefficient r_1 for the given time series.

Solution: We first construct the following table:

S. No.	Y	Y ²	(Y _t - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+1} - $\bar{Y}$)
1	47	2209	-4	
2	64	4096	13	-52
3	23	529	-28	-364
4	71	5041	20	-560
5	38	1444	-13	-260
6	64	4096	13	-169
7	55	3025	4	52
8	41	1681	-10	-40
9	59	3481	8	-80
10	48	2304	-3	-24
Total	$\sum y_i = 510$	$\sum y_i^2 = 27906$		$\sum (Y_t - \bar{Y})(Y_{t+1} - \bar{Y}) = -1497$

From equation (5), we get

$$\bar{y} = \frac{\sum_{i=1}^N y_i}{N} = \frac{510}{10} = 51.0$$

From equation (6), for $k=0$, the autocovariance coefficient is

$$\begin{aligned} c_0 &= \frac{\sum_{i=1}^{N-k} (y_i - \bar{y})^2}{N} = \frac{(\sum y_i^2 - N\bar{y}^2)}{N} \\ &= \frac{(27906 - 26010)}{10} = 189.6 \end{aligned}$$

For $k = 1$,

$$c_1 = \frac{\sum_{t=1}^9 (y_t - \bar{y})(y_{t+1} - \bar{y})}{9} = -166.33$$

From equation (7),

$$r_1 = \frac{c_1}{c_0} = \frac{-166.33}{189.6} = -0.88$$

You may now like to solve a problem to assess your understanding.

E1) Ten successive observations on a stationary time series are as follows:
1.6, 0.8, 1.2, 0.5, 0.6, 1.5, 0.8, 1.2, 0.5, 1.3.
Plot the observations and calculate r_1 .

E2) Fifteen successive observations on a stationary time series are as follows:
34, 24, 23, 31, 38, 34, 35, 31, 29,
28, 25, 27, 32, 33, 30.
Plot the observations and calculate r_1 .

15.4 CORRELOGRAM OF STATIONARY PROCESSES

A useful plot for interpreting a set of autocorrelations coefficient is called a **correlogram** in which the sample autocorrelation coefficients r_k are plotted versus the lag J where $J=1, 2, 3, \dots, k$. This helps us in examining the nature of time series. It is also a very important diagnostic tool for the selection of a suitable model for the process which generates the data. The correlogram is alternatively known as the **sample autocorrelation function (acf)**.

The value of lag J is usually much less than N . For example, a time series of length $N = 200$ given in Fig 15.1a shows the plot of the time series for $N=200$ and Fig. 15.1b shows a plot of the correlogram for a lag up to order 17.

The relatively smooth nature of the time series plot indicates that observations which are close to each other (at smaller lags) are positively correlated. The correlogram suggests that observations with smaller lag are positively correlated and autocorrelation decreases as lag k increases.

In most time series, it is noticed that the absolute value of r_k , i.e., $|r_k|$ decreases as k increases. This is because observations which are located far away are not related to each other, whereas observations lying closer to each other may be positively (or negatively) correlated.

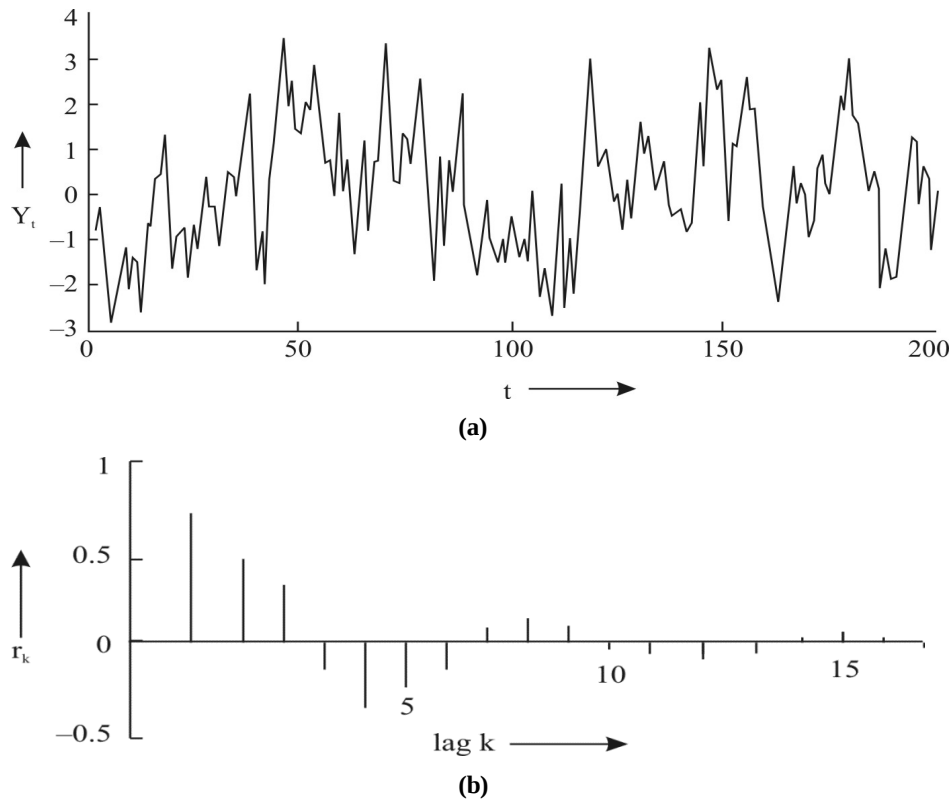


Fig. 15.1: a) Plot of a time series for $N = 200$; b) correlogram for lag $k = 1, 2, \dots, 17$.

15.4.1 Interpretation of Correlogram

Sometimes it is possible to recognise the nature of a time series from its correlogram, though it is not always easy. We shall describe certain types of time series processes and the nature of their correlograms. If the correlogram of a time series is one of these types, it is possible to get a good idea of the process which gave rise to that time series.

Random Series

In case all observations are completely random, i.e., they contain only independent observations, the series is called a **random series**.

This means that $r_k \approx 0$ for all non zero k . The correlogram of such a random series will be oscillating around the axis (zero line). In fact, for such a series, for large N , the values of r_k approximately follow the $N(0, 1/N)$ distribution. Thus, in about 95% cases, the values of r_k lie in a range of $\pm 2/\sqrt{N}$. If the correlogram shows such a behaviour, it is a good indication that the time series is random. However, this behaviour may not always confirm that the time series is random and it may need further examination.

Short-Term Correlation

Stationary time series usually has a few large autocorrelations in absolute value for small lag k . They tend to zero very rapidly with increase in lag k (see Fig. 15.1b). When the first few autocorrelations are positive, the time series is smooth in nature, i.e., if an observation is above mean, it is likely to be followed by an observation above mean and if an observation is below mean, it is likely to be followed by an observation below mean. This gives

an indication of stationary time series with most of the non-zero autocorrelations being either positive or negative.

Alternating Series

If a time series behaves in a very rough and zig-zag manner, alternating between values above and below mean, it is indicated by a negative r_1 and positive r_2 . An alternating time series with its correlogram is shown in Fig.15.2.

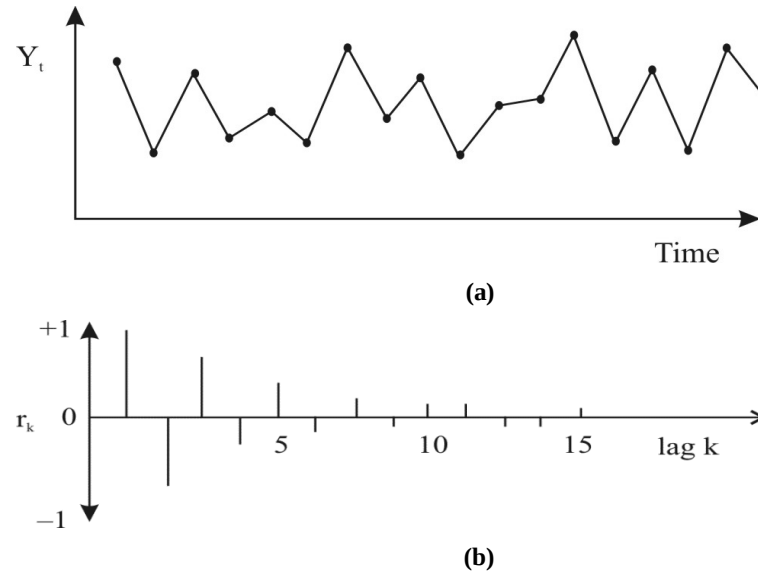


Fig. 15.2: a) Plot of alternating time series; b) correlogram for an alternating series with lag up to 15.

Non-Stationary Time Series

If a time series contains trend, it is said to be **non-stationary**. Such a series is usually very smooth in nature and its autocorrelations go to zero very slowly as the observations are dominated by trend. Due to the presence of trend, the autocorrelations move towards zero very slowly (see Fig. 15.3). One should remove trend from such a time series before doing any further analysis.

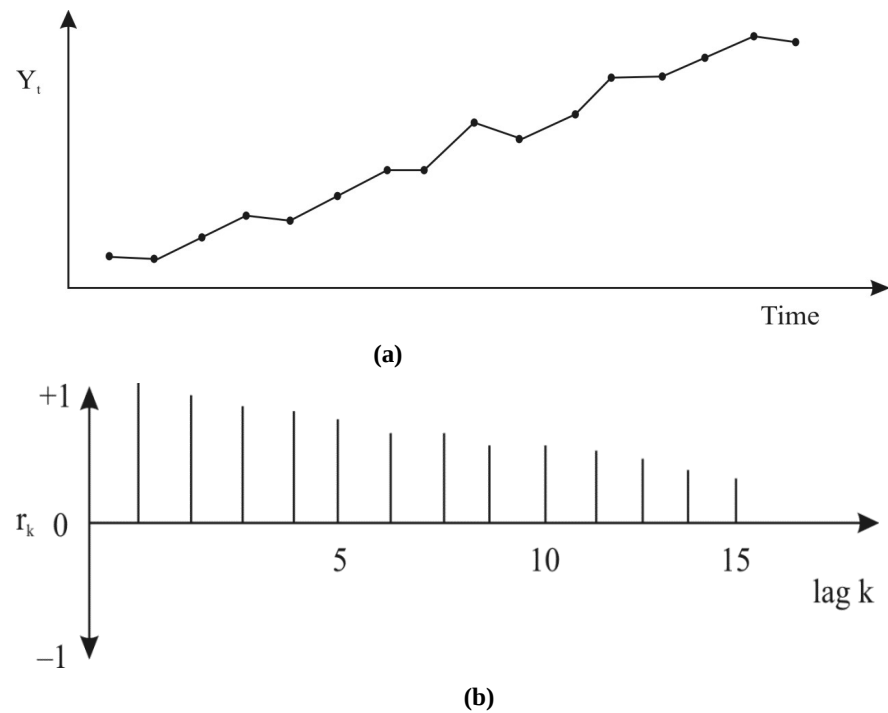


FIG. 15.3: a) Plot of non-stationary time series; b) correlogram for a non-stationary series with lag up to 15.

If a time series has a dominant seasonal pattern, the time plot will show a cyclical behaviour with a periodicity of the seasonal effect. If data have been recorded on monthly basis and the seasonal effect is of twelve months, i.e., $s = 12$, we would expect a highly negative autocorrelation at lag 6 (r_6) and highly positive correlation at lag 12 (r_{12}). In case of quarterly data, we expect to find a large negative r_2 and large positive r_4 . This behaviour will be repeated at r_6, r_8 and so on. This pattern of cyclical behaviour of correlogram will be similar to the time plot of the data.

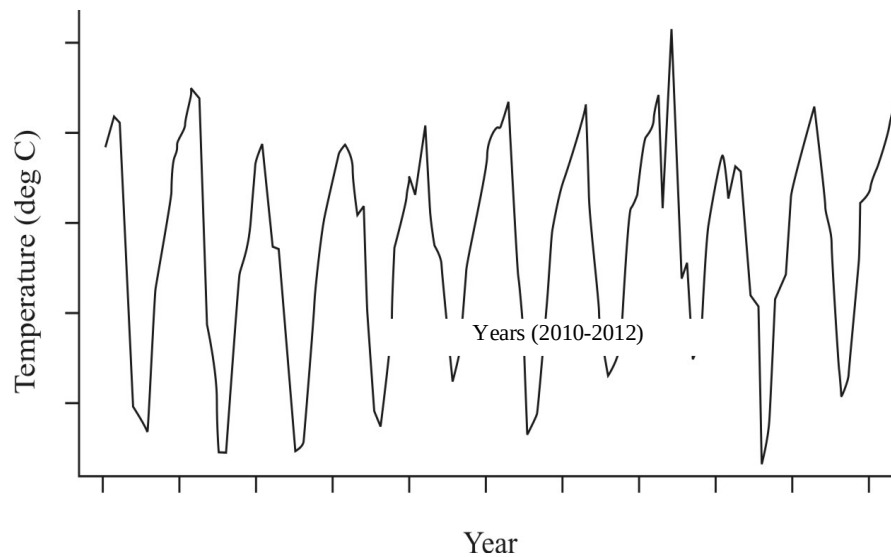


Fig. 15.4: Time plot of the average rainfall at a certain place, in successive months from 2010 to 2012.

Therefore, in this case the correlogram may not contain any more information than what is given by the time plot of the time series.

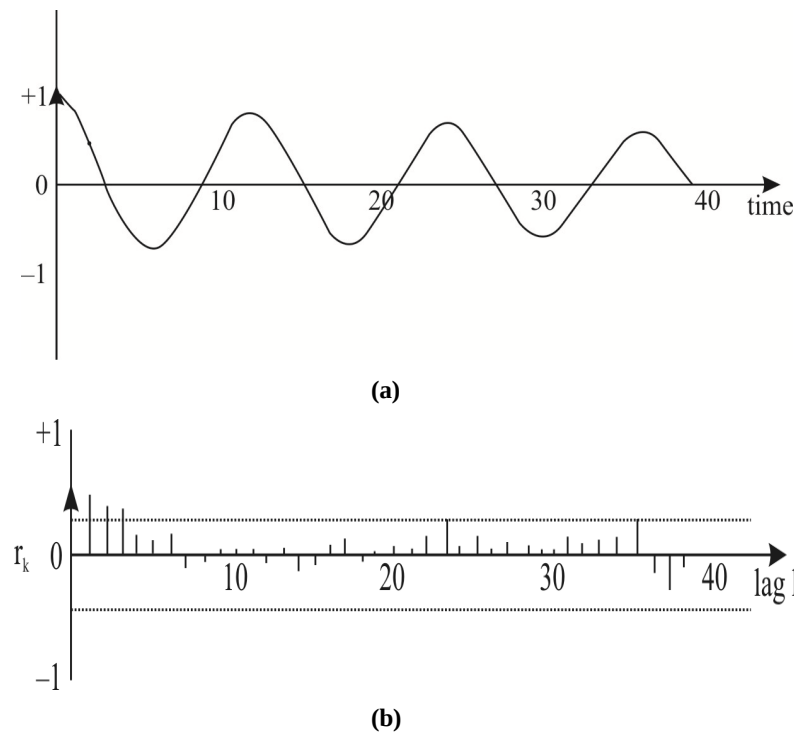


Fig. 15.5: a) Smoothed plot of the average rainfall at a certain place, in successive months from 2010 to 2012; b) correlogram of monthly observations of seasonal time series.

Fig. 15.5a shows a time plot of monthly rainfall and Fig. 15.5b shows the correlogram. Both show a cyclical pattern and the presence of a strong 12 monthly seasonal effect. However, it is doubtful that in such cases the correlogram gives any more information about the presence of seasonal effect as compared to the time plot shown in Fig 15.4.

In general, the interpretation of correlogram is not easy and requires a lot of experience and insight. Estimated autocorrelations (r_k) are subject to sampling fluctuations and if N is small, their variances are large. We shall discuss this in more detail when we consider a particular process. When all the population autocorrelations ρ_k ($k \neq 0$) are zero, as happens in a random series, then the values of r_k are approximately distributed as $N(0, 1/N)$. This is a very good guide for testing whether the population correlations are all zeros or not, i.e., the process is completely random or not.

Example 2: For the time series given in Example 1, calculate r_1, r_2, r_3, r_4 and r_5 and plot a correlogram.

Solution: From Example 1 and its results we have the following:

$$\bar{y} = 51.0, \quad c_0 = 189.6, \quad c_1 = -166.33 \text{ and } r_1 = -0.88$$

Now we form the table for the calculations as follows:

S. No.	Y	Y ²	(Y _t - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+2} - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+3} - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+4} - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+5} - $\bar{Y}$)
1	47	2209	-4				
2	64	4096	13				
3	23	529	-28	112			
4	71	5041	20	260	-80		
5	38	1444	-13	364	-169	52	
6	64	4096	13	260	-364	169	-52
7	55	3025	4	-52	80	-112	52
8	41	1681	-10	-130	130	-200	280
9	59	3481	8	32	104	-104	160
10	48	2304	-3	30	-12	-39	39
Total	510	27906		876	-311	-234	479

We now calculate the autocorrelation coefficients r_2, r_3, r_4 and r_5 as follows:

For $k = 2$, we get

$$\begin{aligned}
 c_2 &= \sum_{t=1}^8 \frac{(y_t - \bar{y})(y_{t+2} - \bar{y})}{8} \\
 &= 876/8 = 109.5 \\
 r_2 &= \frac{c_2}{c_0} = 109.5/189.6 = 0.58
 \end{aligned}$$

For $k = 3$, we get

$$c_3 = \frac{\sum_{t=1}^7 (y_t - \bar{y})(y_{t+3} - \bar{y})}{7}$$

$$= -311/7 = -44.43$$

$$r_3 = \frac{c_3}{c_0} = -44.43 / 189.6 = -0.2343$$

For $k = 4$, we get

$$c_4 = \frac{\sum_{t=1}^6 (y_t - \bar{y})(y_{t+4} - \bar{y})}{6}$$

$$= -234/6 = -39$$

$$r_4 = \frac{c_4}{c_0} = -39 / 189.6 = -0.2057$$

For $k = 5$, we get

$$c_5 = \frac{\sum_{t=1}^5 (y_t - \bar{y})(y_{t+5} - \bar{y})}{5}$$

$$= 479/5 = 95.8$$

$$r_5 = \frac{c_5}{c_0} = 95.8 / 189.6 = -0.5052$$

Thus, we have obtained the autocorrelation coefficients r_1, r_2, r_3, r_4 and r_5 as $r_1 = -0.88, r_2 = 0.58, r_3 = -0.2343, r_4 = -0.2057, r_5 = -0.5052$, respectively.

Now we plot the correlogram for the given time series by plotting the values of the autocorrelation coefficients versus the lag k for $k = 1, 2, \dots, 5$. The correlogram is shown in Fig. 15.6.

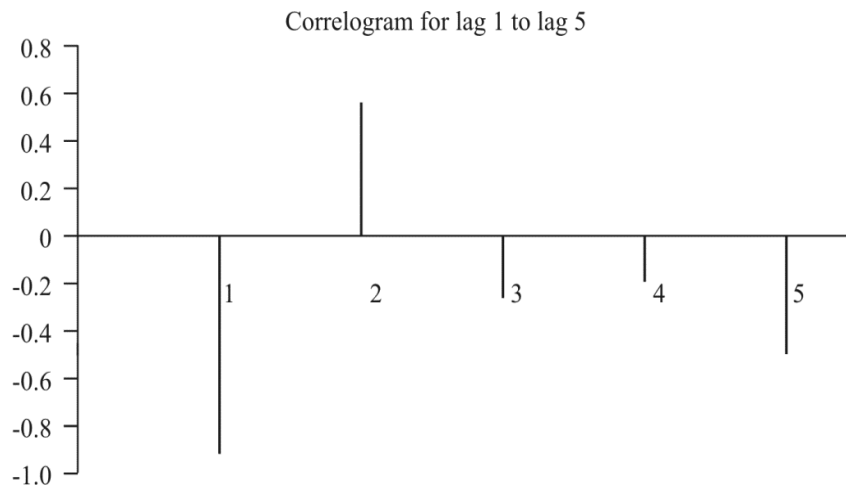


Fig. 15.6: Correlogram for the given time series.

Example 3: A computer generates a series of 200 observations that are supposed to be random. The first 10 sample autocorrelation coefficients of the series are:

$$r_1 = 0.02, r_2 = 0.05, r_3 = -0.09, r_4 = 0.08, r_5 = -0.02, r_6 = -0.07, \\ r_7 = 0.12, r_8 = 0.06, r_9 = 0.02, r_{10} = -0.08$$

Plot the correlogram.

Solution: The correlogram for the given values of autocorrelation coefficients is shown in Fig. 15.7.

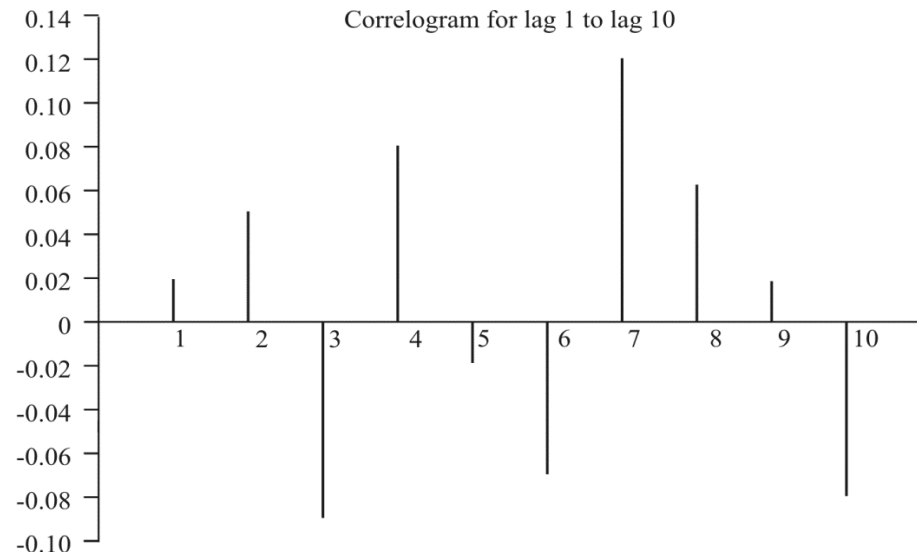


Fig. 15.7: Correlogram for 10 sample autocorrelation coefficients of the series of 200 observations.

Example 4: A random walk (S_t , $t = 0, 1, 2, \dots$) starting at zero is obtained by cumulative sum of independently and identically distributed (i.i.d) random variables. Check whether the series is stationary or non-stationary.

Solution: Since we have a random walk (S_t , $t = 0, 1, 2, \dots$) starting at zero obtained from cumulative sum of independently and identically distributed (i.i.d) random variables, a random walk with zero mean is obtained by defining $S_0 = 0$ and

$$S_t = Y_1 + Y_2 + \dots + Y_t, \quad \text{for } t = 1, 2, \dots$$

where $\{Y_t\}$ is i.i.d. noise with mean zero and variance σ^2 . Then we have

$$E(S_t) = 0, \quad E(S_t^2) = t\sigma^2 < \infty \quad \text{for all } t$$

$$\begin{aligned} \text{Cov}(S_t, S_{t+h}) &= \text{Cov}(S_t, S_t + Y_{t+1} + \dots + Y_{t+h}) \\ &= \text{Cov}(S_t, S_t) = t\sigma^2 \end{aligned}$$

This depends upon t and hence the series S_t is non-stationary.

You may now like to solve the following exercises to assess your understanding about correlogram and stationary processes.

-
- E3)** Calculate r_2, r_3, r_4 and r_5 for the time series given in Exercise 1 and plot a correlogram.
- E4)** Calculate r_2, r_3, r_4 and r_5 for the time series given in Exercise 2 and plot a correlogram.
- E5)** A computer generates a series of 500 observations that are supposed to be random. The first 10 sample autocorrelation coefficients of the series are:

$$r_1 = 0.09, r_2 = -0.08, r_3 = 0.07, r_4 = -0.06, r_5 = -0.05, r_6 = 0.04, \\ r_7 = -0.03, r_8 = 0.02, r_9 = -0.02, r_{10} = -0.01$$

Plot the correlogram.

15.5 SUMMARY

1. A time series is said to be **stationary** if there is no systematic change in mean, variance and covariance of the observations over a period of time. If a time series is stationary, we can model it and draw further inferences and make forecasts. If a time series is not stationary, we cannot do any further analysis and make reliable forecasts. If a time series shows a particular type of non-stationarity and some simple transformations can make it stationary, then we can model it.
2. A **random process** is a statistical phenomenon that evolves in time according to some laws of probability. Mathematically, a random process is defined as the family of random variables which are ordered in time, i.e., a random process $\{Y(t); t \text{ belongs to } T\}$ is a collection of random variables, where T is a set for which all the random variables Y_t are defined on the same sample space.
3. A random process is said to be stationary if the joint distribution of $Y_{t_1}, Y_{t_2}, Y_{t_3}, \dots, Y_{t_k}$ is the same as the joint distribution of $Y_{t_1+J}, Y_{t_2+J}, Y_{t_3+J}, \dots, Y_{t_k+J}$, for all $t_1, t_2, \dots, t_k$ and J . In other words, shifting the origin of time by an amount J has no effect on the joint distribution. This means that it depends only on the interval between $t_1, t_2, \dots, t_k$.
4. If all the finite dimensional distributions of a random process are invariant under the translation of time, it is called a **Strict Sense Stationary process** or **SSS process**. In other words, if the joint distribution of $Y_{t_1}, Y_{t_2}, \dots, Y_{t_k}$ is the same as the joint distribution of $Y_{t_1+J}, Y_{t_2+J}, \dots, Y_{t_k+J}$, for all $t_1, t_2, \dots, t_k$ and $J (> 0)$ for all $k \geq 1$, the random process Y_t is called a Strict Sense Stationary process.
5. A **stationary process is said to have weak stationarity** of order m , if the moments up to order m depend only on the time lag J . If $m = 2$, the stationarity (or weak stationarity) of order 2 implies that moments up to the second order depend only on the time lag J .
6. Of particular interest in the analysis of a time series are the covariance and correlation between Y_{t_1} and Y_{t_2} . Since these are the covariance and correlation within the same time series, they are called **autocovariance** and **autocorrelation**. Some important properties of time series can be studied with the help of autocovariance and autocorrelation. They measure the linear relationship between observations at different time lags. They provide useful descriptive properties of the time series being studied and are important tools for guessing a suitable model for the time series data.
7. A useful plot for interpreting a set of autocorrelation coefficients is called a **correlogram** in which the sample autocorrelation coefficients r_k are plotted versus the lag J where $J=1, 2, 3, \dots, k$. This helps us in examining the nature of time series. It is also a very important diagnostic tool for selection of a suitable model for the process which generates the data. The correlogram is also known as the sample autocorrelation function (acf).
8. In most time series, it is noticed that the absolute value of r_k , i.e., $|r_k|$ decreases as k increases. This is because observations which are located far away are not related to each other, whereas observations lying closer to each other may be positively (or negatively) correlated.

15.6 SOLUTIONS/ANSWERS

E1) We first plot the stationary time series values as shown in Fig. 15.8.

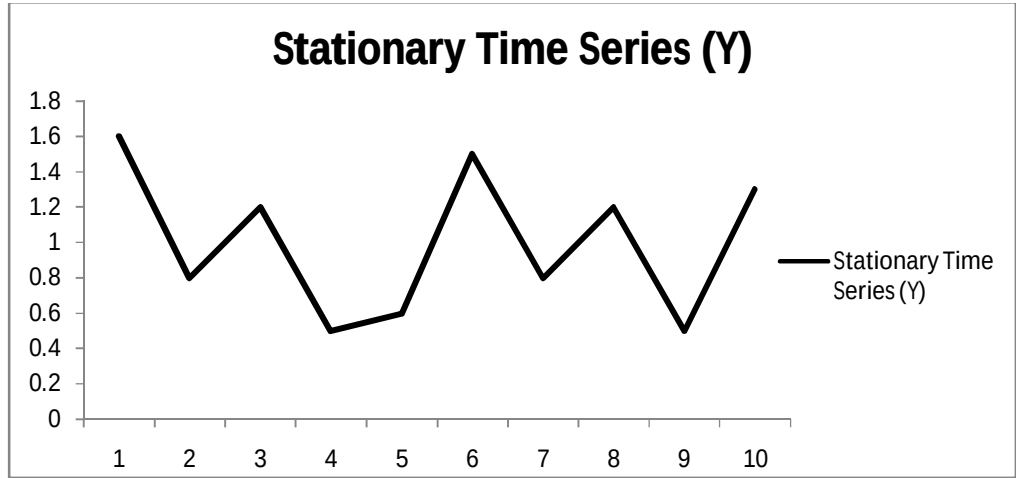


Fig. 15.8: Plot of the given stationary time series.

For calculating r_1 we first calculate the terms given in the table below:

S. No.	Y	Y ²	(Y _t - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+1} - $\bar{Y}$)
1	1.6	2.56	0.6	
2	0.8	0.64	-0.2	-0.12
3	1.2	1.44	0.2	-0.04
4	0.5	0.25	-0.5	-0.1
5	0.6	0.36	-0.4	0.2
6	1.5	2.25	0.5	-0.2
7	0.8	0.64	-0.2	-0.1
8	1.2	1.44	0.2	-0.04
9	0.5	0.25	-0.5	-0.1
10	1.3	1.69	0.3	-0.15
Total	10	11.52		-0.65

$$\bar{y} = \sum_{i=1}^N y_i / 10 = 10 / 10 = 1.0$$

$$c_0 = \sum_{t=1}^{N-k} \frac{(y_t - \bar{y})^2}{N} = \frac{(\sum y_i^2 - N\bar{y}^2)}{N}$$

$$= (11.52 - 10) / 10 = 0.152$$

$$c_1 = \frac{\sum_{t=1}^{N-1} (y_t - \bar{y})(y_{t+1} - \bar{y})}{N-1} = \frac{-0.65}{9} = -0.072$$

$$r_1 = \frac{c_1}{c_0} = -0.072 / 0.152 = -0.475$$

E2) We first plot the stationary time series values as shown in Fig. 15.9.

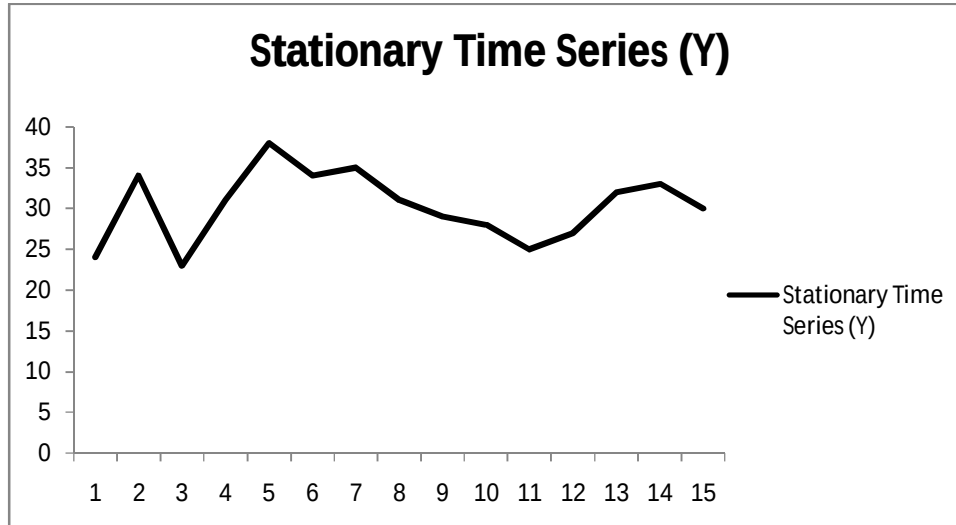


Fig. 15.9: Plot of the given stationary time series.

For the given time series we do the following calculations:

S. No.	Y	Y ²	(Y _t - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+1} - $\bar{Y}$)
1	24	576	-6.267	
2	34	1156	3.733	-23.395
3	23	529	-7.267	-27.129
4	31	961	0.733	-5.329
5	38	1444	7.733	5.671
6	34	1156	3.733	28.871
7	35	1225	4.733	17.671
8	31	961	0.733	3.471
9	29	841	-1.267	-0.929
10	28	784	-2.267	2.871
11	25	625	-5.267	11.938
12	27	729	-3.267	17.204
13	32	1024	1.733	-5.662
14	33	1089	2.733	4.738
15	30	900	-0.267	-0.729
Total	454	14000		1.773

$$\bar{y} = \sum_{i=1}^N y_i / 15 = 454 / 15 = 30.267,$$

$$c_0 = \sum_{t=1}^{N-k} \frac{(y_t - \bar{y})^2}{N} = \frac{(\sum y_i^2 - N \bar{y}^2)}{N}$$

$$= (14000 - 13922.82) / 15 = 5.145$$

$$c_1 = \frac{\sum_{t=1}^{N-1} (y_t - \bar{y})(y_{t+1} - \bar{y})}{N-1}$$

$$= \frac{1.773}{14} = 0.1266$$

$$r_1 = \frac{c_1}{c_0} = 0.1266/5.145 = 0.0246$$

E3) In Exercise 1, we have obtained the following values:

$$\bar{y} = 1.0, c_0 = 0.152, c_1 = -0.072 \text{ and } r_1 = -0.475$$

S. No.	Y	Y ²	(Y _t - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+2} - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+3} - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+4} - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+5} - $\bar{Y}$)
1	1.6	2.56	0.60				
2	0.8	0.64	-0.2				
3	1.2	1.44	0.20	0.12			
4	0.5	0.25	-0.50	0.10	-0.30		
5	0.6	0.36	-0.40	-0.08	0.08	-0.24	
6	1.5	2.25	0.50	-0.25	0.10	-0.10	0.30
7	0.8	0.64	-0.20	0.08	0.10	-0.04	0.04
8	1.2	1.44	0.20	0.10	-0.08	-0.10	0.04
9	0.5	0.25	-0.50	0.10	-0.25	0.20	0.25
10	1.3	1.69	0.30	0.06	-0.06	0.15	-0.12
	10	11.52		0.23	-0.41	-0.13	0.51

We now use the same procedure to calculate the autocorrelation coefficients r_2 , r_3 , r_4 and r_5 as follows:

For $k = 2$,

$$c_2 = \sum_{t=1}^8 \frac{(y_t - \bar{y})(y_{t+2} - \bar{y})}{8}$$

$$= 0.23/8 = 0.02875$$

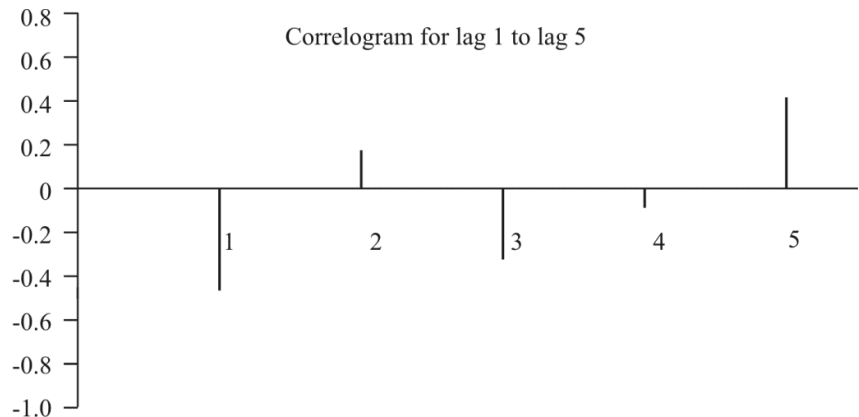
From equation (7),

$$r_2 = \frac{c_2}{c_0} = 0.02875/0.152 = 0.1891$$

Similarly, r_3 , r_4 and r_5 are obtained as

$$r_3 = -0.3371, r_4 = -0.1069, r_5 = 0.4194$$

Now to plot the correlogram for the given time series, we plot the values of the autocorrelation coefficients versus lag k for all $k = 1, 2, \dots, 5$. The correlogram is shown in Fig. 15.10.



E4) In Exercise 2, we have obtained the following values:

$$\bar{y}=30.267, c_0 = 5.145, c_1 = 0.1266 \text{ and } r_1 = 0.0246$$

For the given time series, we do the following calculations:

S. No.	Y	Y ²	(Y _t - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+2} - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+3} - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+4} - $\bar{Y}$)	(Y _t - $\bar{Y}$)(Y _{t+5} - $\bar{Y}$)
1	24	576	-6.267				
2	34	1156	3.733				
3	23	529	-7.267	45.538			
4	31	961	0.733	2.738	-4.595		
5	38	1444	7.733	-56.195	28.871	-48.462	
6	34	1156	3.733	2.738	-27.129	13.938	-23.395
7	35	1225	4.733	36.604	3.471	-34.395	17.671
8	31	961	0.733	2.738	5.671	0.538	-5.329
9	29	841	-1.267	-5.995	-4.729	-9.795	-0.929
10	28	784	-2.267	-1.662	-10.729	-8.462	-17.529
11	25	625	-5.267	6.671	-3.862	-24.929	-19.662
12	27	729	-3.267	7.404	4.138	-2.395	-15.462
13	32	1024	1.733	-9.129	-3.929	-2.195	1.271
14	33	1089	2.733	-8.929	-14.395	-6.195	-3.462
15	30	900	-0.267	-0.462	0.871	1.404	0.604
Total	454	14000		26.502	-9.169	-86.64	-29.511

We now calculate the autocorrelation coefficients r_2, r_3, r_4 and r_5 as follows:

For $k=2$,

$$c_2 = \sum_{t=1}^{14} \frac{(y_t - \bar{y})(y_{t+2} - \bar{y})}{13} = 26.502/13 = 2.0386$$

From equation (7),

$$r_2 = \frac{c_2}{c_0} = 2.0386/5.145 = 0.3962$$

Similarly, r_3, r_4 and r_5 may also be obtained as

$$r_3 = -0.1485, r_4 = -0.72, r_5 = -0.5736$$

Now to plot the correlogram for the given time series, we plot the values of the autocorrelation coefficients versus the lag k for all $k = 1, 2, \dots, 5$. The correlogram is shown in Fig. 15.11

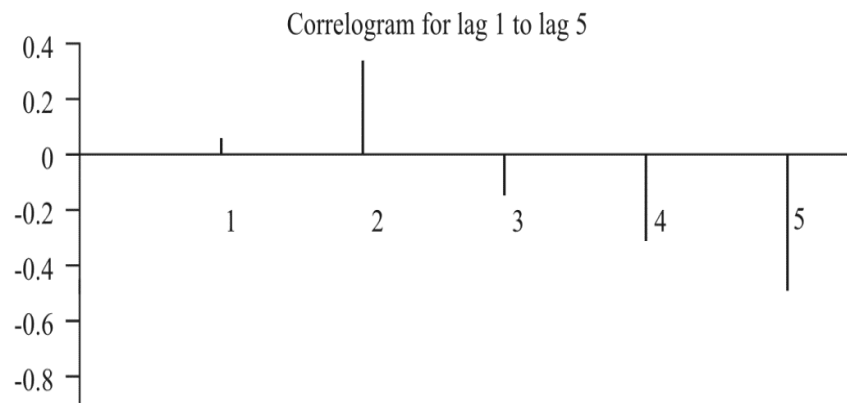


Fig. 15.11: Correlogram for the given time series

- E5)** The correlogram for the given values of autocorrelation coefficients of a stationary process is shown in Fig. 15.12.

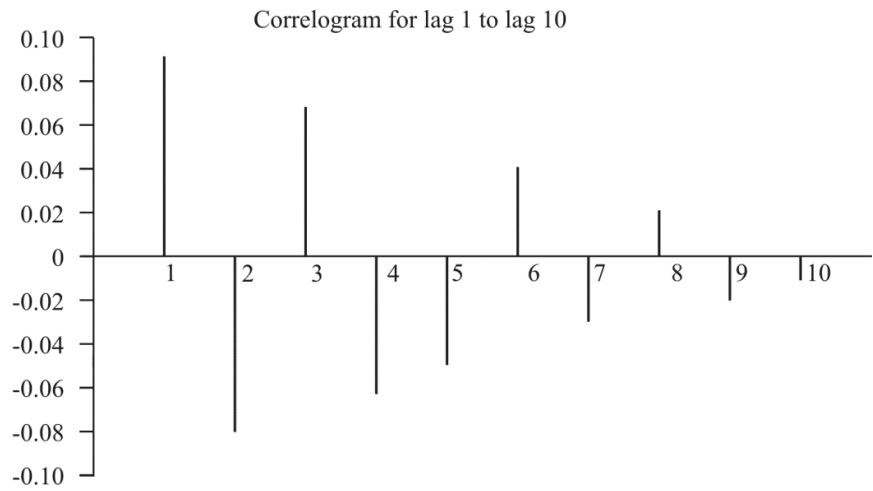


Fig. 15.12: Correlogram for 10 sample autocorrelation coefficients of the series of 400 observations.

UNIT 14 SEASONAL COMPONENT ANALYSIS

Structure

- 14.1 Introduction
 - Objectives
- 14.2 Estimation of Seasonal Component
 - Simple Average Method
 - Ratio to Moving Average Method
 - Ratio to Trend Method
- 14.3 Estimation of Trend from Deseasonalised Data
- 14.4 Forecasting
- 14.5 Summary
- 14.6 Solutions/Answers

14.1 INTRODUCTION

In Unit 13, you have learnt that time series can be decomposed into four components: Trend (T), Cyclic (C), Seasonal (S) and Irregular Component (I). Our aim is to estimate T, C and S components and use them for forecasting. We have already described some methods for smoothing or filtering the time series, namely, the simple moving average method, weighted moving average method and exponential smoothing method in Unit 13. We have also described some methods for estimating Trend and Cyclic components, i.e., the method of least squares and the moving average method in Unit 13.

When time series data do not contain any trend and cyclic components but reflect seasonal variation, we have to estimate the seasonal component S by removing irregular components. In Sec. 14.2 of this unit, we discuss some methods for estimating the seasonal component (S), namely, the simple average method, the ratio to moving average method and the ratio to trend method. If the effect of seasonal variation is not removed from the time series data, the trend estimates are also affected. Therefore, we have to deseasonalise the data by dividing it by the corresponding seasonal indices. Once the data is free from seasonal effects, we estimate the trend equation using the method of least squares as explained in 14.3. In Sec. 14.4, we explain how to use data for forecasting purposes once we have estimated the trend, cyclic and seasonal components of the time series.

In the next unit, we shall discuss the stationary time series and explain the stationary processes, i.e., weak and strict stationary processes. We shall also discuss the autocovariance, autocorrelation function and correlogram of a stationary process.

Objectives

After studying this unit, you should be able to:

- explain the effect of seasonal variation in time series;
- apply the simple average method for estimating seasonal indices;
- apply the ratio to moving average method and ratio to trend method for estimation of seasonal indices;

- describe the method of estimation of trend component from deseasonalised time series data; and
- use trend (T), cyclical (C) and seasonal (S) components for forecasting purposes.

14.2 ESTIMATION OF SEASONAL COMPONENT

If the seasonal variation is substantial, we can express the variation in y_t by additive or multiplicative models:

$$\text{Additive Model:} \quad Y_t = T_t + C_t + S_t + I_t \quad \dots (1)$$

$$\text{Multiplicative Model:} \quad Y_t = T_t \times C_t \times S_t \times I_t \quad \dots (2)$$

In the additive model, the seasonal indices S_t are normalised so that their sum over months in a year is zero. In the multiplicative model, they are normalised. In both cases, the forecast of the yearly output is not affected by the seasonal effect S_t and we can work on yearly totals to estimate the trend. In such cases, we can estimate S_t by working on the annual data. However, in many cases, we may be interested in forecasting monthly (or quarterly) figures. This requires monthly (or quarterly) estimate of the seasonal index S_t .

In many cases it has been found that seasonal effects increase with increase in the mean level of time series. Under these circumstances, it may be more appropriate to use the multiplicative model. If seasonal effects remain constant, the additive model is more appropriate. The classical approach is to consider the multiplicative model and estimate seasonal effect (S_t) for forecasting purposes. In this unit, we use the classical multiplicative model. We describe two methods for estimating seasonal indices based on the ratios of time series observation (Y) and estimated trend and cycle effects:

$$\frac{Y}{(T_t \times C_t)} = \frac{T_t \times C_t \times S_t \times I_t}{(T_t \times C_t)} = S_t \times I_t \quad \dots (3)$$

This ratio gives an estimate of $S_t \times I_t$. We shall estimate the seasonal indices (S_t) by smoothing out the irregular component (I_t).

We now describe the methods of estimating the seasonal indices S_t .

14.2.1 Simple Average Method

The method of simple average is the simplest of all methods. It is used to eliminate the seasonal effect from the given time series data. This method is based on the assumption that the data do not contain any trend and cyclic components and consists of eliminating irregular components by averaging the monthly (or quarterly or yearly) values over years. This assumption may or may not be true since most economic or business time series exhibit trends.

This method consists of the following steps:

Step 1: We arrange the data by years, months or quarters if data are collected on yearly, monthly or quarterly basis.

Step 2: After arranging the time series data, the average $\bar{y}_i$ is calculated for the i^{th} month of the year.

Step 3: After the monthly averages are calculated, we calculate the average of the monthly averages, that is,

$$\bar{\bar{y}} = \frac{\bar{y}_1 + \bar{y}_2 + \dots + \bar{y}_{12}}{12}$$

Step 4: After calculating the average $\bar{\bar{y}}$, we express monthly averages $\bar{y}_i$ as the percentage of the average $\bar{\bar{y}}$. These percentages are known as seasonal indices. Thus,

$$\text{Seasonal index for the } i^{\text{th}} \text{ month} = \frac{\bar{y}_i}{\bar{\bar{y}}} \times 100, \text{ for } i = 1, 2, \dots, 12. \dots (4)$$

Let us consider an example to explain this method.

Example 1: Determine the monthly seasonal indices for the following data of production of a commodity for the years 2010, 2011 and 2012 using the method of simple averages.

Months \ Years	Production in Tonnes		
	2010	2011	2012
January	120	150	160
February	110	140	150
March	100	130	140
April	140	160	160
May	150	160	150
June	150	150	170
July	160	170	160
August	130	120	130
September	110	1360	100
October	100	120	100
November	120	130	110
December	150	140	150

Solution: First of all, we arrange the data as shown in columns 1 to 4 of the table given below:

Table 1: Seasonal indices for the given time series

Months	Production (in Tonnes)			Total	Monthly Avg.	Seasonal Index
	2010	2011	2012			
January	120	150	160	430	143.3	104.886
February	110	140	150	400	133.3	97.566
March	100	130	140	370	123.3	90.247
April	140	160	160	460	153.3	112.205
May	150	160	150	460	153.3	112.205
June	150	150	170	470	156.6	114.620
July	160	170	160	490	163.3	119.524
August	130	120	130	380	126.6	92.662
September	100	130	100	340	113.3	82.928
October	100	120	1000	320	106.6	78.024
November	120	130	110	360	120.0	87.832
December	150	140	150	440	146.6	107.301
Total				4920	1639.5	1200
Average				410	136.625	100

We are given the monthly production of a commodity for 3 years. For calculating the monthly seasonal indices, we first calculate the month-wise total production for the 3 years. Then we calculate the monthly averages for all 12 months. Note from Table 1 that for January, it is 143.3, for February, it is 133.3, ..., etc. Next, we calculate the average of all monthly averages, i.e.

$$\bar{\bar{y}} = \frac{1}{12}(143.3 + 133.3 + \dots + 146.6) = 136.625$$

Now we calculate the seasonal indices by taking the percentage of monthly averages $\bar{y}_i$ to the combined averages $\bar{\bar{y}}$, one at a time, for $i = 1, 2, \dots, 12$:

$$\text{Seasonal Index for January} = \frac{143.3}{136.625} \times 100 = 104.886$$

$$\text{Seasonal Index for February} = \frac{133.3}{136.625} \times 100 = 97.566$$

$$\text{Seasonal Index for March} = \frac{123.3}{136.625} \times 100 = 90.247$$

$$\text{Seasonal Index for April} = \frac{153.3}{136.625} \times 100 = 112.205$$

$$\text{Seasonal Index for May} = \frac{153.3}{136.625} \times 100 = 112.205$$

$$\text{Seasonal Index for June} = \frac{156.6}{136.625} \times 100 = 114.620$$

$$\text{Seasonal Index for July} = \frac{163.3}{136.625} \times 100 = 119.524$$

$$\text{Seasonal Index for August} = \frac{126.6}{136.625} \times 100 = 92.662$$

$$\text{Seasonal Index for September} = \frac{113.3}{136.625} \times 100 = 82.928$$

$$\text{Seasonal Index for October} = \frac{106.6}{136.625} \times 100 = 78.024$$

$$\text{Seasonal Index for November} = \frac{120}{136.625} \times 100 = 87.832$$

$$\text{Seasonal Index for December} = \frac{146.6}{136.625} \times 100 = 107.301$$

14.2.2 Ratio to Moving Average Method

In Sec. 14.2.1, we have discussed the simple average method for calculating seasonal indices. Now we discuss the most widely used method known as the **ratio to moving average method**. It is better because of its accuracy. Also the seasonal indices calculated using this method are free from all the three components, namely, trend (T), cyclic (C) and Irregular variations (I).

As you have learnt in Unit 13, the moving average eliminates periodic variations if the span of period of moving average is equal to the period of the oscillatory variation sought to be eliminated. Therefore, we have to choose the span of time for moving average to be equal to one cycle. For example, if a cycle is completed in 3 months, we calculate the moving

average for 3 months. You may note that for some quarters, three values are available and for some quarters, four values are available. By taking the average over three or four values, we get seasonal indices S_i for $i=1, 2, 3, 4$. We usually normalise them so that their mean is 100 by dividing them by the mean of S_i and multiplying by 100. These normalised S_i have mean 100. Usually, not much difference exists between normalised and non-normalised seasonal indices S_i . When data are monthly, the same procedure will yield twelve monthly seasonal indices S_i . This method of estimating seasonal indices is known as the **ratio to moving average** method.

We have explained the ratio to moving average method for monthly time series data. The same method may be applied for any other periodic data such as quarterly, weekly data, etc. The steps for obtaining seasonal indices using this method are as follows:

- Step 1:** We arrange the data chronologically.
- Step 2:** If the cycle of oscillation is 1 year, we take the 12 months moving average of the 1st year, which will give estimates of the combined effects of trend and cyclic fluctuation. We enter the average value against the middle position, i.e., between the months of June and July.
- Step 3:** We discard the value for the month of January of the first year and include the value for the month of January of the subsequent year. Then we calculate the average of these 12 values and enter it against the middle position, i.e., between July and August. We repeat the process of taking moving averages MA (1) and entering the value in the middle position, till all the monthly data are exhausted.
- Step 4:** We calculate the centred moving average, i.e., MA(2), of the two values of the moving averages MA(1) and enter it against the first value, i.e., the month of July in the first year and subsequent values against the month of August, September, etc.
- Step 5:** After calculating the MA(1) and MA(2) values, we treat the original values (except the first 6 months in the beginning and the last 6 months at the end) as the percentage of the centred moving average values. For this we divide the original monthly values by the corresponding centred moving average, i.e., MA (2) values, and multiply the result by 100. We have now succeeded in eliminating the trend and cyclic variations from the original data. We now have to get rid of the data of irregular variations.
- Step 6:** We prepare another two-way table consisting of the month-wise percentage values calculated in Step 5, for all years. The purpose of this step is to average the percentages and to eliminate the irregular variations in the process of averaging.
- Step 7:** We find the median of the percentages or preliminary seasonal indices calculated in Step 5 month-wise and take the average of the month-wise median. Then we divide the median of each month by the average value and multiply it by 100. Generally, the sum of all medians is not 1200. Therefore, the average of all medians is not equal to 100. Hence, the seasonal indices are subjected to the same operation. We multiply the medians by the ratio of expected total of indices, i.e., 1200 to the actual total as follows:

$$\text{Seasonal Index} = \text{Median} \times \frac{1200}{\text{Total of Indices}}$$

The seasonal index for each month is given in the last column of the table. We calculate the sum of indices.

Let us consider an example to understand this method.

Example 2: Apply the ratio to moving average method to ascertain seasonal indices for the following data:

Year Month	2009	2010	2011	2012
January	500	550	500	600
February	600	550	600	650
March	650	600	550	650
April	750	650	600	750
May	800	700	650	800
June	800	700	750	900
July	850	750	750	1000
August	900	750	850	1000
September	900	750	900	1050
October	950	800	1000	1100
November	1100	900	1100	1200
December	1100	1000	1200	1250

Solution: As described in Sec. 13.5 we shall first eliminate the effect of S_t from time series observations. If the data are quarterly and the period of seasonal effect is one year, then taking a moving average for 4 quarters will eliminate the effect of S_t . In this case, we have to calculate centred moving average by taking the average of two successive moving average values. Table 2 gives the original data and centred moving average values, denoted by MA(2), give an estimate of $T \times C$. Then we calculate seasonal relatives $S \times I$ as percentages. Seasonal relatives in percentages are calculated by the ratio of y_t to MA (2) in percentages:

$$\text{Seasonal relative} = S \times I \times 100 = (y_t / \text{MA}(2)) \times 100$$

These are given for the years 2009-2012 in the following table in bold figures.

Table 2: Calculation of seasonal relatives for the given time series

Year (1)	Month (2)	Sales (3)	Moving Average MA(1) (4)	Centred Moving Average MA (2) (5)	Seasonal Relatives (3) / (5)
2009	January	500			
	February	600			
	March	650			
	April	750			
	May	800			
	June	800			
			825.0		
	July	850		827.00	102.78
			829.0		

**Seasonal Component
Analysis**

	August	900		827.00	108.827
			825.0		
	September	900		823.00	109.356
			821.0		
	October	950		816.75	116.314
			812.5		
	November	1100		808.25	136.096
			804.0		
	December	1100		800.00	137.500
			796.0		
2010	January	550		791.75	69.466
			787.5		
	February	550		781.25	70.4
			775.0		
	March	600		768.75	78.05
			762.5		
	April	650		756.25	85.95
			750.0		
	May	700		741.75	94.371
			733.5		
	June	700		729.25	95.989
			725.0		
	July	750		723.00	103.734
			721.0		
	August	750		723.00	103.734
			725.0		
	September	750		723.00	103.734
			721.0		
	October	800		718.75	111.3
			716.5		
	November	900		714.50	125.96
			712.5		
	December	1000		714.50	139.958
			716.5		
2011	January	500		716.50	69.78
			716.5		
	February	600		720.75	83.246
			725.0		
	March	550		731.25	75.213
			737.5		
	April	600		745.75	80.456
			754.0		
	May	650		762.50	85.246
			771.0		
	June	750		779.25	96.246
			787.5		
	July	750		791.75	94.727
			796.0		

	August	850		798.00	106.516
			800.0		
	September	900		804.25	111.905
			808.5		
	October	1000		814.75	122.737
			821.0		
	November	1100		827.25	132.97
			833.5		
	December	1200		839.75	142.899
			846.0		
2012	January	600		856.25	70.073
			866.5		
	February	650		872.75	74.477
			879.0		
	March	650		885.25	73.425
			891.5		
	April	750		895.75	83.728
			900.0		
	May	800		904.25	88.471
			908.5		
	June	900		910.50	98.846
			912.5		
	July	1000			
	August	1000			
	September	1050			
	October	1100			
	November	1200			
	December	1250			

Once we have obtained $S \times I$, we can take the average to eliminate the effect of irregularity I . This gives seasonal indices, S_i . Now we prepare a two way table which will include the percentage value of column (6) of Table 2 month-wise for every year as follows:

Table 3: Seasonal indices for the given time series

Year Months	2009	2010	2011	2012	Median	Seasonal Indices
January	--	69.466	69.78	70.073	69.78	70.025
February		70.4	83.246	74.477	74.477	74.738
March		78.05	75.213	73.425	75.213	75.477
April		85.95	80.456	83.728	83.728	84.02
May		94.37	85.246	88.471	88.471	88.78
June		95.989	96.246	98.846	96.246	96.584
July	102.78	103.734	94.727		102.78	103.14
August	108.827	103.734	106.516		106.516	106.89
September	109.356	103.734	111.905		109.356	109.74
October	116.314	116.314	122.737		116.314	116.72
November	136.096	125.962	132.97		132.971	133.437
December	137.5	139.958	142.899		139.958	140.446
Total					1195.81	1200
Average					99.65	100

14.2.3 Ratio to Trend Method

This method provides seasonal indices free from trend and is an improved version of the simple average method as it assumes that seasonal variation for a given period is a constant fraction of the trend. The measurement of the seasonal indices by this method consists of the following steps:

- Step 1:** We obtain the trend values by the method of least squares for each period by establishing the trend by fitting a straight line or second degree parabola or a polynomial.
- Step 2:** To express the original time series values as percentages of the trend value, we divide each original value by the corresponding trend value and multiply it by 100. The indices so obtained are free from the trend.
- Step 3:** To obtain the seasonal indices free from the cyclic and irregular variations, we find average (mean or median) of ratio to trend values (or percentages values) for each season for any number of years. It is suggested that median be preferred instead of mean if some extreme values are present, which are not primarily due to seasonal effects. In this way, the irregular variation is removed. If there are a few abnormal values in the percentage values, the mean should be preferred to remove the randomness.
- Step 4:** If the seasonal periods are quarters, the sum of seasonal indices in the case of multiplicative model should be 400 and if the periods are in months, it should be 1200. After calculating the seasonal indices, the expected total is divided by actual total of the indices for quarterly and monthly data, respectively. Most of the time, the sum of all calculated seasonal indices is not exactly the same, as it is expected to be.

By following the above step-wise procedure, we calculate the seasonal indices using this method. This method is based on sound and logical footing and utilises complete information.

Example 3: Compute the seasonal indices for the following time series of sales (in thousand `) of a commodity by the ratio to trend method:

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2008	800	920	880	820
2009	540	760	680	620
2010	400	580	540	480
2011	340	520	500	440
2012	300	400	360	340

Solution: We are given the time series data for 5 years of quarterly sales of a commodity. To compute the seasonal indices, we first determine the trend value for the yearly averages (Y) by fitting a linear trend by the method of least squares. The following table is constructed for fitting the straight line:

$$Y = a + b(X - \bar{X}) = a + bU$$

where $U = (X - \bar{X})$

Table 4: Trend values for the yearly averages

Year X	Yearly Total	Yearly Average (y)	U = X-2010	U×Y	U ²	Trend Value
2008	3400	850	-2	-1700	4	800
2009	2600	650	-1	-650	1	680
2010	2000	500	0	0	0	560
2011	1800	450	1	450	1	440
2012	1400	350	2	700	4	320
Total		2800	0	-1200	10	2800

For the straight line $Y = a + bX$, the normal equations for estimating a and b are:

$$\sum Y = na + b\sum U$$

$$\sum UY = a\sum U + b\sum U^2$$

Now we put the values of $\sum Y$, $\sum U$, $\sum UY$, $\sum U^2$ in the normal equations:

$$5a = 2800 \Rightarrow a = 560$$

$$10b = -1200 \Rightarrow b = -120$$

On putting the optimum value of a and b in the equation of the straight line $Y = a + bX$, we get the trend line for the given time series data as:

$$Y = 560 - 120 U$$

Thus, the trend values for each value of U are obtained as follows:

$$U = -2, \quad Y = 560 - 120 (-2) = 800$$

$$U = -1, \quad Y = 560 - 120 (-1) = 680$$

$$U = 0, \quad Y = 560 - 120 (0) = 560$$

$$U = 1, \quad Y = 560 - 120 (1) = 440$$

$$U = 2, \quad Y = 560 - 120 (2) = 320$$

Since yearly decline in the trend value is -120 , the quarterly increment would be

$$\text{Quarterly increment} = \frac{-120}{4} = -30$$

Now we determine the quarterly trend values as follows:

For 2008, the trend value for the middle quarter, i.e., half of the second quarter and half of the third quarter is 800. Since the quarterly increment is -30 , we obtain the trend value for the 2nd quarter as $800 - (-15)$ and for the 3rd quarter as $800 + (-15)$. Thus, these are 815 and 785, respectively. Consequently, the trend value for the first quarter is $815 - (-30) = 845$ and for the 4th quarter, it is $785 + (-30) = 755$. Similarly, we can get the trend values for other years as we have obtained for all the quarters of the year 2008. After calculating the trend values, we also calculate the seasonal indices for each quarter of every year, which estimates the trend component from the data. This is shown in Table 5.

Table 5: Calculations for seasonal indices

Trend Values					Seasonal Indices (Given value as % of trend values)				Total
Year	1 st Qtr	2 nd Qtr	3 rd Qtr	4 th Qtr	1 st Qtr	2 nd Qtr	3 rd Qtr	4 th Qtr	
2008	845	815	785	755	94.67	112.88	112.1	108.61	428.26
2009	725	695	665	635	74.48	109.35	102.25	97.64	383.72
2010	605	575	545	515	66.11	100.87	99.08	93.2	359.26
2011	485	455	425	395	70.1	114.28	117.64	111.39	413.41
2012	365	335	305	275	82.19	119.4	118.03	123.63	443.25
Total					387.55	556.78	549.1	534.47	2027.9
Average (A. Mean)					77.51	111.356	109.82	106.892	405.08
Adjusted Seasonal Index					76.445	109.824	108.301	105.42	399.99

The average yearly seasonal indices obtained above are adjusted to a total of 400 because the total of the seasonal indices for each quarter is 405.578, which is greater than 400. So we multiply each quarterly index by the ratio

$$K = \frac{400}{\text{Total of Indices}} = \frac{400}{405.08} = 0.986.$$

The adjusted seasonal indices for each quarter are given in the last row of the table.

You may now like to solve the following problems to check your understanding of the three methods explained so far.

-
- E1)** Determine the seasonal indices for the data given below for the average quarterly prices of a commodity for four years:

Years	Quarter I	Quarter II	Quarter III	Quarter IV
2009	554	590	616	653
2010	472	501	521	552
2011	501	531	553	595
2012	403	448	460	480

- E2)** Calculate the seasonal indices for the following data of production of a commodity (in hundred tons) of a firm using the ratio to trend method:

Years	Quarter I	Quarter II	Quarter III	Quarter IV
2001	470	531	500	480
2002	340	450	410	380
2003	270	360	340	310
2004	240	330	320	290
2005	220	270	250	240

- E3)** Apply the ratio to moving average method for calculating the seasonal indices for the time series data given in Example 8 of Unit 13.
-

14.3 ESTIMATION OF TREND FROM DESEASONALISED DATA

In Unit 13, you have learnt the estimation of trend. However, when a substantial seasonal component is present in the data, it is advisable to first remove the seasonal effect from the data. Otherwise, the trend estimates are also affected by seasonal effects, which makes the estimation unreliable. Hence, after estimating the seasonal indices, we deseasonalise the data values by dividing them by the corresponding seasonal indices (S_t). Thus, the deseasonalised values are given by

$$\text{Deseasonalised } Z_t = \frac{Y_t}{S_t} = T_t \times C_t \times I_t$$

Once the data are made free from the seasonal effect, we estimate the trend line by the method of least squares as explained in Sec.13.4 of Unit 13. Thus, we have a reasonably good estimate of T_t , C_t and S_t .

We have not described the estimation of cyclic effect (C_t) separately. A cycle in the time series means a business cycle, which normally exceeds one year in duration. Note that hardly any time series possess strict cycles because cycles are never regular in periodicity and amplitude. This is why the business cycles are the most difficult types of economic fluctuation to measure. To construct meaningful typical cycle indices of curves similar to those that have been developed for trends and seasons is impossible. The successive cycles vary widely in time, amplitude and patterns and are inextricably mixed with irregular factors. Since it is very difficult to distinguish cyclic effects from long term trend effect, and in most cases we assume that either they are not present or they are estimated along with trend, we estimate T_t C_t jointly by the least squares method.

Once we have estimated all components, we shall use them for forecasting purposes as described in the next section.

Example 4: Use the following data and calculate the deseasonalised values $T \times C \times I$. Use these values to estimate the trend line:

Years	Quarter I	Quarter II	Quarter II	Quarter IV
2003	289	241	273	232
2004	336	294	363	274
2005	297	270	263	198
2006	291	209	243	187

Solution: For calculating the deseasonalised values, we first calculate the seasonal indices for the given data. The seasonal indices are given in Table 6. Deseasonalised values are also calculated by dividing the time series values by the corresponding indices S_t .

Table 6: Calculations for obtaining trend values

Year X_t		Y_t	S_t	Deseasonalised values $Z_t = (Y_t/S_t) \times 100$	$(X_t - \bar{X}_t)$	$(X_t - \bar{X}_t)^2$	$Z_t(X_t - \bar{X}_t)$
2003	Q ₁	289	113.13	255.46	1.875	3.515625	478.9875
	Q ₂	241	94.82	254.17	1.625	2.640625	413.02625
	Q ₃	273	107.43	254.12	1.375	1.890625	349.415
	Q ₄	232	84.62	274.17	1.125	1.265625	308.44125

2004	Q ₁	336	113.13	297.00	0.875	0.765625	259.875
	Q ₂	294	94.82	310.06	0.625	0.390625	193.7875
	Q ₃	363	107.43	337.89	0.375	0.140625	126.70875
	Q ₄	274	84.62	323.80	0.125	0.015625	040.475
2005	Q ₁	297	113.13	262.53	-0.125	0.015625	-032.81625
	Q ₂	270	94.82	284.75	-0.375	0.140625	-106.78125
	Q ₃	263	107.43	244.81	-0.625	0.390625	-153.00625
	Q ₄	198	84.62	233.99	-0.875	0.765625	-204.74125
2006	Q ₁	291	113.13	257.23	-1.125	1.265625	-289.38375
	Q ₂	209	94.82	220.42	-1.375	1.890625	-303.0775
	Q ₃	243	107.43	226.19	-1.625	2.640625	-367.55875
	Q ₄	187	84.62	220.99	-1.875	3.515625	-414.35625
Total				4257.58		21.25	298.995

Once the data are deseasonalised, we apply the method of least squares to estimate the trend equation. The following values are calculated (as given in the above table):

$$\bar{X}_t = 2004.5, \quad \sum (X_t - \bar{X}_t)^2 = 21.25 \quad \sum Z_t (X_t - \bar{X}_t) = 298.995$$

$$\hat{a} = \bar{Z} = 266.10 \quad \hat{b} = \frac{\sum Z_t (X_t - \bar{X}_t)}{\sum (X_t - \bar{X}_t)^2} = 14.07$$

The fitted trend equation is:

$$Y_t = 266.10 + 14.07 (X_t - \bar{X}_t)$$

14.4 FORECASTING

Forecasting is one of the main purposes of time series analysis. It is always a risky task as one has to assume that the process will remain the same in future as in the past, at least for the period for which we are forecasting. This assumption is not very realistic and we shall assume that at least for short term forecasting, the process remains the same as in the past.

If a time series plot shows that there is no seasonal effect, or on theoretical basis there is no reason for having a seasonal component (S_t), we can estimate the trend component (T_t) and use the trend equation for forecasting.

If it is empirically observed that there is a significant seasonal effect (S) and on theoretical ground also there is a valid reason for the presence of such a component, we have to take the seasonal effect into account while estimating and forecasting. If data are collected on monthly basis and the period of seasonality is one year, we estimate twelve seasonal indices, one for each month. If data is quarterly, we estimate four seasonal indices. After deseasonalising the data, we fit the trend equation. Then we project the trend for the period for which we have to forecast. Next we adjust it for seasonal effect by multiplying it by the corresponding seasonal index. This gives the final forecast value which has been corrected for the seasonal effect. These steps can be described as follows:

Step 1: Calculate the moving average of suitable order. The order of moving average is taken as the period of the seasonal effect.

Step 2: Calculate the ratio of data to moving average values so that this ratio contains the seasonal component (S_t) and the irregular component (I_t), i.e., $S_t \times I_t$.

Step 3: Determine the seasonal indices by averaging $S_t \times I_t$ for respective seasons. This gives the estimates of the seasonal component (S_t).

Step 4: Obtain deseasonalised values by dividing data values by the corresponding seasonal component (S_t).

Step 5: Fit the trend equation to deseasonalised data. Compute deseasonalised forecast value from the trend equation.

Step 6: Adjust the forecast value for seasonal effect by multiplying it by the corresponding S_t . If the additive model is used, instead of multiplying or dividing we add or subtract the corresponding values. Seasonal effects are normalised so that their sum is equal to zero.

Example 5: Using estimated seasonal indices and the fitted trend equation of Example 4, forecast the value for the first quarter (Q_1) of 2007.

Solution: We have fitted the trend equation in Example 4 as:

$$Y_t = 266.10 + 14.07 (X_t - \bar{X}_t)$$

The projected trend value for Q_1 of 2007 is:

$$\hat{Y}_t = 266.10 + 14.07 (2007.25 - 2004.5)$$

where $\bar{X}_t = 2004.5$

$$\begin{aligned} \text{or } \hat{Y}_t &= 266.10 + 14.07 \times 2.75 \\ &= 304.79 \end{aligned}$$

The season corrected forecast = $(\hat{Y}_t \times S_t) / 100$

$$\begin{aligned} &= (304.79 \times 84.62) / 100 \\ &= 257.9 \end{aligned}$$

You may now like to solve the following exercises to assess your understanding of forecasting.

E4) The following table gives the sales figures (in thousands) of television sets for 16 quarters over four years, coded as 1, 2, 3 and 4:

Quarter Year	Q ₁	Q ₂	Q ₃	Q ₄
1	480	410	600	650
2	580	520	680	740
3	600	560	750	780
4	630	590	800	840

- Calculate the four quarters centred moving average values.
- Compute the seasonal indices for the four quarters.
- Obtain deseasonalised values and estimate the trend line.
- Obtain the season adjusted forecast value (Q_3) of the fifth year.

E5) The following data give production of a certain brand of motor vehicles. Determine indices using the ratio to moving average method for August and September, after calculating the centred moving average for twelve months.

Production (in thousand units)**Seasonal Component
Analysis**

Year	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sep	Oct	Nov	Dec
1985	7.92	7.81	7.91	7.03	7.25	7.17	5.01	3.90	4.64	7.03	6.88	6.14
1986	4.86	4.48	5.26	5.48	6.42	6.82	4.98	2.45	4.51	6.38	7.55	7.59

E6) Given below are the data of production of a company (in lakhs of units) for the years 1973 to 1979:

Year	1973	1974	1975	1976	1977	1978	1979
Production	15	14	18	20	17	24	27

- Compute the linear trend by the method of least squares.
- Compute the trend values for each year.

Let us now summarise the concepts that we have discussed in this unit.

14.5 SUMMARY

- When time series data do not contain any trend and cyclic components but reflect only seasonal variation, we have to estimate the seasonal component S by removing irregular components.
- If the effect of seasonal variation is not removed from the time series data, the trend estimates are also affected. Therefore, we have to deseasonalise the data by dividing it by the corresponding seasonal indices. Once the data is free from the seasonal effects, we estimate the trend equation using the method of least squares.
- If a cycle is completed in 3 months, we calculate the moving average for 3 months. By taking the average over all available values, we get seasonal indices S_i for $i = 1, 2, 3, 4$.
- We usually normalise the seasonal indices so that their mean is 100 by dividing them by mean of S_i for all i and multiplying by 100. The normalised seasonal indices have mean 100. Usually, there is not much difference between normalised and non-normalised seasonal indices.
- The ratio to trend method provides seasonal indices free from trend and is an improved version of the simple average method as it assumes that seasonal variation for a given period is a constant fraction of the trend.
- When substantial seasonal component is present in the data, it is advisable to first remove the seasonal effect from the data. Otherwise, the trend estimates are also affected by seasonal effects, which make the estimation unreliable. Hence, after estimating the seasonal indices, we deseasonalise the data values by dividing it by corresponding seasonal indices (S_t). Thus, the deseasonalised values are given by

$$\text{Deseasonalised } Z_t = \frac{Y_t}{S_t} = T_t \times C_t \times I_t$$

- A cycle in the time series means a business cycle, which normally exceeds one year in length. Note that hardly any time series possess

strict cycles because cycles are never regular in periodicity and amplitude. This is why the business cycles are the most difficult type of economic fluctuations to measure. To construct meaningful typical cycle indices of curves similar to those that have been developed for trends and seasons is impossible. The successive cycles vary widely in time, amplitude and patterns and are inextricably mixed with irregular factors. Since it is very difficult to distinguish cyclic effects from long term trend effect, and in most cases we assume that either they are not present or they are estimated along with trend, we estimate T_t C_t jointly by the least squares method.

14.6 SOLUTIONS/ANSWERS

- E1)** For determining the seasonal indices we first arrange the given data as follows:

Table 7: Seasonal Indices for quarterly price of a commodity

Years Quarters	2009	2010	2011	2012	Total	Quarterly Avgs	Seasonal Index
I	554	472	301	403	1930	482.5	91.51
II	590	501	531	448	2070	517.5	98.21
III	616	521	553	460	2150	537.5	102.01
IV	653	552	595	480	2280	570.0	108.18
Total					8430	2107.5	400
Average					2107.5	526.9	100

We are given quarterly prices of a commodity for 4 years. For calculating the quarterly variation indices, we first calculate the quarter-wise total of the production amount for the 4 years. Then we calculate the quarterly averages for all 4 quarters. For Quarter I, it is 482.5, for Quarter II, 517.5, for Quarter III, 537.5 and for Quarter IV, 570.

After calculating the quarterly averages, we calculate the average of all quarterly averages:

$$\bar{y} = \frac{1}{4}(482.5 + 517.5 + 537.5 + 570) = 526.9$$

Now we calculate the seasonal indices, one at a time, for $i = 1, 2, 3, 4$, as follows:

$$\text{Seasonal index for Quarter I} = \frac{482.5}{526.9} \times 100 = 91.57$$

$$\text{Seasonal index for Quarter II} = \frac{517.5}{526.9} \times 100 = 98.21$$

$$\text{Seasonal index for Quarter III} = \frac{537.5}{526.9} \times 100 = 102.01$$

$$\text{Seasonal index for Quarter IV} = \frac{570}{526.9} \times 100 = 108.18$$

- E2)** We are given the time series data for 5 years of quarterly sales of a commodity. To compute the seasonal indices, we first determine the trend value for the yearly averages (Y) by fitting a linear trend by the

method of least squares. The following table is constructed for fitting the straight line $Y = a + bX$:

Table 8: Calculation of trend values for yearly averages

Year X	Yearly Total	Yearly Average (y)	U = X-2003	U×Y	U ²	Trend Value
2001	1980	495	-2	-990	4	470
2002	1580	395	-1	-395	1	410
2003	1280	320	0	0	0	350
2004	1180	295	1	295	1	290
2005	980	245	2	490	4	230
Total		1750	0	-600	10	1750

For the line $Y = a + b(X - \bar{X}) = a + bU$, the normal equations for estimating a and b are

$$\sum Y = na + b \sum U$$

$$\sum UY = a \sum U + b \sum U^2$$

Now we put the values of $\sum Y$, $\sum U$, $\sum UY$, $\sum U^2$ in the normal equations:

$$5a = 1750 \Rightarrow a = 350$$

$$10b = -600 \Rightarrow b = -60$$

Therefore, by putting the optimum value of a and b in the equation of the line $Y = a + bU$, we get the trend line for the given time series data as:

$$Y = 350 - 60 U$$

Thus, the trend values for different values of U are as follows:

$$U = -2, Y = 350 - 60(-2) = 470$$

$$U = -1, Y = 350 - 60(-1) = 410$$

$$U = 0, Y = 350 - 60(0) = 350$$

$$U = 1, Y = 350 - 60(1) = 290$$

$$U = 2, Y = 350 - 60(2) = 230$$

As the yearly decline in the trend value is -60, the quarterly decline is

$$\text{Quarterly increment} = \frac{-60}{4} = -15$$

We now determine the quarterly trend values as follows:

For 2001, the trend value for the middle quarter, i.e., half of the second quarter and half of the third quarter is 400. Since the quarterly increment is 15, we obtain the trend values for 2nd quarter as $470 - (-7.5)$ and for 3rd quarter as $470 + (-7.5)$, i.e., 477.5 and 462.5, respectively. Consequently the trend value for the first quarter is $477.5 - (-15) = 492.5$ and for the 4th quarter, it is $462.5 + (-15) = 447.5$. We can get the trend values for other years in the same way as for all quarters of the year 2001. After calculating the trend values, we also calculate the seasonal indices for each quarter of every year, which estimate the trend component from the data. These are given in the following table:

Table 9: Calculations for seasonal indices

Trend Values					Seasonal Indices (Given value as % of trend values)			
Year	Qtr I	Qtr II	Qtr III	Qtr IV	Qtr I	Qtr II	Qtr III	Qtr IV
2001	492.5	477.5	462.5	447.5	95.43	110.99	108.1	107.26
2002	432.5	417.5	402.5	387.5	78.61	107.78	101.86	98.06
2003	372.5	357.5	342.5	327.5	72.48	100.7	99.27	94.65
2004	312.5	297.5	282.5	267.5	76.8	110.92	113.27	108.4
2005	252.5	237.5	222.5	207.5	87.13	113.68	112.36	115.66
Total					410.45	544.07	534.86	524.03
Average (A. Mean)					82.09	108.81	106.97	104.80
Adjusted Seasonal Index					81.539	108.081	106.253	104.097

The average yearly seasonal indices obtained above are adjusted to a total of 400 because the total of the seasonal indices for each quarter is 402.67, which is greater than 400. So we multiply each quarterly index by the ratio

$$K = \frac{400}{\text{Total of Indices}} = \frac{400}{402.67} = 0.9933$$

The adjusted seasonal indices for each quarter are given in the last row of the table.

- E3)** As described in Sec. 13.5 we shall first eliminate the effect of S_t from time series observations. Since the data are quarterly and the period of seasonal effect is one year, taking a moving average for 4 quarters will eliminate the effect of S_t . In this case, we have to calculate centred moving average by taking average of two successive moving average values. The following table gives the original data and centred MA (2) values as:

Table 10: Calculation for MA(1), MA(2) and seasonal relatives

Year (1)	Quarter (2)	Sales (3)	Moving Average (4)	Centered Mov. Average (5)	Percentage Values
2001	Qtr 1	935			
	Qtr 2	1215			
			1162.50		
	Qtr 3	1045		1169.375	89.364
			1176.25		
	Qtr 4	1455		1188.75	122.397
			1201.25		
2002	Qtr 1	990		1239.375	79.878
			1277.50		
	Qtr 2	1315		1281.25	102.634
			1285.00		
	Qtr 3	1350		1332.50	101.313
			1380.00		
	Qtr 4	1485		1442.50	102.946
			1505.00		
2003	Qtr 1	1370		1520.00	90.131
			1535.00		

**Seasonal Component
Analysis**

	Qtr 2	1815		1559.125	116.411
			1583.25		
	Qtr 3	1470		1619.75	90.754
			1656.25		
	Qtr 4	1680		1600.00	105.00
			1543.75		
2004	Qtr 1	1160		1510.625	76.789
			1477.50		
	Qtr 2	1365		1448.625	94.227
			1419.75		
	Qtr 3	1205		1340.50	89.89
			1261.25		
	Qtr 4	1445		1275.00	113.33
			1288.75		
2005	Qtr 1	1030		1287.50	80.00
			1286.25		
	Qtr 2	1475		1303.75	113.135
			1321.25		
	Qtr 3	1195		1340.625	89.137
			1360.00		
	Qtr 4	1585		1341.875	118.118
			1323.75		
2006	Qtr 1	1185		1361.875	87.012
			1400.00		
	Qtr 2	1330		1470.00	90.476
			1540.00		
	Qtr 3	1500		1568.125	95.655
			1596.25		
	Qtr 4	2145		1695.00	126.548
			1793.75		
2007	Qtr 1	1410		1845.625	76.396
			1897.50		
	Qtr 2	2120		1845.625	109.95
			1958.75		
	Qtr 3	1915		1928.125	94.948
			2075.00		
	Qtr 4	2390		2016.875	115.007
			2081.25		
2008	Qtr 1	1875		2078.125	89.82
			2093.75		
	Qtr 2	2145		2145.00	100.00
			2196.25		
	Qtr 3	1965		2195.00	89.52
			2193.75		
	Qtr 4	2800		2190.00	127.853
			2186.25		
2009	Qtr 1	1865		2182.50	85.452
			2178.75		
	Qtr 2	2115		2099.375	100.744
			2020.00		
	Qtr 3	1935			
	Qtr 4	2165			

Centred moving average values, denoted by $MA(2)$, give an estimate of $T_t \times C_t$. We calculate seasonal relatives $S_t \times I_t$ in percentages as follows:

$$S_t \times I_t \times 100 = (Y_t / MA(2)) \times 100$$

These are given for the years 1973–1976 in the above table in bold figures. Now we prepare a two way table, which includes the quarterly percentage moving average values year-wise as follows:

Table 11: Calculations for seasonal indices

Years Qtrs	2001	2002	2003	2004	2005	2006	2007	2008	2009	Median	Indices
Qtr 1		79.87	90.13	76.78	80.00	87.01	76.39	89.82	85.45	82.72	84.56
Qtr 2		102.63	116.44	94.22	113.13	90.47	109.95	100.00	100.74	101.68	103.95
Qtr 3	89.364	101.31	90.75	89.89	89.13	95.65	94.95	89.52		90.32	92.33
Qtr 4	122.39	102.94	105.00	113.33	118.12	126.55	115.01	127.85		116.56	119.15
Total										391.29	400
Average											100

Once we have obtained $S \times I$, we can take the average to eliminate the effect of irregularity I . This gives seasonal indices S_i . We note that for some quarters, three values are available and for some quarters, four values are available. By taking the average over three or four values, we get seasonal indices S_i for $i = 1, 2, 3, 4$. We usually normalise them so that their mean is 100. These normalised values of S_i have mean 100. These are obtained in the table given above. Usually not much difference exists between normalised and non-normalised seasonal indices. When the data are collected on monthly basis, the same procedure will yield twelve monthly seasonal indices. This method of estimating seasonal indices is known as **ratio-to-moving average** method.

- E4)** From the given data for sales of television sets, we have the following table:

Table 12: Calculations for seasonal indices and deseasonalised values

Year	Quarter	Sales (y)	Centred MA(MA(1))	$S \times I \times 100 = y/MA$	Seasonal Index S_i	Deseasonalised Value (Z_t)
1	1	480	-	-	93.00	516
	2	410	-	-	83.67	490
	3	600	547.5	109.6	109.14	550
	4	650	573.8	113.3	114.13	570
2	5	580	597.5	97.1	93.06	623
	6	520	618.8	84.0	83.67	621
	7	680	632.5	107.5	109.14	623
	8	740	640.0	115.6	114.13	648

3	9	600	653.8	91.8	93.06	645
	10	560	667.5	83.9	83.67	669
	11	750	676.3	110.9	109.14	687
	12	780	683.8	114.1	114.13	683
4	13	630	693.8	90.8	93.06	677
	14	590	707.5	83.4	83.67	705
	15	800	-	-	109.14	733
	16	840	-	-	114.13	736
	$\bar{X} = 8.5$					

From the above table we get

$$\bar{X}_t = 8.50, \quad \sum (X_t - \bar{X}_t)^2 = 340.00$$

$$\text{and} \quad \sum Z_t (X_t - \bar{X}_t) = 4870.5$$

$$\hat{a} = \bar{Z} = 636 \quad \hat{b} = \frac{\sum Z_t (X_t - \bar{X}_t)}{\sum (X_t - \bar{X}_t)^2} = 14.325$$

The trend equation is

$$\hat{y} = 636 + 14.325 (X_t - 8.50)$$

For Q_3 of fifth year, $X_t = 19.00$. The forecast for trend value is

$$\begin{aligned} \hat{y} &= 636 + 14.325 (19.00 - 8.50) \\ &= 786.4125 \end{aligned}$$

$$\begin{aligned} \text{Season corrected forecast} &= (786.4125 \times 109.14) / 100 \\ &= 858.29 \end{aligned}$$

E5) Values obtained from the data are as follows:

Table 13: Calculation of seasonal indices

	July	Aug.	Sep.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May	June
Centred MA	6.43	6.16	5.91	5.74	5.64	5.59	5.58	5.51	5.45	5.41	5.42	5.50
(y_t/MA)x100	77.92	63.31	78.51	122.47	121.99	109.84	87.10	81.31	96.51	101.29	118.45	124.00
Normalised S_t	79.06	64.23	79.65	124.26	123.77	111.49	88.37	82.49	97.92	102.77	120.18	125.80

The required seasonal indices for August and September are:

$$S_{\text{Aug}} = 64.23, \quad S_{\text{Sep}} = 79.65$$

The seasonal indices have been normalised so that their average is 100.

E6) From the given information, we have calculated:

$$\hat{a} = \bar{Z} = 19.28$$

We also have $\sum (X_t - \bar{X}_t)^2 = 28$ $\sum Z_t (X_t - \bar{X}_t) = 55$

$$\hat{b} = \frac{\sum Z_t (X_t - \bar{X}_t)}{\sum (X_t - \bar{X}_t)^2} = \frac{55}{28} = 1.96$$

$$\hat{Y} = 19.28 + 1.96(X_t - \bar{X}_t)$$

The estimated trend values are:

Years	1973	1974	1975	1976	1977	1978	1979
$\hat{y}_t$	13.40	15.36	17.32	19.28	21.24	23.20	25.16

UNIT 16 TIME SERIES MODELS

Structure

- 16.1 Introduction
 - Objectives
- 16.2 Linear Stationary Processes
 - Moving Average (MA) Process
 - Autoregressive (AR) Process
 - Fitting an Autoregressive Process
 - Determining the Order of an Autoregressive Model
 - Partial Autocorrelation Function (pacf)
- 16.3 Autoregressive Moving Average (ARMA) Models
- 16.4 Autoregressive Integrated Moving Average (ARIMA) Models
- 16.5 Summary
- 16.6 Solutions /Answers

16.1 INTRODUCTION

In Unit 15, you have learnt that there are two types of stationary processes: strict stationary and weak stationary processes. You have also learnt how to determine the values of autocovariance and autocorrelation coefficients, and to plot a correlogram for a stationary process. In this unit, we discuss various time series models.

In Sec. 16.2 of this unit, we introduce an important class of linear stationary processes, known as Moving Average (MA) and Autoregressive (AR) processes and describe their key properties. We discuss Autoregressive Moving Average (ARMA) models in Sec. 16.3. We also discuss their properties in the form of autocorrelations and the fitting of suitable models to the given data. We discuss how to deal with models with trend by considering integrated models, called the Autoregressive Integrated Moving Average (ARIMA) models in Sec. 16.4.

Objectives

After studying this unit, you should be able to:

- describe a linear stationary process;
- explain autoregressive and moving average processes;
- fit autoregressive moving average models;
- describe and use the ARIMA models; and
- explore the properties of AR, MA, ARMA and ARIMA models.

16.2 LINEAR STATIONARY PROCESSES

In Unit 15, we have considered discrete time stationary processes and their properties. Note that the sequences of random variables $\{Y_i\}$ are mutually independent and identically distributed. If a discrete stationary process consists of such sequences of i.i.d. variables, it is called a **purely random process**. Sometimes it is called white noise.

Recall that the random variables are normally distributed with mean zero and variance σ^2 . Similarly, a purely random process has constant mean and variance, i.e.,

$$\gamma_k = \text{Cov}(X_t, X_{t+k}) = \begin{cases} \sigma_Y^2 & k = 0 \\ 0 & k = 1, 2, 3, \dots \end{cases} \quad \dots (1)$$

In this section, we consider some particular cases of a linear process. Let Y_t be a stochastic process with mean μ . We can express it as a weighted sum of previous random noises (shocks). Thus, we have

$$Y_t = \mu + a_t + \psi_1 a_{t-1} + \psi_2 a_{t-2} + \dots \quad \dots (2)$$

Here a_t , ($t = 0, 1, 2, \dots$) represent white noises with mean zero, variance σ_a^2 and ψ_i ($i = 1, 2, \dots$) represent weights. For the linear process to be stationary, the following conditions on weights are required, i.e.,

$$\sum \psi_i^2 < \infty, \quad \sum |\psi_i| < \infty \quad \dots (3)$$

Then, the autocovariance is given by

$$\text{Cov}(a_t, a_{t+k}) = 0 \quad \text{for } k \neq 0 \quad \dots (4)$$

For simplicity, we denote the process by X_t :

$$X_t = Y_t - \mu \quad \dots (5)$$

Therefore, the process X_t has mean zero and we can write it as:

$$X_t = Y_t - \mu = a_t + \psi_1 a_{t-1} + \psi_2 a_{t-2} + \dots \quad \dots (6)$$

Under the above mentioned conditions on weights ψ_i , the model can also be expressed as

$$X_t = a_t + \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \dots \quad \dots (7)$$

Let us now consider two particular cases of the linear stationary processes.

16.2.1 Moving Average (MA) Process

The moving average processes have been often used in econometrics. For example, the economic indicators are affected by many random events such as government decisions, strikes and shortages of raw materials, etc. They have immediate effects as well as effects of lower magnitude in past periods. Such processes have been successfully modelled by moving average processes.

Suppose we write the linear process as

$$X_t = \beta_0 a_t + \beta_1 a_{t-1} + \dots + \beta_q a_{t-q} \quad \dots (8)$$

where β_i , ($i = 0, 1, 2, \dots, q$) are constants. This process is known as the **moving average process** of order q and is abbreviated as MA(q) process. The white noises (a_t) are scaled so that $\beta_0 = 1$. The mean and variance of X_t are given by

$$E(X_t) = 0 \quad \text{and} \quad V(X_t) = \sigma_a^2 \left(1 + \sum_{i=1}^q \beta_i^2 \right) \quad \dots (9)$$

and autocovariance is given as

$$\gamma_k = \text{Cov}(X_t, X_{t+k}) \quad \dots (10)$$

$$\begin{aligned} \gamma_k = \gamma_{-k} &= \text{Cov}(a_t + \beta_1 a_{t-1} + \dots + \beta_q a_{t-q}, a_{t+k} + \beta_1 a_{t+k-1} + \dots + \beta_q a_{t+k-q}) \\ &= 0 \quad \text{for } k > q \\ &= \sigma_a^2 (\beta_k + \beta_1 \beta_{k+1} + \dots + \beta_{q-k} \beta_q) \quad \text{for } k = 1, 2, \dots, q \end{aligned} \quad \dots (11)$$

The autocorrelation function (acf) of the MA(q) process is given by

$$\rho_k = \frac{(\beta_k + \beta_1 \beta_{k+1} + \dots + \beta_{q-k} \beta_q)}{\left(1 + \sum_{i=1}^q \beta_i^2\right)}, \quad k = 1, 2, \dots, q \quad \dots (12)$$

Note that the autocorrelation function (acf) becomes zero, if lag k is greater than the order of the process, i.e., q. This is a very important feature of moving average (MA) processes.

First and Second Order Moving Average (MA) Processes

For the first order moving average {MA (1)} process, we have

$$X_t = a_t + \beta_1 a_{t-1} \quad \dots (13)$$

The mean and variance are obtained for q=1 as

$$E(X_t) = 0, \quad V(X_t) = \sigma_a^2 (1 + \beta_1^2) \quad \dots (14)$$

and the autocorrelation coefficient is obtained for q=1 as

$$\rho_1 = \frac{\beta_1}{(1 + \beta_1^2)} \quad \dots (15)$$

Similarly, for the second order **Moving Average MA(2) process** we have

$$X_t = a_t + \beta_1 a_{t-1} + \beta_2 a_{t-2} \quad \dots (16)$$

For q = 2, the mean and variance are given as

$$E(X_t) = 0, \quad V(X_t) = \sigma_a^2 (1 + \beta_1^2 + \beta_2^2) \quad \dots (17)$$

The autocorrelation coefficients are given as

$$\rho_1 = \frac{(\beta_1 + \beta_1 \beta_2)}{(1 + \beta_1^2 + \beta_2^2)}, \quad \rho_2 = \frac{\beta_2}{(1 + \beta_1^2 + \beta_2^2)} \quad \dots (18)$$

There is no requirement on the constants β_1 and β_2 for stationarity. However, for unique representation of the model, the autocorrelation coefficients should satisfy the condition of **invertibility**, which is satisfied when the roots of

$$\theta(B) = 1 + \beta_1 B + \beta_2 B^2 + \dots + \beta_q B^q = 0 \quad \dots (19a)$$

lie outside the unit-circle, i.e., roots $|B| > 1$.

For MA (1) process, we have

$$\theta(B) = 1 + \beta_1 B = 0 \Rightarrow B = -1/\beta_1 \quad \dots (19b)$$

Therefore, if $|B| > 1$, this implies that $|\beta_1| < 1$. Hence, for invertibility

$$|\beta_1| < 1 \quad \dots (20)$$

Let us consider an example of the moving average process.

Example 1: Consider a time series consisting of 60 consecutive daily over shots from an underground gasoline tank at a filling station. The sample mean and estimate of σ_a^2 with some sample autocorrelations are given as:

$$\text{Sample mean} = 4.0; \quad \sigma_a^2 = 4515.46$$

$$r_1 = -0.5, \quad r_2 = 0.1252, \quad r_3 = -0.2251, \quad r_4 = 0.012, \quad r_5 = 0.0053$$

Check whether a moving average MA (1) process can be fitted to the data and obtain preliminary estimates of the parameters.

Solution: We are given the sample mean of 60 observations as 4.0 and estimate of σ_a^2 , i.e., $\hat{\sigma}_a^2 = 3415.72$.

The MA (1) model is written as

$$Y_t = \mu + a_t + \beta_1 a_{t-1}$$

$$X_t = Y_t - \mu = a_t + \beta_1 a_{t-1}$$

If the process is purely random, all the autocorrelations (r_k) should be in the range of

$$\pm \frac{2}{\sqrt{N}}$$

In this case,
$$\pm \frac{2}{\sqrt{N}} = \pm \frac{2}{\sqrt{60}} = \pm 0.258$$

Here we see that of the given autocorrelations, only r_1 lies outside the range, given by ± 0.258 . This suggests that moving average MA (1) model could be a suitable model since only ρ_1 is significantly different from zero and $\rho_k, k > 1$ lie within the range ± 0.258 .

Equating r_1 to ρ_1 given by equation (15) and using the method of moments, we get

$$r_1 = -0.5 \Rightarrow \frac{\beta_1}{(1 + \beta_1^2)} = -0.5$$

On simplifying the above equation, we get

$$\hat{\beta}_1 = -0.1$$

Hence, the model MA(1) becomes $X_t = a_t - 0.1 a_{t-1}$

Thus, $Y_t = -4.0 + (a_t - 0.1 a_{t-1})$

where a_t is white noise with estimated variance of 4515.46.

You may now like to solve the following exercise to check your understanding about MA processes.

E1) Show that the autocorrelation function of MA(2)

$$X_t = a_t + 0.74 a_{t-1} - 0.19 a_{t-2}$$

is given by

$$\rho_k = \begin{cases} 0.3675 & k = \pm 1 \\ -0.1289 & k = \pm 2 \\ 0 & \text{Otherwise} \end{cases}$$

In Sec.16.2.1, we have considered estimation of parameters $\beta_1, \beta_2 \dots$ by the method of moments, i.e., by equating autocorrelations to their expected values. This method is not a very efficient method of estimation of parameters. For moving average processes, usually the maximum likelihood method is used which gives more efficient estimates when N is large. We do not discuss it here as it is beyond the scope of this course.

16.2.2 Autoregressive (AR) Process

A stationary process Y_t is said to be an autoregressive process of order p , abbreviated as AR (p), if

$$Y_t - \mu = \alpha_1(Y_{t-1} - \mu) + \alpha_2(Y_{t-2} - \mu) + \dots + \alpha_p(Y_{t-p} - \mu) + a_t \dots (21)$$

which is written as

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \dots + \alpha_p X_{t-p} + a_t \dots (22)$$

where $X_t = Y_t - \mu$ and a_t is white noise. It is similar to a multiple regression model, where we regress X_t on its past values and that is why it is called an **autoregressive process**.

A linear stationary process can always be expressed as an autoregressive process of suitable order. Unlike the moving average (MA) process, which puts no restrictions on parameters for stationarity, autoregressive (AR) process requires certain restrictions on the parameters α for stationarity. An autoregressive (AR) process can also be written as

$$(1 - \alpha_1 B - \alpha_2 B^2 - \dots - \alpha_p B^p) X_t = a_t \dots (23)$$

$$\text{or } \phi(\beta) X_t = a_t$$

where B is the backward shift operator, defined as

$$B X_t = X_{t-1}, B^2 X_t = X_{t-2}, \dots, B^p X_t = X_{t-p} \dots (24)$$

For an AR (p) process to be stationary, the roots of

$$\phi(B) = 1 - \alpha_1 B - \alpha_2 B^2 - \dots - \alpha_p B^p = 0 \dots (25)$$

must lie outside the unit circle.

First Order Autoregressive {AR(1)} (Markov) Process

Suppose, we write the linear model as

$$X_t = \alpha_1 X_{t-1} + a_t \dots (26)$$

By repeatedly using this equation you can see that X_t can be expressed as weighted sum of infinite numbers of past noises a_t , i.e.,

$$X_t = a_t + \alpha_1 a_{t-1} + \alpha_2 a_{t-2} + \dots \dots (27)$$

Autocorrelations ρ_k are obtained by multiplying equation (27) by X_{t-k} and taking expectations of the results. Then we get

$$\rho_k = \alpha_1 \rho^{k-1} = \dots = \alpha_1^k \dots (28)$$

Thus, $\rho_0 = 1, \rho_1 = \alpha_1$

$$\text{From equation (27)} \quad \sigma_x^2 = \alpha_1^2 \sigma_x^2 + \sigma_a^2 \dots (29)$$

$$\text{which gives} \quad \sigma_x^2 = \sigma_a^2 / (1 - \alpha_1^2) \dots (30)$$

and σ_x^2 is positive if $|\alpha_1| < 1$. Thus, for stationarity

$$|\alpha_1| < 1 \dots (31)$$

When α_1 is positive and large, the time plot becomes smooth and shows a slow changing trend. When α_1 is large and negative, the time plot shows a very rapid zig-zag movement. It is because of negative autocorrelations. If one value of autocorrelation is above mean, the next value of the autocorrelation is very likely to be below mean, and so on.

Second order Autoregressive {AR (2)} process

This process is obtained by taking $p = 2$ and the model is

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + a_t \quad \dots (32)$$

For stationarity, the following restrictions are placed on the coefficients:

$$\alpha_2 + \alpha_1 < 1; \quad \alpha_2 - \alpha_1 < 1 \text{ and } -1 < \alpha_2 < 1 \quad \dots (33)$$

For autoregressive AR(2) model, the first two autocorrelations ρ_1 and ρ_2 are obtained as follows:

On multiplying equation (32) by X_{t-1} and X_{t-2} and taking expectations and dividing the results by σ_x^2 , we get

$$\rho_1 = \alpha_1 + \alpha_2 \rho_1 \quad \dots (34a)$$

$$\rho_2 = \alpha_1 \rho_1 + \alpha_2 \quad \dots (34b)$$

On simplifying the above equations, we obtain

$$\alpha_1 = \frac{\rho_1(1-\rho_2)}{(1-\rho_1^2)} \quad \alpha_2 = \frac{(\rho_2 - \rho_1^2)}{(1-\rho_1^2)} \quad \dots (35)$$

Similarly, ρ_1 and ρ_2 can be expressed in terms of α_1 and α_2 as

$$\rho_1 = \frac{\alpha_1}{(1-\alpha_2)}, \quad \rho_2 = \alpha_2 + \frac{\alpha_1^2}{(1-\alpha_2)} \quad \dots (36a)$$

$$\sigma_x^2 = \sigma_a^2 / (1 - \rho_1 \alpha_1 - \rho_2 \alpha_2) \quad \dots (36b)$$

Multiplying equation (32) by X_{t-k} , taking expectations and dividing by σ_x^2 gives the autocorrelation function of AR (2) process as

$$\rho_k = \alpha_1 \rho_{k-1} + \alpha_2 \rho_{k-2}, \quad X > 0 \quad \dots (37)$$

We can obtain ρ_k for different values of k by using equation (37) for $k=1, 2, \dots$

Let us consider an example of AR(2) process.

Example 3: Consider an auto regressive AR(2) model

$$X_t = 0.80X_{t-1} - 0.60X_{t-2} + a_t$$

Verify whether the series is stationary.

(i) Obtain ρ_k for $k = 1, 2, \dots, 5$, and (ii) plot the correlogram.

Solution: We have an autoregressive AR (2) model

$$X_t = 0.80 X_{t-1} - 0.60 X_{t-2} + a_t \quad \dots (i)$$

Now from equation (36a), the autocorrelations ρ_1 and ρ_2 are given as

$$\rho_1 = \frac{\alpha_1}{(1-\alpha_2)}, \quad \rho_2 = \alpha_2 + \frac{\alpha_1^2}{(1-\alpha_2)}$$

$$\therefore \rho_0 = 1, \quad \rho_1 = \frac{0.80}{(1+0.60)} = 0.50 \quad \text{and} \quad \rho_2 = -0.50 + \frac{(0.80)^2}{(1+0.60)} = -0.20$$

We obtain the values of the autocorrelations ρ_3, ρ_4 and ρ_5 using equation (37) and get

$$\rho_3 = \alpha_1 \rho_2 + \alpha_2 \rho_1,$$

$$\rho_3 = 0.80 \times -0.20 + (-0.60) \times (0.50) = -0.46$$

$$\rho_4 = 0.80 \times -0.46 + (-0.60) \times (-0.20) = -0.25$$

$$\rho_5 = 0.80 \times -0.25 + (-0.60) \times (-0.46) = 0.076$$

The correlogram of the given AR process is shown in Fig. 16.1.

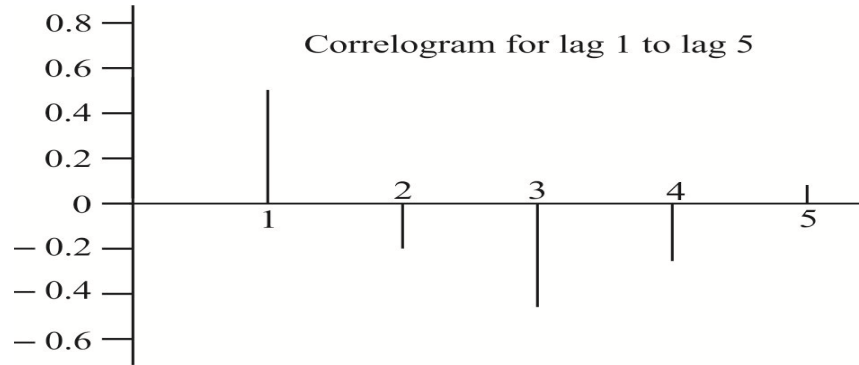


Fig 16.1: Correlogram of the model.

You may now like to solve the following exercises to check your understanding about MA processes.

E2) Consider an AR (2) process given by

$$X_t = X_{t-1} - 0.5 X_{t-2} + a_t$$

Verify whether the series is stationary or not.

a) Obtain ρ_k for $k = 1, 2, \dots, 5$ and b) plot the correlogram

E3) For each of the following processes, write the model using B notations and then determine whether the processes are stationary or not:

a) $X_t = 0.3X_{t-1} + a_t$

b) $X_t = (X_{t-1} + X_{t-2})/12 + a_t$

16.2.3 Fitting an Autoregressive Process

Suppose N observations on a time series $y_1, y_2, \dots, y_N$ are available. We now wish to fit an autoregressive (AR) process of suitable order. Therefore, we need to know the order of autoregressive (AR) process, that is, p .

Suppose, we know the order p . Then we have to estimate parameters $\mu, \alpha_1, \alpha_2, \dots, \alpha_p, \sigma^2_{a_t}$, etc. We calculate the autocorrelations from the data. Usually μ is estimated by $\hat{Y}$. Hence, by subtracting $\hat{Y}$ from Y_t , we calculate

$$X_t = Y_t - \hat{Y} \quad \dots (38a)$$

For the given n_k , we have to calculate parameters $\alpha_1, \alpha_2, \dots, \alpha_p$ of the model:

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \dots + \alpha_p X_{t-p} + a_t \quad \dots (38b)$$

For an autoregressive (AR) process, the least squares estimates of the parameters $\alpha_1, \alpha_2, \dots, \alpha_p$ are obtained by minimising S :

$$S = \sum_{t=p+1}^N (X_t - \alpha_1 X_{t-1} - \alpha_2 X_{t-2} - \dots - \alpha_p X_{t-p})^2 \quad \dots (39)$$

with respect to $\alpha_1, \dots, \alpha_p$, and equating the result to zero. This method provides good estimates.

If $Y_t, t = 1, 2, \dots, N$ is the observed series $X_t = Y_t - \hat{Y}$ are used in equation (39). This looks very similar to multiple regression estimates and by differentiating S with respect to $\alpha_1, \alpha_2, \dots, \alpha_p$ and equating the result to zero, we get a set of k equations

$$R\hat{\alpha} = r \quad \dots (40)$$

where R is a matrix of autocorrelations given by

$$R = \begin{bmatrix} 1 & r_1 & \dots & r_{p-1} \\ r_1 & 1 & \dots & r_{p-2} \\ \dots & \dots & \dots & \dots \\ r_{p-1} & r_{p-2} & \dots & 1 \end{bmatrix} \quad \dots (41)$$

and $r' = (r_1, r_2, \dots, r_p)$ is the row matrix corresponding to the column matrix r .

Thus, $\hat{\alpha}$ is obtained by solving the simultaneous equations (40) using inverse of R matrix denoted by R^{-1} as

$$\hat{\alpha} = R^{-1} r \quad \dots (42)$$

16.2.4 Determining the Order of an Autoregressive Model

For fitting the model, we have to estimate the order of the autoregressive model for the data at hand. For the first order autoregressive model, the autocorrelation function (acf) reduces exponentially as follows:

$$\rho_k = \alpha_1^k \quad \text{as } |\alpha_1| < 1.$$

Hence, for an autoregressive process AR (1), the exponential reduction of autocorrelation function (acf) gives a good indication that the autoregressive process is of order 1. However, this is not true for correlogram of higher orders. For two and higher order autoregressive models, the autocorrelation function (acf) can be a combination of damped exponential or cyclical functions and may be difficult to identify.

One way is to start fitting the model by taking $p = 1$ and then $p = 2$, and so on. As soon as the contribution of the last α_p fitted is not significant, which can be judged from the reduction in the value of residual sum of squares, we should stop and take the order as $p - 1$. An alternative method is to calculate what is called **partial autocorrelation function**.

16.2.5 Partial Autocorrelation Function (pacf)

For an autoregressive AR (p) process, the partial autocorrelation function (pacf) is defined as the value of the last coefficient α_p . We start with $p = 1$ and calculate pacf. Hence, for the AR (1) process, pacf (1) is

$$\alpha_1 = \rho_1 \quad \dots (43a)$$

For AR (2), the pacf is given by

$$\alpha_2 = \frac{(\rho_2 - \rho_1^2)}{(1 - \rho_1^2)} \quad \dots (43b)$$

as described earlier. In this way, we can go on calculating pacf(3) as α_3 and $\alpha_p, p = 4, 5, \dots$. We can estimate these partial autocorrelation functions by

substituting estimated autocorrelations r_k in place of ρ and then test the significance. When partial autocorrelation function (pacf) is zero, its asymptotic standard error is $1/\sqrt{N}$. Hence, we calculate partial autocorrelation functions (pacf) by increasing the order by one every time. As soon as this lies within range of $\pm 2/\sqrt{N}$, we stop and take the order as the last significant partial autocorrelation function (pacf). This is indicated when pacf lies outside the range of $\pm 2/\sqrt{N}$. In the following steps, we give partial autocorrelation functions (pacf) up to autoregressive AR(3) process:

$$\text{pacf}(1) = \rho_1 = \alpha_1; \text{pacf}(2) = \frac{(r_2 - r_1^2)}{(1 - r_1^2)} = \alpha_2 \quad \dots (44a)$$

$$\text{and } \text{pacf}(3) = \frac{\begin{vmatrix} 1 & r_1 & r_2 \\ r_1 & 1 & r_1 \\ r_2 & r_1 & r_3 \end{vmatrix}}{\begin{vmatrix} 1 & r_1 & r_2 \\ r_1 & 1 & r_1 \\ r_2 & r_1 & 1 \end{vmatrix}} \quad \dots (44b)$$

where $|\dots|$ means the determinant of the matrix.

Let us now calculate partial autocorrelation functions for stationary processes.

Example 4: Find the pacf of the AR(2) process:

$$X_t = 0.333X_{t-1} + 0.222X_{t-2} + a_t$$

Solution: For this process, $\alpha_1 = 0.333$ and $\alpha_2 = 0.222$. We use the expressions of ρ_1 and ρ_2 as given in equation (36a) and get

$$\rho_1 = \frac{\alpha_1}{(1 - \alpha_2)} = \frac{0.333}{0.778} = 0.428$$

$$\rho_2 = \alpha_2 + \frac{\alpha_1^2}{(1 - \alpha_2)} = 0.222 + \frac{0.111}{0.778} = 0.365$$

Now, from equations (43a and b),

$$\text{pacf}(1) = \alpha_1 = \rho_1 = 0.428$$

$$\text{and } \text{pacf}(2) = \frac{r_2 - r_1^2}{1 - r_1^2} = \frac{(0.365 - 0.183)}{(1 - 0.183)} = 0.222$$

Also, $\text{pacf}(k) = 0$, for $k \geq 3$.

Example 5: Suppose for a time series of length $N = 100$, the three autocorrelation coefficients are $r_1 = 0.806$, $r_2 = 0.428$, $r_3 = 0.070$. Calculate the pacfs and estimate the order of autoregressive model to be fitted.

Solution: Equating r_k to ρ_k ($k = 1, 2, 3$), from equations (44a and b), we have

$$\text{pacf}(1) = r_1 = 0.806$$

$$\text{pacf}(2) = \frac{(r_2 - r_1^2)}{(1 - r_1^2)} = \frac{(0.428 - 0.650)}{0.350} = -0.634$$

$$\text{pacf}(3) = \frac{\begin{vmatrix} 1 & 0.806 & 0.428 \\ 0.806 & 1 & 0.806 \\ 0.428 & 0.806 & 0.070 \end{vmatrix}}{\begin{vmatrix} 1 & 0.806 & 0.428 \\ 0.806 & 1 & 0.806 \\ 0.428 & 0.806 & 1 \end{vmatrix}} = 0.077$$

and $\text{range} = \pm 2/\sqrt{N} = \pm 2/10 = \pm 0.2$

The partial autocorrelation functions $\text{pacf}(1)$ and $\text{pacf}(2)$ lie outside this range and $\text{pacf}(3)$ lies inside this range. Since the least significant pacf is $\text{pacf}(2)$, the order of the model is 2 and the autoregressive model $\text{AR}(2)$ is suggested for this process.

You may now like to solve the following exercises to check your understanding about MA processes.

E4) For the $\text{AR}(2)$ process

$$X_t = 1.0 X_{t-1} - 0.5 X_{t-2} + a_t$$

calculate ρ_1 and ρ_2 . State whether the model is stationary. Also calculate $\text{pacf}(1)$ and $\text{pacf}(2)$.

E5) For the model

$$X_t = 1.5 X_{t-1} - 0.6 X_{t-2} + a_t$$

obtain ρ_1 and ρ_2 . Is the process stationary?

E6) Find the autocorrelation function (acf) of the process

$$X_t = X_{t-1} - 0.25 X_{t-2} + a_t$$

and obtain ρ_1 and ρ_2 .

E7) The following table gives the number of workers trained during 1980-2010.

(t)	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
y_t	4737	5117	5091	3468	4320	3825	3673	3694	3708	3333
(t)	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
y_t	3367	3614	3362	3655	3963	4405	4595	5045	5700	5716
(t)	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
y_t	5138	5010	5353	6074	5031	5648	5506	4230	4827	3885

Some autocorrelations are given below:

$$r_1 = 0.732, r_2 = 0.661, r_3 = 0.557, r_4 = 0.385, r_5 = 0.272, r_6 = 0.119, r_7 = 0.019, r_8 = -0.139, r_9 = -0.268, r_{10} = -0.375, \bar{y} = 4503.00 \text{ and } \sigma_y = 836.74$$

- Draw the time plot.
- Plot the correlogram.
- Calculate $\text{pacf}(1)$ and $\text{pacf}(2)$ and test their significance.
- Which one of the models, $\text{AR}(1)$ or $\text{AR}(2)$, will be more suitable for this data?
- Fit the suitable model.

16.3 AUTOREGRESSIVE MOVING AVERAGE (ARMA) MODELS

A finite order moving average process can be written as an infinite order autoregressive process. Similarly, a finite order autoregressive process can be written as an infinite order moving average process. We would like to fit a model, which has the least number of parameters. This property is called **parsimony** (most economical). Hence, a combination of autoregressive (AR) and moving average (MA) models may turn out to be the most parsimonious. We represent a combination of AR(p) and MA(q) model as ARMA(p, q) and write

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \dots + \alpha_p X_{t-p} + a_t + \beta_1 a_{t-1} + \beta_2 a_{t-2} + \dots + \beta_q a_{t-q} \quad \dots (45)$$

Using the backward shift operator B, we can write equation (45) as

$$\Phi(B) X_t = \theta(B) a_t \quad \dots (46)$$

where

$$\Phi(B) = 1 - \alpha_1 B - \alpha_2 B^2 - \dots - \alpha_p B^p \quad (\text{AR}) \quad \dots (47a)$$

$$\theta(B) = 1 + \beta_1 B + \beta_2 B^2 + \dots + \beta_q B^q \quad (\text{MA}) \quad \dots (47b)$$

The conditions of stationarity and invertibility are the same as for autoregressive (AR) and moving average (MA) processes, respectively, i.e., the roots of

$$\Phi(B) = 0 \text{ and } \theta(B) = 0 \quad \dots (47c)$$

must lie outside the unit circle. So the modulus of roots of B must be greater than one.

An ARMA (1, 1) model can be written as

$$X_t = \alpha X_{t-1} + a_t + \beta a_{t-1} \quad \dots (48a)$$

which can be written using backward operator B as

$$(1 - \alpha B) X_t = (1 + \beta B) a_t \quad \dots (48b)$$

For a stationary and invertible ARMA (1, 1) process,

$$|\alpha| < 1, \quad |\beta| < 1$$

On multiplying equation (48a) by X_t , X_{t-1} and X_{t-k} and taking expectations, we obtain

$$\gamma_0 = \frac{\sigma_a^2 (1 + \beta^2 + 2\alpha\beta)}{(1 - \alpha^2)} \quad \dots (49a)$$

$$\gamma_1 = \alpha\gamma_0 + \beta\sigma_a^2 \quad \dots (49b)$$

$$\gamma_k = \alpha\gamma_{k-1}, \quad k \geq 2 \quad \dots (49c)$$

We also obtain

$$\rho_1 = \frac{(1 + \alpha\beta)(\alpha + \beta)}{(1 + \beta^2 + 2\alpha\beta)} \quad \dots (49d)$$

$$\rho_k = \alpha\rho_{k-1}, \quad k \geq 2 \quad \dots (49e)$$

Thus, the autocorrelation function decays exponentially from the starting value ρ_1 , which depends on α and β .

Let us take up an example of the ARMA model.

Example 6: Write the following ARMA (1,1) model

$$X_t = 0.5 X_{t-1} + a_t - 0.3 a_{t-1}$$

using backward operator B. Is the process stationary and invertible? Calculate ρ_1 and ρ_2 for the process.

Solution: Since $\alpha = 0.5$ and $\beta = -0.3$, from equation (48b), the model is written using backward operator B as:

$$(1 - 0.5B)X_t = (1 - 0.3B)a_t$$

In this case, from equations (47a and b) we have

$$\Phi(B) = 1 - 0.5B \text{ and } \theta(B) = 1 - 0.3B$$

Therefore, for stationarity and invertibility, from equation (47c), the roots of $1 - 0.5B = 0$ and $1 - 0.3B = 0$ must lie outside the unit circle. The roots of these equations are:

$$B = 1/0.5 = 2.0 \quad \text{and} \quad B = 1/0.3 = 3.33$$

Since both roots lie outside the unit circle, the process is stationary and invertible. From equations (49 d and e),

$$\rho_1 = (1 + \alpha B)(\alpha + \beta) / (1 + \beta^2 + 2\alpha\beta) = 0.215, \text{ and}$$

$$\rho_2 = \alpha\rho_1 = 0.107$$

You may now like to try out an exercise.

E8) Show that the ARMA (1, 1) model

$$X_t - 0.5 X_{t-1} = a_t - 0.5 a_{t-1}$$

can be equivalently written as $X_t = a_t$, which is a white noise model.

16.4 AUTOREGRESSIVE INTEGRATED MOVING AVERAGE (ARIMA) MODELS

In Units 13 and 14, we have discussed that the actual time series often contains trend and seasonal components. In that sense, most of the time series we come across are non-stationary as their mean changes with time. In these units, we have tried to take moving average to remove seasonal component and then we have estimated trend. In this section, we incorporate trend and seasonal effects in the model and then by making suitable operations on the series, transform them to stationary series. Then we apply the methods of stationary models discussed so far.

If a time series is non-stationary because of changes in mean, we can take the difference of successive observations. The modified series is more likely to be stationary. Sometimes more than one difference of successive observations is required to get a modified stationary model. Such a model is called an **integrated** model because the stationary model that is fitted to the modified series has to be summed or integrated to provide a model for the original non-stationary series. The first difference of series X_t is defined as W_t :

$$W_t = \nabla X_t = (1 - B) X_t = X_t - X_{t-1} \quad \dots (50)$$

where ∇ is the **difference operator**. This is called the difference of order 1. We may define a modified series of order d as

$$W_t = \nabla^d X_t = (1-B)^d X_t \quad \dots (51)$$

where d takes values 1, 2,

For d=2, this operation takes differences twice:

$$\begin{aligned} W_t &= \nabla^2 X_t = (1-B)(1-B) X_t = (1-B)(X_t - X_{t-1}) \\ &= X_t - X_{t-1} - (X_{t-1} - X_{t-2}) = X_t - 2X_{t-1} + X_{t-2} \quad \dots (52) \end{aligned}$$

In general, the ARIMA model can be written as:

$$W_t = \alpha_1 W_{t-1} + \alpha_2 W_{t-2} + \dots + \alpha_p W_{t-p} + a_t + \beta_1 a_{t-1} + \beta_2 a_{t-2} + \dots + \beta_q a_{t-q} \quad \dots (53a)$$

or using backward operator B, it can be written as:

$$\Phi(B)W_t = \theta(B)a_t \quad \dots (53b)$$

$$\text{or} \quad \Phi(B)(1-B)^d X_t = \theta(B)a_t \quad \dots (53c)$$

It is denoted by ARIMA (p, d, q). The operator $\Phi(B)(1-B)^d$ has d roots of B equal to 1. For d = 0, the series is an ARMA process. In practice, the first or second difference make the process stationary. A random walk model is an example of the ARIMA model.

Consider the time series

$$X_t = X_{t-1} + a_t \quad \dots (54a)$$

which can be written as

$$(1-B)X_t = a_t \quad \dots (54b)$$

It is clearly non-stationary as one root of

$$\Phi(B) = 1-B = 0 \quad \dots (54c)$$

lies on the unit circle. To make it stationary, we take one difference of X_t , as

$$W_t = X_t - X_{t-1} = a_t$$

So the time series can be written as ARIMA (0,1,0). W_t is a white noise process and stationary.

A plot of the first difference looks like a plot of a stationary process without any trend. The plot of autocorrelations and partial autocorrelations provide the idea of the process.

Example 7: For the model

$$(1-0.2B)(1-B)X_t = (1-0.5B)a_t$$

find p, d, q and express it as ARIMA (p, d, q). Determine whether the process is stationary and invertible.

Solution: We are given the model

$$(1-0.2B)(1-B)X_t = (1-0.5B)a_t$$

a) In this case, from equations (53 b and c), we can write the given model as

$$(1-0.2B)(1-B)^1 X_t = (1-0.5B)a_t$$

which implies that $W_t = (1-B) X_t$, i.e., d = 1 and from equation (53a)

$$X_t - 0.2X_{t-1} = a_t - 0.5a_{t-1}$$

This implies that $p = 1$ and $q = 1$. Hence, the process is ARIMA (1,1,1)

$$b) \quad F(B) = (1 - B)(1 - 0.2B) = 0 \Rightarrow B = 1 \text{ and } B = 5 \text{ and}$$

$$\theta(B) = (1 - 0.5B) = 0 \Rightarrow B = 1/0.5 = 2.0$$

One of the roots of $\Phi(B) = (1 - B)(1 - 0.2B) = 0$ is 1. Hence, the process is non-stationary. However, the root of $\theta(B) = 0$ lies outside the unit circle.

Hence, it is invertible. For the first difference $W_t = (1 - B)X_t$, the process is stationary and invertible.

You may now like to try some more exercises for practice.

E9) Consider the time series

$$X_t = \beta_1 + \beta_2 t + a_t$$

where β_1 and β_2 are known constants and a_t is a white noise with variance σ^2 .

Determine whether X_t is stationary. If X_t is not stationary, find a transformation that produces a stationary process.

E10) Suppose that the correlogram of a time series consisting of 100 observations has

$$r_1=0.31, r_2=0.37, r_3=-0.05, r_4=0.06, r_5=-0.21, r_6=0.11, r_7=0.08, \\ r_8=0.05, r_9=0.12, r_{10}=-0.01$$

Suggest an ARIMA model which may be appropriate for this case.

Let us now summarise the concepts that we have discussed in this unit.

16.5 SUMMARY

1. The sequences of random variables $\{Y_i\}$ are mutually independent and identically distributed. If a discrete stationary process consists of such sequences of i.i.d. variables, it is called a **purely random process**. Sometimes it is called white noise.
2. The moving average processes are used successfully to model stationary time series in econometrics. The MA(q) process of order q is given as

$$X_t = \beta_0 a_t + \beta_1 a_{t-1} + \dots + \beta_q a_{t-q}$$

where β_i , ($i = 0, 1, 2, \dots, q$) are constant.

3. The autocorrelation function (acf) of the MA (q) process is given by

$$\rho_k = \frac{(\beta_k + \beta_1 \beta_{k+1} + \dots + \beta_{q-k} \beta_q)}{\left(1 + \sum_{i=1}^q \beta_i^2\right)}, \quad k = 1, 2, \dots, q$$

It becomes zero if lag k is greater than the order of the process, i.e., q. This is a very important feature of moving average (MA) processes.

4. A linear stationary process can always be expressed as an **autoregressive process** of suitable order. Unlike moving average (MA) process, which puts no restrictions on parameters for stationarity, autoregressive (AR) process requires certain restrictions on the parameter α for stationarity.

5. We can estimate the partial autocorrelation functions by substituting estimated autocorrelations r_k in place of ρ . Then we test the significance. When the partial autocorrelation function (pacf) is zero, its asymptotic standard error is $1/\sqrt{N}$. Hence, we calculate the partial autocorrelation functions (pacf) by increasing the order one at a time. As soon as the order lies within range of $\pm 2/\sqrt{N}$, we stop and take the order as the last significant partial autocorrelation function (pacf).
6. A finite order moving average process can be written as an infinite order autoregressive process. Similarly, a finite order autoregressive process can be written as an infinite order moving average process.
7. If a time series is non-stationary because of changes in mean, we can take the difference of successive observations. The modified series is more likely to be stationary. Sometimes more than one difference is required. Such a modified model is called an **integrated** model because the stationary model that is fitted to the modified series has to be summed or integrated to provide a model for the original non-stationary time series.

16.6 SOLUTIONS/ANSWERS

E1) From equation (8), we are given that

$$q=2, \beta_0=1, \beta_1 = 0.60 \text{ and } \beta_2 = -0.3$$

for the model.

Using equation (12), for $q=2$, we get

$$\rho_1 = \frac{(\beta_1 + \beta_1\beta_2)}{(1 + \beta_1^2 + \beta_2^2)} = \frac{0.6 + (0.6 \times -0.3)}{1 + (0.6)^2 + (-0.3)^2} = 0.29$$

$$\rho_2 = \frac{2\beta_2}{(1 + \beta_1^2 + \beta_2^2)} = \frac{2 \times -0.3}{1 + (0.6)^2 + (-0.3)^2} = -0.4138$$

$$\rho_k = 0, \quad k \geq 3$$

E2) For the series, $\alpha_1=1, \alpha_2 = -0.5$.

a) From equation (33), the stationarity conditions are

$$\alpha_2 + \alpha_1 < 1, \alpha_2 - \alpha_1 < 1, -1 < \alpha_2 < 1$$

$$\Rightarrow \alpha_2 + \alpha_1 = 0.5, \alpha_2 - \alpha_1 = -1.5, -1 < \alpha_2 < 1$$

All three conditions of stationarity are satisfied in this case. Hence, the process is stationary. From equation (36a), we get

$$\rho_2 = \frac{\alpha_1}{(1 - \alpha_2)} = \frac{1}{1.5} = 0.667, \quad \rho_2 = \alpha_2 + \frac{\alpha_1^2}{(1 - \alpha_2)} = 0.167$$

We calculate the other values of ρ_k from equation (37) with $\alpha_1=1, \alpha_2 = -0.5$. Thus,

$$\rho_k = \rho_{k-1} - 0.5\rho_{k-2}$$

$$\Rightarrow \rho_3 = \rho_2 - 0.5\rho_1 = -0.166,$$

$$\rho_4 = \rho_3 - 0.5\rho_2 = -0.250, \text{ and}$$

$$\rho_5 = \rho_4 - 0.5\rho_3 = -0.166$$

b) The correlogram for the process is shown in Fig. 16.2.

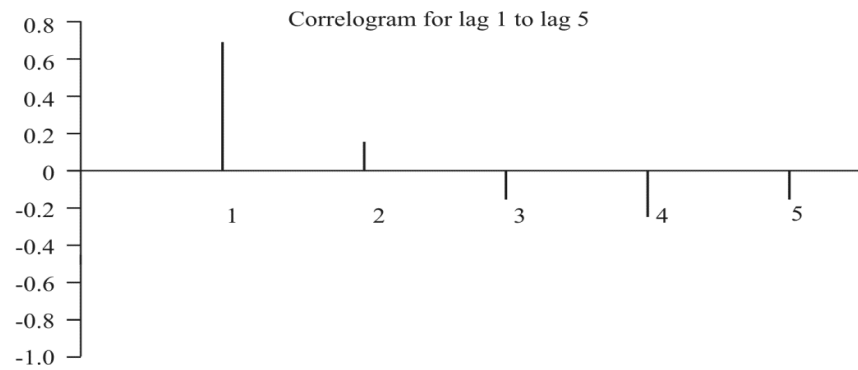


Fig. 16.2: Correlogram for AR(2) Model.

- E3)** a) Here $\alpha_1 = 0.3$. Therefore, using equation (23), we can write the model $X_t = 0.3X_{t-1} + a_t$ in B notation as

$$(1 - 0.3B)X_t = a_t \Rightarrow \phi(B) = (1 - 0.3B)$$

From $\phi(B) = 0$, we get $1 - 0.3B = 0$ or $B = 1/0.3 = 3.333$

which lies outside the unit circle. Hence, the process is stationary.

- c) Here $\alpha_1 = \alpha_2 = 1/12$. Using equation (23), we can write the model $X_t = (X_{t-1} + X_{t-2})/12 + a_t$ in B notation as

$$\{1 - (B + B^2)/12\}X_t = a_t \Rightarrow \phi(B) = \frac{1 - B - B^2}{12}$$

The two roots of $\phi(B) = 1 - (B + B^2)/12 = 0$ are given by

$$B = 3, -4,$$

and for both, $|B| > 1$. Hence, the process is stationary.

- E4)** For the AR (2) process

$$X_t = 1.0X_{t-1} + 0.5X_{t-2} + a_t,$$

we have

$$\alpha_1 = 1.0, \alpha_2 = -0.5$$

Therefore, from equation (36a),

$$\rho_1 = \alpha_1 / (1 - \alpha_2) = 1/1.5 = 0.667$$

$$\text{and } \rho_2 = \alpha_2 + \alpha_1^2 / (1 - \alpha_2) = -0.5 + 1/1.5 = 0.167$$

Since $\alpha_2 + \alpha_1 = 0.5 < 1$, using equation (33), we can say that the process is stationary. From equations (43a and 44a), we have

$$\text{pacf}(1) = \alpha_1 = 1.0, \text{ pacf}(2) = \alpha_2 = -0.5$$

- E5)** For the model $X_t = 1.5X_{t-1} - 0.6X_{t-2} + a_t$

we have $\alpha_1 = 1.5$ and $\alpha_2 = -0.6$.

Multiplying the model by X_{t-1} and X_{t-2} , taking expectation and dividing by σ_a^2 , we get

$$\rho_1 = 1.5 - 0.6\rho_1 \quad \text{and} \quad \rho_2 = 1.5\rho_1 - 0.6$$

Solving the above equations for ρ_1 and ρ_2 , we get

$$\rho_1 = 1.5/1.6 = 0.937 \text{ and } \rho_2 = 0.805$$

E6) We are given the model $X_t = X_{t-1} - 0.25X_{t-2} + a_t$

and we have $\alpha_1 = 1.0$ and $\alpha_2 = -0.25$.

Multiplying by X_{t-k} , taking expectations and dividing by σ_a^2 , we get the autocorrelation function as:

$$\rho_k = \rho_{k-1} - 0.25\rho_{k-2}$$

On putting the values $\alpha_1 = 1$, $\alpha_2 = -0.25$ in equation (36a), we obtain

$$\rho_1 = \alpha_1 / (1 - \alpha_2) = 1 / 1.25 = 0.8, \quad \rho_2 = \alpha_2 + \alpha_1^2 / (1 - \alpha_2) = 0.55$$

E7) We first draw a time plot for the given time series:

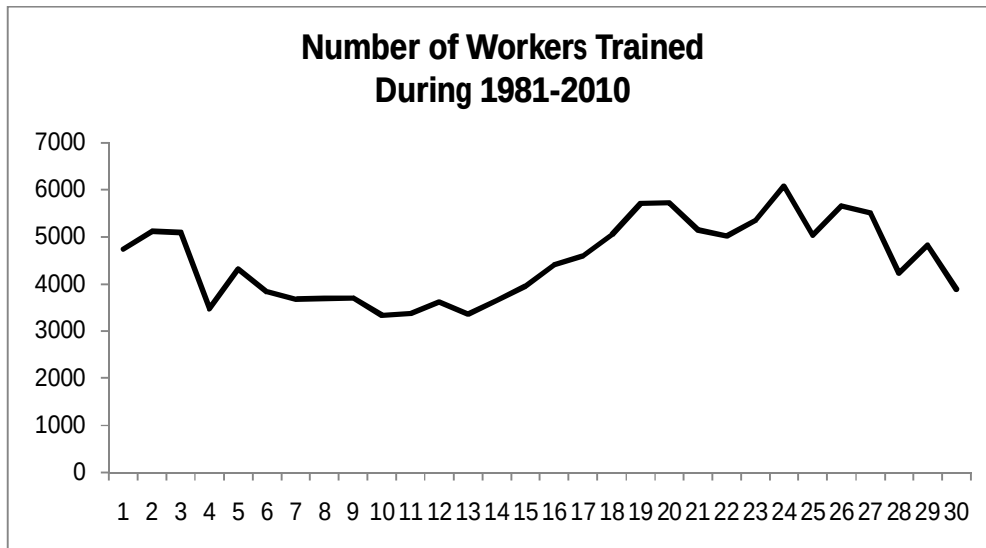


Fig. 16.3: Time plot of the series of number of workers trained during 1981 to 2010.

We are given the following values:

$$r_1 = 0.732, r_2 = 0.661, r_3 = 0.557, r_4 = 0.385, r_5 = 0.272, r_6 = 0.119,$$

$$r_7 = 0.019, r_8 = -0.139, r_9 = -0.268, r_{10} = -0.375, \bar{y} = 4503.00 \text{ and } \sigma_y = 836.74$$

The correlogram for the process is shown in Fig. 16.4.

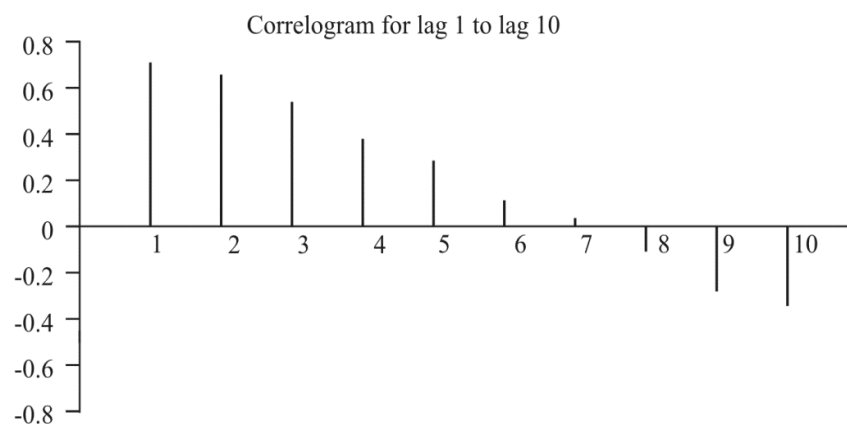


Fig. 16.4: Correlogram for the given series.

Using equation (44a), we get

$$\text{pacf}(1) = r_1 = 0.732,$$

$$\text{and } \text{pacf}(2) = (r_2 - r_1^2) / (1 - r_1^2) = 0.125 / 0.464 = 0.269$$

$$\text{and } \text{range} = \pm 2/\sqrt{N} = \pm 2/\sqrt{30} = \pm 0.365$$

Since p.a.c.f. (1) lies outside the range, while pacf (2) lies within the range, AR (1) will be suitable for this time series.

On putting the value of $\alpha'_1 = r_1 = 0.732$, we get the fitted model as

$$X_t = 0.7332 X_{t-1} + a_t$$

$$\text{or } (Y_t - \bar{Y}) = 0.732 (Y_{t-1} - \bar{Y}) + a_t$$

$$\text{or } (Y_t - 4503.0) = 0.732 (Y_{t-1} - 4503.0) + a_t$$

E8) We are given the ARMA (1,1) model

$$X_t - 0.5 X_{t-1} = a_t - 0.5 a_{t-1}$$

Now we start in reverse order and take

$$X_t = a_t \quad \dots (i)$$

Next we take

$$X_{t-1} = a_{t-1} \quad \dots (ii)$$

We multiply equation (ii) by 0.5. Then we have

$$0.5 X_{t-1} = 0.5 a_{t-1} \quad \dots (iii)$$

On subtracting (iii) from (i), we get the given model as

$$X_t - 0.5 X_{t-1} = a_t - 0.5 a_{t-1}$$

Another way is to write the given model in B form as:

$$(1 - 0.5B) X_t = (1 - 0.5B) a_t$$

$$\text{or } X_t = (1 - 0.5B)^{-1}(1 - 0.5B)a_t \Rightarrow X_t = a_t$$

E9) We are given the time series

$$X_t = \beta_1 + \beta_2 t + a_t \quad \dots (i)$$

Taking expectation of equation (i), we get

$$E(X_t) = \beta_1 + \beta_2 t \quad \dots (ii)$$

Since equation (ii) depends on t and it changes with time, X_t is not a stationary process. Now we consider the first difference series as

$$\begin{aligned} Y_t = \nabla X_t &= (1 - B)X_t = X_t - X_{t-1} \\ &= \beta_1 + \beta_2 t + a_t - \beta_1 - \beta_2 (t-1) - a_{t-1} \\ &= \beta_2 + a_t - a_{t-1} \end{aligned}$$

Now the modified series Y_t has a constant mean β_2 and is a stationary MA(1) process.

E10) We are given a time series consisting of 100 observations, which has

$$r_1=0.31, r_2=0.37, r_3=-0.05, r_4=0.06, r_5=-0.21, r_6=0.11, r_7=0.08, r_8=0.05, r_9=0.12, r_{10}=-0.01$$

The range,

$$\pm 2/\sqrt{N} = \pm 2/\sqrt{100} = \pm 2/10 = \pm 0.20$$

We can see that only r_1 and r_2 are significantly different from zero, and r_5 is marginally significant, which can be ignored as the series is too small ($N = 100$). In this case only r_1 and r_2 are significant and an MA(2) model is suggested. The correlogram for the process is shown in Fig. 16.5.

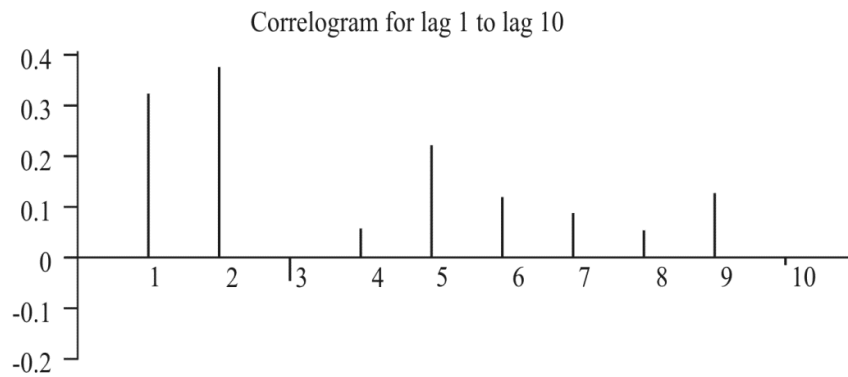


Fig. 16.5: Correlogram of the given time series.